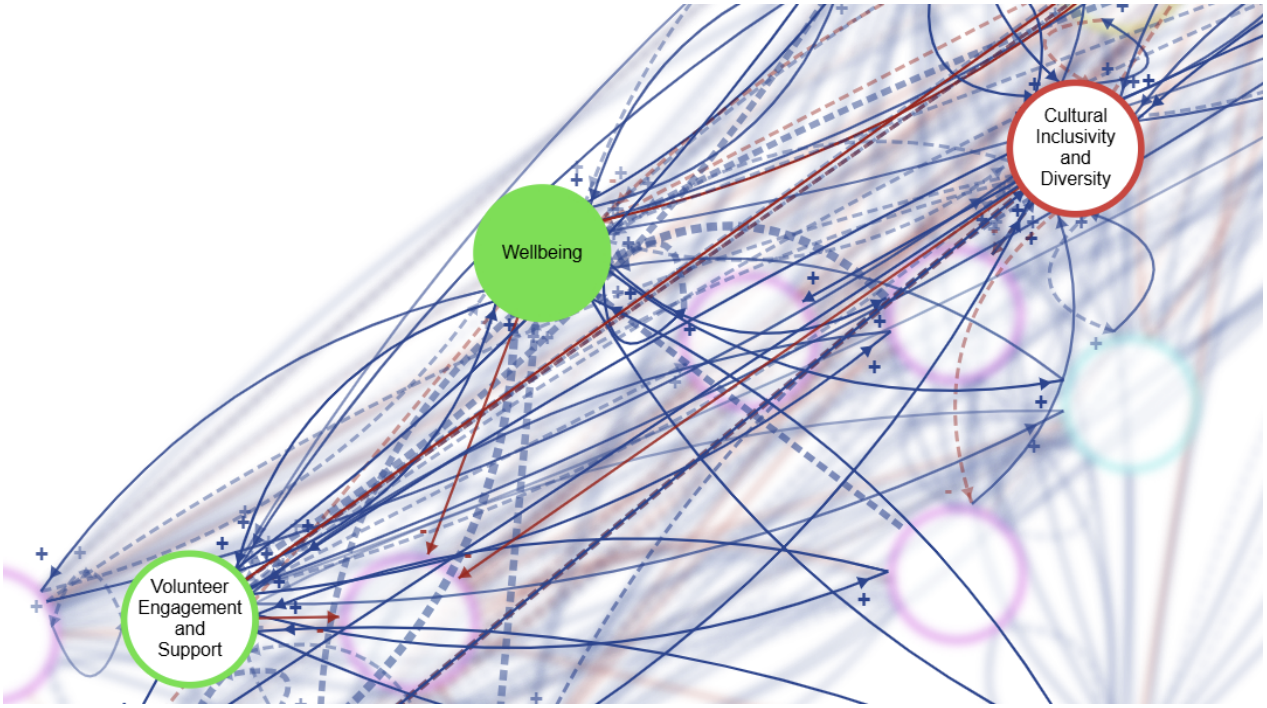
University of Amsterdam

Masters Thesis

Detecting and Mitigating Hallucinations in LLM-Generated Causal Loop Diagrams



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Computational Science

Declaration of Authorship

I, Nitai Nijholt, declare that this thesis, entitled 'Detecting and Mitigating Hallucinations in LLM-Generated Causal Loop Diagrams' and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at the University of Amsterdam.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

The research for this thesis was conducted within a software environment which is property of Causalix, actively using parts of the code and intellectual property developed by others, therefore, a clear definition of scope (what was developed as part of the thesis, what was not) is in order. The main CLD generator algorithm was conceived by Rick Quax with feedback from Cillian Hourican and implemented by Andrew Crossley. Contributions from the author of this thesis (Nitai Nijholt) on the generation algorithm limited to adding a mediation check and conceiving a number of these prompts, in addition to other prompts being contributed by Rick, Andrew and Cillian.

The gathering and transcribing of validation datasets into computational format of the CLD's Depressive Disorder, Social Norms, and Emergency Response datasets—was done by Cillian Hourican.

The underlying software environment contains substantially more features and potential analysis outputs than are required for the research presented in this thesis; accordingly, only a subset of the available functionality is used. Where any algorithm or script executed within this environment produced additional intermediate artefacts or auxiliary outputs, only the artefacts explicitly referenced in the thesis were used in the reported analyses and to support the thesis conclusions.

The contributions presented in this thesis, developed by the author, encompass: the LLM-as-a-Judge framework, the LLM-as-a-Corrector, the Deep Research multi-agent system uncertainty quantification metrics for the generator component, and the simulation framework. Additionally, the author developed the comprehensive evaluation frameworks for the Generator, Deep Research, and Judge components, the generate-then-search methodology, as well as the parallelization architecture, inference time benchmarking, and cost tracking instrumentation. Any other part of the thesis is developed by Nitai Nijholt unless otherwise stated.

Signed:

Date: 3 January 2026

"All models are wrong, but some are useful."

George E. P. Box

UNIVERSITY OF AMSTERDAM

Abstract

Faculty of Science
Informatics Institute

Master of Science in Computational Science

Detecting and Mitigating Hallucinations in LLM-Generated Causal Loop Diagrams

by Nitai Nijholt

Causal loop diagrams (CLDs) can map cause and effect in complex systems but are costly for experts to produce through usual tools such as qualitative interview, literature study and group model building, as edge deliberation cost scales with number of variables squared. LLMs offer a solution by generating initial CLDs, reducing costs, but suffer from hallucinations. LLM-powered text-to-CLD extraction provides grounding but is constrained by source text availability. The alternative, LLM-based CLD generation, loses grounding, increasing the risk of hallucinations.

This thesis tackles this problem by applying established hallucination mitigation methods at test time—through LLM-as-a-judge, LLM-as-a-corrector, and UQ metrics (cosine similarity and logprob-derived measures)—previously untested in CLD generation. We ask: (1) Can LLM-as-a-judge and LLM-as-a-corrector improve CLD F1 versus literature ground truth? (2) Can UQ metrics detect hallucinations? (3) Can adaptive test-time compute, guided by UQ metrics, improve the cost–accuracy trade-off? Findings show, in our dataset of three health CLDs, LLM-based CLD generation performance exceeds a random baseline. In a controlled synthetic hallucination setting (GT Synth), Correctness Judge performance is high, but it drops in the literature ground-truth setting (GT Lit), with only our causal mechanistic prompt beating random baselines. The Citation Judge, which retrieves supporting articles post hoc, does not reliably exceed chance-level discrimination across domains despite substantial agreement with a human rater (80.6%; $\kappa = 0.618$), and high performance on causally unambiguous physics CLDs, suggesting weak discrimination is driven primarily by evidence type and retrieval quality rather than judge malfunction. LLM-as-a-corrector improves judge scores and slightly improves CLD quality vs GT Synth, but not vs GT Lit. UQ metrics (perplexity, min probability, max window entropy, cosine similarity) do not effectively flag hallucinations independently but do as ensemble input to supervised classifiers in-distribution; however, fail to generalise out-of-CLD. As UQ hallucination flagging and Corrector were ineffective, we pilot a multi-agent deep research (DR) system that finds causal evidence for 62.4% of false positives and 42.9% of false negatives. Preliminary human validation suggests 66% causal evidence applicability; however, limited inter-rater agreement (n=13) means extrapolated F1 gains ($0.267 \rightarrow 0.472$) should be interpreted as exploratory. Using a mechanistic Correctness Judge to flag hallucinations for adaptive deployment of DR yields an estimated 15% accuracy-per-dollar increase. Results suggest that LLM-based CLD generation and multi-agent validation-driven research can assist experts in CLD creation; future work includes validating the system across additional CLD domains and conducting component-wise ablation studies of DR to find more efficient compute–accuracy trade-offs.

Keywords: causal loop diagrams, large language models, hallucination detection, LLM-as-a-Judge, LLM-as-a-Corrector, uncertainty quantification, log probability, test-time compute, multi-agent literature search, causal graph generation, health systems modeling

Acknowledgements

I dedicate this work to my mother, Rati Manjari van Sijll, who always stood behind me and gave me a start in life that I remain thankful for every day. Also to my late father, who started and nurtured my love for computers and engineering, and to my grandmother, who through her stroke was thankful for the care I provided her, while being understanding when this thesis required more of my attention to finish.

I also want to thank my supervisor, Dr. Rick Quax, who provided me with the resources and guidance to conduct this research, and taught me many key principles of modelling, simulation, and scientific data analysis which proved invaluable. I also want to thank my examiner, Dr. Roy, who provided clear, actionable feedback which helped in improving the work. I also want to thank all involved including my second assessor Dr. van Lissa, for their understanding of the thesis timeline revision as a result of caring for my grandmother and their accommodation in its assessment.

Lastly, I want to thank Andrew Crossley especially, for building the software framework within which this research was conducted, Cillian Hourican for his advice and help in providing the ground truth datasets, and Mick Scheide (Data Manager, Antoni van Leeuwenhoek) for providing human validation for RQ1.

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Abbreviations

CLD	**C**ausal **L**oop **D**iagram
LLM	**L**arge **L**anguage **M**odel
Judge	**LLM**-as-a-**J**udge
Corrector	**LLM**-as-a-**C**orrector
UQ	**U**ncertainty **Q**uantification
RAG	**R**etrieval-**A**ugmented **G**eneration
DR	**D**eep **R**esearch
GT Lit	**G**round **T**ruth from **Lit**erature
GT Synth	**G**round **T**ruth **Synth**etic
FP	**F**alse **P**ositive
FN	**F**alse **N**egative
TP	**T**rue **P**ositive
CoT	**C**hain-**o**f-**T**hought
BHC	**B**radford **H**ill **C**riteria
GMB	**G**roup **M**odel **B**uilding
LUQ	**L**ogprob **U**ncertainty **Q**uantification
AUC	**A**rea **U**nder the **C**urve
ROC	**R**eceiver **O**perating **C**haracteristic
CI	**C**onfidence **I**nterval
ANOVA	**An**alysis **o**f **Va**riance
API	**A**pplication **P**rogramming **I**nterface
RFE	**R**ecursive **F**eature **E**limination
NLP	**N**atural **L**anguage **P**rocessing

Chapter 1

Introduction

Causal Loop Diagrams (CLDs) are qualitative representations of hypothesized causal structure—directed graphs where nodes represent variables and signed edges represent positive or negative causal influences—that serve as precursors to computational models, enabling intervention simulations and informing new theories and experiments. They make feedback structure explicit (reinforcing and balancing loops), support rapid hypothesis generation about plausible mechanisms, and provide a shared language for interdisciplinary synthesis [1, 2]. CLDs can also be translated into quantitative system dynamics models (e.g., stock–flow and differential-equation simulations), enabling simulation of trajectories over time, counterfactual intervention testing, and analysis of sensitivity and unintended consequences in complex feedback systems [1].

In practice, CLDs are typically produced through a labour-intensive, iterative workflow: practitioners combine literature review with expert interviews and facilitated Group Model Building (GMB) workshops to elicit variables, surface feedback loops, negotiate scope, and converge on a shared causal story that is "good enough" for the model's purpose [1, 2]. Over the last decades, system dynamicists have experimented with approaches to increase stakeholder involvement and learning in the model-building process [3]. Yet a substantial portion of the effort remains qualitative and deliberative: teams must decide not only *whether* a proposed influence exists, but whether it is meaningfully *direct* or better represented via intermediary concepts, and what evidence or rationale supports the claim [4, 5]. This makes CLD construction sensitive to practical constraints (time, coordination costs, and cognitive biases), even before any formal simulation model is built [3].

The information aggregation challenge. The rapid increase in research output brings opportunities for knowledge synthesis and subsequent new knowledge creation, but also significant challenges in efficiently aggregating this information. This aggregation step is crucial to presenting a complete perspective in which interactions between variables are accounted for—a key factor in understanding complex biological systems in the clinic, in research, and for constructing computational models. However, individual researchers cannot keep up with new research given the exponentially increasing volume of publications and the fundamental limits of knowledge accumulation feasible for a single person.

Assuming directed pairwise relationship discussion, CLD construction cost scales as $O(|V|^2)$ for a system with $|V|$ variables, making large-variable systems prohibitively expensive to create—especially through group model-building sessions where expert time is scarce and coordination is challenging (Figure 1.1). Figure 1.1 illustrates this theoretical expert review burden: candidate relationships scale as $O(|V|^2)$; baseline assumes 5 min/pair across 3 experts, while an LLM-assisted workflow assumes 50% valid generation requiring only light validation (1 min/pair) and the remaining 50% requiring full review, yielding the potential time savings indicated by the shaded area. This thesis focuses on health-domain CLDs—specifically models of depressive symptoms, social norms & obesity, and older persons' emergency department visits—where causal modelling directly informs clinical decision-making and intervention design.

To address this limitation and aggregate siloed knowledge, domain expert sessions can be used to exchange and compile specialized knowledge into a single causal loop diagram, which can then be converted into a computational model for use in research, diagnosis, and intervention simulation. However, since expert time is scarce, these sessions are difficult to organise. Expert time is too valuable to be spent reiterating the basics, yet establishing consensus there is often precisely what a large part of these modelling sessions is spent on, when it could be better utilised for adding specialized contributions to the computational model.

If an initial starting model could be provided that aligns with the consensus in scientific literature, experts could spend their limited time contributing specialized knowledge rather than reiterating established findings. This would lead to more comprehensive models built more efficiently.

LLMs as literature synthesis engines. This framing reveals a key analogy and a potential solution. In traditional GMB, domain experts synthesize their knowledge—drawn from years of reading scientific literature, clinical experience, and theoretical training—into a shared causal model. Large language models, trained on vast corpora of scientific text, perform an analogous synthesis: they encode statistical patterns over how causal relationships are described in the literature. When an LLM generates the claim "stress increases cortisol," it reflects not a discovery of causality but a reproduction of how scientists have written about this relationship across thousands of papers.

Given large language models trained on trillions of tokens, a portion of which consists of scientific literature—the primary source and record of most expert knowledge—LLMs have the advantage of effectively compressing orders of magnitude more information than a single expert can process in a lifetime. They also allow for the retrieval of relevant information in negligible time, as well as enabling synthesis across techniques applied in separate, often siloed, branches of science, thereby facilitating knowledge synthesis, which supports and accelerates new knowledge generation.

This analogy suggests a practical value proposition: if LLMs can reliably reproduce expert-hypothesized causal structures at inference time, they can provide a *literature-informed initial hypothesis* for GMB—one that arises from statistical patterns encoded during training on scientific text, and which can subsequently be refined and validated through automated literature search. Rather than replacing expert deliberation, LLM-generated CLDs could serve as initial drafts that experts refine, accelerating the costly elicitation process.

The hallucination problem. However, this paradigm shift is contingent on one critical requirement: the LLM must faithfully reproduce relevant information from the literature. LLMs are known to hallucinate—generating plausible-sounding but factually incorrect content—a challenge that persists despite increases in pretraining compute and data scale [6, 7]. If the starting CLD produces too many spurious connections instead of valid representations of literature findings, the generated CLD will not serve as a useful starting point; it may actively mislead expert deliberation.

Current literature on LLM-based CLD construction focuses primarily on *causal translation*—extracting causal relationships from provided text—which can reduce hallucination through Retrieval-Augmented Generation (RAG). However, this approach assumes availability of a relevant corpus and is fundamentally capped by retrieval quality. An alternative paradigm is *LLM-based CLD generation*, where the model synthesizes causal relationships from its parametric knowledge without requiring an external corpus. This approach is relatively inexpensive and corpus-independent, but suffers from hallucinations that must be detected and mitigated.

Corpus-independent hallucination reduction methods—such as multi-agent LLM-as-a-Judge configurations and low-cost Logprob Uncertainty Quantification (LUQ)—have shown promise for reducing hallucinations in general LLM applications, but remain untested in the CLD generation context. The critical question—which this thesis addresses—is whether we can detect when LLMs diverge from expert consensus (hallucinate) and whether such divergences can be corrected or validated.

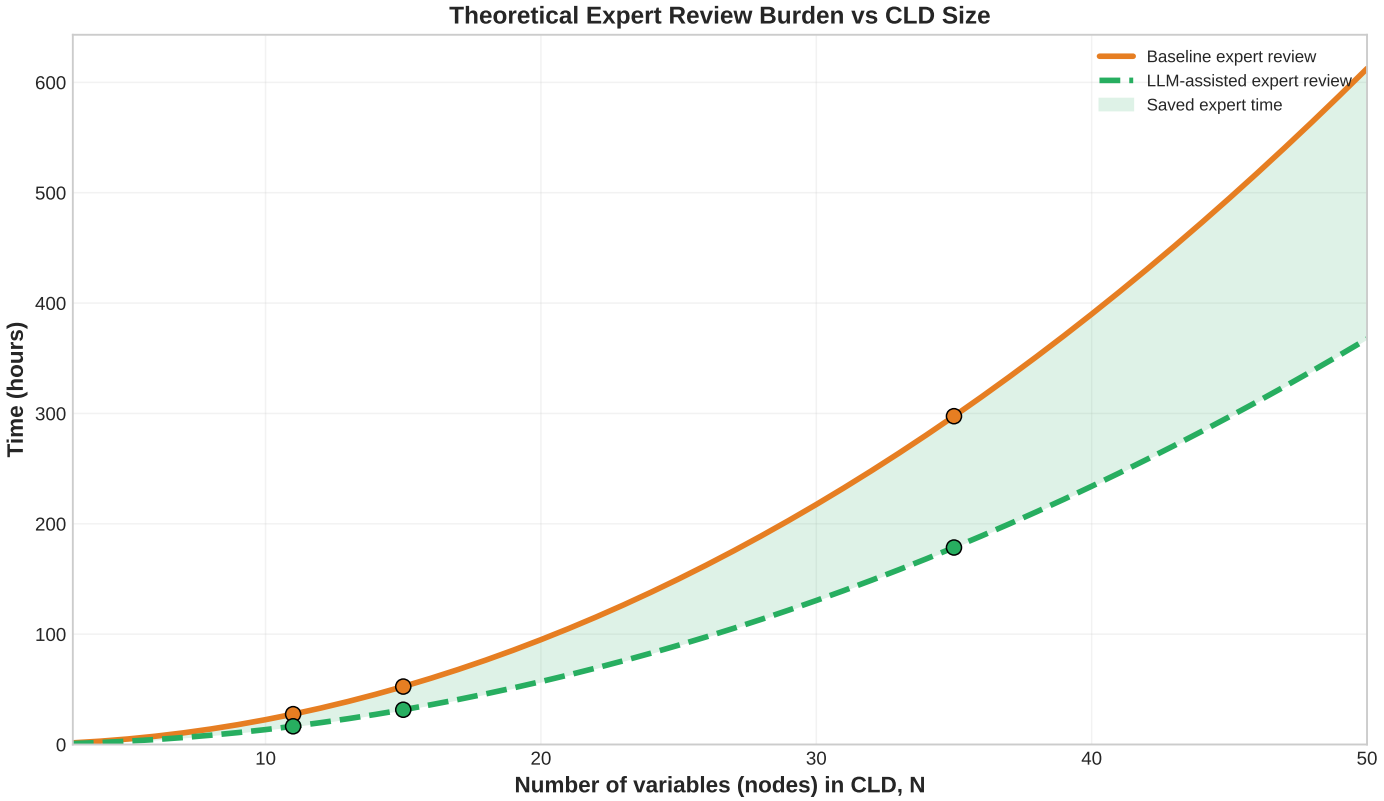


Figure 1.1: Theoretical expert review burden vs CLD size. Candidate relationships scale as $O(|V|^2)$. Baseline assumes 5 min/pair across 3 experts. LLM-assisted assumes 50% valid generation requiring only light validation (1 min/pair); remaining 50% require full review. Shaded area shows potential time savings.

Operationalizing hallucinations and ground truth. Hallucinations in LLMs manifest as confabulations, inconsistencies, or failures to follow instructions [8, 9]. In the CLD context, we operationalise *hallucinations* as incorrectness in causal links: the erroneous presence (false positive) or absence (false negative) of edges, wrong polarity, or invalid causal reasoning. To evaluate LLM-generated CLDs, we require ground truth—CLDs against which generated outputs can be compared.

This thesis uses two complementary ground truth types. *Literature-derived ground truth* (GT Lit) consists of CLDs published in peer-reviewed literature with explicitly defined causal structures (named variables, directed edges, polarities). A generated edge is classified as a *true positive* if it matches an edge in the literature CLD, a *false positive* if it appears in the generated CLD but not the literature CLD, and a *false negative* if it appears in the literature CLD but was not generated. Both false positives and false negatives are considered hallucinations: FPs represent spurious causal claims, while FNs represent failures to capture established causal relationships. *Synthetic ground truth* (GT Synth) provides controlled hallucinations by programmatically corrupting valid LLM-generated edges—introducing spurious edges, flipping polarities, or replacing valid causal reasoning with plausible-but-flawed explanations. Here, hallucination status is known by construction (corrupted = hallucination), enabling precise measurement of detection methods under controlled conditions.

This dual operationalisation enables quantitative evaluation using standard classification metrics (precision, recall, F1). GT Synth provides controlled conditions for method development, while GT Lit tests generalisation to realistic expert-derived ground truth—acknowledging that literature CLDs represent expert-hypothesized relationships rather than empirically validated causal ground truth, a common limitation in complex systems where controlled experiments are infeasible.

Generation: reproduction, synthesis, or new knowledge creation? Can LLMs perform causal inference? This is a key question, and the answer is not apparent from the properties of the autoregressive language modelling process and the next-token prediction objective [10]. The goal of an LLM is to produce the most likely token in a sequence based on its training data. Given this, when prompting an LLM for causal relationships between variables A and B, the model produces an aggregate best guess—the most likely continuation of the sequence given its training data, reflecting patterns learned from related sequences. For reproducing literature consensus or performing synthesis, this may be sufficient, as the LLM has been trained on scientific literature. Although, it may of course not reproduce perfectly, or contain the same biases as the literature, in addition to LLM-architectural biases or biases in its training set. Also, for causal inference in novel settings (for which no relevant data is in its training set), LLMs may not be reliable, as they are known to perform worse out-of-distribution [11, 12], and estimates of out-of-distribution performance may be overstated as benchmarks enter training datasets through retraining processes. This serves as a note of caution against assuming performance transfers to out-of-distribution settings—a concern amplified by the fact that for many commercial LLMs, the training distribution is unknown, making this impossible to verify.

Furthermore, whether, when, and to what extent LLMs can effectively compress and translate the information reflected in their training data to token space such that it leads to correct answers remains a difficult question. State-of-the-art models exhibit unpredictable failure modes and jagged capability frontiers [13]. Established practices exist to identify and mitigate these hallucinations, and this thesis draws from and applies these methods to the CLD generation context.

Beyond reproduction: expert bias mitigation and the possibility of surfacing overlooked relationships. Crucially, it is possible that not all LLM-generated edges that diverge from expert models are errors. Expert groups operate under time constraints, cognitive biases, and disciplinary blind spots; they may omit relationships due to scope decisions, lack of awareness of adjacent literatures, or simple oversight. This problem intensifies in large-variable systems: for a CLD with $|V|$ variables, considering all pairwise relationships scales as $O(|V|^2)$, making exhaustive deliberation impractical (cf. Figure 1.1). An expert group modelling a 50-variable system faces 2,450 potential edges—far more than can be carefully discussed in a typical workshop.

In such settings, LLM-generated edges not found in expert CLDs may represent plausible causal relationships supported by scientific literature that experts simply did not consider. If validated against the literature, these edges could *augment* expert models rather than corrupt them. If evidence supports LLMs' effectiveness in this manner, it could reframe the LLM's role: not merely as a reproducer of expert consensus, but also as a tool for surfacing relationships the literature supports but experts overlooked. The challenge then becomes distinguishing genuine hallucinations (unsupported fabrications) from overlooked-but-valid relationships—a distinction that motivates the Deep Research validation approach explored in this thesis.

The false negative problem. Conversely, *false negatives*—relationships that experts include but LLMs omit—represent a distinct challenge. These gaps may arise from several sources: (1) tacit clinical or practical knowledge that experts hold but is underrepresented in published literature; (2) recent findings not yet incorporated into LLM training data; (3) domain-specific terminology that differs from how relationships are described in text; (4) relationships established through unpublished pilot studies, clinical intuition, or cross-disciplinary insights that experts bring to GMB sessions; or (5) LLM reproduction failure—cases where the relationship *is* well-documented in literature but the LLM fails to retrieve or generate it, representing a form of omission-hallucination distinct from fabrication. False negatives are particularly problematic because they represent genuine expert knowledge that LLM-generated drafts would miss, potentially leading to incomplete models if experts do not carefully review and augment the LLM output.

Mitigating hallucinations. LLM hallucinations—fabricated or inconsistent outputs—undermine reliability for scientific applications [14]. A growing body of work has explored both black-box and white-box mitigation strategies. Black-box approaches include prompting techniques such as chain-of-thought reasoning [15, 16],

retrieval-augmented generation [17, 18], multi-agent orchestration and debate [19, 20], LLM-as-a-Judge evaluation [21], and test-time compute scaling [19, 22, 23]. White-box approaches—which require model weight access—include fine-tuning [24], reinforcement learning from human feedback [25], and activation monitoring or steering [26]. These methods have shown promise for reducing hallucinations across various NLP tasks [27].

LLM-as-a-Judge (Judge) approaches offer the advantage of context-awareness: they can evaluate outputs against nuanced domain requirements without requiring explicit ground truth, based on the parametric knowledge of the Judging LLM. However, Judges remain vulnerable to biases inherent to transformer architectures and can be susceptible to adversarial manipulation [28, 29]. Conversely, more naive hallucination detection UQ metrics based on generator logprobs or retrieval based approaches —such as embedding-based similarity [17, 18], perplexity [27, 30], and minimum generation probability [14]—provide computationally efficient signals that are robust to contextual manipulation but lack semantic understanding of task requirements. The combination of LLM-as-a-Judge with naive UQ metrics remains under-explored in CLD generation, yet represents a promising approach: UQ metrics could flag high-uncertainty generations for judge evaluation, potentially reducing the biases of judge-only systems while enabling more efficient allocation of test-time compute for self-correction. This motivates the research questions of this thesis.

Research Questions

This thesis investigates how LLM-as-a-Judge and UQ metrics can detect and mitigate hallucinations in LLM-generated CLDs, structured around the following research questions:

RQ1: Can LLM-as-a-Judge methods detect hallucinations in LLM-generated CLDs?

- **RQ1a:** How effectively can correctness-based and citation-based judges distinguish valid from hallucinated causal relationships?
- **RQ1b:** Can LLM-as-a-Corrector improve CLD accuracy when guided by judge feedback?

RQ2: Can uncertainty quantification (UQ) metrics predict hallucinations in generated CLDs? Specifically, can logprob-derived uncertainty metrics (perplexity, minimum probability, window entropy) and embedding-based similarity detect hallucinated edges?

RQ3: Can adaptive test-time compute allocation improve CLD accuracy efficiency? Can adaptive deployment of LLM-as-a-Judge & Corrector—triggered by generator uncertainty metrics—provide a more efficient accuracy-per-dollar tradeoff than always-on LLM-as-a-Judge & Corrector?

Significance and Contributions

This research addresses reliability challenges in LLM-based scientific knowledge generation by systematically evaluating hallucination detection methods in the CLD domain. The work is timely: as LLM-based systems are increasingly deployed for knowledge work assistance across institutional and global scales, the accuracy of AI-generated outputs carries significant real-world consequences—particularly in health domains where CLDs inform clinical decision-making and intervention design.

Integrating hallucination detection techniques within a CLD-building system is novel and addresses a critical gap: while LLM-as-a-Judge and LUQ methods have been studied separately in general NLP contexts, their combined application to structured scientific knowledge generation—specifically causal diagrams—remains unexplored. This thesis makes the following contributions:

1. **Systematic evaluation of LLM-as-a-Judge for CLD validation.** We compare correctness-based and citation-based judges across multiple prompt variants and validation settings. We establish that judges achieve substantial agreement with human experts, but exhibit systematic leniency bias that makes them better suited for screening and triage than definitive validation. Performance transfers only partially from synthetic corruptions to expert-derived ground truth, with effectiveness varying by domain and prompt design.

2. **Evaluation of LLM-as-a-Corrector for CLD refinement.** We assess whether LLM correctors, conditioned on judge feedback, can improve CLD quality. We find that correctors yield modest gains on synthetic corruptions but fail to improve accuracy against literature-derived expert ground truth. A key finding is that correctors optimise for judge satisfaction rather than expert consensus—judge scores improve while ground truth alignment worsens—suggesting a limitation of judge-guided correction in our experimental setup.

3. **Analysis of generator uncertainty metrics for hallucination detection.** We evaluate whether generation-time signals (logprob-derived uncertainty and embedding similarity) can predict hallucinated edges without external retrieval. We find that individual metrics provide weak discriminative signal, and while ensemble classifiers achieve strong in-distribution performance, they exhibit poor cross-domain generalisation, limiting their utility as standalone detectors.

4. **Deep Research validation paradigm.** We pilot automated multi-agent literature search as an alternative to correction-based approaches, shifting from *correction* to *validation*. We find that Deep Research retrieves direct causal evidence for a substantial fraction of edges classified as false positives *and false negatives* against expert ground truth. Under strong assumptions about evidence applicability and edge-level construct alignment, this suggests that expert CLDs may systematically underestimate LLM generation quality and that some apparent "hallucinations" (edges not found in literature ground truth CLD's) could represent overlooked-but-valid causal relationships in isolation. However, during second-rater validation, we find low inter-rater reliability, meaning more research is required.

5. **Ground truth enrichment framework.** We develop and evaluate a methodology for enriching expert CLDs with literature-validated LLM-generated edges. Through human validation, we identify causal evidence the Deep Research system finds to be strong and largely applicable to the evaluated causal relationships, suggesting that the ground truth may be incomplete and that the LLM-generated CLDs may be missing causal relationships under strong assumptions (relationship isolation; extrapolation of evidence find rate/quality in the human sample). Including these relationships in the ground truth leads to a substantial improvement in the F1 score of the LLM-generated CLDs. If the effect proves robust after further validation and relaxing of assumptions (a direction for future work), it reframes the evaluation of LLM-generated CLDs: rather than viewing divergence from expert models as error, validated divergences may represent opportunities for model augmentation. However, low inter-rater reliability on human validation of system results strongly cautions against such an optimistic interpretation and warrants further investigation before more validation is done.

The practical implication is that LLM-generated CLDs, when validated through the methods evaluated here, can provide literature-grounded starting points for expert model-building sessions. This allows domain experts to focus on contributing specialized knowledge rather than reiterating established consensus, improving the efficiency of the modelling process and resulting in more comprehensive CLDs for translation into computational models for research, clinical applications, and intervention simulations.

Thesis Structure

Chapter 2 – Background: Formalizes CLDs as signed directed graphs, situates them within the modelling-simulation cycle, gives an overview of LLM architecture, used prompting techniques, hallucination issues, and established solutions from other fields.

Chapter 3 – Literature Review: Contrasts text-to-CLD versus corpus-independent generation paradigms; surveys LLM-as-a-Judge, LUQ metrics, and test-time compute strategies in more detail.

Chapter 4 – System Architecture: Describes the end-to-end system components: generator, correctness and citation judges, the corrector, and the Deep Research (DR) multi-agent literature validation pipeline.

Chapter 5 – Experimental Data: Details evaluation datasets (GT Lit, LLM Predictions, GT Synth), validation and auxiliary datasets, CLD domains, context parameters, and edge structure conventions.

Chapter 6 – Experimental Design: Defines the CLD generation task and hallucination types (FP/FN, polarity, direction, mechanistic validity); specifies the design framework, evaluation metrics, baselines, statistical analysis plan, and master experimental overview across RQ1–RQ3.

Chapter 7 – Experiments and Results: Reports generator baseline comparisons; judge performance on synthetic versus literature ground truth; corrector effects on CLD F1 and judge scores; cross-domain UQ metric generalisation failure; ensemble classifier performance; and DR found causal evidence for 62% of "hallucinations."

Chapter 8 – Discussion: Interprets the GT Synth–GT Lit performance gap; identifies judge and corrector failure mechanisms; discusses UQ metric results and the pivot to DR that reframes many false positives as overlooked-but-valid edges; discusses limitations and offers directions for future work and practitioner recommendations.

Chapter 9 – Conclusion: Positions LLM-generated CLDs as expert-refinable drafts rather than final outputs; highlights literature-supported "hallucinations" as enrichment opportunities.

Chapter 10 – Ethics and Data Management: Addresses ethical considerations, data handling practices, and responsible use of LLM-generated causal models.

Chapter 2

Background

2.1 Modelling, simulation, and why conceptual validity matters

2.1.1 The role of simulations in science.

A simulation can be used to study a model of a system by manipulating its inputs to obtain knowledge about its properties, which may have correlates in the original system or inform new theory development. Simulations enable studying certain phenomena at reduced cost, such as in robot training; at temporal or spatial scales which are difficult or impossible to observe, as in computational astrophysics; or in cases where the consequences of mistakes may be severe, such as in surgery planning. Simulation studies also enable testing what-if scenarios and interventions—for example, examining the effects of vaccines in a pandemic model to assess intervention strategies. Simulations have become so integral to many branches of science that they are often called the third pillar of science, alongside theory and experiment. Nonetheless, properties observed in a model may transfer only partially, or not at all, to the system under study. It is precisely this consideration—determining the extent to which model behaviours generalise—that yields valuable insights.

2.1.2 Model abstraction and computational trade-offs.

A model of a system by definition abstracts away parts of the system, and this information loss inherently limits the transferability of conclusions to the system itself. Thus, all models are inherently flawed representations; however, studying these models can still be highly informative. Making informed choices about which features of the system to include in the model is crucial to reproducing the most important dynamics relevant to the study's context. To reduce information loss and increase the transferability of conclusions, scientists strive to create higher-resolution representations of systems. This often leads to more complex models through increased temporal and spatial resolutions, additional variables, or the relaxation of simplifying assumptions—which may introduce non-linearities and interaction effects. This increase in complexity comes at a computational cost: greater model fidelity typically requires more computational resources, creating a fundamental trade-off between model accuracy and computational tractability. Both computational scientists and domain experts must collaborate to navigate this trade-off efficiently. Domain experts understand which phenomena are most important to capture, while computational scientists possess knowledge of algorithmic optimisations, numerical methods, parallel computing strategies, and complexity analysis. This interdisciplinary collaboration is essential for making intelligent modelling decisions that balance scientific fidelity with computational feasibility. Key considerations include algorithm selection, data structure optimisation, verification and validation procedures, and uncertainty quantification methods.

2.1.3 GMB as a foundation for computational models.

Bringing domain experts together to assimilate their knowledge into a model is a critical preliminary step, especially for complex systems such as those in biology, where subject matter experts may be specialized in only part of the system they wish to model. Convening such groups may present time, cost, and coordination challenges, and the resulting models will inevitably be limited by knowledge gaps and cognitive biases. Nonetheless, these collaborative sessions serve as a valuable basis for model building, and established methods exist to make these sessions more efficient and productive. Within system dynamics—a branch of science concerned with understanding the behaviour of complex systems over time through feedback loops, stocks, and flows—a key type of model is the Causal Loop Diagram (CLD).

The Modelling and Simulation Cycle. The established modelling and simulation (M&S) cycle [31, 32] proceeds through distinct stages: a real system or phenomenon is first mapped to a *conceptual model* through qualitative modelling, then transformed via quantitative modelling into a *computational model*, which is discretized and implemented as a simulation. Results feed back through validation, prediction, and hypothesis testing to refine understanding of the original system. Crucially, validation is required at each stage transition—including the earliest qualitative step. A flawed conceptual model propagates errors downstream: incorrect causal structure at the conceptual stage undermines all subsequent quantitative modelling and simulation efforts, regardless of their numerical sophistication.

Why model and simulate? Ziegler characterises modelling as the systematic organisation of knowledge about a given system [31]. Cellier extends this view, positioning modelling as a unifying activity across scientific and engineering disciplines and arguing that simulation serves both analytical purposes (direct problems) and design applications (inverse problems) [32]. For complex health and social systems—such as those captured by the CLDs in this thesis—simulation is often the *only* feasible approach for several compelling reasons:

- **The physical system is not available for experimentation:** Randomized controlled trials (RCTs) on complex biopsychosocial interventions are often ethically infeasible, prohibitively expensive, or logistically impossible. One cannot experimentally manipulate social norms, community-level obesity prevalence, or healthcare system structures at will, and even if possible, this may raise ethical concerns.
- **Time constants are incompatible with research timelines:** Causal processes in chronic disease and social dynamics unfold over years to decades—far exceeding typical research project durations. Simulation enables exploration of long-term system trajectories within practical timeframes.
- **Control variables may be inaccessible:** In observational health data, key confounders (genetics, lifetime exposures, unmeasured social determinants) cannot be experimentally controlled. Simulation allows systematic exploration of what-if scenarios.
- **To develop deeper understanding and generate hypotheses:** Before investing in expensive empirical studies, simulation-based exploration can identify which causal pathways merit investigation, prioritise intervention targets, and reveal unexpected system behaviours (e.g., counterintuitive feedback effects).
- **What-if scenario simulation:** Simulation enables systematic exploration of hypothetical interventions and policy scenarios that would be impossible or unethical to test in reality. Decision-makers can evaluate the potential consequences of alternative strategies—such as resource allocation changes, prevention programmes, or policy modifications—before committing to costly real-world implementation.

These considerations motivate the value of accurate CLD construction: the qualitative causal structure encoded in a CLD forms the foundation upon which all subsequent quantitative modelling and simulation rests.

2.2 Causal Loop Diagrams (CLDs)

2.2.1 Definition.

A causal loop diagram (CLD) is a qualitative system-dynamics representation of hypothesized causal structure within a bounded system. It is a signed, directed graph whose nodes denote system variables and whose arcs encode *ceteris paribus* monotonic influences: a "+" arc indicates that, relative to a reference level, an increase (decrease) in the source tends to produce an increase (decrease) in the target; a "–" arc indicates the opposite tendency. Salient delays may be marked on arcs to indicate that the dominant effect is time-lagged. Unlike directed acyclic graphs (DAGs), CLDs admit cycles and explicitly emphasise feedback as a first-class structural feature.

Formalization. A CLD can be defined as $G = (V, E, \sigma, \delta)$, where V is a finite set of variables; $E \subseteq V \times V$ is a set of directed arcs; $\sigma : E \to \{+1, -1\}$ assigns each arc a polarity; and $\delta : E \to \{0, 1\}$ flags salient delays. The intended semantics are that for each arc $i \to j \in E$ there exists a (possibly nonlinear) structural relation

$$x_j \;=\; f_j\big(x_{\mathrm{pa}(j)},\, u_j,\, t\big)$$

such that, over the domain of interest and holding other parents approximately constant, $\sigma(i \to j) = +1$ implies $\partial f_j/\partial x_i \geq 0$ and $\sigma(i \to j) = -1$ implies $\partial f_j/\partial x_i \leq 0$.

Feedback loops. A *feedback loop* is any simple directed cycle. Its polarity is the product of its arc signs: $\prod_{e \in C} \sigma(e) = +1$ denotes a *reinforcing* loop (R), which tends to amplify deviations; $\prod_{e \in C} \sigma(e) = -1$ denotes a *balancing* loop (B), which tends toward goal-seeking or stabilization. (Equivalently, reinforcing loops contain an even number of negative arcs, balancing loops an odd number.) Loop polarity characterises qualitative tendencies and does not presume linearity, stationarity, or constant effect sizes.

Scope and use. CLDs deliberately abstract from magnitudes, functional forms, stock-and-flow accounting, and statistical identification of causal effects. They encode theory- or evidence-informed hypotheses about directional dependencies and dominant feedbacks to support elicitation, communication, and high-level reasoning about system behaviour, and they are commonly used as a precursor or complement to quantified stock-and-flow models, experiments, and statistical/causal analyses.

Epistemological status. CLDs encode *hypothesized*—not experimentally validated—causal relationships. In complex biopsychosocial systems, isolating causal effects through controlled experiments is often infeasible or unethical; the systems involve too many interacting variables, long time horizons, and ethical constraints on manipulation. Published CLDs therefore represent expert consensus synthesized from scientific literature, domain knowledge, and group deliberation—not ground-truth causality established through intervention. When we ask whether an LLM can "generate" a CLD, we ask whether it can reproduce expert-hypothesized causal structures, and when we detect "hallucinations," we identify divergence from expert models rather than departure from proven causal facts.

2.3 Group Model Building (GMB) and the CLD-to-Computational Model Pipeline

The systematic construction of CLDs typically occurs through *Group Model Building* (GMB)—a participatory method where domain experts collaboratively develop shared mental models of complex systems [5]. This

process transforms tacit expert knowledge into explicit causal structures that can subsequently inform computational models.

The CLD-to-SDM pipeline. Crielaard et al. [5] formalize the transition from conceptual to computational models through an *annotated CLD* (aCLD) that captures additional information required for quantification. The key steps are:

1. **Research question and context of validity:** Define the specific research question and establish the spatial/temporal scales over which the model should be valid. The context determines which variables are stocks (accumulating over time), constants (fixed within the temporal scale), or auxiliaries (instantaneously computed).
2. **Variable selection via scale separation map:** Identify all relevant variables and organise them by temporal and spatial scale. Variables acting on smaller temporal scales than the variable of interest can be averaged out; variables on larger scales can be treated as constants.
3. **Causal link identification:** For each pair of variables, determine whether a causal link exists. Links are classified as:
   - **Present (with consensus):** Experts agree the relationship exists
   - **Uncertain:** Experts disagree or lack evidence
   - **Known-to-be-absent:** Explicitly ruled out (critically different from simply unrecorded)
4. **Intermediary variable documentation:** Record mediating variables underlying each causal link. If two proposed links share an intermediary variable, it should be added to the model to avoid spurious independence assumptions.
5. **Functional form specification:** For each link, determine the mathematical relationship (linear, sigmoidal, polynomial, etc.). Sources include expert elicitation, literature review, or computational fitting.
6. **Interaction term identification:** Identify many-to-one relationships where multiple variables jointly affect an outcome (AND gates, OR gates, conditional effects).
7. **Evidence annotation:** Document the source and quality of evidence for each link using frameworks like GRADE [33] (high/moderate/low/very low).

Uncertainty quantification. A key contribution of the Crielaard et al. framework is explicit treatment of uncertainty at each stage:

- **Structural uncertainty:** Uncertainty about which links exist and their functional forms, leading to a "model space" of possible model topologies.
- **Parameter uncertainty:** Uncertainty about parameter values within a given model structure.
- **Propagation:** Both sources of uncertainty should be propagated through simulation to obtain confidence intervals on predictions.

Implications for LLM-based CLD generation. Crielaard's systematic framework [5] highlights what LLMs would need to replicate to construct comprehensive aCLDs: not merely variable-relationship pairs, but also consensus levels, intermediary variables, functional forms, and evidence quality. However, a simpler starting

point—accurately providing variables, relationships reflecting literature consensus, and their references—would already reduce expert time requirements. This is especially valuable if achievable in a source-text-corpus-independent manner, as corpus dependence constrains the currently most prevalent text-to-CLD methods in the literature.

Current LLM-based approaches focus on core causal structure (variables, links, polarity) without capturing these richer annotations. Some approaches like Theoraizer [34] are more closely aligned with the method in this thesis, as they also perform CLD generation while capturing uncertainty quantification through logprobs and references to CLDs—but provide no clear measurement indicating the hallucination reduction potential of these methods.

This thesis aims to validate multi-agent test-time compute methods such as LLM-as-a-Judge and Corrector, alongside Logprob Uncertainty Quantification (LUQ) and embedding-similarity-based hallucination reduction methods. These approaches are well-established outside the field of CLD generation but have not been systematically validated within it.

The gap between LLM-generated CLDs and fully annotated aCLDs—which also include functional forms and rigorous consensus assessment—represents a direction for future work, and underscores why expert validation remains essential even if LLMs provide useful initial CLD drafts. The intended contribution of this thesis is that, by establishing where LLMs can and cannot be useful in constructing initial CLDs, expert time can be spent more efficiently in creating aCLDs. These aCLDs, in turn, offer scientific value by informing theory, can be translated into computational models in which interventions can be simulated, and through this, help clarify the role of LLMs in contributing to the scientific process.

2.4 Large language model (LLM) background

2.4.0.1 Architectural details of large language models

Tokenization and embeddings. Modern large language models (LLMs) operate on *tokens*: discrete units (characters, subwords, or words) produced by a tokenizer from raw text. Each token index is mapped to a continuous *embedding* vector, yielding a sequence of vectors $x_{1:L} \in \mathbb{R}^{L\times d}$. In this representation space, semantic relatedness is often operationalised via cosine similarity,

$$\mathrm{sim}(u, v) = \frac{u^\top v}{\|u\|\|v\|} \in [-1, 1],$$

which is also the basis of embedding-based retrieval: by retrieving vectors nearest to a query embedding and injecting the associated text as context, Retrieval-Augmented Generation (RAG) can ground generation in external evidence [35].

Figure 2.1: Tokenization maps raw text into a sequence of discrete tokens (including special tokens such as `[UNK]` and padding), which are then embedded as vectors. Reproduced (cropped) from Amidi and Amidi [36], used under the MIT License.

The attention mechanism. The dominant architecture underlying LLMs is the Transformer [37]. Its central mechanism is (scaled dot-product) attention, which computes content-dependent mixing weights over

the input:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V,$$

where $X \in \mathbb{R}^{L \times d_{\text{model}}}$ is the matrix of input token representations (one row per token) and the query, key, and value matrices are computed by learned linear projections $Q = XW_Q$, $K = XW_K$, and $V = XW_V$. Here $W_Q, W_K \in \mathbb{R}^{d_{\text{model}} \times d_k}$ and $W_V \in \mathbb{R}^{d_{\text{model}} \times d_v}$ are trainable parameter matrices, so $Q, K \in \mathbb{R}^{L \times d_k}$ and $V \in \mathbb{R}^{L \times d_v}$; the scaling factor d_k is the key/query dimensionality. Because attention computes all pairwise query–key interactions, *dense* self-attention scales quadratically in sequence length L (forming an $L \times L$ attention pattern). In common (non-IO-aware) implementations that materialize the attention matrix, this leads to $\mathcal{O}(L^2)$ memory usage and $\mathcal{O}(L^2)$ work (up to factors of the embedding dimension), and the attention distribution can become increasingly diffuse for long contexts (each position receiving weight on the order of $1/L$ in the extreme). To preserve order information in a permutation-invariant attention mechanism, Transformers incorporate positional information (e.g., learned or fixed positional embeddings) added to token embeddings prior to attention.

This quadratic cost motivates two common optimisation directions. *Sparse attention* restricts each token to attend only to a subset of positions (e.g., local windows or structured patterns), reducing query–key interactions and yielding sub-quadratic time and memory growth in L (with scaling determined by the sparsity factorization) [38]. In contrast, *FlashAttention* preserves *exact dense* attention but makes it IO-aware, improving wall-clock performance without changing the $\mathcal{O}(L^2)$ scaling of dense attention [39].

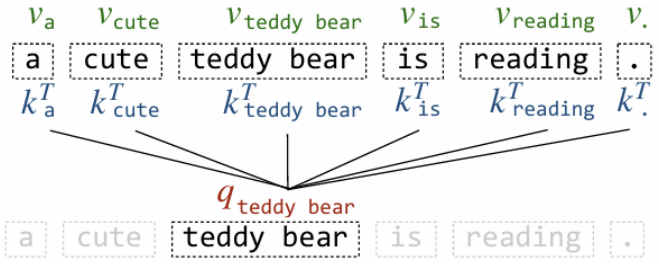


Figure 2.2: Attention computes a query-dependent weighted combination over keys/values, selecting which tokens should influence the representation at a given position. Reproduced (cropped) from Amidi and Amidi [36], used under the MIT License.

Multi-head attention. In practice, Transformers use *multi-head attention*, computing several attention operations in parallel and combining them with a learned output projection. This allows different heads to specialize in different relational patterns in the sequence.

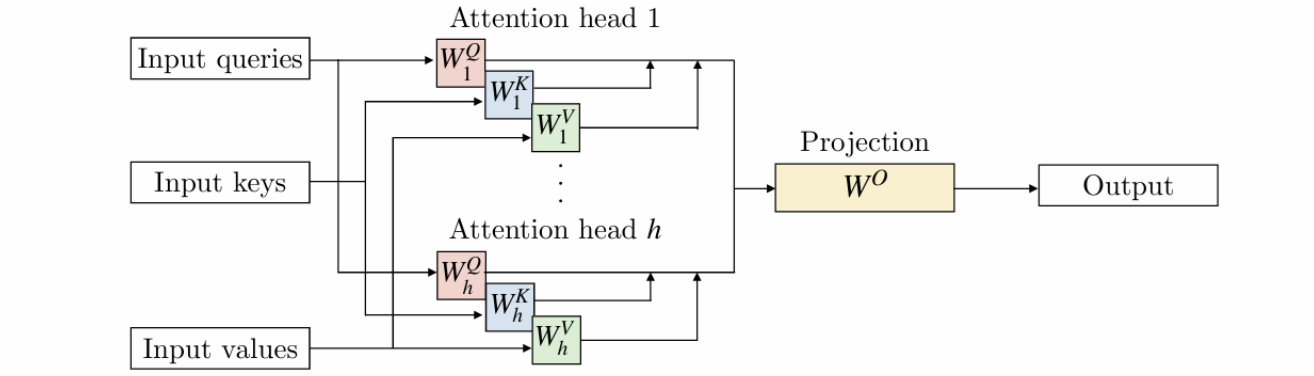


Figure 2.3: Multi-head attention: multiple sets of W_Q, W_K, W_V are applied in parallel, then concatenated and projected to the model dimension. Reproduced (cropped) from Amidi and Amidi [36], used under the MIT License.

Feed-forward networks. Each Transformer layer includes a *position-wise feed-forward network* (FFN) applied independently to each token representation. The FFN typically consists of two linear transformations with a non-linear activation (commonly ReLU or GELU) in between:

$$\mathrm{FFN}(x) = W_2 \cdot \sigma(W_1 x + b_1) + b_2,$$

where $W_1 \in \mathbb{R}^{d_{\text{model}} \times d_{\text{ff}}}$ projects to a larger hidden dimension d_{ff} (typically $4 \times d_{\text{model}}$), and $W_2 \in \mathbb{R}^{d_{\text{ff}} \times d_{\text{model}}}$ projects back. This expansion-contraction pattern allows the model to learn complex, non-linear transformations of individual token representations after the attention layer has mixed information across positions.

Residual connections and layer normalization. Transformer layers employ *residual connections* (also called skip connections) around both the attention and feed-forward sublayers, followed by *layer normalization*. For a sublayer function f (either attention or FFN), the output is computed as:

$$\mathrm{LayerNorm}(x + f(x)),$$

where the residual connection $x + f(x)$ adds the sublayer's input directly to its output. This design serves two purposes: (1) residual connections facilitate gradient flow during training, enabling effective optimisation of deep networks; and (2) layer normalization stabilizes activations by normalizing across the feature dimension, reducing sensitivity to the scale of inputs. These "Add & Norm" blocks, visible in Figure 2.6, are applied after every attention and feed-forward sublayer throughout the Transformer stack.

Encoder–decoder architecture. The original Transformer was introduced in an encoder–decoder setting [37]: an encoder produces contextual representations of an input sequence, and a decoder generates an output sequence while attending to the encoder via cross-attention.

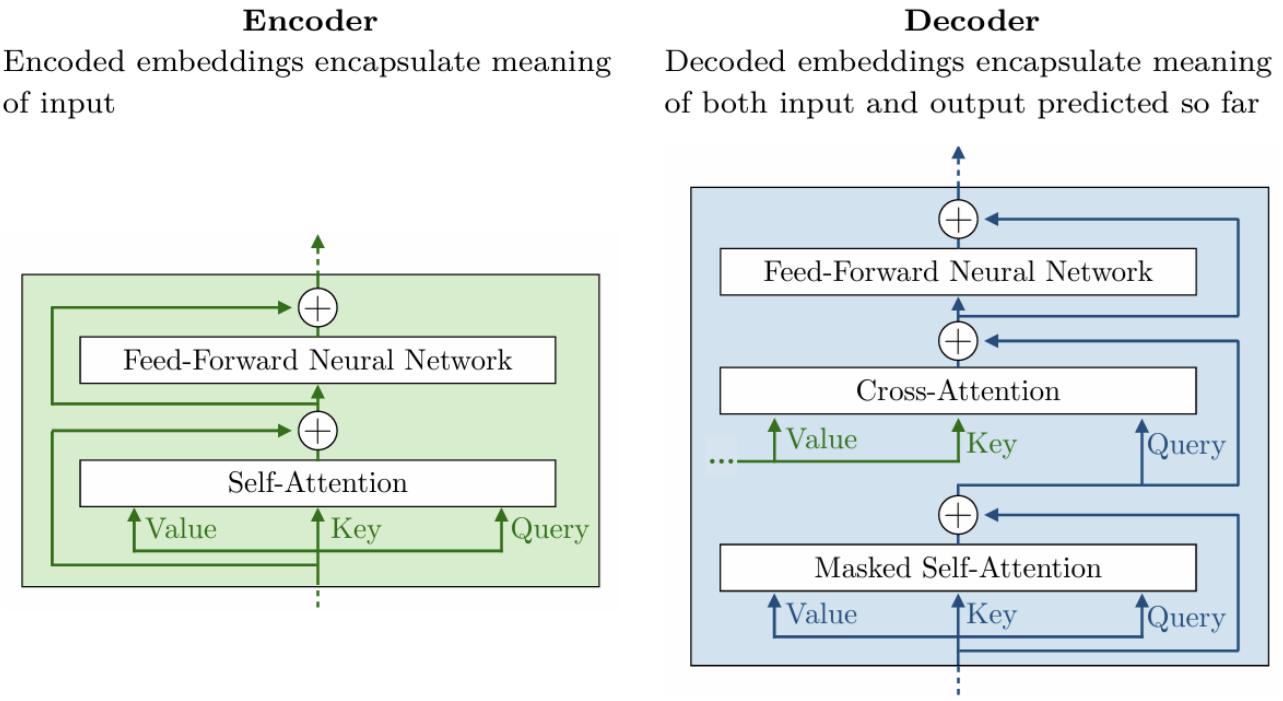


Figure 2.4: Encoder and decoder building blocks. The decoder includes masked self-attention (for causality) and cross-attention (to condition on encoder outputs). Reproduced (cropped) from Amidi and Amidi [36], used under the MIT License.

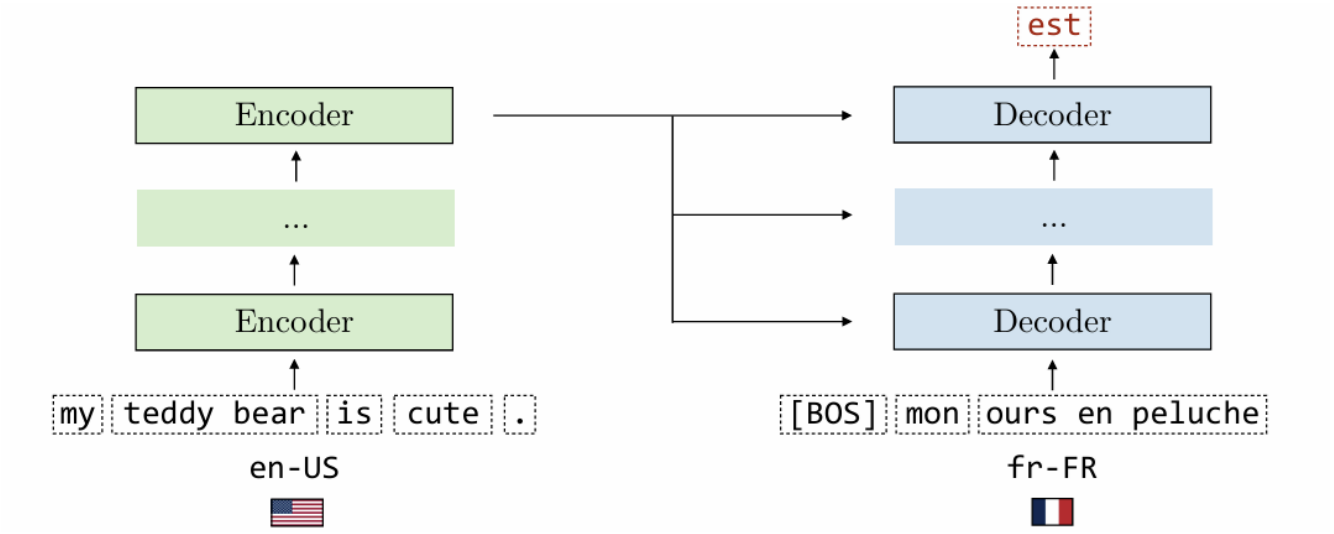


Figure 2.5: Encoder–decoder example (e.g., translation): the encoder reads the input sequence, and the decoder generates the output sequence autoregressively. Reproduced (cropped) from Amidi and Amidi [36], used under the MIT License.

Decoder-only models and autoregressive generation. Many widely used LLMs are *decoder-only* Transformers (the GPT family) trained with a next-token prediction objective [16]. Figure 2.6 shows the original encoder–decoder Transformer architecture from which decoder-only models are derived.

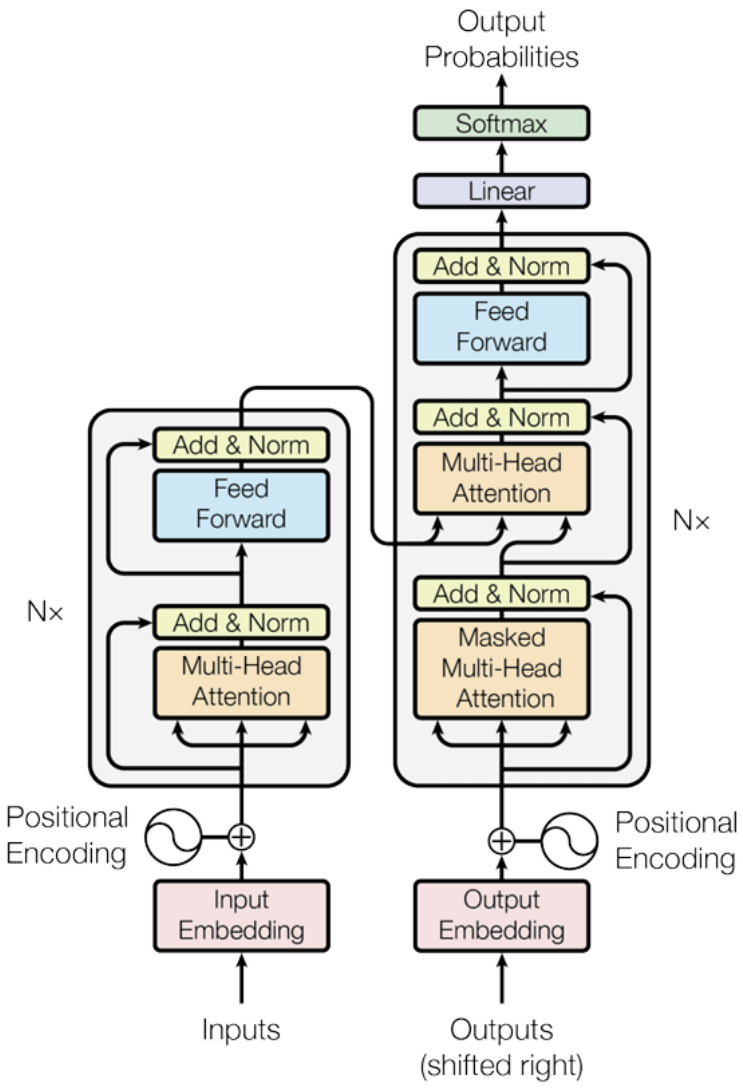


Figure 2.6: The Transformer architecture, consisting of stacked encoder (left) and decoder (right) blocks. Each encoder layer comprises multi-head self-attention and a feed-forward network with residual connections and layer normalization. The decoder additionally includes masked self-attention to enforce causality and cross-attention over encoder outputs. Adapted from Vaswani et al. [37].

Training typically uses *teacher forcing*: at position t, the model is given the ground-truth prefix $x_{<t}$ and is optimised to assign high probability to the next token x_t. Concretely, parameters are learned by minimizing the negative log-likelihood (equivalently, cross-entropy) over a corpus,

$$\mathcal{L} = -\sum_{t=1}^{T} \log P\big(x_t \mid x_{<t}\big).$$

The architectural mechanism that enforces this conditional structure inside the Transformer is *masked* self-attention: a causal (typically lower-triangular) attention mask prevents each position from attending to any future tokens $x_{>t}$, ensuring that information flows only from past to present.

This yields an autoregressive factorization of the sequence likelihood:

$$P(x_{1:T}) = \prod_{t=1}^{T} P(x_t \mid x_{<t}),$$

and generation proceeds one token at a time, conditioning strictly on previously produced tokens; once a token is emitted it becomes part of the immutable context for subsequent predictions.

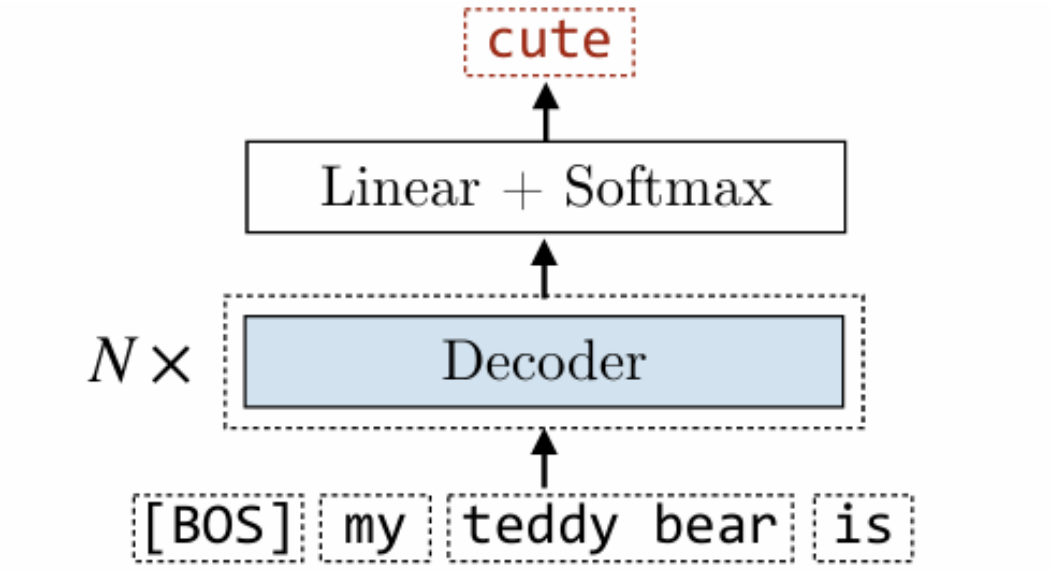


Figure 2.7: Decoder-only LLMs stack masked self-attention blocks and predict the next token via a linear layer and softmax. Reproduced (cropped) from Amidi and Amidi [36], used under the MIT License.

Decoding strategies. For fixed parameters and fixed context, the forward pass produces a deterministic next-token distribution via a softmax over logits. Apparent stochasticity in model outputs arises primarily from *decoding*: sampling from this distribution using choices such as temperature scaling, top-k truncation, or nucleus (top-p) sampling [40]. Consequently, repeated generations with identical prompts can vary, and controlling the sampling strategy is a key lever for trading off diversity versus determinism.

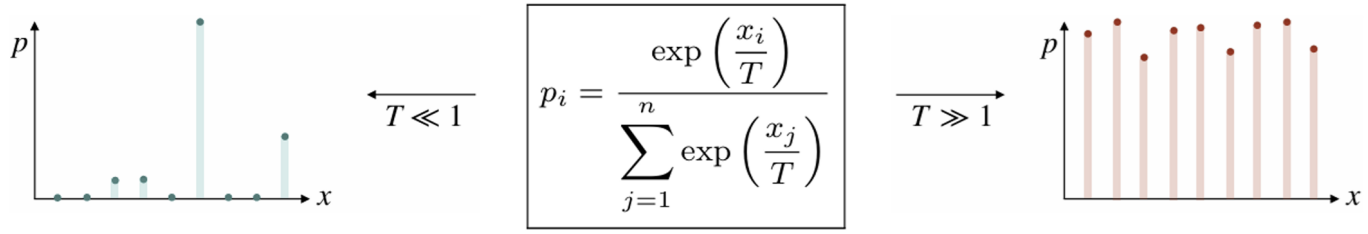


Figure 2.8: Temperature scaling reshapes the next-token distribution: low T concentrates probability mass (more deterministic), while high T flattens it (more diverse). Reproduced (cropped) from Amidi and Amidi [36], used under the MIT License.

Autoregressive generation and hallucination mechanisms. Modern LLMs generate text through an *autoregressive* language modelling process: tokens are generated sequentially, conditioned on a context c (e.g., the prompt and any provided documents), such that the probability of a token sequence $x_{1:T}$ factorizes as:

$$P(x_{1:T} \mid c) = \prod_{t=1}^{T} P(x_t \mid x_{<t}, c) \tag{2.1}$$

At each generation step, the model produces a probability distribution over its vocabulary $\mathcal{V}$, and a next token is sampled from this distribution. For vocabulary item $v \in \mathcal{V}$ with logit s_v, the corresponding probability is given by the softmax:

$$P(v \mid x_{<t}, c) = \frac{\exp(s_v)}{\sum_{j \in \mathcal{V}} \exp(s_j)} \tag{2.2}$$

where $\{s_j\}_{j \in \mathcal{V}}$ are the model's logits for the next-token distribution at step t.

This sequential, generation process has structural properties that contribute to hallucination [8]:

Hallucination snowballing. A form of error propagation that occurs once an incorrect token is generated, the model conditions subsequent generation on this error, compounding inaccuracy. Zhang et al. [41] demonstrated that LLM hallucinations can "snowball" - an initial minor error leads the model to generate increasingly confident but fabricated content to maintain coherence with its erroneous premise. This creates a failure mode where the model appears fluent and confident while being factually wrong.

Decoding-induced degeneration. The choice of decoding strategy affects hallucination rates. Holtzman et al. [40] showed that maximization-based decoding (beam search) can lead to repetitive, degenerate text, while sampling from the full distribution introduces randomness that may select low-probability but plausible-sounding tokens - potentially hallucinations. This tension between coherence and diversity is inherent to autoregressive generation.

Three key parameters control the sampling process:

- **Temperature** (T): Scales logits before softmax, with $P(v) \propto \exp(s_v/T)$. Lower temperatures ($T < 1$) sharpen the distribution toward high-probability tokens, increasing determinism but risking repetition. Higher temperatures ($T > 1$) flatten the distribution, increasing diversity but also the probability of sampling unlikely (potentially hallucinated) tokens.
- **Top-k sampling**: Restricts sampling to the k highest-probability tokens, redistributing probability mass among them. This prevents sampling from the long tail of implausible tokens, but a fixed k may be too restrictive for peaked distributions or too permissive for flat ones. Therefore, nucleus sampling is often used over top-k for its dynamic properties.
- **Top-p (nucleus) sampling**: Dynamically selects the smallest set of tokens whose cumulative probability exceeds threshold p (e.g., $p = 0.9$). This adapts to the shape of each distribution - sampling from few tokens when the model is confident, more when uncertain [40].

Each parameter represents a trade-off: more restrictive settings reduce hallucination risk but may produce less natural text, while more permissive settings increase fluency at the cost of factual reliability.

2.5 Hallucinations in LLMs (general)

Recent work from OpenAI supports LLM hallucinations arise naturally from the pretraining objective—by reducing generation to an "Is-It-Valid" binary classification problem, cross-entropy–trained base models will still produce errors (notably on arbitrary facts or under model misspecification), even with error-free data [6]. Moreover, prevalent 0–1 graded benchmarks penalize abstentions and uncertainty, so models are effectively rewarded for confident guesses, which sustains overconfident hallucinations. Under the Open World Assumption, hallucination is a direct consequence of generalisation in an unbounded environment: finite training data can never guarantee correctness in all future situations [7].

2.5.1 Taxonomy of Hallucinations

A comprehensive survey on LLM hallucinations [9] establishes a principled taxonomy distinguishing two main types: *factuality hallucinations* (contradictions with real-world facts, fabrication, overclaims, and unverifiable content) and *faithfulness hallucinations* (instruction inconsistency, context inconsistency, and logical inconsistency). In the context of CLD generation, most errors map to *relation-error factuality hallucinations* (incorrect causal relationships) and *logical inconsistency faithfulness hallucinations* (invalid feedback loop structures or polarity assignments).

The survey strongly supports the view that LLM hallucinations arise when the model operates outside its *knowledge boundary*—which may be incomplete (long-tail knowledge), outdated (temporal cutoff), or legally restricted (copyright). CLDs inherently involve niche domain variables, implicit causal relations, and theory-level abstractions—exactly the type of long-tail and underspecified knowledge where hallucinations are most likely.

These considerations motivate the exploration of *corpus-independent* hallucination mitigation methods that do not rely solely on retrieval quality, as detailed in Section 3.7.

2.5.2 Hallucination Mitigation Methods

Methods for combating hallucinations can be grouped into *black-box* approaches (no access to model weights required) and *white-box* approaches (modifying or inspecting model internals).

2.5.2.1 Black-box methods.

Prompt-based. Prompt engineering specifies the model's role and constraints via system prompts, uses Chain-of-Thought (CoT) prompting to scaffold reasoning [15], and provides one or more input–output exemplars (few-/multi-shot learning) to anchor the desired behaviour [16]; see Section 3.5 for a detailed review of prompting strategies.

Retrieval-Augmented Generation. RAG conditions generation on external knowledge by embedding the query and a corpus into a shared vector space and retrieving the most relevant context using distance metrics such as cosine similarity or L_2 distance; retrieved sources can come from vector databases or structured stores such as knowledge graphs [17]. In *reverse retrieval* settings, the model output and corpus are embedded in a vector space and the documents most similar to the output are retrieved, identifying which sources best support a given response.

System orchestration. Verification agents assign LLMs explicit checking roles, e.g., LLM-as-a-judge to assess another model's output and LLM-as-a-corrector to reflect and repair detected errors. More generally, multi-agent ensembles can be arranged serially or in parallel with distinct instruction sets to "divide and conquer" the reasoning and verification process [19]. This principle is supported by research on decomposition-based prompting, which demonstrates that breaking complex tasks into smaller subproblems—solved sequentially with earlier solutions informing later ones—significantly improves LLM performance on multi-step reasoning tasks [42]. Relatedly, *test-time compute* increases inference-time computation—often via multi-step reasoning, agentic reflection, and/or selection among candidates via verifiers/reward models—to improve answer quality and reduce hallucinations [22, 23, 43].

Corpus-independent metrics. *Corpus-independent* metrics do not require a researcher-provided text corpus to retrieve from or compare against at inference time. Hallucinations can be flagged using statistical signals that do not require deep semantic interpretation: (i) low cosine similarity between generated text and retrieved/cited

sources suggests unsupported content, (ii) perplexity or log-likelihood anomalies can indicate low-confidence generation [14], (iii) elevated semantic entropy (uncertainty over meaning) correlates with hallucination risk [44], and (iv) instability in multi-hop reasoning trajectories—e.g., analysed via Hamiltonian interpretations of reasoning paths—can signal unreliable chains [45].

2.5.2.2 White-box methods.

Fine-tuning and RLHF. Weight updates using supervised fine-tuning and reinforcement learning from human feedback (RLHF) improve task adherence and can reduce hallucinations [25]; parameter-efficient methods such as LoRA reduce training cost and preserve generality by adapting low-rank components and/or fine-tuning only subsets of layers [24], with fine-tuning improving performance on hallucination detection tasks [46].

Test-time compute. Beyond orchestration as a black-box strategy, test-time compute can also be viewed as a model-level paradigm shift: when pre-training gains diminish with scale, allocating compute at inference (e.g., multiple rollouts, reflection, RL-guided selection) becomes a primary driver of improved reasoning and reduced errors [47].

Activation monitoring and steering. Monitoring internal activations can associate hallucinations with specific neurons or layers and mitigate them by steering activations; related work suggests that suppressing undesirable neuron behaviours (e.g., overly high sensitivity) via interventions such as dropout during fine-tuning can reduce hallucination propensity [14, 26].

Taken together, these mitigation strategies reveal different approaches along different axes of hallucination reduction exist, and that not a single technique but a design space spanning prompt design, retrieval and external grounding, agentic verification pipelines, inference-time scaling, and weight- or activation-level interventions is available. With a broad overview of different types of hallucination detection and mitigation methods given as well as background CLD's and LLMs the literature review will focus specifically around the related work to our questions, focussed on use of LLMs in constructing CLDs, how hallucinations manifest there, and mitigation methods as well as adaptive test-time compute.

Chapter 3

Literature Review

Most CLD work in system dynamics remains grounded in manual and participatory practice. Practitioners construct CLDs through interviews, literature synthesis, and facilitated workshops because diagram quality depends on careful construct definition, loop reasoning, and—critically—judgments about directness, mediation, and evidentiary support [1, 2, 4, 5]. The effectiveness of such participatory approaches has been systematically reviewed: Rouwette et al. [3] synthesise assessment studies and conclude that GMB can be valuable for involvement, communication, and shared understanding, but remains resource-intensive and context-dependent.

3.1 Standard CLD construction paradigms

In practice, CLDs are most commonly constructed through expert- and stakeholder-driven workflows rather than fully automated pipelines. These paradigms vary in the balance between participatory elicitation (capturing tacit knowledge, scope assumptions, and stakeholder mental models) and literature synthesis (capturing published evidence and consensus), but share a broadly iterative structure: variable elicitation and refinement, causal link elicitation (direction and polarity), feedback loop identification, and repeated revision to improve construct clarity and model usefulness for the stated purpose [1, 2, 4].

3.1.1 Participatory group model building (GMB)

GMB workshops combine facilitated sessions with domain experts and stakeholders to elicit variables, propose and negotiate causal links, and converge on a shared causal narrative that is "good enough" for downstream use (e.g., communication, policy discussion, or subsequent formalisation into a simulation model) [1, 2]. Empirical reviews indicate that GMB can improve involvement, communication, and shared understanding, but outcomes are context-dependent and the process remains resource-intensive and difficult to scale to large variable sets [3].

3.1.2 Literature-driven expert synthesis

An alternative (or complement) to workshops is expert-led construction grounded in literature review: practitioners define constructs and link rationales based on prior studies, then iteratively refine the diagram as new evidence and scope constraints are incorporated [4, 5]. While this can strengthen evidentiary support, it still requires substantial manual effort to assess directness vs. mediation, reconcile conflicting findings, and document assumptions consistently across many candidate links.

3.1.3 Hybrid workflows and progression to quantitative models

Many applied projects combine these approaches: a literature-informed draft is used to structure workshop discussion, or workshop outputs are subsequently validated and extended through targeted evidence gathering. In system dynamics practice, CLDs are often treated as intermediate artefacts that can be translated into more formal representations (e.g., stock–flow structures and parameterised simulation models) once scope, feedback structure, and key mechanisms are sufficiently specified [1].

Scaling of manual synthesis work. A core reason the workflow is costly is combinatorial: for a CLD with $|V|$ variables there are $O(|V|^2)$ ordered candidate links to consider, even before accounting for polarity, delays, and scope decisions. If the modelling goal requires assessing *directness*—i.e., for each candidate $X \to Y$, asking whether the influence is plausibly direct or better represented via one (or more) mediators among the remaining $|V|-2$ concepts—the deliberative workload can increase substantially. In the worst case, considering mediation pathways for each ordered pair introduces an additional factor that scales with $|V|$, pushing the upper bound of the review task toward $O(|V|^3)$ checks, though in practice modellers use heuristics and domain knowledge to prune the search space [4, 5].

This cost motivates automating part of CLD construction, for which large language models are natural candidates given their demonstrated capabilities in causal reasoning and structured knowledge synthesis (Chapter 1). The following sections review prior work on employing LLMs in a causal discovery role and summarise their reported results to assess suitability for this purpose.

3.2 Automated CLD construction paradigms (text-to-CLD vs. generation)

Current literature on LLM-based CLD construction can be organised into two paradigms. First, *text-to-CLD* approaches (also described as *causal translation*) extract variables and causal links from provided documents, and can reduce hallucination risk through retrieval grounding, but assume availability of a relevant corpus and are capped by retrieval quality. Second, *corpus-independent CLD generation* synthesizes candidate causal structures from the model's parametric knowledge without requiring an external corpus (i.e., we directly prompt the LLM to create a CLD or set of causal relationships and take its output to be the CLD); this is inexpensive and broadly applicable, but lacks a grounding corpus to reduce risk of hallucinations.

Large language models show promise for automating Causal Loop Diagram construction but exhibit systematic failure modes that demand specialized uncertainty quantification approaches. This review synthesizes 2022–2025 literature connecting general hallucination detection metrics to the specific challenge of generating reliable causal structures from text, distinguishing between benchmarks that evaluate (a) causal extraction from text, (b) causal reasoning over given graphs, and (c) causal structure discovery from statistical data.

The remainder of this chapter reviews related work and evidence for each paradigm, then synthesizes a CLD-specific taxonomy of failure modes and verification signals, motivating the hybrid approach evaluated in this thesis (RQ1–RQ3).

3.3 LLM-Based Causal Loop Diagram Generation

Notably, generating causal loop diagrams directly is relatively under-researched in the literature found in this review, with most approaches focusing on giving the LLM text, or a causal graph, from which the causal structure is to be extracted or reasoned over, with hallucination mitigation methods largely coming down to

different forms of retrieval-augmented generation. To understand the capabilities and limitations of LLMs for CLD-related tasks, we first survey benchmark studies that evaluate different forms of LLM-powered causal reasoning, before examining specific CLD generation and extraction approaches.

3.3.1 From LLM-Powered Causal Pattern Matching to Causal Loop Diagrams

Benchmark studies reveal highly varying performance across causal task types using LLMs, though each evaluates a distinct form of causal reasoning:

COPA (Choice of Plausible Alternatives) tests commonsense causal judgement: given a premise (e.g., "The man broke his toe") and two alternatives, models select which is more plausibly the cause or effect. This multiple-choice format with everyday scenarios yields near-perfect GPT-4 accuracy but requires no graph construction or variable extraction—it tests recognition of familiar causal patterns rather than systematic causal inference.

CLadder [11] tests causal query answering over provided graphs: models receive a causal graph with binary variables and must answer queries at three levels of Pearl's ladder—associational ("What is P(Y—X)?"), interventional ("What happens if we set X=1?"), and retrospective what-if ("Would Y have occurred if X had been different?"). GPT-4 achieves only ∼62% accuracy, with performance degrading sharply on interventional and retrospective what-if queries. CLadder evaluates reasoning *with* a given causal structure, whereas CLD generation requires synthesizing a plausible causal structure from a task prompt or domain specification without being given the causal graph as input. However, polarity errors in CLD generation may reflect similar failures in interventional reasoning.

The **CaLM benchmark** [48] provides the most comprehensive evaluation, testing 28 models across 92 causal targets organised into four modules. Tasks range from causal attribution (identifying causes of events) to effect prediction and what-if reasoning. The key finding—"as complexity of causal reasoning increases, accuracy progressively deteriorates, eventually falling almost to zero"—suggests that CLD generation involving multi-step causal chains will face compounding errors. This is especially relevant for CLDs, where feedback loops and longer cause–effect chains imply that errors in individual edges can propagate through downstream reasoning about the system. It also motivates edge-level error correction to prevent compounding failures, for example via LLM-as-a-judge and LLM-as-a-corrector pipelines (if these components are empirically shown to work in this setting).

CausalBench evaluates three subtasks: correlation identification, causal skeleton identification (undirected graph structure), and causality identification (directed edges). LLMs significantly lag behind traditional algorithms on networks exceeding 50 nodes and struggle with collider structures while excelling at chains.

Darvariu et al. [49] frame LLMs as low-cost sources of background knowledge for causal graph discovery: by eliciting structural priors from an LLM, the hypothesis space over candidate graphs can be reduced before running downstream discovery algorithms. They propose metrics to assess LLM causal judgements independently of any specific discovery method and show that LLM-derived priors are most useful for edge directionality (e.g., 84% correctly oriented edges for LLaMA2-70B when combined with observational data).

Corr2Cause [50] tests inferring causal graph structure from correlation statements: given statements like "A is correlated with B" and "B is correlated with C," models must determine whether "A causes B" or identify the correct DAG structure. Testing 17 models on 200K+ samples, all achieved near-random performance (∼29% F1). This benchmark isolates pure causal structure inference without natural language complexity. The near-random performance suggests low success for text-to-CLD systems in settings where causal structure must be inferred from limited signals. In our CLD generation task, we similarly prompt the model to answer

"Does A cause B?" (surrounded by prompt scaffolding) at the edge level; Corr2Cause results suggest this edge-level decision may be challenging, although this remains to be seen since Corr2Cause conditions on correlation statements rather than the domain-grounded prompts used in this thesis.

However, evidence closer related to our use case shows that, for causal graph discovery from domain knowledge, LLMs show complementary strengths and weaknesses. Jiralerspong et al. [51] demonstrate that querying LLMs about pairwise causal relationships using breadth-first search can achieve state-of-the-art results on real-world causal graphs (typically DAG-oriented discovery). This is adjacent to CLD generation in that it elicits edge-level causal hypotheses from domain knowledge, but it does not, by itself, target signed feedback-loop structures as CLDs do.

In the text-to-CLD causal extraction task, an example is the System Dynamics Bot [52], which extracts causal variables and directed causal links from natural language text to construct feedback diagrams and outputs structured CLDs. Evaluated against human-coded ground truth from system dynamics literature and participant responses, the bot recovers only 59% of causal links and 66% of feedback loops using GPT-4.

Liu and Keith [53] use curated prompting techniques extracting CLDs from dynamic hypothesis texts, comparing four prompt designs: zero-shot, few-shot, guided instructions, and a two-stage prompt that first extracts variables then maps signed causal links. On 44 textbook DH–CLD pairs (expert CLDs hand-encoded in DOT), curated prompting—especially the two-stage setup, which the generator algorithm employed in this thesis also uses—produces CLDs closest to expert diagrams for simple structures, but errors persist (missed loops, wrong polarities, and weak handling of exogenous/implicit relationships) as complexity grows.

Complementing these point evaluations, the SD-AI benchmark [54] proposes a standardised test suite and public leaderboard for CLD *causal translation* (i.e., text-to-CLD extraction) and conformance, showing substantial variation across LLMs and motivating continued benchmarking of instruction-following and technical correctness in AI-assisted system dynamics tooling.

An approach exploring CLD generation comes from **Theoraizer** [34], which is closest to the approach we explore in this thesis. Theoraizer operationalises CLD construction as a stepwise, edge-level elicitation pipeline in which an LLM functions as a "digital extension" of an expert group rather than a replacement. Users provide (or optionally generate) a variable list, and the system then queries the LLM over all ordered variable pairs to (i) classify whether a causal relation exists, (ii) infer directionality, and (iii) assign polarity (+/−). A key design choice is to force single-token outputs (e.g., "C/N", "yes/no") and use the model's log probabilities as "model-implied" confidence scores for each edge, enabling thresholding to trade off sparsity vs. coverage based on the logprob implied uncertainty. To reduce prompt sensitivity, Theoraizer uses prompt ensembling (multiple prompt variants aggregated), and keeps humans in the loop by exposing intermediate tables/plots and encouraging iterative editing of variables and edges.

Methodologically, Theoraizer is closely aligned with CLD generation workflows that decompose structure construction into repeated pairwise decisions (e.g., prompting "Does A cause B?"), and it explicitly targets scalability: evaluating $|V|(|V|-1)$ candidate edges becomes computational rather than purely cognitive labour. In validation, Theoraizer reports that LLM–human agreement on causal judgements is comparable to inter-expert agreement, suggesting LLMs can act as plausible "stand-in" raters for initial candidate CLDs, and supporting this by finding comparable LLM–human and human–human inter-rater agreement. At the same time, the paper emphasises limitations that mirror hallucination risks in causal structure generation: judgements are made pairwise and in isolation (encouraging dense graphs and "short-circuit" links that may actually be mediated), outputs reflect linguistic priors rather than epistemic truth, and biases/hallucinations necessitate expert review.

3.3.1.1 Preliminary positioning

Contrasting Theoraizer's approach with this thesis, we add a mediating-variable check to address this limitation of reviewing links in isolation, and we apply hallucination detection and correction methods established in the broader field of LLM hallucination detection to CLD generation (e.g., LLM-as-a-judge, LLM-as-a-corrector, and uncertainty-quantification signals). In terms of using the logprobs-based uncertainty quantification, our approach slightly differs, as our generating model also produces a causal narrative through which variables are related (and which can also be used for post-hoc validation). Each token of the causal narratives carries possible top-k alternatives over which the probabilities are distributed. This allows uncertainty quantification over this whole narrative, which is what this thesis will explore. All of this of course relies on the premise that these uncertainties reflect a model's tendency to hallucinate; this question along with the hallucination reduction techniques used will be discussed later in this review.

3.3.2 Causal parrots: pattern retrieval versus genuine reasoning

However, firstly we still want to address a broader question about what such edge-level pipelines recover from LLMs: are they primarily reproducing patterns that are well represented in training data, or do LLM outputs constitute causal reasoning? Work from the "causal parrots" hypothesis [10] offers a theoretical framework unifying these findings: LLMs encode causal facts within "meta-SCMs" (structural causal models describing relationships between causal facts in training data) but cannot perform genuine interventional or what-if reasoning. When LLMs succeed at causal tasks, they retrieve correlations between causal statements seen during training.

For our purpose, namely LLMs providing initial CLDs that reflect the literature, simply reproducing an accurate reflection of consensus on the causal relationships found in their literature may be enough. For genuine causal reasoning this model may break down and implies that models may accurately reproduce causal structures from well-documented domains that are part of training data while fabricating plausible-sounding but incorrect links in novel domains. The difficulty with this then becomes knowing what is part of the training data and what is not, which is often unclear, and also separate from the equally important questions of whether the model can faithfully reproduce that training data, or if the claims found in that data are true. All of these issues directly motivate hallucination mitigation approaches accompanying LLM-based CLD construction pipelines.

3.4 How Hallucinations Manifest in Causal Structure Generation

Hallucinations in CLD generation take distinct forms compared to free-text generation, and related work treats these failures through different benchmarks, annotations, and mitigation strategies. To synthesise this literature, we organise findings per hallucination type. Across the surveyed work, we summarise five recurring edge/graph-level error categories reported across benchmarks and expert analyses:

Fabricated causal links represent the most direct hallucination type—generating relationships between variables without textual grounding. The ReCAST benchmark [55] directly evaluates multi-node causal graph construction from academic text—a task closely aligned with text-to-CLD causal translation. Unlike synthetic benchmarks, ReCAST uses unstructured, naturally written academic texts of varying length and complexity. Models must identify causal variables and construct directed graphs capturing the causal relationships described. The best-performing model achieves $F_1 = 0.477$, with common errors including fabricated links (e.g., "loan usage flexibility → cash crop cultivation" when no such relationship exists), difficulty with implicit causal information, and failure to connect causally relevant information spread across lengthy texts.

Polarity errors involve incorrect +/− assignment to causal links. This manifests when LLMs state A increases B when it actually decreases B—a failure to apply "ceteris paribus" reasoning correctly. Unlike free-text hallucinations, CLDs require internally consistent sign patterns around feedback loops [54].

Spurious variable introduction creates variables not grounded in source text through over-generalisation. Reinholtz et al. [56] document examples like "Macroeconomic and finance conditions" with unknown polarity, or "Global technology learning"—concepts too vague to support directional interpretation. Their study uses Pipeline Algebra to construct CLDs from text, with expert refinement identifying systematic LLM errors as hallucination mitigation.

Missing causal links represent faithfulness failures where explicit source relationships are omitted. The System Dynamics Bot study [52] found that missing links often occur when "authors assumed readers would know" implicit information—highlighting the challenge of implicit versus explicit causal knowledge in text-to-CLD extraction.

Direction reversal and **DAG violations** constitute structural hallucinations unique to graph outputs. LLMs may reverse cause and effect or create cycles violating directed acyclic graph constraints. Jiralerspong et al. [51], in their pairwise causal query approach to graph discovery, document that LLMs sometimes predict bidirectional causation or inconsistent edge directions when queried about the same variable pair in different orderings. Also in CLD construction, where feedback loops (or lack of them) have a crucial effect on the system's dynamics, this failure mode may be problematic.

3.4.0.1 Formal CLD Hallucination Taxonomies

The **SD-AI benchmark** [54] provides a systematic hallucination taxonomy for text-to-CLD tasks, maintaining a public leaderboard comparing model capabilities.[1] The taxonomy distinguishes: (1) *lack of conformance* (structural/formatting failures), (2) *variable extraction failures*, (3) *spurious edges* (false positives), (4) *missing edges* (false negatives), (5) *polarity errors*, and (6) *wrong causal narratives*. This thesis adopts categories 3–6 as the primary focus, with conformance addressed through structured outputs (JSON schema constraints) with the Judge being a qualitative check on conformance, variable generation not being the focus of this study, with exploratory variable-level analysis relegated to Appendix G. Main experiments use a ground truth given node set to isolate performance on edges to make the direct mapping to ground truths feasible by removing node-level differences from the comparison. The operational definitions and ground truth specifications used in our experiments appear in Section 6.2.

Expert evaluation at MIT [56] produced a complementary practical taxonomy with five categories: (A) ambiguous/overly broad variables, (B) duplicate/syntactically odd labels, (C) mis-specified linkage or polarity, (D) redundant elements, and (E) process-level limitations. These taxonomies provide actionable categories for automated detection systems in text-to-CLD pipelines, and although no comparable analysis for CLD generation approaches has been found (with closest work being the aforementioned), we adopt the SD-AI taxonomy to aid in making evaluation results comparable to text-to-CLD approaches.

3.4.1 CLD Hallucination Mitigation Approaches

Approaches that aim to reduce hallucinations through Retrieval-Augmented Generation show promise but remain constrained by retrieval quality. Zhang et al. [18] introduce LACR, an LLM-assisted causal discovery method that uses RAG over scientific papers to recover causal graphs: for each variable pair it retrieves

[1] SD-AI Leaderboard: https://ub-iad.github.io/sd-ai/#/leaderboard/cld

relevant literature, prompts the LLM to infer (in)dependence and direct vs. indirect association in a constraint-based manner, aggregates these decisions to build the graph skeleton, and then uses the LLM to orient edges. Experiments on benchmark datasets show improved graph recovery and sensitivity to newer literature that may contradict outdated "ground truth" graphs.

Another approach integrates a causal graph into RAG. Causal-graph RAG (CausalRAG) [57] shows promise for reducing hallucinations by retrieving causally relevant context beyond traditional RAG limitations like chunking, but they use the LLM for causal-path inference mainly post-retrieval, and either approach remains constrained by the coverage and quality of the underlying corpus/graph.

3.4.1.1 RAG and the constraints of corpus dependence

There are several failure modes when conditioning an LLM's generation on retrieved evidence. First, generation quality is fundamentally bounded by retrieval: the necessary evidence may be absent from the corpus (coverage/availability), inaccessible due to licensing or indexing constraints, or fragmented by preprocessing choices such as chunking and metadata loss. Second, even when the evidence exists, it may not be retrieved due to query–document mismatch (lexical or semantic mismatch, long-tail terminology, synonymy), embedding or retriever bias, suboptimal query formulation, or limitations of top-k recall under tight context budgets. Third, retrieval can return partially relevant or misleading context—e.g., semantically similar but causally irrelevant passages, outdated or contradictory sources, or redundant snippets that crowd out decisive evidence—so the model's response is conditioned on noise rather than signal. Finally, post-retrieval steps (reranking, summarization, graph/community aggregation) can further distort evidence by compressing away qualifiers and uncertainty, exacerbating omission and bias in the generated answer.

3.4.1.2 Arguing for corpus independence from a practicality standpoint

In CLD construction, if an expert must provide the corpus that is used as input to the CLD pipeline, the role of the LLM is constrained to a translation engine. The texts selected by the expert are presumably already, to some degree, familiar to the researcher, which can limit the model's ability to reduce bias—for example by suggesting hypotheses or search directions grounded in its broader training corpus that the researcher may be unfamiliar with. In contrast, if the model can be used to propose an initial causal hypothesis or search direction, this can save time and may yield new insight, conditional on the approach being reliable enough to improve discovery efficiency on average. Before discussing corpus-independent methods that could increase reliability of LLM-based causal discovery (including hallucination reduction) in more detail in the next part of the literature review, we first conclude with an overview of the surveyed approaches.

3.4.2 Comparison of LLM-Based CLD Approaches

Table 3.1 summarises prior work on LLM-based CLD construction, distinguishing approaches by their input requirements, hallucination mitigation strategies, and evaluation methods.

Table 3.1: Comparison of LLM-based CLD Generation and Validation Approaches

Approach	Method	Input	Corpus Req.	Hallucination Mitigation
SD-Bot [52]	Text-to-CLD extraction via LLM	Text documents	Yes	Conditioning on source text
SD-AI [54]	Benchmark suite (causal translation + conformance)	Text prompts / test cases	Yes	Evaluation benchmark (not a mitigation method)
Theoraizer [34]	Variable-pair querying; multiple candidates	User-defined variables	No	Logprob-based UQ; candidate generation; human in the loop
Liu & Keith [58]	Curated prompting techniques	Dynamic hypothesis text	Yes	Conditioning on source text
Pipeline Algebra [56]	Typed operators; iterative prompting	Abstract + web search	Partial	Expert refinement step
CausalRAG [57]	RAG-enhanced causal discovery	Scientific literature	Yes	RAG grounding (92.9% context precision)
Susanti & Holsm. [59]	Validation (not generation); fine-tuning vs prompting	Existing causal graphs	Yes	Fine-tuning (+20.5 F1 over prompting)
Zhang et al. [18]	RAG-based causal graph discovery	Scientific literature	Yes	RAG grounding
This thesis	Corpus-independent generation + post-hoc validation	Domain spec. + GT Lit node set (V)	**No**	LLM-as-a-Judge; UQ metrics; Deep Research

Overall observations:

Causal translation/extraction approaches outnumber corpus-independent generation-only approaches in the reviewed literature. This review identifies several gaps that inform the thesis approach:

- **Corpus dependency:** Most approaches require an external corpus (text-to-CLD extraction or RAG), limiting applicability when relevant corpora are unavailable. Among the CLD-focused approaches surveyed in this review, Theoraizer [34] and this thesis are notable for operating corpus-independently and for constructing CLD-level outputs, in contrast to adjacent pairwise causal discovery work that primarily targets causal graphs (often DAGs) rather than signed feedback-loop CLDs.
- **Hallucination mitigation:** Prior work largely neglects systematic hallucination detection outside of RAG, especially corpus-independent methods such as LLM-as-a-Judge. Theoraizer [34] uses logprobs for uncertainty quantification but does not report a systematic evaluation of hallucination detection performance. RAG-based approaches (CausalRAG [57], Zhang et al. [18]) rely on retrieval grounding but are capped by retrieval quality. This thesis systematically evaluates multiple corpus-independent detection methods.

- **Evaluation scope:** SD-Bot [52] reports 60% link recall and 66–83% loop recall on controlled datasets. Liu & Keith [58] demonstrate expert-comparable quality on simple structures. Pipeline Algebra [56] presents qualitative case studies. No prior work operationalises multiple definitions of LLM hallucinations in a CLD context (literature ground truth vs. synthetic CLD-specific hallucination types) and compares these systematically while controlling for prompt variation and LLM stochasticity.

Positioning of our approach: Given the prevalence of corpus-dependent text-to-CLD/causal translation pipelines and their reliance on retrieval quality, and the understudy of the (corpus-independent) alternative, CLD generation, combined with recent CLD-specific hallucination taxonomies, provides an opportunity for both quantifying hallucination rates of CLD generation and positioning the approach relative to causal translation. Also, this provides a platform to study whether established corpus-independent hallucination mitigation primitives—introduced later in this review, including LLM-as-a-Judge, LLM-as-a-Corrector, and logprob-derived uncertainty signals—not studied systematically in CLD generation, can reliably detect and remediate CLD hallucination types.

These CLD-specific failure modes and benchmark findings motivate an edge-level generation pipeline paired with post-hoc validation. The generator design evaluated in this thesis is described in Methods (Section 4.2).

Scope disclaimer. While comprehensive ablation of the generator component is beyond the scope of this thesis, we report baseline performance metrics and share selected generator system experimentation conducted as part of the preliminaries to contextualize hallucination detection experiments; however, these results carry their own limitations as detailed in the appendix. It should also be noted that the generator design builds on intellectual contributions from Rick Quax, with implementation contributions from Cillian Hourican and Andrew Crossley, as well as contributions from Nitai Nijholt (as stated in the claim of authorship).

3.5 Prompting strategies for LLM-based causal reasoning

Prompts have been found to be highly influential in determining LLM outputs [15, 16]. Considering the autoregressive mechanism, where each token is generated conditioned on all previous tokens including the prompt, this makes intuitive sense: the prompt establishes the context and reasoning trajectory that the model follows during generation. Even subtle variations in prompt wording, structure, or examples can lead to substantially different outputs, making prompt engineering a critical component of LLM-based systems.

3.5.1 Dominant Prompting Paradigms.

Three dominant paradigms have emerged in the prompting literature:

3.5.2 Few-Shot Prompting.

Few-shot prompting provides the LLM with a small number of input-output example pairs that demonstrate the desired task format and reasoning pattern [16]. By conditioning generation on these examples, models can adapt to novel tasks without parameter updates. Research has shown that few-shot prompting can improve performance by 20–50% on classification tasks, with effectiveness depending on the quality, diversity, and ordering of examples provided. The key insight is that examples serve as implicit task specifications, allowing models to infer the desired behaviour from demonstrated patterns rather than explicit instructions.

3.5.3 Chain-of-Thought (CoT) Prompting.

Chain-of-thought prompting encourages LLMs to generate intermediate reasoning steps before producing a final answer [15]. By decomposing complex problems into sequential reasoning traces, CoT enables models to tackle multi-step reasoning tasks that would otherwise exceed their capabilities. Kojima et al. [60] demonstrated that even zero-shot CoT—achieved by simply appending "Let's think step by step" to prompts—can dramatically improve reasoning performance: accuracy on the MultiArith dataset increased from 17.7% to 78.7%, and on GSM8K from 10.4% to 40.7%. This finding suggests that LLMs possess latent reasoning capabilities that can be elicited through appropriate prompting, without requiring task-specific examples. Structured CoT variants, which impose explicit frameworks for breaking down tasks, have shown additional improvements—up to 16.8% increases in faithfulness to grounding documents.

3.5.4 Role-Based Prompting.

Role-based prompting assigns specific personas or expert roles to guide model behaviour (e.g., "You are a medical expert" or "Act as a causal reasoning specialist"). This approach has been shown to improve zero-shot reasoning capabilities, particularly in domain-specific tasks where role context activates relevant knowledge encoded during pretraining [61, 62]. Role assignment can be combined with other paradigms—for example, instructing a model to "act as a causal scientist and think step by step" combines role-based and CoT approaches.

3.5.5 Prompting in Causal Discovery Contexts.

In the specific context of causal discovery with LLMs, these same paradigms have been employed with varying success. Kıcıman et al. [13] provide a comprehensive overview of LLM capabilities in causal reasoning, demonstrating that prompting strategies significantly affect performance on causal inference tasks. Their work shows that LLMs can leverage domain knowledge encoded during pretraining to make causal judgements, but that this capability is highly sensitive to prompt formulation.

The "statistical causal prompting" (SCP) approach [63] combines statistical causal discovery with knowledge-based causal inference facilitated by LLMs, embedding domain expert knowledge into algorithms through carefully designed prompts. Experiments demonstrate that augmenting statistical causal discovery with LLM-derived knowledge approaches ground truth more closely than methods without prior knowledge integration. Similarly, the Autonomous LLM-Augmented Causal Discovery Framework (ALCM) [64] synergizes data-driven algorithms with LLMs through structured prompting, automating the generation of more accurate and interpretable causal graphs.

However, research has also identified systematic pitfalls. Joshi et al. [12] show that LLMs are prone to fallacies in causal inference, such as inferring causality based on the order of entity mentions or temporal sequences rather than genuine causal structure. These findings suggest that while prompting can improve LLM performance on causal tasks, careful prompt design is essential to avoid eliciting spurious causal claims.

For CLD generation specifically, Liu and Keith [58] demonstrate that curated prompting techniques can produce CLDs of comparable quality to expert-constructed diagrams on simple structures. Their approach uses few-shot examples of CLD construction combined with structured output requirements. However, this work focuses on generation quality rather than hallucination detection, and does not systematically evaluate prompt variants for verification tasks.

3.5.6 Prompting for Hallucination Detection and Correction.

Prompting for hallucination detection represents a distinct challenge from prompting for generation or causal reasoning. While generation prompts optimise for producing coherent, relevant outputs, detection prompts must elicit critical evaluation of potentially erroneous content—a fundamentally different cognitive stance.

Recent work has developed several prompting strategies specifically for hallucination detection:

- **Self-verification prompting:** Methods like Chain-of-Verification (CoVe) [65] prompt models to generate verification questions about their own outputs, answer these questions independently (to avoid confirmation bias), and then revise responses based on verification findings. This approach has shown significant hallucination reduction in free-text generation tasks.
- **Self-contradiction analysis:** By prompting models to generate deliberately contradictory responses and analysing inconsistencies, systems can identify claims where the model exhibits uncertainty or conflicting knowledge—potential hallucination indicators [9, 66].
- **Claim decomposition:** Breaking complex claims into atomic sub-claims enables more granular verification. Each sub-claim can be independently assessed for plausibility, with inconsistencies across sub-claims signalling potential errors [65].
- **Multi-agent critique:** Du et al. [67] show that prompting multiple LLM instances to debate and critique each other's responses reduces hallucinations and improves factual accuracy, as errors are more likely to be identified through adversarial review.

However, none of these approaches have been specifically adapted to the causal discovery setting, where verification requires assessing whether a proposed causal relationship is scientifically plausible given domain knowledge. This gap motivates the development of domain-specific prompting strategies for CLD verification.

3.5.7 Bradford Hill Criteria (BHC) as a Prompting Framework.

To address the gap between general hallucination detection and CLD-specific verification, this thesis proposes a prompting method based on the BHC (BHC) for causal assessment [68]. Originally developed for epidemiological research, the BHC provide a systematic framework for evaluating evidence for causal relationships. The nine criteria are: (1) *Strength*—the magnitude of association; (2) *Consistency*—reproducibility across studies; (3) *Specificity*—specific cause leads to specific effect; (4) *Temporality*—cause precedes effect; (5) *Biological gradient*—dose-response relationship; (6) *Plausibility*—biological or mechanistic plausibility; (7) *Coherence*—alignment with existing knowledge; (8) *Experiment*—experimental evidence supports association; and (9) *Analogy*—similar associations observed elsewhere.

Shimonovich et al. [68] conducted a systematic review of causal inference frameworks, finding that a subset of these criteria—particularly plausibility, temporality, and experiment—are ubiquitous across diverse causal assessment frameworks in epidemiology, social science, and systems thinking. We adapt these findings for CLD verification by focusing on four key criteria:

- **Plausibility** assesses whether the proposed causal mechanism is scientifically reasonable given domain knowledge.
- **Temporality** verifies that the causal direction is appropriate (cause must precede effect).

- **Strength** evaluates the magnitude and significance of the association.
- **Experiment** checks for experimental evidence supporting the association. In cases where direct experimental design is not available (e.g., general knowledge verification), we operationalise **Coherence** as a proxy to check alignment with established theory and empirical findings.

Building on this framework, we develop a "Mechanistic" prompt variant that structures LLM evaluation around these criteria, providing explicit guidance for assessing causal claims. This contrasts with baseline prompts (direct evaluation without scaffolding) and pure CoT prompts (step-by-step reasoning without Shimonovich-specific criteria).

3.5.8 Application to Judge and Corrector Roles.

We apply these prompting principles across different roles and tasks in the CLD verification pipeline:

3.5.9 LLM-as-a-Judge.

For hallucination detection, the judge receives a generated causal edge (including source variable, target variable, polarity, and causal narrative) and must assess its validity. We evaluate three prompt variants:

1. **Baseline**: Direct evaluation instruction without reasoning scaffolding.
2. **CoT**: Chain-of-Thought prompting with step-by-step reasoning instruction [60] and systematic evaluation guidance.
3. **Mechanistic**: Structured multi-criteria evaluation using BHC-aligned criteria (Plausibility, Temporality, Strength, Coherence). In this variant, *Coherence* is operationalised as a proxy for experimental evidence, as direct experimental design details are often not accessible in the general knowledge setting.

For citation judging (assessing whether cited sources support claimed causal relationships), we additionally evaluate a **Mechanistic-Lit** variant. This variant explicitly incorporates the **Experiment** criterion from the BHC framework, replacing Coherence, as the presence of cited literature allows for the direct assessment of study design and experimental evidence.

3.5.10 LLM-as-a-Corrector.

For hallucination correction, the corrector receives edges flagged as potentially problematic by the judge and must determine appropriate remediation actions (retain, revise motivation, change polarity, or remove). Prompt variants follow a cumulative design:

1. **Baseline**: Examples only (control condition).
2. **CoT**: Examples plus step-by-step CoT reasoning instructions (1.45× baseline length).
3. **Mechanistic**: Examples plus CoT plus Bradford Hill causal criteria (1.91× baseline length).

This design enables systematic evaluation of whether domain-specific prompting (BHC) provides value beyond general reasoning scaffolding (CoT) for CLD verification tasks. The experimental results (Chapter 7) compare these variants on both detection accuracy and correction effectiveness.

3.6 LLM-as-a-Judge: Automated Evaluation of LLM Outputs

The **LLM-as-a-Judge** paradigm employs large language models to evaluate the outputs of other LLMs, providing scalable automated assessment where human evaluation is prohibitively expensive or time-constrained [21]. This approach has emerged as a primary evaluation methodology for open-ended generation tasks where traditional metrics (BLEU, ROUGE) fail to capture semantic quality.

3.6.1 Foundations and Empirical Validation

The paradigm was systematically established by Zheng et al. [21], who introduced MT-Bench—a multi-turn benchmark of 80 challenging questions—and Chatbot Arena, a crowdsourced platform for pairwise model comparison. Their key finding was that strong LLM judges (GPT-4) achieve over 80% agreement with human preferences on pairwise comparisons, approaching the level of inter-annotator agreement among humans themselves. This high concordance established LLM-as-a-Judge as a viable proxy for human evaluation at scale.

Subsequent studies have confirmed and extended these findings. Recent comprehensive surveys indicate that LLM-as-a-Judge-derived accuracy metrics align more closely with human evaluations than traditional UQ metrics, achieving up to 80% agreement with human evaluators across diverse tasks [69]. This makes LLM judges strong candidates for evaluating context-aware outputs—such as causal loop diagram edges—especially when expert human evaluation time is scarce.

Reference-free vs. reference-based evaluation. LLM-as-a-judge work has converged on a useful distinction between reference-free and reference-based evaluation [70, 71]. A *correctness judge* is intentionally context-independent: it evaluates whether an edge narrative is internally coherent and mechanistically plausible even when no external evidence is available or when evidence is incomplete/ambiguous. This aligns with intrinsic verification approaches that assess logical consistency and step validity without retrieval—such as critique-and-correction setups [72] and deductive verification of chain-of-thought reasoning [73]. In a causal discovery setting, this judge targets reasoning quality—directionality, polarity, conditionality, and alternative explanations—because semantic topical match alone can miss these causal specifics, and because some plausibility criteria (e.g., Bradford Hill–style coherence/plausibility) are not reducible to "is there a matching quote."

Citation judge as evidence-based verifier. Complementarily, the literature supports a citation judge as a retrieval-augmented, evidence-based verifier that makes judgements about support, not just about relevance. Evidence-grounded paradigms like decompose-then-verify [74] and search-augmented verification [75] operationalise this by checking atomic claims against retrieved passages, while citation-focused work uses natural language inference to evaluate whether cited text actually entails the stated proposition [76]. This maps cleanly onto a judge that operates on retrieved quotes rather than model priors.

3.6.2 Systematic Biases in LLM Judges

However, using one LLM to evaluate another introduces structural biases inherent to the transformer architecture and training process. Research has identified several systematic biases that affect judge reliability:

Position Bias. LLM judges exhibit systematic preferences based on the *order* in which candidate responses are presented. In pairwise comparison settings, judges may favour responses presented first (primacy bias) or last (recency bias), independent of actual quality [21]. This positional preference can flip evaluation outcomes when response order is reversed, introducing noise unrelated to content quality.

Verbosity Bias. LLM judges tend to prefer longer, more detailed responses even when concise answers are equally or more accurate [21, 69]. This bias is particularly problematic for scientific evaluation, where precision and accuracy may be better served by focused claims rather than elaborate explanations. In CLD evaluation, this could manifest as preference for edges with verbose justifications over those with succinct but accurate mechanistic explanations.

Self-Enhancement Bias. LLM judges exhibit a tendency to rate outputs from their own model family more favourably than outputs from other models [69]. This self-preference introduces systematic bias when the same model (or model family) is used for both generation and evaluation, creating a form of circular validation.

Susceptibility to Adversarial Manipulation. LLM-as-a-Judge has been shown to be susceptible to adversarial attacks that exploit these biases [29, 69]. Adversarial prompts can be crafted to inflate judge scores without improving actual output quality, raising concerns about robustness in high-stakes evaluation settings.

3.6.3 Evaluation Modes: Pointwise vs. Pairwise

LLM judges can operate in two primary modes, each with distinct trade-offs:

Pointwise Evaluation. The judge assesses a single response against defined criteria, producing an absolute score (e.g., 0–1 scale) or categorical verdict (correct/incorrect/partially correct). This mode is computationally efficient and enables direct comparison across different evaluation sessions, but requires well-calibrated scoring criteria.

Pairwise Evaluation. The judge compares two candidate responses and selects the better one, often with explanation. This relative comparison is more robust to calibration issues—the judge need only rank, not score—but requires $O(n^2)$ comparisons for n candidates and cannot produce absolute quality estimates.

For CLD hallucination detection, pointwise evaluation is more practical: each edge requires an independent assessment of validity rather than comparison against alternatives. The challenge lies in establishing reliable scoring criteria and thresholds that distinguish hallucinations from valid relationships.

Taken together, this literature supports judge-style verification (and judge-then-correct patterns) as scalable hallucination mitigation primitives. The specific judge and corrector designs evaluated in this thesis are detailed in Methods (Sections 4.4.2, 4.4.3, and 4.6).

Connection to test-time compute. Conceptually, LLM-as-a-Judge can be viewed as a *test-time compute* primitive: it allocates additional inference-time computation to validate (and potentially reject) a candidate output that was produced by a cheaper first-pass generation step. In this thesis, judges are therefore not only evaluation instruments but also building blocks for compute-scaled validation pipelines. We discuss the broader test-time compute perspective—including serial "generate–verify–revise" patterns and parallel sampling/selection strategies that subsume judge-style verification—in more detail in Section 3.10.

3.6.4 LLM-as-a-Corrector: Beyond Detection to Remediation

Extending the judge paradigm, **LLM-as-a-Corrector** uses language models to not only identify errors but actively remediate them [19, 65]. In this setup, a judge first identifies problematic outputs, then a corrector LLM receives the original output along with the judge's critique and generates an improved version.

This judge-then-correct pipeline has shown promise in iterative refinement settings. Chain-of-Verification (CoVe) [65] applies this pattern by: (1) generating a baseline response, (2) planning verification questions,

(3) answering questions independently to avoid bias, and (4) generating a corrected response incorporating verification findings. Similar approaches include Reflexion [19], which maintains a memory of past errors to guide future corrections. Similarly, Self-Refine [77] enables LLMs to improve their own outputs through an iterative feedback-refinement loop using a single model for both generation, feedback, and refinement, reporting average performance gains of ∼20% across diverse tasks without requiring supervised training data.

However, correction introduces its own failure modes. Correctors may introduce new errors while fixing detected ones, particularly when the underlying knowledge is genuinely uncertain. In CLD contexts, a corrector might "fix" a valid but uncommon causal relationship by replacing it with a more stereotypical one, trading one form of error for another. Also, since the corrector receives the Judge's feedback as input, if the Judge makes an error, this will likely propagate to the corrector, which is instructed to adhere to the Judge's instructions.

Connection to test-time compute. LLM-as-a-Corrector instantiates *serial* test-time compute: additional inference-time steps are used to iteratively revise an initial output, conditioned on critique signals (here: judge verdicts/messages). This places the judge–corrector loop within a larger family of compute-scaled refinement methods (e.g., generate–verify–edit and debate-style revision), and clarifies why correction cost should be treated as a resource to allocate selectively rather than uniformly. The broader design space of serial vs. parallel test-time compute strategies—and how these motivate our later Deep Research validation design—is reviewed in Section 3.10.

3.6.5 Implications for CLD Hallucination Detection

The LLM-as-a-Judge and LLM-as-a-Corrector paradigms offer scalable approaches to CLD validation. In the CLD setting, prior work suggests several practical considerations and potential mitigations:

1. **Bias mitigation:** Position and verbosity biases can be partially addressed through prompt engineering (randomizing presentation order, penalizing excessive length) or ensemble methods (aggregating multiple judge calls with varied conditions) [21, 69].
2. **Complementary validation:** Given systematic judge biases and susceptibility to adversarial manipulation, LLM judges are best treated as one signal among others and combined with complementary checks (e.g., intrinsic uncertainty signals and retrieval/embedding-based grounding) rather than used in isolation [29, 69].
3. **Conservative thresholding:** Because high scores can reflect superficial plausibility rather than correctness, edges flagged as low-confidence warrant serious scrutiny, while edges receiving high scores may still benefit from additional validation in high-stakes settings [69].
4. **Domain-specific calibration:** Judge reliability and score calibration can vary with task framing and domain; thresholds and interpretation may therefore require domain-specific tuning and periodic revalidation [69].
5. **Error propagation in correction:** When correctors receive judge feedback as input, judge errors can propagate downstream—an incorrect verdict may lead the corrector to "fix" a valid edge or preserve an invalid one. This coupling implies that corrector effectiveness is bounded by judge accuracy and that correction benefits are context-dependent [19, 65, 77].
6. **Correction-induced errors:** Correctors may introduce new errors while addressing detected ones, particularly when the underlying knowledge is genuinely uncertain [19, 65]. In CLD contexts, a corrector might replace a valid but uncommon causal relationship with a more stereotypical one, trading one error type for another; when available, evidence-grounding can help reduce such failures.

3.6.6 Limitations

Several fundamental limitations constrain the applicability of judge–corrector pipelines for CLD validation:

- **Circular validation risk:** Using the same model (or model family) for generation, judgement, and correction creates potential for circular validation, where systematic biases are reinforced rather than detected [69].
- **Knowledge boundary constraints:** Without access to external evidence (e.g., retrieval or citations), judges and correctors share the same parametric knowledge limitations as generators, and therefore cannot reliably verify claims that fall outside the model's knowledge boundary [6, 7, 9].
- **Causal reasoning limitations:** LLMs exhibit systematic weaknesses in causal reasoning, including sensitivity to variable ordering and susceptibility to spurious correlations [12, 13]. These limitations affect judges assessing causal validity and correctors attempting to repair causal claims.
- **Computational cost:** Multi-step judge–corrector pipelines multiply inference costs. For large-scale CLD generation, this overhead may be prohibitive without selective application guided by uncertainty signals.
- **Evaluation complexity:** Assessing judge and corrector performance requires ground-truth labels, which are expensive to obtain for CLD edges and may themselves reflect expert disagreement rather than objective truth.

Taken together, this literature supports hybrid detection approaches that combine LLM-based judges and correctors with complementary uncertainty and grounding signals, rather than relying on any single mechanism in isolation.

3.7 Logprob Uncertainty Quantification (LUQ) Metrics for Hallucination Detection

In the Background chapter we reviewed autoregressive decoding (including temperature, top-k, and top-p sampling) and the general mechanisms by which autoregressive generation can amplify errors. Here, we focus on *corpus-independent* predictors and grounding signals that can be computed from the generation process and lightweight comparisons, motivating the metrics evaluated in this thesis.

Observing the assigned probabilities over an LLM's output tokens can provide computationally inexpensive uncertainty signals derived directly from the generation process without requiring external knowledge bases or retrieval systems. Relying on LLM-generated signals to quantify uncertainty requires that this model uncertainty is reflected in the model's internal state at some level—i.e., that models 'know what they don't know'—in a manner that is measurable through either the internal state or the outputs, a premise supported by Kadavath et al. [78]. Yet, hallucination is associated with a model asserting something factually incorrect in a confident manner [79], so this uncertainty may not be reflected in the output text itself. It may, however, be reflected in the LLM's output log-probability distribution over tokens, which has been shown to be directly related to hallucinations [14, 80].

While these metrics are directly accessible with white-box model access when running models locally or via open-source model providers, the proprietary OpenAI LLM APIs also expose token-level log-probabilities during

generation, enabling computation of Logprob Uncertainty Quantification (LUQ) metrics using state-of-the-art commercial models (at time of writing, GPT-4.1).

These signals contain useful information that can be employed for, e.g., knowledge distillation, where a student model is trained to replicate a teacher model's output token probability distributions [81], thereby improving the student's performance. Other use cases include autocomplete and evaluation for RAG. The utility of the information contained in output distribution logprobs may also explain why some providers of proprietary models (i.e., Anthropic) do not expose them, whether for data/privacy reasons [82] or to prevent model stealing [83] or distillation. Notably, OpenAI exposes only a truncated probability distribution over the vocabulary, limited to the top k tokens. Nonetheless, logprobs remain available for open-source alternatives and are thus feasible to compute in many cases. As noted, because logprobs are generated as part of the sequence generation process, they are computationally inexpensive and have the additional benefit of providing a measure of uncertainty that is independent of external knowledge bases or retrieval systems. Consequently, if logprobs prove to be useful signals for hallucination detection in a CLD generation context, they would constitute a desirable signal to employ.

3.7.1 The Uncertainty–Hallucination Link.

Crucially, the probability distribution at each generation step provides a window into the model's epistemic state. When the model "knows" the correct continuation, the distribution tends to be peaked; when uncertain, probability mass spreads across alternatives. Xiao and Wang [79] empirically confirmed that higher predictive uncertainty - measured through the token distribution - corresponds to higher hallucination rates. This insight motivates LUQ metrics: by examining the shape of these distributions, we can identify generation steps where the model may be confabulating.

3.7.2 LUQ Metrics: From Sequence to Token Granularity

Token-level probability distributions, accessible through model logits or log-probabilities, provide direct insight into the LLM's confidence during generation. Varshney et al. [14] demonstrated that low-confidence generations correlate with hallucinated content, proposing validation of outputs based on minimum token probability. Their "stitch in time" approach identifies potentially hallucinated spans by detecting tokens where $P(x_i|x_{<i}) < \tau$ for a calibrated threshold τ, enabling early intervention before erroneous content propagates.

Given these token-level probability distributions, we can derive uncertainty features at multiple granularities. We present three complementary metrics, ordered from coarsest (whole-sequence) to finest (single-token) aggregation:

3.7.2.1 Perplexity (Sequence-Level).

Perplexity aggregates uncertainty across the *entire generated sequence*, providing a holistic measure of how "surprising" the output was to the model. Defined as the exponentiated average negative log-likelihood with *length normalization*:

$$\text{PPL} = \exp\left(-\frac{1}{N}\sum_{i=1}^{N}\log P(x_i|x_{<i})\right) \tag{3.1}$$

Length normalization (division by N) ensures comparability across sequences of different lengths, as shorter sequences generally have higher joint probability [80]. Higher perplexity indicates the model found the sequence surprising, potentially signalling confabulation. However, as a whole-sequence average, perplexity can be diluted

by confident tokens surrounding a localized hallucination, and can be inflated by domain-specific terminology or rare but valid tokens.

3.7.2.2 Maximum Window Entropy (Subsequence-Level).

To detect *localized* uncertainty spikes that whole-sequence averaging would obscure, Sriramanan et al. [80] proposed windowed aggregation. They observe that the entire sentence may not be hallucinatory, and that length-average scores may therefore be insufficient for accurate hallucination detection, motivating a windowed approach sensitive to short, localized hallucination spans.

At each token position t, entropy is computed over the top-k alternative tokens:

$$H_k(t) = -\sum_{j=1}^{k} \tilde{p}_j \log \tilde{p}_j \tag{3.2}$$

where $\tilde{p}_j$ represents the normalized probability of the j-th most likely token. We then apply a sliding window of size w to compute rolling averages, returning the maximum:

$$H_{\text{max-win}} = \max_i \left(\frac{1}{w} \sum_{t=i}^{i+w-1} H_k(t) \right) \tag{3.3}$$

This identifies the highest-uncertainty *subsequence* in the generated text - particularly suited for detecting hallucination onset, where uncertainty may spike locally before the model commits to a fabricated claim and in the generated sequence contingent on that claim the model may rise in confidence again.

3.7.2.3 Minimum Probability (Token-Level).

At the finest granularity, minimum probability captures the single "weakest link" in the generation chain:

$$P_{\min} = \min_{i \in [1,N]} P(x_i | x_{<i}) \tag{3.4}$$

Varshney et al. [14] demonstrated that this metric effectively identifies hallucination onset - a single highly uncertain token often marks where fabrication begins. Their "stitch in time" approach uses this signal for early intervention, detecting tokens where $P(x_i|x_{<i}) < \tau$ for a calibrated threshold τ.

3.7.3 Worked Example: Applying LUQ Metrics

To illustrate these metrics, consider an LLM generating the causal claim: *"Stress increases cortisol levels"*. At each token position t, the API returns the probability of the selected token along with probabilities for top-k alternatives. Figure 3.1 visualises a hypothetical generation with $k = 3$:

Position 3 ("cortisol") exhibits lower confidence ($P = 0.45$) and higher distributional uncertainty. Applying the LUQ metrics at each granularity:

- **Perplexity** (Sequence): $\exp\left(-\frac{1}{4}[\log 0.92 + \log 0.88 + \log 0.45 + \log 0.95]\right) \approx 1.60$ - the uncertain token at $t = 3$ is partially masked by confident neighbours.

$t = 1$	$t = 2$	$t = 3$	$t = 4$
Stress	**increases**	**cortisol**	**levels**
$P = 0.92$	$P = 0.88$	$P = 0.45$	$P = 0.95$
Top-k alternatives (normalized $\tilde{p}$):			
Stress (0.85)	increases (0.80)	cortisol (0.50)	levels (0.90)
Chronic (0.10)	elevates (0.15)	adrenaline (0.30)	production (0.07)
Acute (0.05)	raises (0.05)	hormone (0.20)	release (0.03)

Figure 3.1: Token probabilities during LLM generation. Top row: selected tokens with probabilities $P(x_t|x_{<t})$. Below: top-k alternatives with normalized probabilities. Position $t = 3$ shows high uncertainty - alternatives "adrenaline" and "hormone" have substantial probability mass.

- **Max Window Entropy** (Subsequence): With $w = 2$, the window spanning $t = 3, 4$ would show elevated entropy due to position 3, localizing the uncertainty spike.
- **Min Probability** (Token): $\min(0.92, 0.88, 0.45, 0.95) = 0.45$ at $t = 3$ - directly identifying the weakest link.

This example illustrates the complementary nature of the three granularities: perplexity provides a global summary, windowed entropy localizes problematic subsequences, and min probability pinpoints the exact token of concern.

3.7.4 Types of Uncertainty in the Logprob Signal

The uncertainty captured by LUQ metrics is not monolithic. Following the classical decomposition in machine learning [84], uncertainty in autoregressive language models can be categorized into two fundamentally different types:

3.7.4.1 Uncertainty sources in autoregressive generation.

In the context of autoregressive language modelling, these uncertainties arise from distinct sources [85]:

- *Epistemic*: Out-of-distribution inputs, rare token combinations, knowledge gaps, extrapolation beyond training distribution
- *Aleatoric*: Genuinely ambiguous contexts, multiple valid completions, underspecified prompts
  - *Lexical* (sub-case): Synonym choices, paraphrase options, stylistic variation, tokenization alternatives

3.7.4.2 Aleatoric Uncertainty (Data Uncertainty).

This is inherent, irreducible uncertainty arising from ambiguity in the task itself. In language generation, aleatoric uncertainty manifests when multiple valid continuations exist - for instance, completing "The cat sat on the..." could legitimately yield "mat," "sofa," or "windowsill." This uncertainty reflects genuine linguistic variability rather than model ignorance. High entropy at such positions is *not* indicative of hallucination; it reflects the creative freedom inherent in language.

A notable sub-case is *lexical uncertainty* (surface-form variation), where multiple tokens or phrasings convey the same semantic content. When a model shows uncertainty between "large" and "big," or "however" and "nevertheless," this reflects stylistic optionality rather than factual doubt. This distinction is central to semantic entropy approaches [44], which cluster semantically equivalent outputs to separate surface-level lexical uncertainty from deeper semantic uncertainty about factual content. For hallucination detection, lexical uncertainty is largely noise - what matters is whether the model is uncertain about *what to claim*, not *how to phrase it*.

3.7.4.3 Epistemic Uncertainty (Model Uncertainty).

This is reducible uncertainty arising from the model's lack of knowledge - gaps in training data or limitations in model capacity. When a model is asked about events after its knowledge cutoff, obscure facts, or domains underrepresented in training, the resulting uncertainty reflects genuine ignorance. Xiao and Wang [79] demonstrated that epistemic uncertainty is more indicative of hallucination than aleatoric or total uncertainty, as it captures precisely those cases where the model is "guessing" rather than expressing legitimate ambiguity.

The challenge for hallucination detection is that LUQ metrics capture *total* uncertainty without distinguishing its source. A flat probability distribution may indicate either "I don't know" (epistemic - hallucination risk) or "many options are valid" (aleatoric - creative freedom). This conflation is a fundamental limitation of single-generation uncertainty metrics, motivating approaches like semantic entropy [44] that attempt to separate these components through multiple sampling.

3.7.5 Limitations of LUQ Metrics

Despite their theoretical grounding, LUQ metrics face fundamental limitations:

Calibration assumptions. LUQ metrics assume that model confidence correlates with factual accuracy - that uncertainty signals epistemic doubt. However, this calibration is not guaranteed by standard training objectives [78]. Models can be confidently wrong, particularly on memorized but incorrect patterns.

Domain shift. Metric distributions vary with domain. A model may exhibit high perplexity on valid but specialized terminology (e.g., medical or legal jargon), producing false positives. Conversely, fluent hallucinations in the model's "comfort zone" may show low uncertainty.

Confident hallucinations. The hallucination snowballing phenomenon [41] means that once a model commits to a fabricated premise, subsequent tokens may be generated with high confidence as the model maintains internal coherence - despite being factually wrong.

Uncertainty conflation. As discussed in Section 3.7, the logprob distribution conflates epistemic, aleatoric, and lexical uncertainty into a single signal. A high-entropy distribution may indicate genuine ignorance (hallucination risk), legitimate ambiguity (multiple valid continuations), or mere stylistic optionality (synonym choices). LUQ metrics cannot distinguish these cases from a single generation. This conflation means that high uncertainty is a necessary but not sufficient condition for hallucination - the model may be uncertain for benign reasons. Approaches like semantic entropy [44] attempt to address this by sampling multiple generations and clustering semantically equivalent outputs, thereby isolating epistemic uncertainty. However, such methods require multiple forward passes and are computationally expensive compared to single-generation LUQ metrics.

These limitations motivate hybrid approaches combining LUQ metrics with context-sensitive validation (e.g., retrieval-augmented verification) and ensemble classification over multiple uncertainty signals.

3.8 Embedding-Based Grounding Metrics

Complementing LUQ metrics, embedding-based similarity between a candidate text span t and an external reference s provides a context-sensitive grounding check. More generally, for any text span whose support is being assessed (e.g., an LLM output, a human-written claim, or an intermediate system-generated hypothesis), the cosine similarity between the candidate text t and a chosen reference text s can serve as a soft grounding signal:

$$\text{sim}(\mathbf{e}_t, \mathbf{e}_s) = \frac{\mathbf{e}_t \cdot \mathbf{e}_s}{\|\mathbf{e}_t\|\|\mathbf{e}_s\|} \tag{3.5}$$

where $\mathbf{e}_t$ and $\mathbf{e}_s$ are embedding vectors of the candidate text span t and the chosen reference text s, respectively. Low similarity may indicate that the candidate text is weakly supported by (or inconsistent with) the chosen reference—including cases where a claim is paired with an irrelevant or mismatched reference ("attribution hallucination") [27, 86].

3.8.1 Design choices: reference text and input/query text.

The usefulness of embedding-based grounding depends on what is selected as the reference text s and what is embedded as the input/query x (when retrieval is involved). Valid choices for the *reference text* s include: (i) an explicitly cited span or paragraph, (ii) retrieved passages provided as context (e.g., RAG evidence), (iii) a gold-standard evidence snippet in an evaluation dataset, (iv) a full document section (trading precision for coverage), or (v) other machine-generated text intended as evidence (e.g., a judge's retrieved quote). Likewise, the *input/query text* x can be the original user prompt, the model's full answer, an extracted atomic claim, or a structured hypothesis (e.g., a causal edge narrative). In many practical pipelines, x is derived directly from the candidate span t (e.g., $x = t$ or a structured reformulation of t) when the goal is to retrieve evidence for that candidate claim. These choices induce different failure modes (e.g., dilution when embedding long answers, overconstraint when using retrieval context as the only reference) and therefore matter when comparing grounding approaches later in this thesis.

Unlike LUQ metrics, which are purely intrinsic to the generation process, embedding-based grounding requires access to external reference material. This represents both a strength (verification against source text becomes feasible) and a limitation (access to reference material is required). Furthermore, proprietary model providers do not provide access to the training corpus of their language models, making direct source attribution infeasible in such cases, as the mapping between a generated sequence and its original training sources is unavailable.

3.8.2 Retrieval-Augmented Generation

The canonical Retrieval-Augmented Generation (RAG) approach [87] uses an input sequence x to retrieve passages z from a knowledge base $\mathcal{K}$ according to a retrieval objective, typically based on embedding similarity:

$$z^* = \arg\max_{z \in \mathcal{K}} \text{sim}(\mathbf{e}_x, \mathbf{e}_z) \tag{3.6}$$

where $\mathbf{e}_x$ and $\mathbf{e}_z$ denote the embeddings of the query x and candidate passage z, respectively. The retrieved passage(s) are then provided as context to a generator, producing an output sequence y conditioned on both x and the retrieved evidence (e.g., $p_\theta(y \mid x, z^*)$), with retrieval typically parameterized as $p_\eta(z \mid x)$ [87].

Modern LLM web interfaces from providers such as Anthropic, OpenAI, Google, and Perplexity incorporate search-augmented generation, combining iteratively refined web searches with RAG to produce grounded responses [88]. This approach aims to anchor generation in retrievable sources, reducing the risk of hallucination and providing direct attribution.

3.8.3 Limitations of Retrieval-Constrained Generation

A notable limitation of the RAG paradigm is that retrieval may *overconstrain* generation, leading to a failure mode where the model, instructed to condition its response solely on retrieved content, fails to assert a true claim simply because the retrieval process did not surface supporting evidence [89]. This can occur even when an unconstrained model—drawing on its parametric knowledge—might have correctly inferred the relationship, but the retrieved context does not contain (or even contradicts) the relevant support [88].

This failure mode becomes more likely as the source that would confirm the claim may not appear in search results, with the probability of this occurrence inversely related to the depth and breadth of the search. Furthermore, evidence suggests that LLM responses may synthesize knowledge from multiple sources [90, 91], such that a valid claim may not be directly attributable to any single retrievable document. This synthesis property of LLMs exacerbates the recall limitations of RAG-constrained systems, as the supporting evidence for a synthesized claim may be distributed across multiple sources that are not jointly retrievable.

In contrast, unconstrained generation (standard LLM inference without retrieval constraints) does not suffer this recall limitation but is more susceptible to false positives (hallucinations), as the model may generate plausible-sounding but unsupported claims.

3.8.4 A Hybrid Approach: Generate-then-Search

Given the complementary strengths and weaknesses of unconstrained generation and retrieval-augmented validation, we propose a hybrid approach that maximises recall through unconstrained generation while boosting precision through post-hoc retrieval-augmented validation. This approach proceeds as follows:

1. **Hypothesis Generation:** An LLM generates a hypothesized causal link with a narrative justification (e.g., "A causes B through mechanism M").
2. **Evidence Retrieval and Validation:** The hypothesis is submitted as a query to a retrieval system (e.g., web search). An LLM-as-a-judge evaluates whether the retrieved evidence supports the causal narrative.
3. **Iterative Correction:** If the judge validates the claim, the hypothesis is accepted. If not, a corrector revises the causal narrative accordingly before re-evaluation.

This generate-then-search paradigm separates the creative generation phase (optimised for recall) from the critical validation phase (optimised for precision), potentially yielding higher F1 scores with respect to ground truth than either approach alone.

3.8.5 Limitations of the Hybrid Approach

Several limitations of this hybrid approach warrant acknowledgment:

Retrieval dependency. The quality of LLM-as-a-judge validation remains contingent on retrieval system accuracy and the availability of relevant source documents, and even the position of those source texts in the context window [89]. Claims for which no supporting documentation exists in the search corpus cannot be validated, regardless of their truth value.

Confirmation bias. As formulated, the system queries only for *supporting* evidence and does not actively seek counter-evidence. This asymmetry may introduce confirmation bias, where claims are accepted based on partial supporting evidence without consideration of contradictory information [92].

Semantic gap. Embedding-based similarity may fail to capture nuanced semantic relationships. A generated claim and its purported source may have high embedding similarity while differing in critical details, or conversely, may express the same fact in sufficiently different language to register low similarity.

Despite these limitations, when a reasonable match is found between a generated sequence and a source text, this provides supporting evidence for the generated claim—a desirable starting point in scientific contexts, from which further verification or attempts at falsification can proceed.

3.9 Synthesis: Integrating Uncertainty Quantification Approaches

The preceding sections established that LUQ metrics and embedding-based grounding have complementary strengths and limitations: LUQ metrics are corpus-independent but conflate uncertainty types and cannot distinguish confident hallucinations from genuine knowledge, while embedding-based grounding provides external validation but inherits retrieval coverage constraints and cannot capture causal semantics from topical similarity alone. This section articulates how these complementary signals can be combined in a unified hallucination detection framework.

The key insight motivating our evaluation strategy is that neither approach alone suffices for CLD validation. LUQ metrics may flag uncertainty but cannot verify factual correctness; embedding similarity may confirm topical relevance but cannot validate causal direction or polarity. We therefore treat both as lightweight features that complement—rather than replace—more expensive context-sensitive validation. Specifically, we pair these metrics with correctness-based and citation-based generate-then-search judging to assess whether retrieved passages provide causal evidence for an edge narrative.

This layered approach enables adaptive test-time compute allocation: given that UQ metrics are computationally cheaper than LLM-as-a-judge or corrector pipelines, if effective, UQ signals could flag likely hallucinations and trigger deployment of more expensive verification only where needed. The experiments in this thesis quantify how much each component (LUQ, embedding grounding, retrieval, and judging) contributes to hallucination detection and, where effective, to adaptive allocation of test-time verification compute.

The concrete UQ feature set and its use in our experiments are specified in Methods (Section 4.2.1).

3.10 Test-time compute and multi-agent validation

RQ3 investigates whether adaptive test-time compute allocation—triggered by uncertainty signals—can improve CLD accuracy and efficiency. Prior work demonstrates that test-time strategies can reduce hallucinations in reasoning and knowledge-intensive tasks, and that compute-optimal allocation per-prompt yields up to $\sim 4\times$ efficiency gains over uniform sampling [43]. This section synthesizes the main paradigms—serial refinement and parallel sampling—and examines their potential application to CLD generation.

3.10.1 Serial Compute: Iterative Refinement

Serial test-time compute arranges LLM calls sequentially, enabling "divide and conquer" reasoning where each step can reflect on and correct errors from preceding stages. Several serial strategies have demonstrated hallucination reduction in reasoning tasks.

3.10.1.1 ReAct: Reasoning and Acting.

Yao et al. [93] introduced a paradigm that interleaves chain-of-thought reasoning with actions querying external tools (e.g., search APIs, knowledge bases). Unlike pure chain-of-thought prompting—which relies solely on parametric knowledge and is prone to hallucination—ReAct grounds reasoning in retrieved evidence, allowing models to verify claims against external sources. The framework demonstrates that reasoning and action are mutually beneficial: reasoning traces guide action selection, while action observations refine subsequent reasoning, enabling iterative error correction. Empirically, ReAct reduced hallucination rates on knowledge-intensive tasks compared to reasoning-only baselines. The Deep Research pipeline developed in this thesis (Section 4.7) instantiates a ReAct-style approach adapted specifically for causal claim validation.

3.10.1.2 Multi-Agent Debate.

Du et al. [67] introduced a method where multiple LLM instances engage in multi-round debates to enhance response accuracy. Each model proposes and critiques responses over several rounds, leading to a consensus. The authors report that this approach significantly enhances mathematical and strategic reasoning across a number of tasks and improves the factual validity of generated content by reducing fallacious answers and hallucinations. The approach applies directly to existing black-box models using uniform procedures and prompts across different tasks.

3.10.1.3 Chain-of-Verification (CoVe).

CoVe [65], detailed in Section 3.6, exemplifies the generate-verify-correct pattern central to serial test-time compute. While CoVe reduces hallucinations through iterative self-verification, it has not been adapted to verify causal claims—for instance, generating verification questions about whether a proposed relationship $X \rightarrow Y$ is supported by evidence or whether the assigned polarity is correct. This thesis explores whether analogous Judge–Corrector primitives can reduce hallucinations in CLD generation (RQ1).

3.10.1.4 Misconception Refutation (MR).

Estornell and Liu [94] propose editing responses rather than only selecting among them: (1) prompt an LLM to identify errors, misconceptions, and inconsistencies, then (2) prompt it again to minimally correct the response. This approach draws on the insight that LLMs are often more skilled at evaluating answers than at directly providing them. The authors prove theoretically that misconception refutation increases the probability of debate converging to the correct answer. MR is particularly valuable when combined with quality and diversity pruning in multi-agent debate settings.

3.10.2 Parallel Compute: Sampling and Selection

Parallel test-time compute generates multiple candidate outputs simultaneously, then selects or aggregates results.

3.10.2.1 Self-Consistency.

Wang et al. [66] introduced self-consistency, a decoding strategy that samples diverse reasoning paths and selects the most consistent answer by marginalizing over sampled paths. The key insight is that complex reasoning problems typically admit multiple different ways of thinking that lead to the correct answer. Unlike greedy decoding which commits to a single path, self-consistency generates multiple candidates in parallel and aggregates via majority voting. This achieved significant improvements on arithmetic and commonsense reasoning benchmarks: GSM8K (+17.9%), SVAMP (+11.0%), AQuA (+12.2%), StrategyQA (+6.4%), and ARC-challenge (+3.9%).

3.10.2.2 Best-of-N Sampling.

This general strategy generates N candidate responses in parallel and selects the best according to a reward model or verifier. Efficiency gains are possible through compute-optimal allocation strategies that adapt N based on problem difficulty [43]. Extensions include uncertainty-weighted aggregation, where candidate selection incorporates confidence signals such as logprobs. Whether such parallel sampling strategies can improve hallucination detection in CLD generation remains unexplored.

3.10.3 Adaptive Compute Allocation

A key efficiency question is whether expensive verification can be deployed *selectively* based on signals that predict when it is needed. Snell et al. [43] demonstrate that compute-optimal strategies—adapting verification effort per-prompt—yield up to 4× efficiency gains over uniform allocation. RouteLLM [95] routes queries between smaller and larger models based on predicted difficulty, achieving significant cost savings while maintaining quality. More recently, BEST-Route [96] extends this to test-time optimal compute allocation across model cascades. These approaches predict query difficulty to route compute, but do not use generator-side uncertainty metrics (e.g., logprobs, perplexity) as routing signals. This thesis investigates whether such UQ metrics from CLD generation can serve as triggers for selective deployment of expensive verification—whether Judge, Corrector, or Deep Research—addressing RQ3.

Collectively, this literature motivates adaptive test-time compute for hallucination mitigation but reveals that the core primitives (Judge, Corrector) have not been validated for CLD generation, and UQ-triggered selective verification remains unexplored in this domain. This thesis first validates these primitives (RQ1–RQ2) before exploring adaptive allocation strategies (RQ3).

3.11 Synthesis and bridge to methods

Prior work provides many ingredients (LLM-based extraction/generation, judge-style evaluation, uncertainty proxies, retrieval grounding), which have been developed and tested to some level of sophistication and success

in hallucination reduction in other contexts. However, the literature lacks a systematic evaluation for the specific task of hallucination detection in CLD generation.

Rather than choosing breadth, we choose an approach that first thoroughly validates the core agent primitives of LLM-as-a-Judge and LLM-as-a-Corrector in a single-agent setting. If validated individually and together, these primitives form a natural building block for multi-agent systems, supporting an observe-act-react cycle that allows for iterative error correction. While established in the literature, these have not yet been validated in the CLD context; knowledge of how these building blocks function can then inform the design of more complex multi-agent systems, chaining together generation, evaluation, and retrieval as the CLD generation task demands.

Conditional on LLM-as-a-Judge and LLM-as-a-Corrector being effective, and hallucination detection using (L)UQ showing discriminative signal, we can explore allocating expensive verification (judges, correctors, literature search) *selectively*—using these computationally cheap LUQ + cosine signals—rather than applying it uniformly across all candidate edges. The original RQ3 design proposed using UQ metrics as triggers for corrector deployment; the pivot to Deep Research explores an alternative: allocating test-time compute to external evidence validation rather than internal correction.

The relative contribution of LLM-as-a-Judge, LLM-as-a-Corrector, and each metric class—LUQ metrics, embedding-based grounding, and retrieval-augmented validation—to hallucination detection in a CLD generation context remains an empirical question. This thesis addresses this question through systematic evaluation across multiple experimental conditions.

Chapter 4

System Architecture

This section documents all system components: the generator, judges (correctness and citation), corrector, corruptor, and Deep Research system.

4.1 Pipeline Overview

Figure 4.1 presents the complete CLD generation, judging, correction, and validation pipeline. The system operates in four phases:

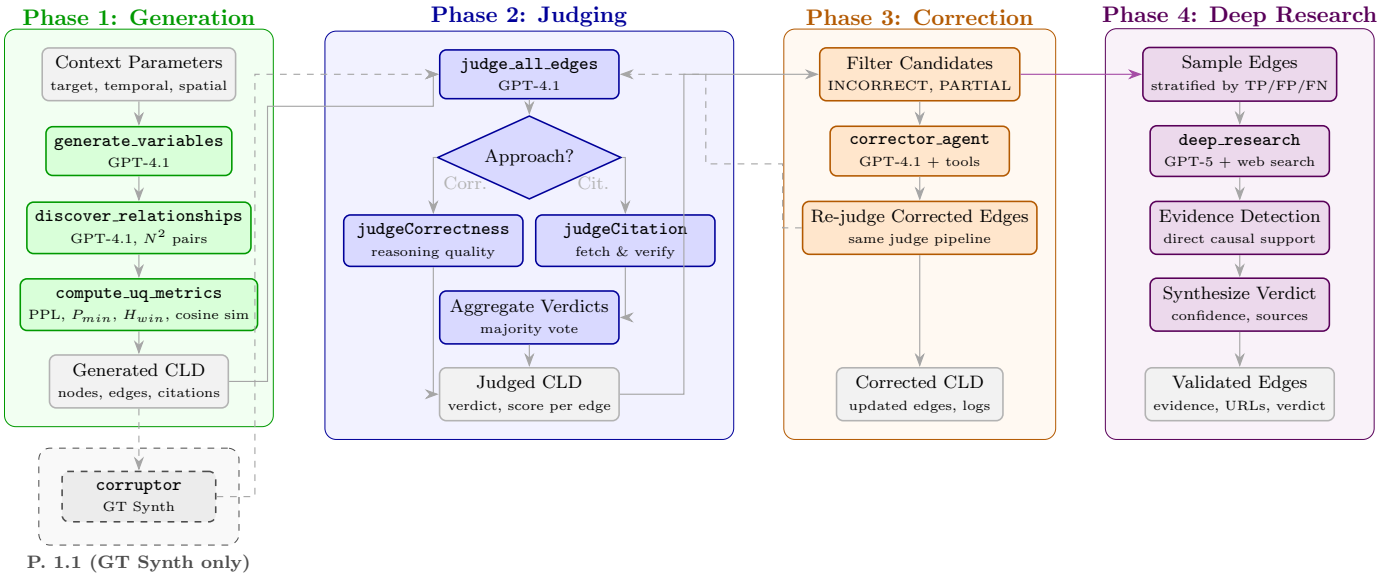


Figure 4.1: Complete CLD pipeline showing four phases: **Generation** (green) creates variables and edges with UQ metrics capture; **Judging** (blue) evaluates edge quality via correctness or citation-based approaches with majority voting; **Correction** (orange) applies LLM-guided fixes to flagged edges followed by re-judging; **Deep Research** (purple) deploys a multi-agent system (GPT-5 + Brave web search) to find direct causal evidence in academic literature—edges are stratified by TP/FP/FN, searched iteratively with claim decomposition, and synthesized into confidence-scored verdicts with supporting URLs. **Optional (GT Synth):** a corruption step (dashed) can be applied after Generation and before Judging to produce `corrupted=True/False` labels for synthetic ground truth.

Phase 1: Generation: The LLM generator (GPT-4.1) constructs CLDs through iterative node and edge generation, capturing uncertainty quantification (UQ) metrics for downstream hallucination detection.

Phase 1.1: (GT Synth only) Corruption: A Corruptor can be applied after Generation and before Judging to introduce synthetic perturbations and emit `corrupted=True/False` labels for known-ground-truth hallucinations.

Phase 2: Judging: An LLM judge evaluates each edge using either correctness-based reasoning assessment or citation-based verification, producing per-edge verdicts.

Phase 3: Correction: An LLM corrector applies tool-based fixes to flagged edges, followed by re-judging to validate corrections.

Phase 4: Deep Research: A multi-agent system (GPT-5 + web search) performs intensive literature validation on stratified edge samples, synthesizing evidence-based verdicts with confidence scores and source URLs.

4.2 Generator

Design rationale: The generator decomposes CLD construction into edge-level decisions with mediation checks and UQ metric capture. For full motivation of the pairwise generation approach, relationship constraints, and connection to prior CLD-to-SDM pipelines, see Appendix F.7. **Scope note:** The generator algorithm design choices (aside from UQ metrics capture) are out of scope for this thesis, as they represent the intellectual contribution of collaborators acknowledged in the Acknowledgements section.

4.2.1 Generator implementation

Input: Domain specification (target phenomenon, spatial/temporal scale, scope), optional seed node set.

Process: The Generator constructs CLDs through iterative node and edge generation (see Appendix I.1 for algorithmic details):

- **Node generation:** Proposes system variables relevant to the domain
- **Edge generation:** For each variable pair, proposes relationship type (`POSITIVE`, `NEGATIVE`, `NONE`) with causal narrative
- **UQ capture:** Records token-level logprobs during generation for downstream uncertainty quantification

Output: Complete CLD (nodes + edges with causal narratives) and per-edge UQ metrics.

Note: In all RQ1 and RQ2 experiments, nodes were seeded with the validation CLD node sets; node generation was disabled to isolate edge-level performance.

Configuration: GPT-4.1, Nitai_C prompt, T=0.7—selected via preliminary studies (Section D.1).

Algorithm reference: Appendix I (Algorithm I.1). **Input parameters (prompt):** Appendix F.8. **Input parameters (configuration):** Appendix C.4.10 (Table C.34).

UQ Metrics Capture

UQ metrics are computed solely from the generator LLM's internal signals during edge generation—they do not require external context. We capture four UQ metrics from generator logprobs and embeddings (see Appendix I.6 for algorithmic details):

Design rationale: UQ metrics are treated as cheap, corpus-independent signals that complement retrieval-based grounding. For the full motivation of why we combine LUQ metrics with embedding/retrieval signals and how they are intended to be used (e.g., flagging/triggering more expensive verification), see Section 3.9.

Table 4.1: UQ metrics captured during edge generation. Conceptual formulas shown; implementation uses capping for numerical stability (see Section 6.5.3).

Metric	Formula	Expected
Perplexity	$\text{PPL} = \exp\left(\frac{1}{n}\sum_{i=1}^{n}\text{nll}_i\right)$	$\uparrow \Rightarrow$ halluc
Min Probability	$P_{\min} = \min_{i\in\{1,\ldots,n\}} P(x_i)$	$\downarrow \Rightarrow$ halluc
Max Window Entropy	$H_{\max\text{-}win} = \max_i \frac{1}{w}\sum_{t=i}^{i+w-1} H_k(t)$	$\uparrow \Rightarrow$ halluc
Cosine Similarity	$\text{sim} = \max_c \cos(\mathbf{e}_{\text{narr}}, \mathbf{e}_c)$	$\downarrow \Rightarrow$ halluc

Where: $\text{nll}_i = -\log P(x_i)$ with $P(x_i) = \exp(\text{logprob}_i)$; $H_k(t) = -\sum_{j=1}^{k} \tilde{p}_j(t) \log \tilde{p}_j(t)$ is entropy over top-k=5 tokens with $\tilde{p}_j(t) = \text{softmax}(\text{logprob}_j(t))$; w=5 is the sliding window size; $\mathbf{e}$ denotes embeddings (768-dim; see Section 6.5.3 for model details).

4.2.2 UQ metrics implementation

Input: Token-level logprobs from generator LLM response (when available via API).

Process: Four metrics are computed from the generator's completion tokens (see Appendix I.6 for algorithmic details):

- **Perplexity:** Average token-level uncertainty with two-stage capping (token NLL capped at 20.0, average capped at 10.0) to prevent outliers from dominating
- **Min Probability:** Confidence floor—minimum token probability across the sequence, capturing worst-case uncertainty
- **Max Window Entropy:** Highest-uncertainty region detected via sliding window (w=5) over per-token entropies (top-k=5 alternatives)
- **Cosine Similarity:** Semantic alignment between narrative and citation chunks (citation experiments only; see Section 6.5.3)

Output: Per-edge UQ feature vector for downstream hallucination classification.

Configuration: Logprobs from OpenAI Chat Completions API (`logprobs=True`, `top_logprobs=5`); embedding model details in Section 6.5.3.

Full implementation details: Exact formulas with numerical stability measures (capping thresholds, sliding window parameters, chunking strategy) are provided in Section 6.5.3.

Algorithm reference: Appendix I (Algorithm I.6).

Token sequence note. When available (OpenAI Chat Completions with `logprobs=True` and `top_logprobs=5`), logprob-derived UQ metrics are computed over the model's *completion* token stream exposed as `choice.logprobs.cont` (a list of per-token entries with fields `token`, `logprob`, and `top_logprobs`). This token stream corresponds to the full assistant completion for the edge call, i.e., the entire JSON string `{"Relationship": "...", "Explanation": "..."}` including JSON syntax and the `Relationship` label. Prompt/input tokens are not included in this API output, so these metrics reflect uncertainty in the generated output string rather than the prompt. For providers/endpoints without logprobs support (e.g., non-OpenAI models or the OpenAI Responses API), logprob-derived metrics are omitted.

4.3 Corruptor (Synthetic corruption for GT Synth)

Input: An LLM-generated base edge table over a fixed node set V (ordered edges (A, B) with relationship label and narrative), a corruption seed, and a target corruption rate $r \in [0, 1]$.

Process: Using an LLM-based hallucination injection methodology inspired by HaluEval [97] for motivation corruption, applying the main edge-level hallucination types found in SD-AI's taxonomy [54] and applying spurious edges and polarity flips programmatically, in 1/3 equal proportions at a corruption rate of 0.3 (meaning we apply corruption to 30% of the edges):

- **Motivation corruption:** replace a valid mechanistic narrative with plausible-but-flawed reasoning while keeping the edge fixed.
- **Spurious edges:** convert `NONE` → `POSITIVE` or `NEGATIVE` to create false positive links.
- **Polarity flips:** invert `POSITIVE` ↔ `NEGATIVE`.

For reproducibility, corruption is seeded; the output table preserves both original and corrupted fields for auditing.

Output: A corrupted edge table with a per-edge binary label `corrupted` and the corresponding corrupted content (relationship label and/or narrative). This dataset is used as GT Synth in RQ1a to evaluate judge detection under known injected errors.

Configuration used in this thesis: `model=gpt-4.1`, `temp=0.7`, `top_p=1`, `rate=0.3`. **Algorithm reference:** Appendix I (Algorithm I.4). **Reproducibility (run commands):** Appendix C.4.

4.4 Judges

4.4.1 Design rationale: LLM-as-a-Judge

We operationalise two judge types—Correctness and Citation—based on prior work on LLM-based evaluation [21]. This section explains the specific scaffolding choices we adopt for CLD edge verification.

The LLM-as-a-Judge system evaluated here incorporates principles from both the hallucination detection literature and causal inference methodology.

We operationalise two judge types: Correctness and Citation-based judging.

Correctness Judge. The correctness judge evaluates whether an edge narrative is internally coherent, and is by design independent of external context. To address the gap between general hallucination detection and CLD-specific verification—where assessment requires evaluating scientific plausibility of causal mechanisms rather than surface-level coherence—we adapt the Bradford Hill Criteria (BHC) from epidemiological causal assessment frameworks [68] as structured evaluation scaffolding. The mechanistic prompt variant operationalises four key BHC criteria (plausibility, temporality, strength, and coherence), combined with Chain-of-Thought prompting to elicit systematic reasoning [15, 60].

Citation Judge. For citation-based judging, we incorporate self-verification patterns from Chain-of-Verification [65] and retrieval-augmented evidence assessment [98], grounding causal claims against retrieved literature. When judging evidence retrieved from multiple sources, this thesis evaluates citations *one at a time* rather than concatenating all citations into a single context window. This keeps the judging context clean and reduces ambiguity: each verdict is conditioned on a single citation's content, avoiding cross-document interference and allowing failure cases to be attributed to specific sources. These separate judge verdicts are then aggregated via majority voting, choosing the mean score as a starting point to reflect average support in the retrieved documents, although alternatives (median, top-k/max, or explicit reranking/selection of the best citation by a verifier model) may be more robust depending on the evidence distribution and retrieval noise (see Discussion).

Following established LLM-as-a-Judge methodology [21], we employ pointwise scoring with thresholds for binary hallucination classification.

4.4.2 Correctness Judge implementation

Input: Variables, relationship type, causal narrative, prompt type.

Process: LLM-as-a-judge [99] assesses edge correctness by verifying logical/scientific coherence of the causal claim (see Appendix I.2 for algorithmic details).

Output Scores: 0=Incorrect, 0.5=Partially correct, 1=Correct.

Configuration: GPT-4.1, T=0 for reproducibility [21, 100].

Prompt Variants

1. **Baseline** (1500 max output tokens): Direct evaluation without scaffolding
2. **Mechanistic** (1500 max output tokens): Structured multi-criteria evaluation using Bradford Hill-aligned framework [68] (Plausibility, Temporality, Strength, Coherence)
3. **CoT** (10000 max output tokens): Chain-of-Thought prompting with explicit step-by-step reasoning [60] and systematic evaluation of causal logic

Full prompt texts are provided in Appendix F.1.

Algorithm reference: Appendix I (Algorithm I.2). **Reproducibility (run commands):** Appendix C.4 (see also Table C.5) and Appendix C.4.1.

4.4.3 Citation Judge implementation

Input: Variables, relationship type, causal narrative, prompt type.

Process: The Citation Judge implements a **generate-then-search** pipeline: it first uses Brave Search to retrieve candidate article URLs for the edge narrative (Algorithm I.8), then fetches the corresponding full text (ContentScraper-only in the thesis configuration; Algorithm I.9), and finally applies an LLM-as-a-judge [99] to determine whether each fetched source supports the causal narrative (see Appendix I.3 for algorithmic details).

Output Scores: 0=Not supported, 0.5=Partially supported, 1=Fully supported.

Aggregate Judge Score: Each retrieved citation is judged independently with a categorical verdict that is mapped to a scalar score:

$$\textsc{Supported} \to 1.0, \quad \textsc{Partially Supported} \to 0.5, \quad \textsc{Contradicted/Unaddressed} \to 0.0.$$

For an edge with C retrieved citations, the aggregate citation-judge score is computed as the mean:

$$\text{aggregate\_score} = \frac{1}{C} \sum_{c=1}^{C} \text{score}_c.$$

Note (mean aggregation vulnerabilities): The mean can be brittle: it is sensitive to the number of retrieved citations, can be dominated by a single strong/weak source, and can be skewed by retrieval/fetch failures (missing-at-random is not guaranteed). It also implicitly treats citations as exchangeable and independent, which may not hold when sources are redundant or correlated.

Qualitative Aggregation: In addition to the numeric score, an LLM aggregator synthesizes a qualitative verdict from the per-citation results (see Algorithm I.3, line 19). This thesis uses only the numeric aggregate scores for quantitative analysis.

Configuration: GPT-4.1 (or GPT-5-mini for cost reduction in GT Synth), T=0 for reproducibility.

Prompt Variants

1. **Baseline** (1500 max output tokens): Direct evaluation without scaffolding
2. **Mechanistic** (1500 max output tokens): Bradford Hill-aligned framework
3. **CoT** (10000 max output tokens): Chain-of-Thought prompting with explicit step-by-step reasoning
4. **Mechanistic-Lit** (1500 max output tokens): Mechanistic prompt enhanced with explicit literature grounding requirements

Full prompt texts are provided in Appendix F.1.

Algorithm reference: Appendix I (Algorithm I.3). **Fetcher algorithm/config:** Appendix I (Algorithm I.9).

Retrieval Mechanism

Step 1 (Search): Brave Search is used to obtain up to 5 candidate citation URLs per edge, optionally filtered by a trusted-domain allowlist (Algorithm I.8).

Step 2 (Fetch): For each URL, full text is fetched using the citation fetcher. In the experiments reported in this thesis, fetching was run in **ContentScraper-only** mode (BeautifulSoup-based fetching with PDF-capable extraction). See Appendix I.9 for the full fetcher pipeline and configuration details.

Retrieval policy (compact summary).

- **Domain filtering (search):** Citation URLs are obtained via Brave Search and filtered with a trusted-domain allowlist (up to 5 URLs retained per edge; Appendix I.8).

- **Query construction:** For each edge, the search query concatenates the `source`, `relationship`, and `target` variables with the generated causal narrative/claim text; queries are truncated to Brave API limits (50 words, 400 characters).
- **Paywalls / partial text:** Full-text fetching uses ContentScraper with local PDF extraction; sources that fail to fetch or yield short/error content are discarded. If no valid full text can be fetched, the edge is assigned a `NO_CITATION` outcome for citation-based judging.

Retrieval Strategy: Generate-then-Search

The Citation Judge follows a **generate-then-search** paradigm: the generator produces edges without retrieval constraints, and the judge subsequently searches for supporting literature. This decouples generation recall from retrieval coverage—the generator leverages its full parametric knowledge, while retrieval serves as post-hoc validation rather than a constraint on generation. See Section 3.8.4 for motivation and limitations of this approach.

4.5 Citation Search (Brave)

Input: An edge claim (source variable, relationship type, target variable, and causal narrative), plus an optional trusted-domain allowlist and a Brave Search API key.

Process: The citation search component issues a web search query for each edge to obtain candidate citation URLs. Queries concatenate `source`, `relationship`, `target`, and the edge narrative text, and are truncated to Brave API limits (50 words, 400 characters). Results are optionally filtered by a trusted-domain allowlist (e.g., academic/government domains) to bias toward credible sources. In this codebase, Brave search is used during citation filling when no usable citations are returned directly by an LLM (see Algorithm I.8).

Output: Up to 5 candidate citation URLs per edge, stored as `r.citations` for downstream citation fetching and citation-based judging.

Algorithm reference: Appendix I (Algorithm I.8).

4.6 Corrector

4.6.1 Design rationale: LLM-as-a-Corrector

The corrector is implemented as a post-hoc remediation stage conditioned on judge feedback, allowing detection and remediation to be measured separately [19, 65].

In this paragraph, we motivate how prior judge–corrector and iterative refinement frameworks inform the design of the corrector component implemented in this thesis. Motivated by these judge-then-correct frameworks [19, 65, 77], we implement the corrector as a post-hoc remediation phase that is explicitly conditioned on judge feedback: candidate edges are selected from a Neo4j session based on low judge verdicts or low aggregate scores, and the corrector receives the edge $(A \rightarrow B)$, its current relationship type and motivation (causal narrative), and the judge's critique (verdict + message). Crucially, we explicitly separate judging and correction into distinct phases, rather than having a single model perform both tasks simultaneously. This separation follows the principle that task decomposition improves LLM performance on complex reasoning tasks [42], and allows us to measure detection and remediation performance independently—an important methodological requirement when evaluating whether hallucinations are being correctly identified versus successfully repaired. The corrector

then executes one of a small set of auditable actions—either *revise* the motivation while preserving the edge type, or *change_type* by switching the relationship label (e.g., `POSITIVE`/`NEGATIVE`↔`NONE`) while providing a new motivation—mirroring the "generate then refine" pattern in Self-Refine [77] and the final correction step in CoVe [65] while keeping the action space constrained to reduce uncontrolled edits. To mitigate the risk of introducing new errors, we optionally re-run the judge after corrections to re-score the updated edge set, closing a lightweight verify–correct loop. Unlike Reflexion [19], we do not maintain cross-run memory; instead, correction decisions are logged back into the graph (e.g., `corrected`, `correction_action`, `corrector_message`) for reproducibility and downstream analysis. Full algorithmic details and the exact prompt variants used for the corrector are provided in Appendix I (Algorithm I.5) and Appendix F.1.

4.6.2 Corrector implementation

Input: Variables, relationship type, causal narrative, judgement from LLM-as-a-judge.

Process: Changes an edge by (see Appendix I.5 for algorithmic details):

- **a) Changing relationship type:** e.g., `None`→`Positive`, `Negative`→`None`
- **b) Revising motivation:** Rewriting causal narrative with improved reasoning/evidence

Configuration: GPT-4.1, T=0.7 for exploration [77]. Full prompt variants (Baseline, CoT, Mechanistic) are provided in Appendix F.1.

Direction Correction Mechanism Because all pairs are evaluated bidirectionally ($A \to B$ and $B \to A$), direction errors are corrected via type changes:

Example: If $A \to B$ is erroneous (should be $B \to A$): $A \xrightarrow{\texttt{NONE}} B$ (remove wrong) + $B \xrightarrow{\texttt{POS}} A$ (add correct)

Algorithm reference: Appendix I (Algorithm I.5). **Input parameters (prompts):** Appendix F.1.

Temperature defaults. Other temperature defaults (judge T=0.0) were chosen based on established recommendations from the LLM-as-a-Judge literature [21, 100]. The corruptor temperature default was chosen to be the same as the generator temperature (T=0.7) to minimise stylistic distributional shifts, ensuring that hallucinations must be detected based on content rather than stylistic artefacts. The corrector's temperature default was also chosen to be T=0.7 to align with the generator, in line with the Self-Refine framework which also selects similar temperatures for generator and corrector [77].

4.7 Deep Research System

The Deep Research (DR) system is a multi-agent pipeline that iteratively searches scientific literature for evidence supporting or refuting causal relationships.

4.7.1 Design rationale: Deep Research

Deep Research is designed as a multi-agent, test-time compute pipeline for external evidence validation of edge claims. This section motivates the key design choices (decomposition, parallel search, reranking, and quoting) in relation to prior multi-agent and verification patterns.

In this subsection, we motivate how the multi-agent and test-time compute literature informs the Deep Research design choices evaluated in this thesis. The Deep Research system evaluated here is motivated by several recurring principles in the LLM multi-agent design literature. First, it uses iterative refinement to improve search behaviour over time, in the spirit of self-improvement loops such as Self-Refine [77]. Second, it decomposes an edge claim into multiple independent search directions—including a verbatim query of the causal claim—and executes these directions in parallel, consistent with claim-decomposition patterns (e.g., CoVe) [65] and compute-scaling strategies that benefit from parallel exploration [43]. Third, it incorporates an explicit reranking step, which is common in literature search and retrieval systems and benefits from strong embedding models [101, 102]. In our implementation, we initially operate over Brave Search result snippets as lightweight evidence proxies—leveraging the provider's built-in retrieval and ranking via the Brave Search API [103]—and then perform full-text article fetching only for the highest-rated candidates most likely to contain direct causal evidence. Fourth, it adopts an orchestrated agent architecture with judge/corrector style control flow, implemented using typed agent tooling (PydanticAI) [104] and aligned with broader test-time compute perspectives [43]. Finally, it requires explicit passage quoting to support evidence claims, in line with the SAFE framework, which has been shown to reduce hallucinations in retrieval-augmented settings [86]. Collectively, these choices prioritise breadth over minimality, maximizing the probability of finding direct causal evidence and enabling an initial test of whether automated literature search merits further investigation for CLD validation. Full system architecture and algorithmic details are described in Section 4.7.

4.7.2 Deep Research implementation

Input: Causal claim (source variable, target variable, relationship type, causal motivation/narrative).

Process: Multi-agent literature search pipeline with iterative refinement (see Appendix I.11 for algorithmic details):

- **Hypothesis evaluation:** Produces a focused search plan (3–4 queries) plus keywords and confounders; additionally inserts the *verbatim claim* as a search query.
- **Parallel web search:** Runs the queries in parallel; within each query thread, the search agent iteratively calls a web-search tool with a hard search budget (max 5 searches per query thread). Retrieval uses Brave Search with a trusted-domain priority list and query augmentation (appending "scientific study research evidence").
- **Evidence scoring:** Scores each retrieved snippet/source on `relevance_score` and `quality_score` (1–10) with brief methodological justification.
- **Article selection & full-text fetch:** Selects up to 2–3 candidate articles likely to contain direct causal evidence and fetches full text via HTML parsing (BeautifulSoup), truncating to a fixed length cap to avoid token overload.
- **Direct causal evidence detection:** Applies strict criteria for *direct* causal support (e.g., RCTs/experiments, natural experiments, and meta-analyses/systematic reviews confirming causality); requires a verbatim quoted passage from the source.
- **Judgement, refinement, synthesis:** Judges aggregate support (and can propose a modified claim for a subsequent iteration), then synthesizes the final structured verdict.

Output: Final verdict (`supported`/`partially_supported`/`unsupported`/`invalid`), confidence score (1–10), `direct_causal_found` flag, strongest-evidence fields (`strongest_causal_evidence_url`, `..._passage`, `..._study_type`, `..._confidence`), and a list of retrieved sources (`evidence_urls`).

Multi-Agent Architecture

The system is implemented using the PydanticAI framework [104] with typed agent definitions and structured outputs. Seven primary LLM agents are coordinated by a Python orchestrator class in a multi-iteration pipeline (see Appendix I.11 for algorithmic details); the codebase also defines an optional history summariser for token-budget management. The full multi-agent pipeline is documented in Table I.10 in Appendix I.10.

Models: The Deep Research system uses a two-tier model architecture:

- **Heavy model (default: GPT-5):** Used for complex reasoning tasks—hypothesis evaluation, article selection, direct causal evidence detection, evidence judgement, and verdict synthesis.
- **Light model (default: GPT-5-mini):** Used for high-volume tasks—iterative web search and evidence scoring (and optional history summarization).

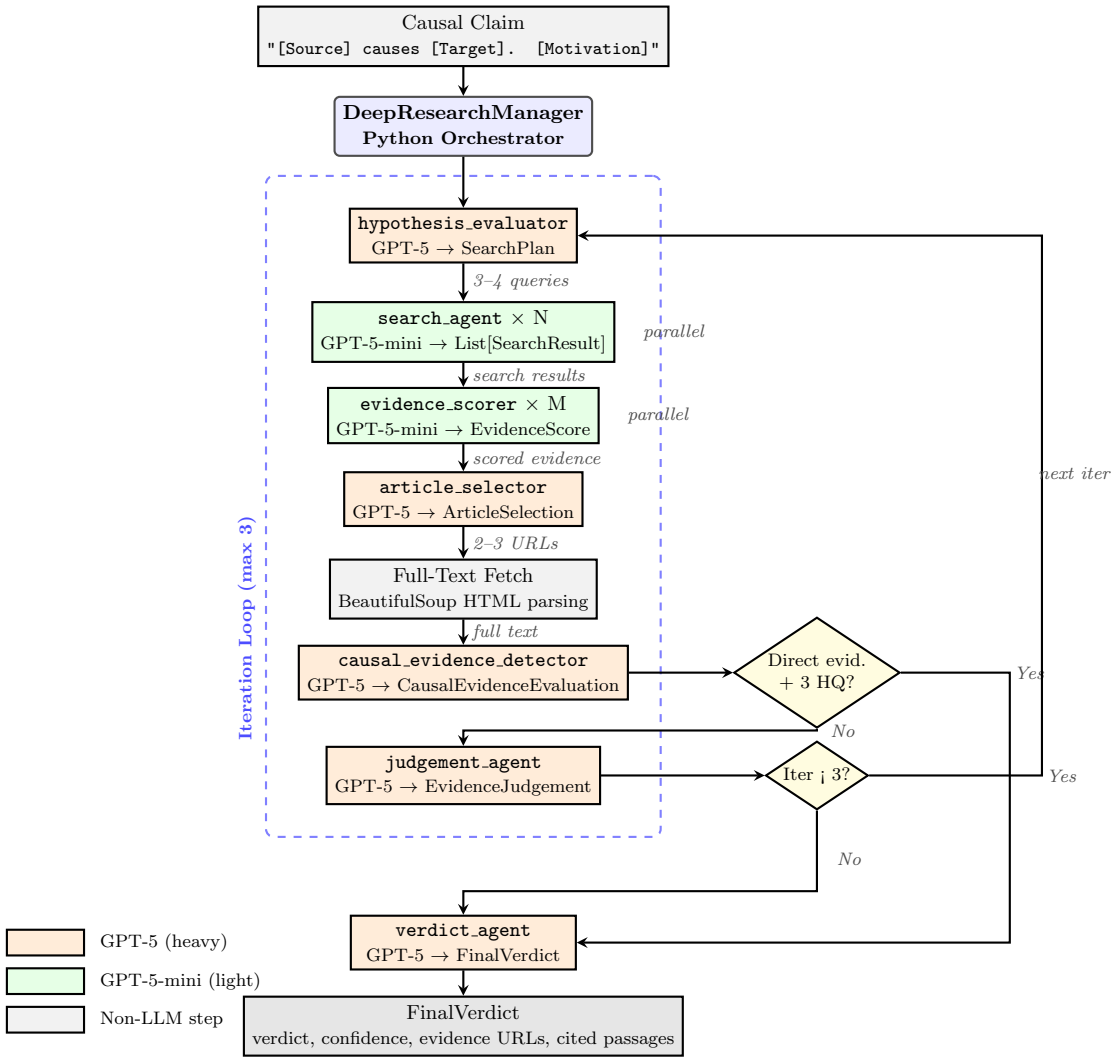


Figure 4.2: Deep Research multi-agent pipeline execution flow. The DeepResearchManager orchestrates seven LLM agents across up to three iterations. Parallel execution (dashed annotations) is used for search and scoring. Early stopping occurs when direct causal evidence is found with 3+ high-quality sources. Orange nodes indicate GPT-5 (heavy model) for complex reasoning; green nodes indicate GPT-5-mini (light model) for high-volume tasks.

Search Infrastructure

Search Provider: Brave Search API with academic domain filtering [103].

Query Enhancement: All queries are automatically appended with "scientific study research evidence" to bias toward academic content.

Retry Strategy: Retry-enabled HTTP sessions for Brave API calls and full-text fetching. Timeouts: 10s connect, 40s read (Brave); 15s for full-text fetch.

Algorithm reference: Appendix I (Algorithm I.11). **Agent components:** Appendix I.10 (Table I.10). **Input parameters (configuration):** Appendix C.4.8 (Table C.26).

Early Stopping Criteria

The system terminates early when any of the following conditions are met:

- **Strong evidence found:** Direct causal evidence detected AND $\geq$3 high-quality sources (relevance ≥ 8, quality ≥ 7)
- **High confidence support:** Judgement supported with confidence ≥ 8
- **Time limit:** 900 seconds (15 minutes) elapsed
- **Max iterations:** Configured iteration limit reached

Chapter 5

Experimental Data

This section documents the CLD domains, context parameters, and validation datasets used across all experiments.

5.1 Evaluation Datasets (GT Lit, LLM Predictions, GT Synth)

Across experiments we distinguish between (i) **ground-truth datasets** that define evaluation labels and (ii) **prediction artefacts** produced by the model that are scored against those labels.

Ground truth datasets

- **GT Lit (literature ground truth):** The published CLDs (nodes and edges) extracted from peer-reviewed sources (Appendix C.1, Table C.1). For each domain, GT Lit defines the validation node set V and the validation edge labels used for scoring (see edge definition below). In GT Lit experiments, hallucinations are operationalised as FP/FN relative to this literature CLD.
- **GT Synth (synthetic corruption ground truth):** A controlled dataset constructed by taking an LLM Prediction edge table produced by Phase 1: Generation (Section 4.2; Figure 4.1) and applying the Corruptor (Section 4.3) to introduce known errors at a fixed corruption rate. As in all edge-level experiments, the node set is fixed to the validation node set V, so corruption is defined over the same ordered-edge universe. Ground truth is the per-edge `corrupted` flag (rather than agreement with GT Lit). GT Synth is used to validate judge detection under fully known, injected error labels.

Model predictions

- **LLM Predictions (generated edges over GT Lit nodes):** For each domain and run seed, we generate a CLD by fixing the node set to the GT Lit validation variables V and generating edges only. These generated edge tables are produced by the Phase 1: Generation pipeline (Section 4.2; Figure 4.1) and are the primary predictions evaluated in RQ1–RQ3 (Figure 6.1). We compute accuracy/FP/FN metrics by comparing these predictions against either (i) GT Lit labels (deviation from literature) or (ii) GT Synth corruption labels when synthetic corruption is applied. Per-RQ output directories for these run artefacts (under `final_runs/`) are listed in Appendix C.1 (Table C.4); file types and naming conventions are in Appendix C.5 (Table C.42).

5.2 Validation and Auxiliary Datasets

In addition to the core CLD datasets above, several auxiliary datasets are used for baselines, validation, and robustness checks:

1. **TruthfulQA verification set:** General-domain hallucination benchmark (1,634 Q&A pairs) used as a judge sanity check prior to CLD evaluation (RQ1; Section D.3). *Location:* Appendix C.1, Table C.2.
2. **Uleman's Alzheimer's CLD [105]:** Neurodegenerative disease model (38 variables, 150 edges) mapping the multicausality of Alzheimer's disease. Constructed via Group Model Building (GMB) with 17 domain experts from neuroscience, geriatrics, and epidemiology. Each edge includes bibliographic citations from the original supplementary materials. Used as external validation for citation-judge reliability and search infrastructure comparison (Section D.6). *Location:* Appendix C.1, Table C.2.
3. **Human–LLM agreement set:** A small human-annotated edge set (72 edges) used to assess agreement between human judgements and citation-judge scores (Section D.6.2). *Location:* Appendix C.1, Table C.2.
4. **Physics CLDs:** Four curated CLDs (26 nodes, 31 edges total) derived from unambiguous physical laws, used as a sanity check for judge behaviour under clear causal structure (Section D.6.3):
   - *Thermostat Heating* (7 nodes, 8 edges): Newton's Law of Cooling [106], First Law of Thermodynamics [107]
   - *Predator-Prey* (6 nodes, 8 edges): Lotka-Volterra equations [108, 109]
   - *RC Circuit* (7 nodes, 8 edges): Ohm's Law [110], Kirchhoff's Laws [111]
   - *Water Tank* (6 nodes, 7 edges): Pascal's Law [112], Torricelli's Law [113]

   Location: Appendix C.1, Table C.3.
5. **Deep Research human validation sample:** A stratified subset of Deep Research outputs (50 edges) used for applicability calibration and threshold selection (RQ3 validation; Section D.9). *Location:* Appendix C.1, Table C.2.

Dataset filenames and locations for all auxiliary datasets are summarised in Appendix C.1 (Table C.2); per-experiment output locations are in Table C.4.

5.3 CLD Domains

Three domain-specific Causal Loop Diagrams (CLDs) representing distinct scientific disciplines:

1. **Depressive Symptoms [114]:** Biopsychosocial model (14 variables, 34 edges) capturing feedback loops between cognitive, behavioural, and physiological factors in depressive states. Constructed via triangulation of group model building, literature review, and causal discovery.
2. **Social Norms [115]:** Public health model (10 variables, 12 edges) representing interplay between social norms, body weight perception, and behavioural dynamics. Developed using system dynamics modelling with the HELIUS cohort.
3. **Emergency Department [116]:** Healthcare systems model (34 variables, 66 edges) of factors influencing older persons' ED utilisation patterns. Constructed through group model building with interdisciplinary clinical experts.

The exact ground-truth CLD datasets (including filenames and locations) are listed in Appendix C.1 (Table C.1).

Note on "edges" and validation edges used in evaluation. Throughout Results, an **edge** is defined as an ordered (source, target) pair with a relationship label in {`POSITIVE`, `NEGATIVE`, `NONE`}, i.e., the full directed

pair set with $Edges = |V|(|V| - 1)$ represented by the `All Edges` table. The effective number of ground-truth validation edges used for scoring is the count of rows with `In Validation Graph = Yes` (so $TP+FN$ equals the validation-edge count under this representation).

5.4 Context Parameters for LLM Generation

Each CLD was configured with domain-specific context parameters extracted from the source Excel files. These parameters are provided to the LLM generator to guide causal reasoning within appropriate temporal, spatial, and domain boundaries:

Table 5.1: LLM Context Parameters by CLD (Extracted from Validation Excel Files)

Parameter	Value
Depressive Symptoms CLD	
Target	How biopsychosocial factors interact to influence the development and persistence of depressive symptoms in healthy adults
Spatial Scale	Psychological, social, behavioural, and biological variables
Temporal Scale	Medium- to long-term processes, capturing changes over weeks to months as stressors accumulate and symptoms evolve
Scope	Individual-level stress response mechanisms
Social Norms CLD	
Target	Variations in social norms and cultural perceptions of body weight across different ethnic and gender groups
Spatial Scale	Amsterdam, focusing on specific sociocultural groups within the city
Temporal Scale	Monthly time steps
Scope	Adults aged 18+ from six sociocultural groups in Amsterdam: Dutch, Moroccan, and South-Asian Surinamese men and women
Emergency Department CLD	
Target	Why individuals aged 65 and older in Amsterdam visit the emergency department
Spatial Scale	Local healthcare context and experiences of older adults within this urban setting
Temporal Scale	Hours to days (inferred from healthcare decision-making timeline)
Scope	Patient-level factors, healthcare system factors, social environment, and socioeconomic/contextual influences

For repository paths to the underlying ground-truth Excel files from which these parameters are extracted, see Appendix C.1 (Table C.1).

Following system dynamics principles [1], variables acting on smaller temporal scales than the variable of interest can be averaged out, while variables on larger scales can be treated as constants. By specifying explicit temporal boundaries, the LLM generator is guided to focus on causal mechanisms operating within the relevant timeframe—e.g., acute stressor responses (weeks–months) for depressive symptoms versus monthly social dynamics for obesity prevalence. Similarly, spatial scale constrains whether causal relationships operate at the individual, community, or healthcare system level. The scope parameter further constrains the domain of variables the LLM should consider.

5.5 CLD Selection Rationale

Ground truth CLDs were selected from peer-reviewed publications with explicitly defined causal structures (named variables, directed edges, polarities). The three CLDs span distinct subdomains within health—biopsychosocial health, public health behaviour, and healthcare systems—which aligns with the thesis's *Mechanistic* (Bradford Hill–aligned) prompting framework for evaluating causal claims in health literature [68].

CLD sizes range from 12 to 66 edges (validation sets: Social Norms 12, Depressive Symptoms 34, Emergency Department 66), providing variation in system complexity.

Limitation: All three CLDs fall within the health domain, limiting generalizability to other fields (e.g., economics, ecology, engineering). Additionally, these CLDs represent expert-hypothesized relationships rather than empirically validated causal effects—a common limitation in complex systems where controlled experiments are infeasible [114–116]. Evaluation therefore measures reproduction of published expert models, which may or may not be causally correct.

Construct validity supplement: To address the limitation that health CLDs reflect expert-hypothesised rather than empirically established causality, we include three physics-based CLDs (Thermostat Heating, RC Circuit, Water Tank; Section D.6.3) as supplementary validation cases where causal relationships are unambiguously grounded in mathematical laws. Unlike the health domain CLDs, these systems offer *known* causal structure derivable from first principles—if the judge rejects edges in these "easy" cases, it would indicate fundamental judge failure rather than domain complexity. This design provides a lower bound on expected judge performance and helps disentangle judge functioning from the inherent causal ambiguity present in complex socio-biomedical systems. Due to cost constraints, these physics CLDs were not included in the main RQ1–RQ3 experiments but serve as auxiliary validation benchmarks.

5.6 Edge Structure

Each CLD edge contains: **source/target variables** (named entities with definitions), **relationship polarity** (`POSITIVE`/`NEGATIVE`/`NONE`), **causal motivation** (natural language mechanism explanation), and **scientific citations** (peer-reviewed references). The standardised output artefacts produced per run (including judged Excel files, JSON metrics, and naming conventions) are documented in Appendix C.5 (Table C.42).

Example: Edge row. Table 5.2 shows a single edge as stored in the evaluation spreadsheets, including source/target variables, relationship type, classification against the validation CLD, judge verdict, causal motivation, and retrieved citations.

Source	Target	Type	Class.	Judge	Motivation (excerpt)	Citation(s) (excerpt)
Language barriers	Health literacy	`NEG`	`TP`	Fully supp.	Language barriers reduce access to health information...	`pmc.../PMC9205365`, `pmc.../PMC5091931`, ...

Table 5.2: Example edge row from the evaluation spreadsheets (Emergency Department; Mechanistic-Lit citation judge).

Example: Corrector columns (same row, continued). Table 5.3 shows the corrector-related columns for the same edge. When the judge flags an edge as potentially problematic, the corrector either revises the motivation or changes the relationship type.

Source	Target	...	Action	Original motivation (excerpt)	Corrected motivation (excerpt)
Language barriers	Health literacy	...	revise	Language barriers reduce access to health information...	Language barriers impede comprehension of medical instructions, reducing health literacy...

Table 5.3: Corrector columns for the same edge row (continued). The "..." indicates columns from Table 5.2.

Example: UQ metric columns (same row, continued). Table 5.4 shows the generation-time UQ metrics for the same edge. These logprob-derived signals capture token-level uncertainty: *perplexity* measures average surprisal; *min probability* captures the least confident token; *max window entropy* reflects local uncertainty spikes; and *cosine similarity* measures semantic alignment between the motivation and retrieved citations.

Source	**Target**	**...**	**Perplexity**	**Min Prob**	**Max Entropy**	**Cosine Sim.**
Language barriers	Health literacy	...	1.242	0.041	0.630	0.840

Table 5.4: UQ metric columns for the same edge row (continued). The "..." indicates columns from Tables 5.2–5.3.

Chapter 6

Experimental Design

This chapter defines the experimental task, operationalises hallucination for evaluation, and presents the experimental design for each research question.

6.1 Task, Scope and Input

Task & Scope The primary task evaluated is detecting and mitigating hallucinations in CLD edge predictions. Hallucination detection assigns each edge a binary label (hallucinated vs. non-hallucinated) under a specified ground-truth construct (GT Lit vs. GT Synth). Mitigation mechanisms (corrector / Deep Research) attempt to improve alignment. Generation itself is treated as a given input process and is outside (detailed) evaluation scope. By fixing the node set V to the GT Lit variables and evaluating downstream methods on the resulting edge tables, we isolate detection/mitigation performance from confounds due to differing node sets or generation variance.

Input Experiments operate on ordered edges (A, B) over a fixed node set V obtained from ground-truth literature CLDs. For each edge, an LLM generator outputs a relationship label in {`POSITIVE`, `NEGATIVE`, `NONE`} plus a short mechanistic narrative (and, where applicable, citations). These edge tables constitute the input to all RQ1–RQ3 experiments. Full generation details are in Appendix I (Algorithm I.1); data specifications are in Appendix C.1.

6.2 Hallucination Operationalisation

In this thesis, hallucinations manifest as incorrectness in causal links: erroneous absence (FN) or presence (FP) of links, wrong direction, or wrong causal narrative. We operationalise hallucination under two complementary ground-truth constructs, summarised in Table 6.1:

- **GT Lit** (literature ground truth): Hallucination is defined as FP or FN with respect to a peer-reviewed expert CLD. This evaluates alignment with domain expert consensus.
- **GT Synth** (synthetic ground truth): Hallucination is defined as edges marked `corrupted=True` by an LLM corruptor that introduces controlled perturbations at a 30% rate. This provides fully known error labels for controlled evaluation, connecting to broader hallucination benchmarks such as HaluEval [97].

Results across GT Lit and GT Synth are complementary: GT Synth validates controlled detection of induced errors, while GT Lit evaluates alignment with expert reference CLDs in domain-specific settings. The corruptor component is described in Section 4.3; the full algorithm is in Appendix I.

Table 6.1: Two uses of TP/FP/FN in this thesis (generator scoring vs. judge scoring).

Use	**Positive class (y=1)**	**Ground truth label and interpretation**
Generator edge recovery (vs. GT Lit)	Edge present (`POS`∨`NEG`)	GT Lit provides an expert edge set over fixed nodes V. We score edge existence in ordered slots: FP/FN/TN/TP are with respect to whether an edge is present in the literature CLD. Polarity/narratives are evaluated separately.
Judge hallucination detection	Hallucination	GT Synth: y=1 iff `corrupted=True`. GT Lit: y=1 iff the generator output is misaligned with the expert CLD (i.e., FP∨FN w.r.t. GT Lit).

Table 6.1 provides this clarification because the same TP/FP/FN terminology is used in two distinct evaluation contexts throughout this thesis: generator performance (edge recovery against expert CLDs) and judge performance (hallucination detection against ground-truth labels). For example, when the generator predicts an edge that does not exist in the literature CLD, this is a generator FP (w.r.t. GT Lit). If the judge correctly flags this edge as hallucinated, this constitutes a judge TP (w.r.t. hallucination labels)—the judge correctly detected the generator's false positive.

6.3 Chapter Overview

This chapter follows a top-down structure: we first present a helicopter view of the end-to-end pipeline (Figure 6.1), then describe the experimental design framework (Section 6.4), evaluation metrics (Section 6.5), and statistical analysis procedures (Section 6.6). We provide a master experimental overview table summarising the main components of each experiment (Table 6.2, Section 6.7), followed by reproducibility details (Section 6.9) and validity considerations (Section 6.10). Per-RQ experimental specifications are in Appendix D.4–D.9, placed there to aid readability given the large volume of experiments. Dataset locations are in Appendix C.1.

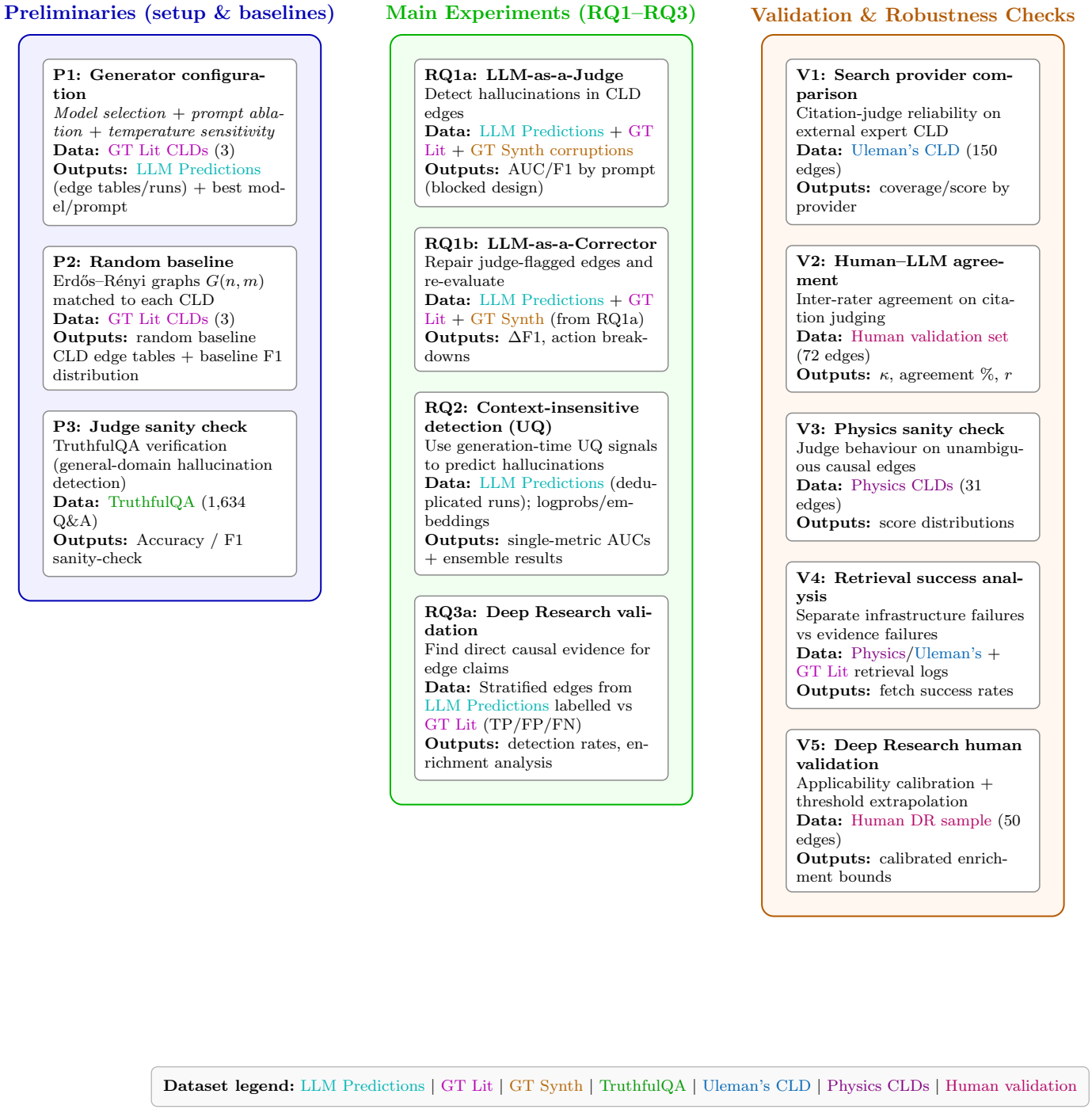


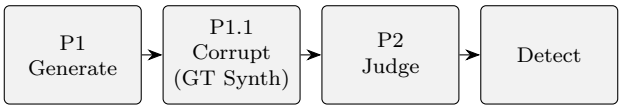
Figure 6.1: End-to-end experimental pipeline overview. The workflow is organised into three phases: **Preliminaries** (configuration selection and baselines), **Main experiments** (RQ1–RQ3), and **Validation** (RQ reliability/robustness checks). Gray boxes indicate datasets and where they enter the pipeline. **Direct dataset links:** GT Lit CLD files are listed in Appendix C.1 (Table C.1); validation/auxiliary datasets (TruthfulQA, Uleman's, Physics, human validation) are in Appendix C.1 (Table C.2); and the per-RQ run outputs (including LLM Prediction edge tables generated in Phase 1 and used throughout RQ1–RQ3) are mapped in Appendix C.1 (Table C.4).

6.4 Experimental Design Framework

Per-RQ Experimental Design (RQ1a, RQ1b, RQ2, RQ3)

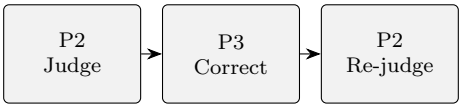
The design framework is instantiated in four main experiments (RQ1a, RQ1b, RQ2, RQ3). We summarise the **unit of analysis**, **treatments**, **datasets**, and **outputs** here to make the per-RQ design explicit (full specifications appear in Sections D.4, D.7, D.8, and D.9). The blocking structure and unit of inference for the main repeated-measures experiments are defined immediately below (Section 6.4).

RQ1a (Judge detection).



Goal: measure whether LLM judges detect hallucinated edges.
Design: a **2×2 factorial design** crossing **Ground Truth type** $\in \{\text{GT Lit}, \text{GT Synth}\}$ with **Judge type** $\in \{\text{Correctness}, \text{Citation}\}$, yielding four experimental cells. Within each cell we use a repeated-measures design with 9 blocks (3 CLDs $\times$ 3 runs), treating prompt variant as the within-block factor.
Unit of Analysis: Block-level (CLD×run) means ($N = 9$).
Datasets: **GT Lit** and **GT Synth** (see Section 5.1). Hallucination: FP/FN (GT Lit) or corrupted status (GT Synth).
Outputs: per-edge judge scores and binary hallucination labels; block-level (CLD×run) aggregates for inference.

RQ1b (Corrector efficacy).



Goal: test whether a corrector improves edge quality on judge-flagged edges.
Design: repeated-measures design with 9 blocks (3 CLDs $\times$ 3 runs); within-block factor = corrector prompt.
Unit of Analysis: Block-level (CLD×run) means ($N = 9$).
Datasets: judge-flagged edges from RQ1a (see Section 5.1 for base datasets).
Outputs: pre/post correction edge tables and re-judged scores; block-level improvements ΔPrecision/ΔRecall/ΔF1 and Δjudge score.

RQ2 (Context-insensitive detection via UQ).

Goal: evaluate whether generator-side uncertainty metrics predict hallucinations without external context.
Design: blocked aggregation with blocks = CLD×run ($N = 9$); inference uses block means.
Unit of Analysis: Block-level (CLD×run) means ($N = 9$).
Datasets: deduplicated **GT Lit** runs (see Section 5.1). Hallucination labels defined as FP/FN w.r.t. GT Lit.

Outputs: per-edge UQ metrics (perplexity, min probability, max window entropy; plus cosine similarity for citation runs), per-file and per-block AUCs, and (optionally) ensemble classifier performance under group-aware splits.

Analysis pipeline (Phases 1–3: single-metric). Phase 1 computes **per-file** AUC-ROC for each UQ metric by treating the raw metric as a univariate classifier score (hallucinated vs. correct edges). Phase 2 aggregates those per-file AUCs into **block-level** AUCs (blocks = CLD×run, $N = 9$). Phase 3 tests above-chance discrimination using a one-sample **Wilcoxon signed-rank** test on (AUC − 0.5) across the 9 blocks and reports mean ± 95% CI over blocks. (Phases 4–6 extend this to multivariate ensembles under group-aware splits; see below and Appendix D.8.0.3.)

RQ3 (Deep Research literature validation).

Goal: test whether the Deep Research system can distinguish valid from hallucinated causal claims by finding **direct causal evidence** in the scientific literature.
Design: exploratory pilot study (single-run, no replication) due to computational cost; analysis is descriptive with confirmatory tests limited to class-wise differences in detection rates (see Section D.9).
Unit of Analysis: Edge-level (descriptive statistics and proportion tests).
Datasets: a stratified sample of edges from RQ1a GT Lit experiments, labelled by GT Lit validation class (TP/FP/FN).
Outputs: per-edge Deep Research verdicts (`direct_causal_found`, confidence, evidence list/URLs) and class-wise detection rates (TP/FP/FN) with associated uncertainty and discrimination tests.

Design Rationale and Unit of Inference

- **GT Lit experiments:** CLD generation varies (seeds 10, 20, 30) → tests judge generalisation across CLD instances
- **GT Synth experiments:** Corruption varies (seeds 1, 2, 3), base CLD fixed → isolates corruption detection ability
- **Seed separation:** Methodological independence between paradigms (10/20/30 vs 1/2/3)

Controlling for confounds. The seed structure explicitly controls for potential confounds. In GT Synth experiments, fixing the base CLD while varying only the corruption seed ensures that performance differences reflect the judge's ability to detect different corruption patterns—not differences in underlying CLD structure. In GT Lit experiments, CLD structure variation is intentional: we deliberately test whether the judge generalises across different valid CLD realizations. The seed separation (10/20/30 for generation vs. 1/2/3 for corruption) maintains methodological independence.

Experiments use a **repeated-measures design** with 9 independent blocks (3 CLDs × 3 runs; seeds 10, 20, 30 for generation; 1, 2, 3 for corruption). Each block receives all treatment levels (prompt variants), enabling within-block comparisons. This structure:

- **Controls for CLD-level variance:** Performance is confounded by domain difficulty (Depressive Symptoms vs. Social Norms vs. Emergency Department); blocking isolates treatment effects.
- **Controls for run-to-run variance:** Different seeds produce different CLD instances; blocking treats each (CLD, run) pair as a subject.

- **Enables within-subject analysis:** Prompt variants or methods are compared within the same block, increasing statistical power.

Unit of inference: The block-level (CLD×run) summary statistic is the unit of analysis for meta-level statistical tests. Where multiple prompt variants exist within a block, prompt is treated as a within-block factor (paired comparisons); for pooled summaries we use the block mean. This yields $N = 9$ independent observations for hypothesis testing.

6.5 Metrics

Definitions and Output Metrics

This section consolidates all metrics used across experiments.

6.5.1 RQ1a Metrics (Judges)

Classification Metrics (Judge Evaluation; RQ1a). To evaluate the discriminative performance of LLM judges, we adopt standard binary classification metrics. The area under the receiver operating characteristic curve (AUC-ROC) quantifies the judge's ability to rank hallucinated edges below correct edges across all possible score thresholds; an AUC of 0.5 corresponds to chance-level discrimination. Precision measures the proportion of edges flagged as hallucinations that are true positives:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{6.1}$$

Recall (sensitivity) quantifies the proportion of actual hallucinations that the judge correctly identifies:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{6.2}$$

The F1 score provides the harmonic mean of precision and recall, balancing both error types:

$$\text{F1} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \tag{6.3}$$

To assess the linear relationship between continuous judge scores and binary hallucination labels, we compute the point-biserial correlation coefficient:

$$r_{pb} = \frac{\bar{s}_1 - \bar{s}_0}{s_s} \sqrt{pq} \tag{6.4}$$

where $\bar{s}_1$ and $\bar{s}_0$ denote the mean scores for hallucinated (y=1) and correct (y=0) edges respectively, s_s is the pooled standard deviation of scores, and $p = \Pr(y{=}1)$, $q = \Pr(y{=}0)$ represent the class proportions. For binary classification, we apply a threshold of ≤ 0.5 on judge scores to predict hallucination. The confusion matrix terms are defined as: $TP = \#\{y{=}1, \hat{y}{=}1\}$ (true positives), $FP = \#\{y{=}0, \hat{y}{=}1\}$ (false positives), $FN = \#\{y{=}1, \hat{y}{=}0\}$ (false negatives), and $TN = \#\{y{=}0, \hat{y}{=}0\}$ (true negatives).

Threshold justification. Scores are ordinal and discrete by design (0.0/0.5/1.0): **0.0** indicates an explicit error, **0.5** indicates a detected partial flaw/uncertainty, and **1.0** indicates full correctness/support. Under our operational definition of hallucination as *edge-level incorrectness*, any detected error (i.e., `INCORRECT` or `PARTIALLY_CORRECT` / `NOT_SUPPORTED` or `PARTIALLY_SUPPORTED`) implies the edge should be flagged as hallucinated, which yields the natural cutpoint ≤ 0.5. We additionally report ROC curves from a discrete threshold

sweep over the only meaningful operating points induced by these three score levels (Appendix J; Figures J.3–J.4, J.1–J.2); for judges that exceed chance discrimination (AUC>0.5), the ≤ 0.5 operating point corresponds to the intended "any detected error" decision rule used throughout the main thresholded Precision/Recall/F1 analyses.

Judge Output Scores. Both judge types produce ordinal scores on a three-point scale. The Correctness Judge assigns per-edge scores of 0.0 (Incorrect), 0.5 (Partially Correct), or 1.0 (Correct), reflecting the degree of logical and scientific coherence in the causal narrative. The Citation Judge evaluates each retrieved citation independently, assigning scores of 0.0 (Not Supported or Contradicted), 0.5 (Partially Supported), or 1.0 (Supported). For edges with multiple citations, we aggregate per-citation verdicts by computing the arithmetic mean. Specifically, for an edge with C successfully retrieved citations, the edge-level score is:

$$\text{score}_{\text{edge}} = \frac{1}{C} \sum_{c=1}^{C} \text{verdict}_c \tag{6.5}$$

where $\text{verdict}_c \in \{0.0, 0.5, 1.0\}$ corresponds to the categorical judgement for citation c.

6.5.2 RQ1b Metrics (Corrector)

Correction Metrics (RQ1b). To evaluate corrector efficacy, we measure the change in classification performance between pre- and post-correction edge sets. The metrics ΔF1, ΔPrecision, and ΔRecall quantify the improvement (or degradation) in alignment with ground truth following correction. Additionally, we compute the change in mean judge score, defined as ΔJudge Score = mean(corrected edges) − mean(original edges), which indicates whether the corrector succeeds in increasing judge confidence independent of ground-truth alignment.

6.5.3 RQ2 Metrics (Uncertainty Quantification)

UQ Metrics (RQ2). We employ four uncertainty quantification metrics derived from generator-side signals, designed to operate without access to external context. Perplexity summarises average token-level uncertainty based on capped negative log-likelihood values. Min Probability identifies the confidence floor by extracting the minimum token probability across the generated sequence. Max Window Entropy detects localised uncertainty spikes using a sliding window over per-token entropies. Finally, Cosine Similarity measures semantic alignment between the generated causal narrative and retrieved citation texts (available only in citation experiments). The computational details for each metric are specified below.

Perplexity computation. Two-stage capping improves robustness and avoids numerical instability: (1) individual token NLL values are capped at 20.0 before averaging,

$$\text{nll}_i = \min(-\log P(x_i),\ 20.0)\,,$$

preventing single outlier tokens ($P < 2 \times 10^{-9}$) from dominating; (2) the average NLL is capped at 10.0, and perplexity is computed as:

$$\text{PPL} = \exp\left(\min\left(\frac{1}{n}\sum_{i=1}^{n} \text{nll}_i,\ 10.0\right)\right),$$

bounding perplexity at $e^{10} \approx 22{,}026$ to prevent extreme values from overwhelming comparisons.

Min Probability. The minimum token probability across the entire generated sequence:

$$P_{\min} = \min_{i \in \{1,\ldots,n\}} P(x_i),$$

representing the confidence floor of the single least-confident token prediction. Low minimum probability indicates at least one token was generated with high uncertainty, which we treat as a candidate indicator of localized uncertainty. This metric is complementary to perplexity: while perplexity captures average uncertainty, min probability captures worst-case uncertainty.

Max Window Entropy. Two-step aggregation detects localized uncertainty spikes: (1) at each token position, compute entropy over top-k=5 alternatives returned by the API,

$$H_k(t) = -\sum_{j=1}^{k} \tilde{p}_j(t) \log \tilde{p}_j(t), \qquad \tilde{p}_j(t) = \text{softmax}(\text{logprob}_j(t));$$

(2) apply a sliding window of size w=5 to compute rolling averages (via `np.convolve`), then return the maximum:

$$H_{\text{max-}win} = \max_i \frac{1}{w} \sum_{t=i}^{i+w-1} H_k(t).$$

This identifies the highest-uncertainty region in the generated text.

Cosine Similarity. Embeddings are computed locally using `all-mpnet-base-v2` [101] (768-dim, 384-token limit). Citation texts are chunked with an adaptive window: `chunk_size` $= \max(256, \min(4000, |\text{motiv.}|))$ characters, 20% overlap, de-duplicated. This adaptive sizing ensures chunks remain within the model's 384-token context (motivations average 56 tokens, max 137; implied chunks 57–176 tokens). For each edge, sim $= \max_c \cos(\mathbf{e}_{\text{narr}}, \mathbf{e}_c)$ over all citation chunks. Available only for citation experiments.

Embedding Model Selection. We selected `all-mpnet-base-v2` based on benchmarking against Qwen3-Embedding-8B (Appendix H; Tables H.1 and H.2): 22× faster (3.0 vs. 67.3 ms/doc) in our setup. Because cosine similarity scoring is throughput-bound in our pipeline, we prioritise latency; published MTEB results are included in Appendix H for context only (not as a direct cross-model comparison).

RQ2 Model Interpretation Metrics (UQ ensembles). To identify which UQ features contribute most to ensemble classifier performance, we employ two complementary model interpretation techniques. Recursive Feature Elimination (RFE) selection frequency measures the proportion of cross-validation folds in which a given feature is retained by the RFE procedure, serving as a stability diagnostic for feature importance. Permutation importance quantifies the decrease in test-set AUC when a single feature is randomly shuffled while all other features are held fixed; larger AUC drops indicate greater classifier reliance on that feature.

Cross-Domain Generalisation Gap (RQ2). When comparing in-distribution vs. leave-one-CLD-out evaluation, we report the AUC drop:

$$\Delta\text{AUC} = \text{AUC}_{\text{in}} - \text{AUC}_{\text{out}}. \tag{6.6}$$

6.5.4 RQ3 Metrics (Deep Research)

Deep Research Metrics (RQ3). The primary outcome for Deep Research evaluation is the detection rate, defined as the proportion of edges for which the system identifies direct causal evidence in the scientific

literature. For an edge class $A \in \{\text{TP}, \text{FP}, \text{FN}\}$, the detection rate is computed as:

$$\text{DR}_A = \frac{1}{|E_A|} \sum_{e \in E_A} \mathbb{I}\{\texttt{direct\_causal\_found}(e) = \texttt{True}\},$$

where E_A denotes the set of edges in class A. To assess discriminative ability, we report pairwise detection rate gaps (in percentage points) between edge classes—specifically, $\text{DR}_{\text{TP}} - \text{DR}_{\text{FP}}$, $\text{DR}_{\text{TP}} - \text{DR}_{\text{FN}}$, and $\text{DR}_{\text{FP}} - \text{DR}_{\text{FN}}$—where gaps exceeding 20 percentage points are interpreted as indicating meaningful discrimination. The Deep Research system also produces a self-assessed confidence score on a 1–10 scale, which we report as class- and CLD-level means. Uncertainty in detection rate estimates is quantified using Wilson score 95% confidence intervals, which provide appropriate coverage for proportions (see Appendix E). Formal discrimination tests employ Fisher's exact test on 2×2 contingency tables for each class pair, with Bonferroni-adjusted p-values to control the family-wise error rate. Effect sizes are reported as Cohen's h for proportion differences and as odds ratios with 95% confidence intervals derived from the contingency tables.

6.5.5 Validation and Auxiliary Metrics

Human Validation Metrics (RQ1a Validation, RQ3). To assess the reliability of LLM judgements relative to human expert opinion, we employ established inter-rater agreement statistics. Cohen's kappa (κ) provides a chance-corrected measure of categorical agreement:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{6.7}$$

where p_o represents observed agreement and p_e denotes expected agreement under chance. Following Landis and Koch [117], we interpret κ values as: <0.20 slight agreement, 0.21–0.40 fair, 0.41–0.60 moderate, 0.61–0.80 substantial, and >0.80 almost perfect agreement. We also report the raw agreement rate (exact match percentage) prior to chance correction. For continuous scores, Pearson's r quantifies the linear correlation between LLM judge scores and human ratings, assessing score-level rather than categorical alignment. When evaluating inter-rater reliability on continuous applicability scores in RQ3, we compute the intraclass correlation coefficient ICC(2,1) under a two-way random effects model with absolute agreement:

$$\text{ICC} = \frac{\sigma^2_{\text{between}}}{\sigma^2_{\text{between}} + \sigma^2_{\text{within}}} \tag{6.8}$$

Following Koo and Li [118], ICC values are interpreted as: <0.50 poor, 0.50–0.75 moderate, 0.75–0.90 good, and >0.90 excellent reliability.

RQ3 Human Validation Metrics (Applicability-Based Calibration). For Deep Research validation, human annotators assign an applicability score on a continuous $[0, 1]$ scale reflecting how closely the retrieved evidence matches the specific edge claim. The scoring rubric is as follows: 1.0 indicates an exact match to the claimed causal relationship; 0.8–0.9 indicates the same relationship with different operationalisations of the variables; approximately 0.7 indicates partial coverage of the claim; approximately 0.6 indicates proxy constructs that do not directly address the claimed variables; and ≤0.5 indicates weak or non-causal evidence. Given these scores, we define coverage at threshold t as the proportion of edges with applicability at or above t:

$$\text{Coverage}(t) = \frac{|\{e : \text{applicability}(e) \geq t\}|}{|E|} \tag{6.9}$$

The correct-causal rate quantifies the proportion of edges passing the applicability threshold that are additionally labelled as genuinely causal by the human validator:

$$\text{Correct-causal}(t) = \frac{|\{e : \text{applicability}(e) \geq t \wedge \text{verdict}(e) = \texttt{causal}\}|}{|\{e : \text{applicability}(e) \geq t\}|} \tag{6.10}$$

Together, coverage and correct-causal rate characterise the precision–recall trade-off as a function of the applicability threshold.

Applicability threshold selection (elbow/gradient rule). To select a single operating-point threshold $t^\star$ for extrapolation, we identify the knee of the trade-off between coverage and correct-causal rate. Let $\mathcal{T} = \{t_1, \ldots, t_m\}$ be a grid of candidate thresholds with adjacent pairs ordered from stricter to looser ($t_h > t_\ell$). For each adjacent segment $t_h \to t_\ell$, define the marginal trade-off score:

$$S(t_h \to t_\ell) = \frac{\text{Coverage}(t_\ell) - \text{Coverage}(t_h)}{|\text{Correct-causal}(t_\ell) - \text{Correct-causal}(t_h)| + \epsilon}, \tag{6.11}$$

where $\epsilon > 0$ is a small constant to avoid division by zero. We select $t^\star$ as the lower (looser) threshold t_ℓ that maximises S, corresponding to the largest gain in coverage per unit loss in correct-causal rate (diminishing returns point).

Aggregation Methods (Generator Evaluation). When aggregating classification metrics across multiple CLDs, we employ two complementary strategies. Micro-averaging pools all true positives, false positives, and false negatives across CLDs before computing Precision, Recall, and F1, thereby assigning equal weight to each edge regardless of which CLD it belongs to. Macro-averaging computes Precision, Recall, and F1 separately for each CLD and then averages these per-CLD scores, giving equal weight to each CLD irrespective of its size. The choice between micro- and macro-averaging affects interpretation when CLD sizes differ substantially. To contextualise random baseline performance, we report graph density—the fraction of possible directed edges that are present in the ground-truth CLD:

$$\text{Density} = \frac{|E|}{|V| \times (|V| - 1)} \tag{6.12}$$

where $|E|$ denotes the edge count and $|V|$ denotes the node count. Denser graphs yield higher random-baseline F1 scores, making density an essential reference point for interpreting generator performance.

Accuracy (TruthfulQA Verification).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{6.13}$$

Used to verify judge implementation on the TruthfulQA benchmark before CLD-specific experiments.

Citation Retrieval Metrics (Validation; Citation Judge). Because citation-based judging depends on successful full-text fetching, we report metrics that distinguish retrieval infrastructure performance from judge evaluation quality. The retrieval success rate (fetch coverage) quantifies the proportion of edges for which at least one citation text was successfully obtained:

$$\text{RetrievalSuccess} = \frac{|\{e : C_e \geq 1\}|}{|E|} \tag{6.14}$$

where C_e denotes the number of usable fetched citations for edge e. The complementary NO_CITATION rate equals $1 - \text{RetrievalSuccess}$, indicating the fraction of edges where retrieval failed entirely. When comparing fetcher configurations, we report the score difference as:

$$\Delta\text{Score} = \bar{s}_A - \bar{s}_B \tag{6.15}$$

representing the difference in mean citation-judge scores between configurations A and B. For retrieval diagnostics, we report an *approval* rate defined as the proportion of edges whose citation-judge aggregate verdict indicates non-zero support (`FULLY_SUPPORTED` or `PARTIALLY_SUPPORTED`, i.e., score ≥ 0.5). To isolate judge behaviour from infrastructure limitations, we additionally report the conditional approval rate computed only over edges with at least one successfully retrieved citation ($C_e \geq 1$). (This approval metric is used only for retrieval-validation reporting and should not be conflated with the hallucination decision threshold, which flags scores ≤ 0.5.)

Computational Efficiency Metrics (Supplementary Analysis). To characterise the cost–accuracy trade-off and parallelisation benefits, we report several efficiency metrics. The ΔF1 per dollar metric quantifies accuracy improvement relative to compute expenditure, typically scaled as ΔF1 per \$100 for interpretability. Relatedly, cost per +0.1 F1 extrapolates the dollar expenditure required to achieve a 0.1-point increase in expected F1 under a specified scaling model. For parallelisation analysis, we measure wall-clock time as the elapsed runtime in seconds for a fixed workload under a given worker count W. Speedup is defined as $\text{Speedup} = T_1/T_W$, where T_1 denotes single-worker runtime and T_W denotes runtime with W workers. Parallel efficiency normalises speedup by the worker count: $\text{Efficiency} = (\text{Speedup}/W) \times 100\%$, indicating how effectively additional workers are utilised.

6.6 Statistical Analysis

Non-Parametric Testing Framework (Main Results)

For **prompt sensitivity** (RQ1a, RQ1b), we employ non-parametric repeated-measures tests because the number of independent replicates is small ($n = 9$ blocks) and distributional assumptions are difficult to verify. Moreover, several primary outcomes are **bounded** (e.g., AUC $\in [0, 1]$, Precision/Recall/F1 $\in [0, 1]$) and judge scores are **discrete** (0.0/0.5/1.0), which can induce non-normality and ceiling/floor effects under small-n aggregation; rank-based tests reduce sensitivity to these issues. The Friedman test serves as the non-parametric analogue of repeated-measures ANOVA, ranking observations within each block and testing whether rank sums differ significantly across treatments (see Appendix E). Post-hoc pairwise comparisons between specific prompts within matched blocks are conducted using Wilcoxon signed-rank tests. Effect sizes are reported as Kendall's W, interpreted as negligible (<0.1), small (0.1–0.3), medium (0.3–0.5), or large (≥0.5). Confidence intervals (95% CIs) that exclude the null value (e.g., 0 for differences, 0.5 for AUC, 1 for odds ratios) indicate statistical significance at $\alpha = 0.05$.

This framework is robust to small sample sizes, non-normality, and outliers. The trade-off is that it primarily answers *whether prompts differ* (in a rank-based sense), but it does not yield the same kind of variance decomposition and sensitivity measures as parametric repeated-measures models.

Sensitivity analysis (supplementary RM-ANOVA). For sensitivity-style interpretation (e.g., "how much variability is explained by prompts"), we also report **Repeated-Measures ANOVA** (RM-ANOVA) with variance decomposition (partial η_p^2; Appendix A). The trade-off is the opposite of the non-parametric approach: RM-ANOVA is more interpretable for sensitivity, but it relies on stronger distributional assumptions

that are hard to validate convincingly at $n = 9$. We therefore treat RM-ANOVA as a *secondary* sensitivity summary (reporting F, df, p, and η_p^2), and keep the non-parametric results as the primary inferential evidence for prompt differences.

RQ2 (single-metric UQ detection): block-level hypothesis tests. For each UQ metric, we treat the raw metric as a univariate classifier score, compute block-level AUC-ROC (blocks = CLD×run, $n = 9$), and test whether discrimination exceeds chance. To remain robust under small-n aggregation, the primary test is a one-sample **Wilcoxon signed-rank** test on (AUC − 0.5) against H_0 : median(AUC − 0.5) = 0 (see Appendix E). This yields **one p-value per metric** (a single test across the 9 block values), not one p-value per block. For uncertainty reporting, we additionally report mean ± 95% CI using the t-distribution over blocks.

For within-domain distribution diagnostics, we compare hallucinated versus correct edge distributions using Mann–Whitney U tests and report rank-biserial r as an effect size; these edge-level results are treated as descriptive diagnostics rather than confirmatory inference. Correlation analyses use Pearson's r (equivalently point-biserial correlation for binary labels) computed at the file level and then summarised at the block level (mean across files within each CLD×run block); block-level correlations are aggregated across blocks using the Fisher z-transformation, and any correlation significance markers likewise correspond to one block-level test per metric (Appendix E).

RQ2 (ensemble UQ detection): group-aware predictive evaluation (Phases 4–6). For multivariate detection, we fit supervised classifiers on the UQ feature vector and evaluate performance under group-aware splits designed to reduce pseudo-replication. Phase 5 uses GroupKFold/GroupShuffleSplit with blocks = CLD×run to estimate in-distribution performance; Phase 6 uses leave-one-CLD-out evaluation to estimate cross-domain generalisation. We report AUC-ROC as a threshold-free ranking metric, and fixed-threshold F1 as an operational metric.

Threshold policy (comparability). Because UQ classifier scores do not have intrinsic semantics, we do not treat 0.5 as a meaningful default threshold under domain shift. Instead, we select a fixed threshold t^* per classifier on Phase 5 **training blocks only** by maximizing out-of-fold F1, and then freeze t^* for Phase 6 evaluation.

No Phase 6 chance test. We do not perform a formal hypothesis test of Phase 6 AUC vs. chance because the leave-one-CLD-out procedure reuses the same CLDs across folds; blocks within a held-out CLD share the same trained model; and training sets overlap heavily across folds. These dependencies violate i.i.d. assumptions, so p-values would be easy to over-interpret. Accordingly, Phase 6 results are interpreted as predictive performance estimates rather than confirmatory hypothesis tests.

RQ3 hypothesis tests (Deep Research discrimination). RQ3 compares **detection rates** (Found vs. Not Found for `direct_causal_found`) between edge classes. The primary test employs Fisher's exact test on 2×2 contingency tables for the TP–FP, TP–FN, and FP–FN class pairs (see Appendix E); p-values are Bonferroni-adjusted within this family. Uncertainty in detection-rate proportions is quantified using Wilson score 95% confidence intervals. Effect sizes are reported as Cohen's h for proportion differences and as odds ratios with 95% confidence intervals derived from the contingency tables.

Multiple Comparison Corrections

Bonferroni corrections control the family-wise error rate, with correction families grouped by research question (see Appendix E). For RQ1a, four omnibus Friedman tests (one per judge×GT setting) yield $\alpha_{\text{adj}} = 0.0125$; post-hoc prompt comparisons within each setting use $\alpha_{\text{adj}} = 0.0167$ for the Correctness Judge ($k = 3$ prompts,

3 pairwise comparisons) and $\alpha_{\text{adj}} = 0.0083$ for the Citation Judge ($k = 4$ prompts, 6 comparisons). For RQ1b, two omnibus Friedman tests yield $\alpha_{\text{adj}} = 0.025$, with post-hoc pairwise comparisons at $\alpha_{\text{adj}} = 0.0167$. For RQ2, four single-metric tests are corrected at $\alpha_{\text{adj}} = 0.0125$. For RQ3, three Fisher's exact tests are corrected at $\alpha_{\text{adj}} = 0.017$.

Scope of correction families. Unless stated otherwise, we apply Bonferroni corrections *within each research question* (RQ) and within the corresponding family of statistical tests listed above. For RQ1a and RQ1b prompt-sensitivity analyses, corrections apply to post-hoc prompt comparisons within each experiment/metric (e.g., a given judge×GT setting and metric such as F1 or AUC). We do not apply a single global correction across all RQs because each RQ tests a distinct hypothesis and is reported as a separate confirmatory family.

Statistical Hypothesis and Assumption Testing

For transparency and reproducibility, the formal hypotheses, assumption diagnostics, and interval/correction procedures for all statistical tests used throughout the thesis are provided in Appendix E.

Effect Size Reporting

Following Cohen [119], all significant findings are accompanied by standardised effect sizes to facilitate interpretation beyond statistical significance. For ANOVA-based analyses, we report Cohen's f, interpreted as negligible (<0.10), small (0.10–0.25), medium (0.25–0.40), or large (≥0.40). For paired differences, we use Cohen's d with thresholds of negligible (<0.20), small (0.20–0.50), medium (0.50–0.80), and large (≥0.80). Cohen's h for proportion differences follows the same interpretive thresholds as d. For correlations, Pearson's r is interpreted as negligible (<0.10), small (0.10–0.30), medium (0.30–0.50), or large (≥0.50).

Uncertainty Reporting Rule

For repeated measurements (across CLDs, runs/seeds, or CV folds), we report **mean ± 95% CI** using the t-distribution with $df = N - 1$. This accounts for variance uncertainty in small samples (e.g., $N = 3$ runs or $N = 9$ blocks). For proportions (e.g., detection rates), we use Wilson score 95% intervals.

Normal t-based confidence interval. For a sample of N observations with mean $\bar{x}$ and sample standard deviation s, the 95% confidence interval is:

$$\bar{x} \pm t_{0.975,\,N-1} \cdot \frac{s}{\sqrt{N}} \tag{6.16}$$

where $t_{0.975,\,N-1}$ is the critical value from the t-distribution with $N - 1$ degrees of freedom. This interval is appropriate for continuous metrics (e.g., AUC, F1) when block-level means are approximately normal.

Wilson score confidence interval. For a proportion $\hat{p} = k/n$ based on k successes in n trials, the Wilson score 95% interval [120] is:

$$\frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}} \tag{6.17}$$

where $z = 1.96$ for 95% confidence. Unlike the normal approximation (Wald interval), the Wilson interval avoids overshoot beyond $[0, 1]$ and provides better coverage for proportions near 0 or 1, making it suitable for detection rates and accuracy metrics.

Interpretation of confidence intervals: Readers should interpret small-N intervals ($N < 10$) as approximate rather than precise; confirmatory conclusions rely on block-based non-parametric tests (Friedman/Wilcoxon) rather than visual CI overlap. CV fold scores are not independent draws (folds share training data), so fold-based intervals should not be interpreted as strict confidence intervals.

Statistical Power. Given the small number of independent blocks ($N = 9$), our main repeated-measures analyses have limited sensitivity to small effects; null results should therefore not be over-interpreted as evidence of no effect. We emphasise standardised effect sizes alongside p-values (Table 7.24, Results).

Computational Efficiency Analysis

Time and cost scaling characterise: (1) CLD size vs. computational requirements, (2) correctness vs. citation cost trade-off, and (3) model efficiency (e.g., GPT-4.1 vs. GPT-5-mini). Since each edge is judged independently, time complexity is $O(E)$ where E is edges; for dense CLDs, $O(N^2)$ w.r.t. nodes.

Cost estimates use the OpenAI API pricing assumed in our cost analysis scripts (December 2025): GPT-4.1 costs \$2.00 per million input tokens and \$8.00 per million output tokens; GPT-5 costs \$1.25/\$10.00 per million input/output tokens; and GPT-5-mini costs \$0.25/\$2.00 per million input/output tokens. These differences make the choice of provider and model tier a major driver of end-to-end cost.

Full scaling results in Appendix C.4.14.

6.7 Master Experimental Overview

Table 6.2 provides a complete overview of all experimental configurations, statistical methods, and analysis scripts across research questions.

Table 6.2: Master Experimental Configuration: Design, Statistical Methods, and Analysis References

RQ	Task	Dataset	GT	Varies	Fixed	Model	Design	N	Unit	Metric	Test	Details	Result Output	Hypothesis
Preliminary & Validation Studies														
Pre	Generator selection	GT CLDs	Lit	Model (3)	3 CLDs, 3 runs	3 LLMs	3C×3r×3m	27	Run	F1/Prec./Rec.	Descr.	§D.1	Tab. 7.1	Best F1 + logprobs
Pre	TruthfulQA verification	TruthfulQA	TQA	Q&A pairs	sj=42	4.1	817q×2v	1634	Q&A	Acc./F1	Descr.	§D.3	Tab. 7.4	Acc > 80%
Val	Uleman's provider study	Uleman's CLD	Expert	Prov., fetcher	150 edges	4.1	4prov×2fetch×150e	1200	Edge	Score, cov.	Descr.	§D.6	Tab. 7.7	Infra. limits
Val	Human–LLM agreement	Human val. set	Human	Rater	72 edges	4.1	72e×2raters	72	Edge	κ, agree.%	κ	§D.6.2	Tab. 7.5	$\kappa > 0.6$
Val	Physics sanity check	Physics CLDs	Phys.	4 CLDs	Laws fixed	4.1	4C×31e	31	Edge	Score	Descr.	§D.6.3	Tab. C.39	Sanity pass
Main Experiments (RQ1–RQ3)														
1a	Correctness judge (Lit)	LLM preds + GT CLDs	Lit	CLD s∈{10,20,30}	Tj=0; sj=42; cr=0%	4.1	3C×3r×3p	27	CLD×run	AUC,F1	Fried.+Wilc.	§D.4.0.1	Fig. 7.4–7.5	H1: Low score⇒halluc.
	Correctness judge (Synth)	LLM preds + GT Synth corruptions	Synth	Corr s∈{1,2,3}	base CLD; cr=30%; Tj=0; sj=42	4.1	3C×3r×3p	27	CLD×run	AUC,F1	Fried.+Wilc.	§D.4.0.2	Fig. 7.2–7.3	
	Citation judge (Lit)	LLM preds + GT CLDs	Lit	CLD s∈{10,20,30}	Tj=0; sj=42; cr=0%	4.1	3C×3r×4p	36	CLD×run	AUC,F1	Fried.+Wilc.	§D.4.0.3	Fig. 7.8–7.9	H2: Low score⇒halluc.
	Citation judge (Synth)	LLM preds + GT Synth corruptions	Synth	Corr s∈{1,2,3}	base CLD; cr=30%; Tj=0; sj=42	5m	3C×3r×4p	36	CLD×run	AUC,F1	Fried.+Wilc.	§D.4.0.4	Fig. 7.6–7.7	
1b	Correctness corrector (Synth)	Judge-flagged LLM preds + GT Synth	Synth	Prompt (3)	baseline judge: Tj=0; sj=42	4.1	3C×3r×3p	27	CLD×run	ΔF1	Fried.+Wilc.	§D.7.0.1	Tab. 7.10	H3: Corr.↑F1
	Correctness corrector (Lit)	Judge-flagged LLM preds + GT CLDs	Lit	Prompt (3)	baseline judge: Tj=0; sj=42	4.1	3C×3r×3p	27	CLD×run	ΔF1	Fried.+Wilc.	§D.7.0.2	Tab. 7.12	
Sens.	Prompt sensitivity analysis	LLM preds + GT CLDs	Lit	Prompt (k=3–4)	blocks=CLD×run	4.1	9blk×kp	9	CLD×run	η_p^2	RM-ANOVA	§6.6	Fig. A.1	Sensitivity
2	UQ (logprobs)	LLM preds (RQ2 dedup)	Lit	CLD s∈{10,20,30}	Tj=0; sj=42	4.1	3C×3r×(3–4)p	63	CLD×run	AUC vs. 0.5	Wilc. (blocks)	§D.8.0.1	Fig. 7.10; Tab. 7.14	H4: ↑PPL/↓Pmin/↑Hwin⇒halluc.
	CosSim (embeddings)	LLM preds (RQ2 dedup)	Lit	CLD s∈{10,20,30}	mpnet	mpnet	3C×3r×4p	36	CLD×run	AUC vs. 0.5	Wilc. (blocks)	§D.8.0.2	Fig. 7.10; Tab. 7.14	H5: Low sim.⇒halluc.
	Ensemble UQ (Phases 4–6)	LLM preds (RQ2 dedup)	Lit	CLD s∈{10,20,30}	blocks=CLD×run; LOCO	4.1	GroupKFold/LOCO	9 blk / 3 CLDs	Block/CLD	AUC, F1@t^*	Pred. eval	§D.8.0.3	Fig. 7.11; Tab. 7.16	—
3	Deep Research	DR outputs	Lit	TP/FP/FN strata	1 iter, 5 src	5/5m	3 CLDs × TP/FP/FN	285	Edge	Det. rate	Fisher	§D.9.0.1	Tab. 7.17; Fig. 7.12	H6: $\text{Det}_{\text{TP}} > \text{Det}_{\text{FP}}$
	Human validation	Val. sample	DR+	$t \in \{0.5–0.9\}$	Stratified	Human	strat. (TP/FP/FN×CLD)	50	Edge	Applic., valid.	Wilson	§D.9.0.5	Fig. 7.13; Tab. 7.21	H7: Thresh⇒valid.
	Smart trigger	DR outputs	Lit	Judge score	mech. ≤ 0.5	4.1+5/5m	triggered DR subset	142	Edge	ΔF1/$	Extrap.	§D.9.0.7	Fig. 7.14	H8: Sel. DR↑eff.

Legend. **RQ:** Pre=Preliminary, Val=Validation, Sens.=Sensitivity. **Models:** 4.1=GPT-4.1, 5m=GPT-5-mini, 5/5m=GPT-5 + GPT-5-mini mix (Deep Research), mpnet=all-mpnet-base-v2, Human=human rater. **GT:** Lit=literature GT, Synth=synthetic corruption (30%), TQA=TruthfulQA, Expert=Uleman's GMB, Phys.=physics laws, DR+=DR output with human applicability. **Datasets:** LLM preds=LLM Prediction edge tables (Phase 1 outputs) used throughout RQ1–RQ3; RQ2 dedup=deduplicated RQ1a GT Lit LLM preds; DR outputs=Deep Research edges; Val. sample=stratified validation sample. **Varies/Fixed:** sj=judge seed (fixed at 42), Corr seed=run number, Tj=judge temperature, cr=corruption rate. **Unit:** CLD×run = block (3 CLDs × 3 runs = 9 independent replicates); Edge = individual causal relationship. **Metrics:** AUC=AUC-ROC, PPL=Perplexity, Pmin=Min Prob, Hwin=Max Window Entropy, κ=Cohen's kappa, η_p^2=partial eta-squared, Det.=Detection Rate. **Tests:** Friedman=omnibus; Wilcoxon=post-hoc (Bonferroni); RM-ANOVA=repeated-measures; t-test=one-sample on block means; Fisher's=exact test; Wilson CI=score interval (see Appendix E). **Multiple-comparison families:** Bonferroni corrections are defined per research question (RQ); for prompt-sensitivity analyses (RQ1a/RQ1b), post-hoc prompt comparisons are corrected within each experiment/metric family (e.g., per judge×GT setting and metric). **Design:** C=CLDs, r=runs, p=prompts, m=models, q=questions, v=variants, e=edges, blk=blocks, prov=providers, fetch=fetchers; strat.=stratified. **Task:** Prov.=Provider. **Symbols:** ↑/↓=increase/decrease; ⇒=implies. **Common:** Generator T=0.7, Judge T=0.0, Corruptor T=0.7. halluc.=hallucination (FP∨FN w.r.t. GT). **Details:** section with experiment design/procedure (see Appendix C for scripts/data). **Result Output:** key Result figure/table links. **Seed structure:** GT Lit uses CLD seeds 10/20/30 (varies generation); GT Synth uses corruption seeds 1/2/3 with base CLD fixed (isolates corruption detection). **Data access:** All datasets, run artefacts, and analysis code are in the project repository (`https://github.com/causalix/causalix.ai`); see also `final_runs/DATA_README.md`.

Companion table: For a consolidated summary of effect sizes, statistical test outcomes, and hypothesis conclusions across all experiments, see Table 7.24 (Results Chapter).

6.8 Detailed Experimental Specifications

For readability, the complete, reproducible specifications for every experiment corresponding to each row of Table 6.2 are provided in Appendix D. Each experiment specification follows a consistent structure:

- **RQ & Goal / Hypotheses:** What the experiment tests and the expected direction of effects.
- **Source Data:** Datasets used and ground truth label definitions (GT Lit vs. GT Synth).
- **Simulation Parameters:** What varies vs. what is fixed (models, prompts, seeds, temperatures).
- **Simulation/Analysis Procedure:** Operational steps (generation → judging → correction).
- **Outputs:** The artefacts produced and where they enter downstream analyses.
- **Metrics / Statistical Analysis:** Reported metrics, unit of analysis, and inference procedure.
- **Reproducibility:** Exact scripts and entry points to rerun the experiment.

Each experiment in the Results chapter (Chapter 7) references its corresponding specification in Appendix D, enabling readers to trace any reported finding back to its full experimental protocol.

6.9 Reproducibility

All analysis scripts and experimental result data required to reproduce the thesis outputs (figures and tables) are available in a public reproduction repository.[1]

- **Unified entry points:** Appendix C.4.1 (Table C.7)
- **Data availability:** Appendix C.1 (Tables C.1, C.2, C.4)
- **Environment setup:** Appendix C.3 (Tables C.9, C.10)
- **Execution commands:** Appendix C.4 (simulation scripts; requires private platform access)

Analysis scripts regenerate all Results-chapter figures and tables from included experimental data without API keys. Re-running the original generation/judging experiments requires access to the private platform repository (see footnote) and OpenAI/Brave Search API keys; see Section 10 (Data Management) for access details and data availability summary.

6.10 Validity Considerations

Our data are highly structured: many edges come from the same generation scenario, and multiple prompt variants are evaluated within the same (CLD×run) block. To mitigate pseudo-replication and leakage, we adopt block-aware design choices: (i) treating (CLD×run) as the independent unit for prompt comparisons ($N = 9$ blocks; Section 6.4); (ii) using group-aware cross-validation splits that keep entire blocks together (GroupKFold / GroupShuffleSplit; Section D.8.0.3); and (iii) treating edge-level results as descriptive diagnostics rather than inferential claims.

Statistical power. With $N = 9$ blocks, statistical power is limited; non-significant results should not be interpreted as evidence of no effect. We prioritise effect sizes and confidence intervals over p-values alone, and interpret null findings conservatively.

[1] Public reproduction repository: https://github.com/NitaiNijholt/cld-hallucination-detection (branch: `thesis-reproduction`). Private platform repository (for re-running generation/judging experiments): https://github.com/CausalixAI/platform (branch: `thesis-nitai`).

A comprehensive discussion of validity threats—including construct validity (what "hallucination" measures in GT Lit vs. GT Synth), external validity (domain, model, and retrieval dependence), conclusion validity (statistical power and multiple comparisons), and reliability concerns (LLM stochasticity and judge biases)—is provided in Section 8.11.

Chapter 7

Results

For reproduction commands and a complete mapping from thesis results to the executable analysis pipelines, see Appendix C (especially Appendix C.4, Appendix C.4.1, and Table C.7).

7.1 Preliminary: Generator Model Comparison

Experimental design: Section D.1.0.1.

Before establishing baseline performance, we compare three frontier LLMs for CLD edge generation: GPT-4.1 (OpenAI), Sonar-Pro (Perplexity), and Claude Sonnet 4 (Anthropic). All models were evaluated on the same three validation CLDs using identical prompts (Nitai_C, T=0.7), with three independent runs per CLD. Full configuration details (including run scripts and parameters) are provided in Appendix C.4.10.

Table 7.1: Generator Model Comparison: Edge Recovery Performance. Bold indicates highest F1 per CLD; in aggregate row, bold indicates highest Precision, Recall, and F1 separately. **Edges processed** refers to the number of ordered (source, target) edge slots evaluated (including `NONE`).

Model	CLD	Edges processed	TP	FP	FN	Precision	Recall	F1
GPT-4.1	Social Norms	270	17	24	16	0.415	0.515	**0.459**
GPT-4.1	Depressive	546	80	106	22	0.430	0.784	**0.556**
GPT-4.1	Emergency Dept	3168	19	356	80	0.051	0.192	0.080
Sonar-Pro	Social Norms	268	14	34	19	0.292	0.424	0.346
Sonar-Pro	Depressive	540	71	162	31	0.305	0.696	0.424
Sonar-Pro	Emergency Dept	3095	65	1184	34	0.052	0.657	**0.096**
Claude-Sonnet-4	Social Norms	270	30	74	3	0.288	0.909	0.438
Claude-Sonnet-4	Depressive	546	96	329	6	0.226	0.941	0.364
Claude-Sonnet-4	Emergency Dept	3168	92	1970	7	0.045	0.929	0.085
Micro-averaged (pooled TP/FP/FN across all CLDs):								
GPT-4.1	Total	3984	116	486	118	**0.193**	0.496	**0.278**
Sonar-Pro	Total	3903	150	1380	84	0.098	0.641	0.170
Claude-Sonnet-4	Total	3984	218	2373	16	0.084	**0.932**	0.154

Note. All models used identical prompts (Nitai_C variant) and temperature (T=0.7). Results pool TP/FP/FN counts across 3 independent runs per CLD (same configuration; stochastic sampling). Nodes (variables) were directly retrieved from the **GT lit CLDs**. **Edges processed** denotes the number of ordered (source, target) edge slots evaluated (including `NONE`), i.e., ideally $|V|(|V|-1)$; it can be slightly lower if some rows are missing due to parsing/processing errors.
Interpretation: GPT-4.1 achieves the best aggregate F1 (0.278).
Claude-Sonnet-4 shows the highest aggregate recall (0.932), while GPT-4.1 has the highest aggregate precision (0.193). Based on best micro-average F1 score, **GPT-4.1 was selected** as the generator model for all subsequent experiments. Additionally, GPT-4.1 is the only model that provides token-level log probabilities (logprobs), enabling confidence-based edge filtering which is necessary for RQ2 experiments.

Table 7.1 shows the edge recovery performance of the three generator models. GPT-4.1 achieves the best aggregate F1 (0.278) through balanced precision-recall. Sonar-Pro performs competitively and wins the Emergency Department CLD (F1=0.096), but its micro-average aggregate F1 is still lower than GPT-4.1 due to substantially more false positives overall. Claude-Sonnet-4 shows highest recall (0.932) but lowest precision (0.084), generating many false positives and yielding a lower aggregate F1. Based on best micro-average F1 score, **GPT-4.1 was selected** as the generator model for all subsequent experiments. Additionally, GPT-4.1 is the only model that provides token-level log probabilities (logprobs), enabling confidence-based edge filtering which is necessary for RQ2 experiments.

7.2 Preliminary: Generator Baseline Validation

Experimental design: Sections D.2 and D.1.0.3.

Having selected GPT-4.1 based on the model comparison above, we now establish its baseline performance. These results show that the CLD generator recovers substantially more ground-truth edges than a structural random baseline (under our GT Lit operationalisation) using the Nitai_C prompt at $T = 0.7$.

7.2.1 Generator Performance

Experimental design: Section D.2.

To confirm the generator produces meaningful causal structure rather than random edge configurations, we compare LLM-generated edges against a Monte Carlo random baseline. Full random-baseline configuration details are provided in Appendix C.4.11.

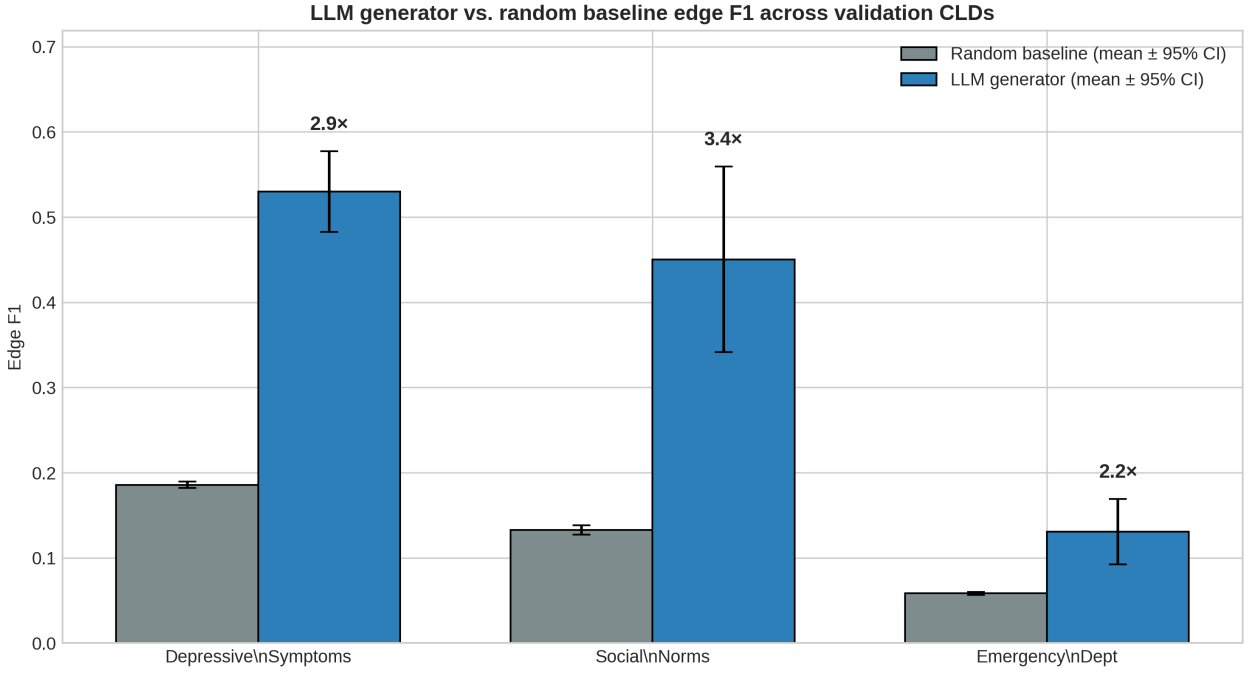


Figure 7.1: LLM generator (GPT-4.1, T=0.7, Nitai_C prompt) vs. random baseline edge F1 across all three validation CLDs. Error bars show mean ± 95% CI across 3 runs (t-distribution, $df = 2$).

Table 7.2: Random Baseline Comparison for Generator Edge Recovery

CLD	$\|V\|$	$\|E\|$	Density	Random F1 (mean ± 95% CI)
Depressive Symptoms	14	34	0.187	0.186±0.004
Social Norms	10	12	0.133	0.133±0.006
Emergency Department	34	66	0.059	0.058±0.002

Note. Random F1 computed via Monte Carlo simulation (1000 iterations) and reported as mean ± 95% CI for the Monte Carlo mean.
Density = $|E|/(|V| \times (|V| - 1))$.
LLM generator must significantly exceed these baselines to demonstrate causal reasoning.

Figure 7.1 and Table 7.2 show the LLM consistently outperforms random chance (2.2–3.4× improvement), supporting that it captures non-random structure aligned with the published CLDs. Performance varies by domain complexity: Social Norms shows the largest relative improvement (3.4×) while Emergency Department shows the smallest (2.2×) due to lower absolute F1 against a Monte Carlo random baseline which wires edges given the same variables with similar density.

7.2.2 Temperature Robustness

Experimental design: Section D.1.0.3.

To validate the generator temperature choice (T=0.7), we tested whether temperature significantly affects edge recovery performance across all three validation CLDs.

Table 7.3: Generator Temperature Sensitivity: Edge F1 Score by CLD

Temperature	Edge F1 Score (Mean ± 95% CI)			
	Social Norms	**Depressive**	**Emergency Dept**	**Overall**
T=0.0	0.325 ± 0.099	0.455 ± 0.055	0.113 ± 0.031	0.298 ± 0.116
T=0.3	0.373 ± 0.087	0.445 ± 0.014	0.125 ± 0.043	0.314 ± 0.113
T=0.5	0.314 ± 0.102	0.480 ± 0.051	0.105 ± 0.012	0.299 ± 0.126
T=0.7	0.348 ± 0.148	0.446 ± 0.014	0.104 ± 0.032	0.299 ± 0.120
T=1.0	0.381 ± 0.054	0.444 ± 0.027	0.105 ± 0.013	0.310 ± 0.120

Note. Edge F1 = $\frac{2 \cdot \text{TP}}{2 \cdot \text{TP} + \text{FP} + \text{FN}}$ comparing generated edges against literature ground truth.
Design. 5 temperatures × 3 runs × 3 CLDs = 45 total experiments. Seeds: 10, 20, 30 per temperature.
Per-CLD columns: Mean ± 95% CI across 3 runs per CLD at each temperature.
Overall column: Mean ± 95% CI computed across all 9 individual runs (3 CLDs × 3 runs) per temperature. The larger CI reflects variability both within and between CLDs.
Per-CLD ANOVA (temperature effect): Social Norms $F(4, 10) = 1.49$, $p = 0.28$; Depressive $F(4, 10) = 3.08$, $p = 0.07$; Emergency Dept $F(4, 10) = 1.76$, $p = 0.21$. None significant at $\alpha = .05$.

Temperature (T) = 0.7 was chosen for the generator LLM as initial default to strike a balance between diversity and determinism in sampled outputs. Subsequent sensitivity analysis shown in Table 7.3 shows that temperature does not significantly affect Edge F1: one-way ANOVA $F(4, 40) = 0.035$, $p = 0.998$, $\eta^2 = 0.004$ (negligible effect). Shapiro-Wilk [121] normality and Levene's [122] homogeneity assumptions were satisfied (all $p > .05$). Temperature does not significantly affect Edge F1 ($p > 0.05$), justifying T=0.7 as a reasonable default.

7.3 Preliminary: Judge Verification (TruthfulQA)

Experimental design: Section D.3.

Before evaluating the LLM-as-a-Judge approach on CLD edges, we sanity-check our correctness-based judge implementation on TruthfulQA—a commonly used benchmark for distinguishing truthful from false-but-plausible answers—to verify that the evaluation pipeline behaves as expected on a controlled setting.

Table 7.4: TruthfulQA Judge Verification (n=1634)

Metric	Value	
Accuracy	0.876	
Precision	0.890	
Recall	0.859	
F1 Score	0.874	
	Pred: Halluc.	**Pred: Truthful**
Actual: Halluc.	702 (TP)	115 (FN)
Actual: Truthful	87 (FP)	730 (TN)

Note. Judge: GPT-4.1, temp=0.0, seed=42, baseline correctness prompt. Dataset: 1634 samples (817 hallucinated, 817 truthful; 50% base rate). Classification threshold: judge_score $< 0.5 \Rightarrow$ Hallucinated.

Table 7.4 shows the performance of the correctness judge on the TruthfulQA dataset using the baseline prompt. Classification performance across both Recall (85.9%) and Precision (89.0%) are both high, indicating the judge neither over-flags nor under-flags. These results support that our correctness judge implementation works correctly on this benchmark. The subsequent RQ1a experiments will reveal whether this capability transfers to the more challenging domain of CLD edge validation.

7.4 RQ1a: LLM-as-a-Judge for Hallucination Detection

RQ1a evaluates whether LLM-based judges can reliably detect hallucinated edges in generated CLDs. The experimental design follows a 2×2 factorial structure crossing two factors: (1) Ground Truth type—GT Synth (synthetic corruptions introduced into LLM-generated CLDs) versus GT Lit (deviation from expert-curated literature CLDs)—and (2) Judge type—Correctness (evaluates causal plausibility without external sources) versus Citation (retrieves and evaluates supporting literature post hoc). This yields four experimental cells, each reported in a dedicated subsection below. Within each cell, we test multiple prompt variants (Baseline, CoT, Mechanistic, and for the Citation Judge also Mechanistic-Lit) using a repeated-measures design with 9 blocks (3 CLDs × 3 runs).

The objective is to determine (i) whether LLM judges can distinguish hallucinated from valid edges, (ii) which judge type and prompt variant performs best, and (iii) whether performance generalises from controlled corruptions (GT Synth) to real-world literature ground truth (GT Lit). Following the four factorial cells, we present external validation on an independent expert CLD (Uleman's) and human validation of the Citation Judge to assess reliability beyond the main experimental CLDs.

7.4.1 Operational Definitions: GT Lit, GT Synth, and Hallucination

RQ1a uses two distinct operationalisations of "ground truth," which yield different hallucination constructs. **Ground Truth from Literature (GT Lit)** treats published expert CLDs as the reference edge set over a fixed variable set: an edge *present* in the expert CLD is reference-positive and an edge *absent* is reference-negative. Under this operationalisation, we define "hallucination" as disagreement with the expert CLD: **FP** = generated edge absent from the expert CLD, and **FN** = expert edge the generator missed. **Ground Truth from Synthetic Corruption (GT Synth)** introduces controlled corruptions (polarity flips, spurious edges,

and motivation changes) into LLM-generated CLDs, yielding known error labels (`corrupted=True`) with high internal validity.

Interpretation caveat. Treating expert CLDs as absolute ground truth is an acknowledged oversimplification: expert models can encode scope decisions, construct/variable mappings, and literature incompleteness. An edge absent from an expert CLD (making an LLM generation a false positive) may therefore reflect genuine non-causality *or* a scope omission. This issue is explored further in the RQ3 Deep Research validation and enrichment analyses (Section 7.7).

7.4.2 GT Synth × Correctness Judge

Factorial cell: GT Synth (synthetic corruption ground truth) × Correctness Judge.

Experimental design: Section D.4.0.2.

7.4.2.1 Performance Metrics Overview

This section presents hallucination detection performance across all CLDs and prompt variants using the Correctness Judge. Ground truth is established by generating CLDs with an LLM and then introducing controlled corruptions. A *true positive* (TP) occurs when the judge correctly flags a corrupted edge, while a *false positive* (FP) occurs when an uncorrupted edge is incorrectly flagged as corrupted. A *false negative* (FN) is a corrupted edge that the judge fails to detect, and a *true negative* (TN) is an uncorrupted edge that the judge correctly does not flag. The classification threshold is set at 0.5: edges receiving a judge score ≤ 0.5 are classified as hallucinations. **Methodological note:** judge aggregate scores are discrete (0, 0.5, 1), so ROC curves represent interpolation between only a small number of operating points; we report AUC as a compact summary metric. Figure 7.2 shows the performance metrics overview for the correctness-based corruption detection task. A sensitivity analysis of judge prompt variants consolidated across experiments (effect sizes and assumption checks across all RQ1a prompt ablations) is provided in Appendix A.1.

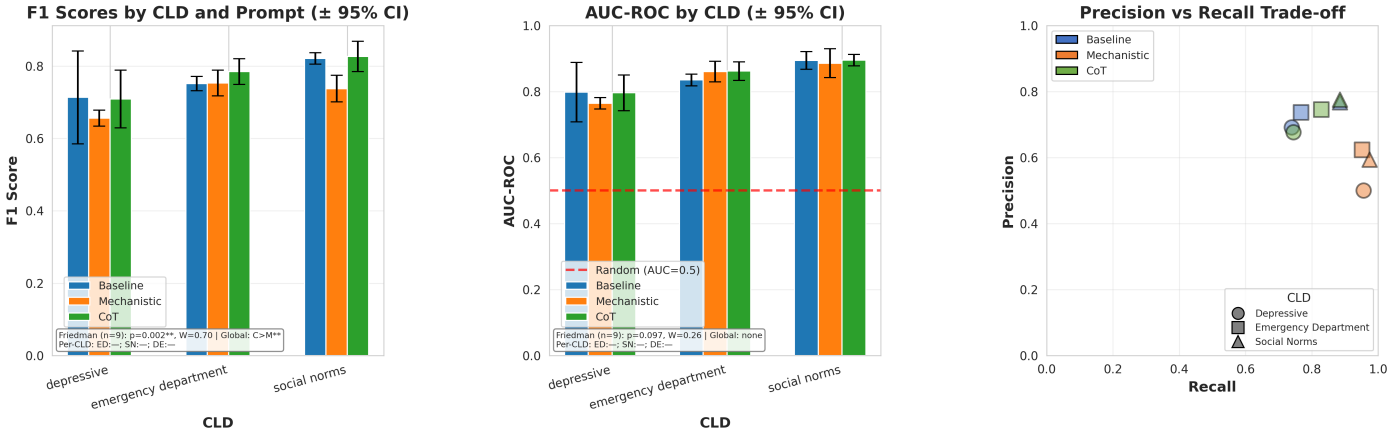


Figure 7.2: Performance metrics overview for corruption detection (Correctness Judge). Shows F1 scores (Left), ROC-AUC with random (AUC=0.5) baseline (Middle), and aggregate performance (Right) by prompt. Friedman test (see Appendix E) (below each panel) tests prompt effect with (CLD, run) as blocks; per-CLD Wilcoxon tests (see Appendix E) show which pairs differ within each CLD (e.g., "DE:M>B" = Depressive Symptoms: Mechanistic>Baseline). W = Kendall's W. DE=Depressive Symptoms, ED=Emergency Department, SN=Social Norms; B=Baseline, M=Mechanistic, C=CoT. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$ (unadjusted; Bonferroni handled in text).

The leftmost plot of Figure 7.2 shows F1 scores in the range 0.65–0.82 across CLDs and prompt variants. Across CLDs, Social Norms shows highest F1 on average, with Emergency Department intermediate, and Depressive Symptoms lowest, indicating domain-dependent difficulty in corruption detection. Prompt-to-prompt differences are generally modest relative to the 95% CIs (N=3 runs), though Social Norms shows a clearer separation between CoT and Mechanistic; we do not treat this visual separation as inferential because the plotted 95% CIs are descriptive (not a hypothesis test) and per-CLD comparisons have low power ($n = 3$) and multiple-comparison risk, so inferential claims are based on the blocked Friedman/Wilcoxon tests over CLD×run blocks instead. The Friedman omnibus test over CLD×run blocks ($n = 9$) rejects the null of no prompt effect (prompt ranks equal within blocks), remaining significant under the RQ1a omnibus Bonferroni threshold (Bonferroni over 4 RQ1a omnibus tests: 2 judge types × 2 ground-truth settings; $\alpha_{\text{adj}} = 0.05/4 = 0.0125$). Post-hoc paired Wilcoxon signed-rank tests across the same $n = 9$ blocks indicate CoT outperforms Mechanistic (consistent with the plot annotation "Global: C>M"), significant under the post-hoc Bonferroni threshold for 3 prompt pairs ($\alpha_{\text{adj}} = 0.0167$). However, within-CLD Wilcoxon tests ($n = 3$ runs per CLD) show no significant pairwise differences, plausibly due to low per-CLD power. Taken together, this indicates a global prompt effect for corruption detection under the correctness judge (GT Synth), with post-hoc evidence that CoT tends to outperform Mechanistic (C>M) across CLD×run blocks; no other pairwise prompt comparisons are significant after Bonferroni correction (i.e., CoT does not significantly outperform Baseline). Overall, these results indicate that prompt engineering can change behaviour (e.g., CoT improves over Mechanistic) but do not yield reliable improvement over baseline, with the dominant driver of performance being the CLD/domain. For brevity, subsequent RQ1a figure interpretations that use this same testing framework report only the corresponding omnibus/post-hoc conclusions (rather than repeating each step of the full framework each time).

The middle plot of Figure 7.2 shows the AUC-ROC vs. a random (AUC = 0.5) baseline, indicating that the correctness judge can discriminate corrupted from non-corrupted edges in each CLD and prompt. AUC ranges from 0.77 at the lowest end (Mechanistic prompt in Depressive Symptoms) to 0.90 at the highest end (Baseline prompt in Social Norms). The highest AUC on average is obtained for Social Norms, followed by Emergency Department, with Depressive Symptoms the lowest. 95% CIs indicate relatively low run-to-run variability overall, with the widest intervals in Depressive Symptoms (especially Baseline, and to a lesser extent CoT) and a secondary widening in Social Norms under the Mechanistic prompt. Under the blocked Friedman test over CLD×run blocks ($n = 9$), we do not find evidence of a global prompt effect on AUC (Friedman $p = 0.097 > \alpha_{\text{adj}} = 0.0125$ under the RQ1a omnibus Bonferroni correction).

The right plot of Figure 7.2 shows the precision–recall trade-off per prompt variant and CLD. The Mechanistic prompt achieves the highest recall across CLDs, while Baseline and CoT achieve higher precision. In the Emergency Department CLD, CoT has slightly higher precision and higher recall than Baseline. In this setting, mechanistic prompting is preferable when the goal is to maximise detection sensitivity (high recall) for corrupted edges, whereas Baseline/CoT are preferable when false positives are more costly (higher precision); CoT offers a modest middle-ground in the Emergency Department CLD by improving both precision and recall relative to Baseline.

7.4.2.2 Detailed Analysis

In-depth analysis of corruption type detection patterns and score discrimination ability.

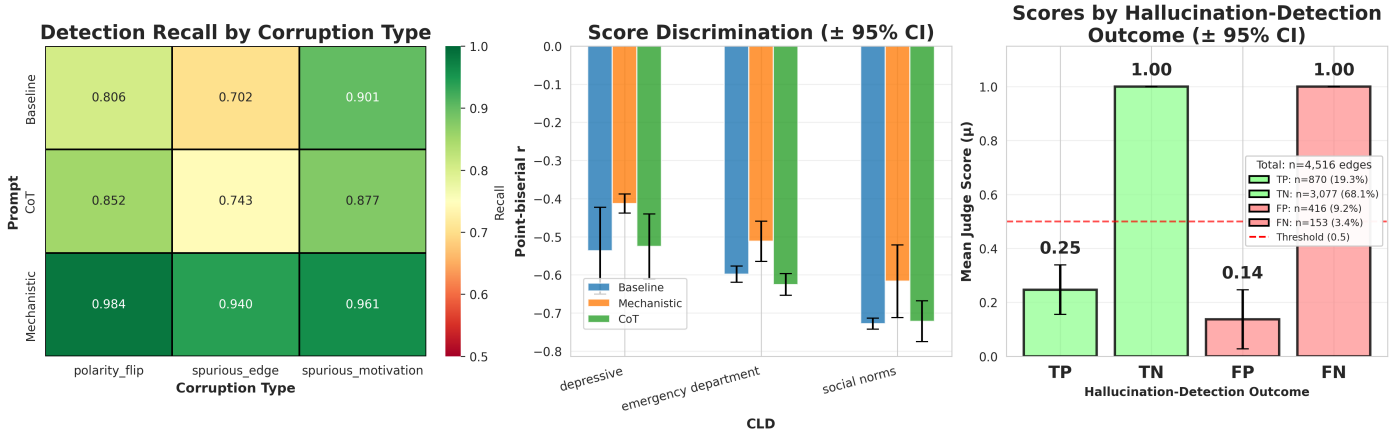


Figure 7.3: Detailed analysis for correctness-based corruption detection. Recall by corruption type with colour scale from 0.5 to 1.0 (Left). Score discrimination showing point-biserial correlation between ground truth hallucinations (corruption=True) and judge scores (Middle). Judge scores by hallucination-detection outcome (Right): mean judge scores (μ) and 95% confidence intervals, with correct outcomes shown in green and incorrect outcomes shown in red.

Figure 7.3 shows the detailed analysis of the correctness judge's performance. The leftmost plot of Figure 7.3 shows the recall by corruption type aggregated across all CLDs and 3 runs, and shows that the correctness judge with Mechanistic prompt operates with higher recall for each corruption type than Baseline and CoT prompts, with CoT outperforming Baseline on spurious and polarity flip corruption types but underperforming Baseline on spurious motivation, while still scoring less than Mechanistic overall. If the objective is to maximise sensitivity to all corruption types (including harder spurious-edge and spurious-motivation cases), the Mechanistic prompt is the most reliable choice; Baseline/CoT trade off substantially lower recall—especially for spurious edges—so they are better suited to settings where reducing false positives (precision) is prioritised over catching every corruption.

The middle plot of Figure 7.3 shows the point-biserial correlation (Pearson r) between the ground truth corruption label (`corrupted=True`) and the continuous judge aggregate score. Negative correlations are the expected direction because lower judge scores should be associated with corrupted edges under the ≤ 0.5 decision rule. We report these correlations as descriptive diagnostics. Mechanistic shows slightly lower-magnitude (less negative) correlations on Depressive Symptoms relative to Baseline/CoT. Overall, correlations are large and in the expected direction (corruptions associated with lower scores), ranging from roughly -0.55 to -0.7 for CoT and Baseline and roughly -0.4 to -0.6 for Mechanistic. The main finding is that all prompt variants have correlations of considerable magnitude and in the expected direction.

The rightmost plot of Figure 7.3 shows mean judge score (μ) by *hallucination-detection outcome* (TP/FP/TN/FN), using the same classification label (`corruption=True`) and score-threshold decision rule (≤ 0.5) used to compute the F1 metrics above. As expected under thresholding, outcomes predicted as "clean" (TN and FN; score > 0.5) have higher mean scores than outcomes predicted as "hallucination" (TP and FP; score ≤ 0.5). The substantive failure mode is the FN bar: a non-trivial fraction of corrupted edges are missed yet still receive high (often near-maximal) judge scores, indicating that some synthetic corruptions are incorrectly classified as non-hallucinations under correctness judging; however, the FN class remains relatively minor (3.4% of total edges). Finally, TP and FP mean scores are close (0.25 vs. 0.14) with substantial overlap in uncertainty, indicating that the judge score provides only weak separation between *correct* vs. *incorrect* hallucination flags within the predicted-hallucination set (i.e., conditional on score ≤ 0.5).

Full ROC curves are provided in Appendix J.

7.4.3 GT Lit × Correctness Judge

Factorial cell: GT Lit (literature ground truth) × Correctness Judge.

Experimental design: Section D.4.0.1.

For this experiment, the variables from ground truth from literature are passed and LLM generated edges are judged. In this scenario, ground truth from literature (GT Lit.) is used, such that a hallucination is defined as an FP/FN with respect to the ground truth.

7.4.3.1 Performance Metrics Overview

Detailed view of core performance metrics across all CLDs and prompt variants.

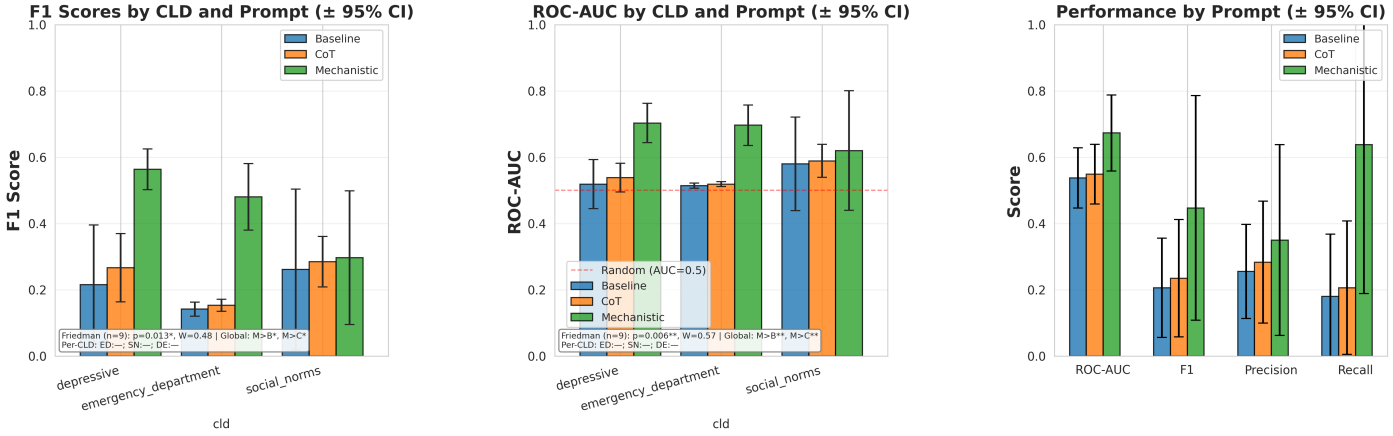


Figure 7.4: Performance metrics overview for ground truth correctness validation. Shows F1 scores (Left), ROC-AUC with random (AUC=0.5) baseline (Middle), and aggregate performance (Right) by prompt. Friedman test (see Appendix E) (below each panel) tests prompt effect with (CLD, run) as blocks; per-CLD Wilcoxon tests (see Appendix E) show which pairs differ within each CLD (e.g., "DE:M>B"). W = Kendall's W. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$ (unadjusted; Bonferroni handled in text).

The leftmost plot of Figure 7.4 shows the correctness judge F1 scores in the range of 0.15 to 0.55 (notably lower than in the corruption-detection task). Mechanistic achieves higher F1 than Baseline and CoT for Depressive Symptoms and Emergency Department, with a clearer separation between prompt means relative to run-to-run uncertainty; in Social Norms, prompt means are closer together and uncertainty is larger, so the plot does not show a stable prompt ordering. Using the blocked Friedman/Wilcoxon framework (CLD×run blocks, $n = 9$), the omnibus Friedman test reports $p = 0.013$ with Kendall's $W = 0.48$. This is marginally above the RQ1a omnibus Bonferroni threshold ($\alpha_{\text{adj}} = 0.0125$), so we treat the prompt effect on GT Lit F1 as suggestive rather than confirmatory under the corrected criterion. The plot's global post-hoc summary is based on paired Wilcoxon signed-rank tests across the same $n = 9$ blocks (Bonferroni-adjusted over 3 prompt pairs, $\alpha_{\text{adj}} = 0.0167$) and indicates Mechanistic>Baseline and Mechanistic>CoT, while the per-CLD Wilcoxon summaries show no significant within-CLD prompt differences ($n = 3$ runs per CLD).

The middle panel of Figure 7.4 reports ROC–AUC relative to a chance baseline (AUC = 0.5). By point estimate, the Mechanistic prompt attains the highest AUC in Depressive Symptoms and Emergency Department, whereas in Social Norms the prompt estimates are closer together with wide uncertainty, so no stable ordering is visually apparent. In absolute terms, Baseline and CoT confidence intervals include 0.5 across CLDs, and Mechanistic exceeds 0.5 in Depressive Symptoms and Emergency Department but not in Social Norms. At

the block level (CLD×run blocks, $n = 9$), the Friedman test indicates a significant prompt effect on AUC ($p = 0.006 < \alpha_{\text{adj}} = 0.0125$), suggesting that differences in AUC are driven primarily by the Mechanistic prompt rather than consistent above-chance discrimination across all prompts and domains.

The right plot of Figure 7.4 shows the aggregate performance by prompt across all CLDs and all runs. While the Mechanistic prompt achieves the highest F1 and ROC-AUC, uncertainty at this aggregation level remains large relative to between-prompt mean gaps, so the plot does not show clear separation between prompts across CLDs. Precision and recall are also highest for Mechanistic, with CoT intermediate and Baseline lowest, though differences are small relative to uncertainty.

7.4.3.2 Detailed Analysis

In-depth analysis of precision-recall trade-offs, score distributions, and point-biserial correlations.

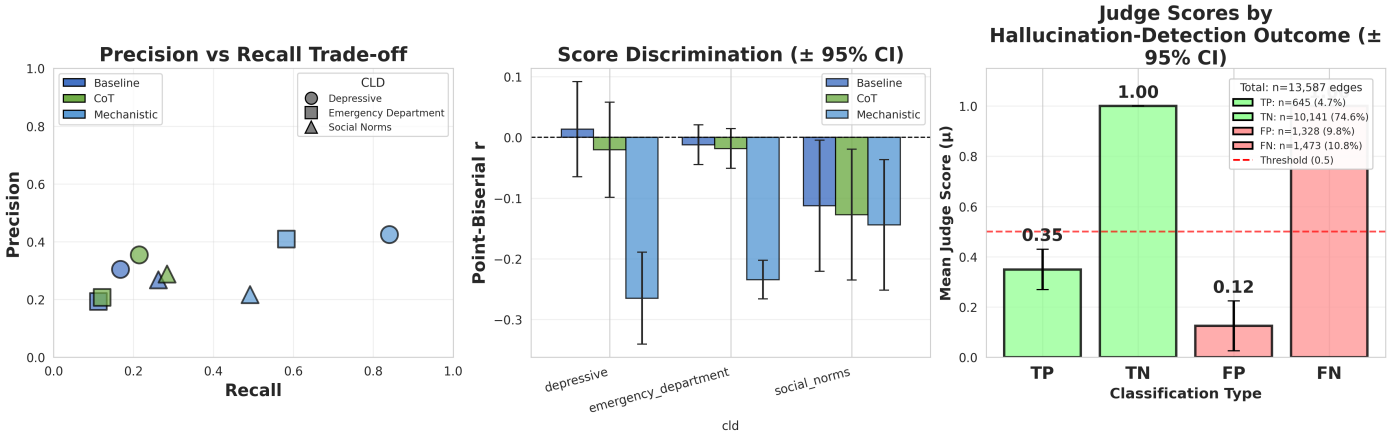


Figure 7.5: Detailed analysis for ground truth correctness validation. Precision-recall trade-off (Left). Point-biserial correlations showing discrimination ability (Middle). Judge scores by hallucination-detection outcome (Right): mean judge scores (μ) and 95% confidence intervals for TP/FP/TN/FN under the hallucination label (GT Lit: FP∨FN) and score-threshold decision rule used for the F1 plots.

The leftmost plot of Figure 7.5 shows the precision–recall trade-off for each prompt variant and CLD. The Mechanistic prompt achieves the highest recall and slightly higher precision than CoT and Baseline; Baseline and CoT are broadly similar, with CoT performing slightly better in terms of both recall and precision. Overall, the Mechanistic prompt therefore performs best on this plot, consistent with the F1 and AUC results.

The middle plot of Figure 7.5 shows the point-biserial correlation (Pearson r) between the GT Lit hallucination label (FP∨FN) and the continuous judge aggregate score (threshold judge score ≤ 0.5). Across CLDs, Baseline and CoT correlations are close to zero in Depressive Symptoms and Emergency Department (weak score–label association), while the Mechanistic prompt shows a stronger negative correlation between aggregate judge score and hallucinations in Depressive Symptoms and Emergency Department. Overall, the correlations are substantially smaller in magnitude than in the GT Synth setting (and in some cases near-zero), consistent with weaker discrimination of GT Lit hallucinations by judge scores. In Social Norms, prompt differences are not substantial given the uncertainty; by contrast, Mechanistic remains the strongest variant in the other two CLDs.

The rightmost plot of Figure 7.5 shows the mean (± 95% CI) judge score distributions by *hallucination-detection outcome* (TP/FP/TN/FN), using the same hallucination label (GT Lit: FP∨FN) and the same score-threshold decision rule (≤ 0.5) used to compute the F1 metrics above. As expected under thresholding, outcomes

predicted as "clean" (TN and FN) have higher mean scores than outcomes predicted as "hallucination" (TP and FP). As observed in the GT Synth setting, the main failure mode remains the large FN mass with high scores: many GT Lit hallucinations (FP/FN relative to the expert CLD) are not flagged by the judge and retain high scores, consistent with the weak point-biserial correlations and low-to-moderate F1 outside the Mechanistic prompt. Notably, among edges that the judge flags as hallucinations (TP and FP; score ≤ 0.5), false positives receive *lower* mean scores than true positives, indicating overconfident false alarms (i.e., the judge is more certain on some incorrect flags than on correct hallucination detections). This TP–FP score separation does not translate into better thresholded classification performance because both groups are still on the same side of the decision boundary and the dominant error remains missed hallucinations with high scores (FN), which depress recall and F1.

7.4.4 GT Synth × Citation Judge

Factorial cell: GT Synth (synthetic corruption ground truth) × Citation Judge.

Experimental design: Section D.4.0.4.

7.4.4.1 Performance Metrics Overview

Detailed view of core performance metrics of Citation Judge on GT synth (corruption detection) across all CLDs and four prompt variants (Baseline, Mechanistic, CoT, Mechanistic-Lit) across 3 runs.

Compute budget note (model confound). Due to experimental budget limits, this citation-judge experiment was run with **GPT-5-mini** ("5m") to achieve an approximate **90% cost reduction** relative to GPT-4.1 for citation validation (Appendix Table 7.25). As a result, prompt effects here may be partially confounded with model-specific behaviour. **Methodological note:** judge aggregate scores are discrete (0, 0.5, 1), so ROC curves represent interpolation between only a small number of operating points; we report AUC as a compact summary metric.

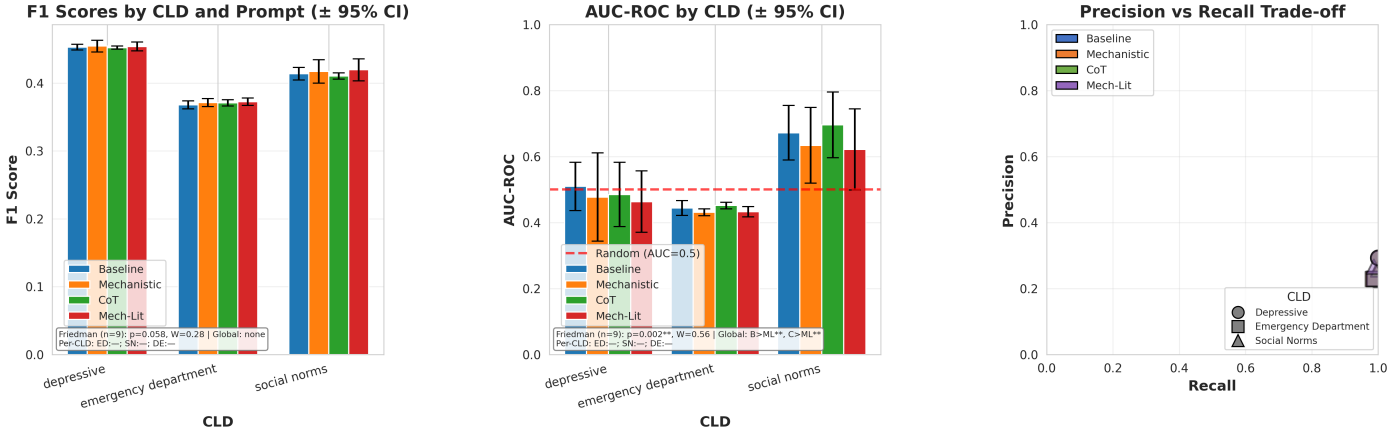


Figure 7.6: Performance metrics overview for citation-based corruption detection. Shows F1 scores (Left), AUC-ROC with chance baseline (Middle), and aggregate performance (Right). Friedman test (see Appendix E) (below each panel) tests prompt effect; per-CLD Wilcoxon tests (see Appendix E) show which pairs differ within each CLD. 4 prompts × 3 CLDs × 3 runs = 36 experiments. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$ (unadjusted; Bonferroni handled in text).

The leftmost plot of Figure 7.6 shows the F1 scores for each prompt variant and CLD. F1 varies primarily by CLD (Emergency Department lowest at ∼0.35, Depressive Symptoms ∼0.45, Social Norms ∼0.40), while prompt-to-prompt differences within each CLD are small. Under the same blocked Friedman/Wilcoxon framework as above, we do not detect a Bonferroni-significant global prompt effect on F1 for the citation judge (Friedman $p = 0.058 > \alpha_{\text{adj}} = 0.0125$), and no pairwise prompt comparisons survive Bonferroni correction (6 prompt pairs; $\alpha_{\text{adj}} = 0.0083$). Overall, prompt engineering does not yield a reliable F1 improvement in this setup; performance is dominated by the CLD/domain.

The middle plot of Figure 7.6 shows the AUC-ROC with a chance baseline (AUC = 0.5). Under the same blocked framework, the omnibus prompt effect on AUC-ROC is Bonferroni-significant (Bonferroni over 4 RQ1a omnibus tests: 2 judge types × 2 ground-truth settings; $\alpha_{\text{adj}} = 0.05/4 = 0.0125$; Friedman $p = 0.0017 < 0.0125$, $W = 0.56$). Post-hoc Wilcoxon tests indicate the effect is driven by **Mechanistic-Lit** underperforming other prompts, with both CoT vs. Mech-Lit ($p = 0.0039$) and Baseline vs. Mech-Lit ($p = 0.0078$) remaining significant under the post-hoc Bonferroni threshold for 6 prompt pairs ($\alpha_{\text{adj}} = 0.05/6 \approx 0.0083$). Descriptively, Depressive Symptoms AUCs cluster near/below 0.5 for most prompts, Emergency Department is low with substantial uncertainty, and Social Norms is consistently above 0.5 across prompts.

The rightmost plot of Figure 7.6 shows the precision–recall operating points per prompt and per CLD. In this setting, the markers largely overlap (recall near-ceiling with similarly low precision), indicating that prompt variants do *not* meaningfully separate on the precision–recall trade-off: they behave similarly under the fixed score-threshold decision rule, and any apparent differences are negligible relative to the plot resolution.

7.4.4.2 Detailed Analysis

In-depth analysis of corruption type detection patterns and score discrimination ability.

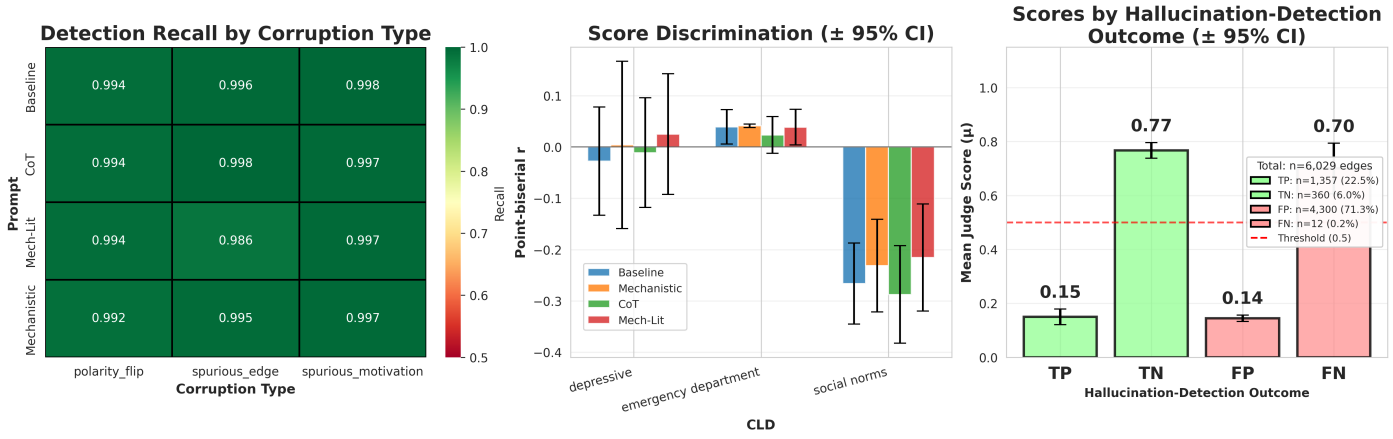


Figure 7.7: Detailed analysis for citation-based corruption detection. Recall by corruption type with colour scale from 0.5 to 1.0 (Left). Score discrimination showing point-biserial correlation between ground truth (corruption=True) and judge scores (Middle). Judge scores by hallucination-detection outcome (Right): mean judge scores (μ) and 95% confidence intervals, with correct outcomes shown in green and incorrect outcomes shown in red.

The leftmost plot of Figure 7.7 shows recall by corruption type with a colour scale from 0.5 to 1.0. Recall is consistently near-ceiling across prompts (approximately 0.98–1.00), indicating that the citation judge rarely misses synthetic corruptions under the ≤ 0.5 decision rule. When comparing this to the correctness-judge corruption results (Figure 7.3), note that this citation-judge experiment was run with **GPT-5-mini** for budget reasons, whereas the correctness-judge experiments used **GPT-4.1**. This means we cannot attribute the higher

recall observed here to either the *judge task type* (citation vs. correctness) or the *model*; both factors differ simultaneously.

The middle plot of Figure 7.7 shows the point-biserial correlation (Pearson r) between hallucinations (`corrupted=True` in GT Synth) and judge scores. We treat these correlations as descriptive: Depressive Symptoms and Emergency Department show correlations near zero (weak score–label association) across prompt variants, while Social Norms shows a moderate negative association (lower scores associated with corruption), indicating more consistent score discrimination in that CLD.

The rightmost plot of Figure 7.7 shows that true positives and false positives receive similar scores (0.15 and 0.14, respectively), while TN and FN are much higher on average (0.77 and 0.70). The TP–FP separation here is only 0.01 (the smallest TP–FP separation observed across the RQ1a score-by-outcome plots), indicating that the judge does not differentiate true vs. false positives in its scoring even when conditioning on "hallucination" predictions (score ≤ 0.5).

Full ROC curves are provided in Appendix J.

7.4.5 GT Lit × Citation Judge

Factorial cell: GT Lit (literature ground truth) × Citation Judge.

Experimental design: Section D.4.0.3.

7.4.5.1 Performance Metrics Overview

Detailed view of core performance metrics of Citation Judge on GT Lit (hallucination is FN/FP in GT lit.) across all CLDs and prompt variants across 3 runs.

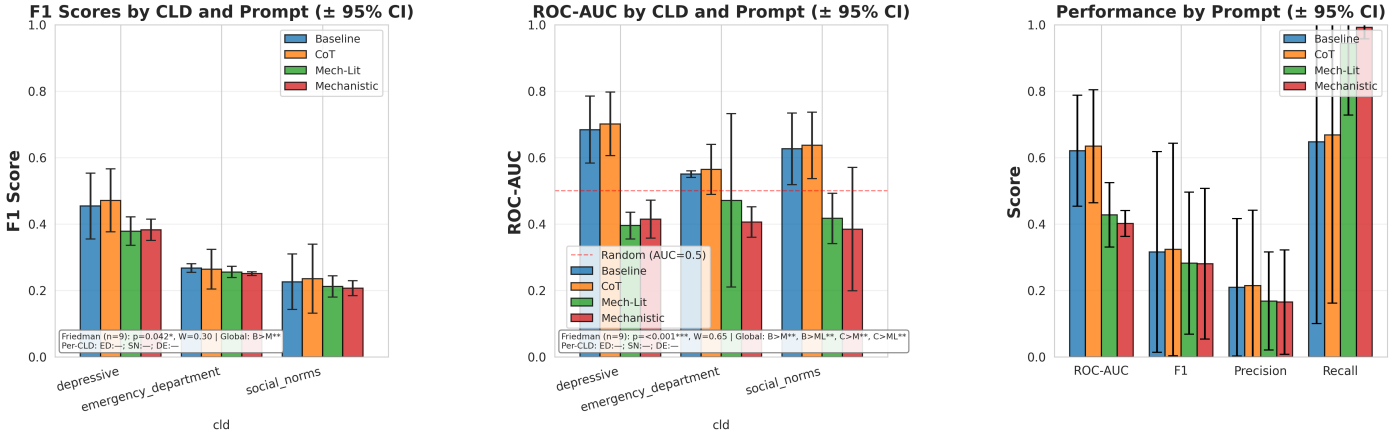


Figure 7.8: Performance metrics overview for ground truth citation validation. Shows F1 scores (Left), ROC-AUC with random baseline (Middle), and aggregate performance (Right) by prompt. Friedman test (see Appendix E) (below each panel) tests prompt effect; per-CLD Wilcoxon tests (see Appendix E) show which pairs differ within each CLD. W = Kendall's W. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$ (unadjusted; Bonferroni handled in text).

The leftmost plot of Figure 7.8 shows the F1 scores for each prompt variant and CLD. F1 is around 0.2 for Social Norms and 0.22 for Emergency Department, with prompt means roughly similar within each CLD;

Depressive Symptoms is higher overall, with Baseline and CoT near ∼0.46–0.48 and both mechanistic variants near ∼0.38. Under the blocked Friedman test over CLD×run blocks ($n = 9$), the omnibus prompt effect on F1 is weak ($p = 0.042$) and does not meet the panel-level Bonferroni threshold shown in the plot ($\alpha_{\text{adj}} = 0.016$; Kendall's $W = 0.30$). The plot's global paired Wilcoxon summary indicates Baseline>Mechanistic (*) across blocks, while other prompt pairs are not marked as significant.

The middle plot of Figure 7.8 shows the ROC–AUC relative to a chance baseline (AUC = 0.5). Baseline and CoT are above chance in Depressive Symptoms and Social Norms, whereas both mechanistic variants are below chance; in Emergency Department, Baseline is above chance while the other prompts are closer to chance and some intervals straddle 0.5. Under the blocked Friedman test over CLD×run blocks ($n = 9$), the omnibus prompt effect on AUC is significant under the RQ1a omnibus Bonferroni threshold (Bonferroni over 4 RQ1a omnibus tests: 2 judge types × 2 ground-truth settings; $\alpha_{\text{adj}} = 0.05/4 = 0.0125$; Friedman $p = 0.0005 < 0.0125$, $W = 0.65$). For global post-hoc paired Wilcoxon tests, using Bonferroni over 6 prompt pairs ($\alpha_{\text{adj}} = 0.05/6 \approx 0.0083$), we obtain Baseline>Mechanistic ($p = 0.0039 < 0.0083$), CoT>Mechanistic ($p = 0.0039 < 0.0083$), Baseline>Mechanistic-Lit ($p = 0.0078 < 0.0083$), and CoT>Mechanistic-Lit ($p = 0.0078 < 0.0083$); Baseline vs. CoT is not significant ($p = 0.074 > 0.0083$). Overall, this confirms that mechanistic prompting underperforms on discrimination in this setting (worse-than-chance AUC) across all CLDs.

The rightmost plot of Figure 7.8 shows the aggregate metric scores across all CLDs and 3 runs, with larger variance. While mechanistic variants show higher recall, it does not offset their lower precision, which results in lower ROC-AUC and F1 overall. At this aggregation level, uncertainty remains large relative to between-prompt mean gaps.

7.4.5.2 Detailed Analysis

In-depth analysis of precision-recall trade-offs, score distributions, and point-biserial correlations.

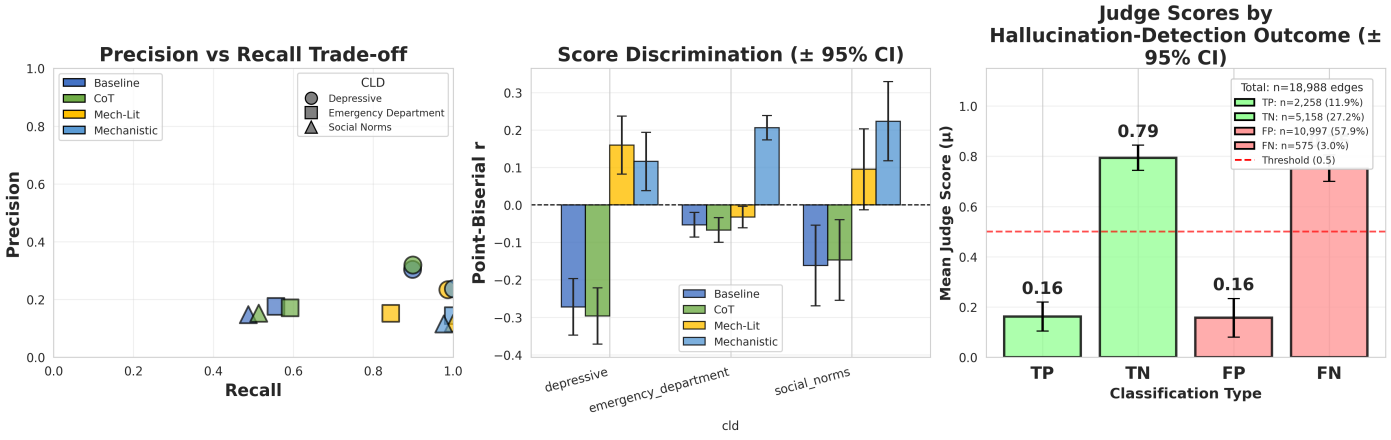


Figure 7.9: Detailed analysis for ground truth citation validation. Precision-recall trade-off (Left). Point-biserial correlations showing discrimination ability (Middle). Judge scores by hallucination-detection outcome (Right): mean judge scores (μ) and 95% confidence intervals for TP/FP/TN/FN under the hallucination label (GT Lit: FP∨FN) and score-threshold decision rule used for the F1 plots.

The leftmost plot in Figure 7.9 shows the precision-recall trade-off per prompt and per CLD, showing highest recall for Mechanistic-Lit, followed by Mechanistic across CLDs, where Baseline and CoT score similarly but worse on recall than the mechanistic variants and slightly better on precision.

The middle plot of Figure 7.9 shows the point-biserial correlation (Pearson r) between hallucination (FP∨FN in GT Lit) and judge scores. Descriptively, Baseline and CoT exhibit negative correlations (higher judge scores tend to align with lower hallucination rates), with small-to-moderate magnitudes that vary by CLD. In contrast, the mechanistic prompt variants show positive correlations in multiple CLDs, indicating a counterintuitive association where higher citation-judge scores coincide with a higher likelihood of GT Lit hallucination; this is consistent with a failure mode where citation scoring tracks topical relevance rather than faithful verification of the specific causal claim.

The rightmost plot of Figure 7.9 shows the score distributions by *hallucination-detection outcome* (TP/FP/TN/FN), using the same hallucination label (GT Lit: FP∨FN) and the same score-threshold decision rule (≤ 0.5) used to compute the F1 metrics above. Similar to the correctness case, TN and FN outcomes have higher mean scores than TP and FP due to thresholding. The substantive issue is again the FN class: a large fraction of GT Lit hallucinations are assigned high scores (missed by the judge), contributing to weak discrimination despite high recall for some prompt variants. Notably, unlike the GT Lit correctness case (Figure 7.5) where FP scores were lower than TP scores, here TP and FP mean scores are nearly identical ($\sim$0.16) with substantial overlap in uncertainty, indicating that the citation judge provides essentially no score-based separation between correct and incorrect hallucination flags—once an edge is flagged (score ≤ 0.5), the score itself carries no additional signal about whether the flag is a true or false positive.

7.4.5.3 Rationale for validation studies

Taken together, the four factorial RQ1a cells show that LLM-as-a-judge can be effective in a controlled setting (GT Synth), but performance does not reliably transfer to literature ground truth (GT Lit) for CLD edge validation. In particular, citation-based judging on GT Lit shows weak and sometimes counterintuitive discrimination (e.g., near-chance or worse-than-chance AUC for some prompt variants, positive score–hallucination correlations, and minimal TP–FP score separation), consistent with citation scoring tracking topical relevance rather than faithfully verifying the specific causal claim. This ambiguity can arise from (i) judge failure (miscalibration or incorrect causal reasoning), (ii) retrieval failure (queries/fetchers surfacing uninformative or incomplete evidence), and/or (iii) domain/evidence dependence (available literature is correlational, hedged, or only indirectly applicable to the edge claim). We therefore run targeted diagnostic validation studies that isolate these factors: human–LLM agreement tests whether judge scores track an expert rater on the retrieved evidence; Uleman's expert CLD tests performance when citations are curated (reducing the "wrong evidence found" explanation); and physics CLDs provide an "easy-domain" sanity check where causal relationships are unambiguous. Together, these studies help interpret whether weak citation-judge discrimination primarily reflects judge malfunction, retrieval limitations, or evidence-type/domain constraints.

7.4.6 Validation: Human–LLM Agreement

Experimental design: Section D.6.2.

To validate LLM judge reliability, we measured inter-rater agreement between citation-based LLM judges and a domain expert (Mick) across 72 causal edges from 3 independent datasets.

7.4.6.1 Per-Dataset and Combined Agreement

Table 7.5: Human Validation: LLM Judge vs. Expert Inter-Rater Agreement

Dataset	Generator	n	Cohen's κ	Agreement	Pearson r
File 1 (Depressive Symptoms)	GPT-4.1	30	0.435	70.0%	0.527**
File 2 (Depressive Symptoms)	Claude 4 Sonnet	30	0.923	96.7%	0.926***
File 3 (Social Norms)	GPT-4.1	12	0.461	66.7%	0.798**
Combined	**2 models**	**72**	**0.618**	**80.6%**	**0.732*** **
Human baseline	N/A (expert)	13	∼0.60	80.0%	—

Note. Inter-rater reliability between LLM citation judge (GPT-4.1, temp=0.0) and domain expert.
Verdicts: Not supported (0), Partially supported (0.5), Fully supported (1).
κ interpretation [117]: <0.20 slight, 0.21–0.40 fair, 0.41–0.60 moderate, 0.61–0.80 substantial, >0.80 almost perfect.
Human baseline from prior GMB validation studies on comparable expert tasks.
Significance: ** $p < 0.01$, *** $p < 0.001$.

Table 7.5 shows agreement ranging from moderate (File 3, $\kappa = 0.461$) to almost perfect (File 2, $\kappa = 0.923$). The combined agreement ($\kappa = 0.618$) indicates substantial reliability, comparable to the cited human baseline ($\kappa \approx 0.60$). This provides evidence that the citation judge's ordinal scores align with a human rater on this sample, supporting its use as a screening/triage signal; however, agreement varies by dataset and the Social Norms subset is small ($n = 12$), so generalisation should be treated cautiously.

7.4.6.2 Disagreement Analysis

Table 7.6: Confusion Matrix: LLM Judge vs. Human Expert (Combined, n=72)

	LLM: Not supported	LLM: Partially	LLM: Fully
Human: Not supported	37 (51.4%)	11 (15.3%)	0 (0%)
Human: Partially	0 (0%)	21 (29.2%)	3 (4.2%)
Human: Fully	0 (0%)	0 (0%)	0 (0%)

Note. Rows = human expert verdicts; columns = LLM judge verdicts.

Table 7.6 shows 14 disagreements (19.4%), all with LLM more lenient than human: 11 cases where LLM said "Partially" when human said "Not supported", and 3 cases where LLM said "Fully" when human said "Partially". Notably, there were zero false negatives—the LLM never under-estimated evidence support.

All 14 disagreements showed systematic leniency bias, with the LLM more permissive than the human expert. Agreement varied by domain and judge model, ranging from moderate (Social Norms, $\kappa = 0.461$) to almost perfect (Claude-generated Depressive Symptoms, $\kappa = 0.923$). This pattern is *suggestive* that judge reliability may depend not only on domain but also on the underlying judge model. However, this comparison is **not** a controlled head-to-head model evaluation: while the Claude and GPT-4.1 figures refer to the same underlying CLD domain, they were computed on different (small) sampled sets of edges, so sampling variation (and any differences in the judged edge set) could plausibly explain part of the observed gap. We therefore interpret the Claude-vs.-GPT agreement difference as a preliminary indication of potential model dependence, not definitive evidence. On this sample, the citation-based LLM judge achieves inter-rater agreement comparable to the cited

human baseline when aggregating across generators. In both cases the judge model is GPT-4.1. The consistent leniency bias (zero false negatives) indicates that LLM judges provide permissive screening—appropriate for identifying low-confidence edges requiring human review, but insufficient for strict validation without expert oversight.

7.4.7 Validation: Uleman's External CLD

Experimental design: Sections D.6.1 and D.6.4.

To validate the citation-based judge on an independently expert-validated dataset, we evaluated the judge on the Uleman's Alzheimer's disease CLD—a 150-edge causal model constructed through Group Model Building with 17 domain experts. This external validation tests whether the judge performance generalises beyond our three primary CLDs and assesses the impact of search provider and citation fetcher choices.

7.4.7.1 Search Provider Comparison

Experimental design: Section D.6.1.

We compared four academic search providers (OpenAlex, Brave, Perplexity, PubMed) combined with two citation fetching methods (Jina AI Reader, ContentScraper) to assess infrastructure impact on judge scores.

Table 7.7: Search Provider Comparison on Uleman's Expert CLD (150 edges)

Provider	Jina AI Avg Score	ContentScraper Avg Score	Jina Edges w/ Citations	CS Edges w/ Citations	Jina Coverage	CS Coverage
OpenAlex	0.232	0.381	150/150	59/150	100.0%	39.3%
Brave (no whitelist)	0.296	0.326	150/150	134/150	100.0%	89.3%
Brave (whitelist)	0.290	0.329	150/150	134/150	100.0%	89.3%
Perplexity	0.278	0.335	150/150	120/150	100.0%	80.0%
PubMed	0.306	0.283	127/150	120/150	84.7%	80.0%

Note. Jina AI Reader = paywall-bypassing document fetcher; ContentScraper = standard web scraper.
Coverage = percentage of edges with successfully fetched citation content.
Score Difference = ContentScraper Avg Score − Jina AI Avg Score.

Table 7.7 reveals four findings: (1) Jina AI achieves 100% coverage (vs. 39–89% for ContentScraper) by bypassing paywalls; (2) ContentScraper Citation Judge scores are 10–15% higher when it succeeds; (3) PubMed achieves highest Jina score (0.306) but lowest coverage (84.7%); (4) All scores are low (<0.4). The Uleman's expert CLD receives uniformly low judge scores (0.23–0.38) across all search provider and fetcher configurations. Recall that weak discrimination (low F1, near-chance AUC) in RQ1a GT Lit citation judging arose because the judge assigned low scores to both valid and invalid edges, making them indistinguishable. Uleman's confirms this pattern on an independently expert-constructed CLD with bibliographic references: even edges known to be valid receive low citation scores. This relative infrastructure-invariance suggests that the weak discrimination is not solely an artefact of a single retrieval backend in our pipeline; however, this does not rule out retrieval limitations more broadly, and Uleman's (Alzheimer's) is a domain where direct causal evidence is often scarce and much of the literature is correlational or hedged.

This experiment was added to test a "best-effort retrieval" setting with multiple providers/fetchers; even under this setup, judge scores remain low on expert-provided edges. This is consistent with the interpretation that evidence strength/specificity (not just retrieval coverage) is a binding constraint for citation-based validation in complex biomedical domains.

This argument can of course only be made if judge scores correlate with expert judgement; we assess this in the Human–LLM Agreement study.

7.4.8 Validation: Physics CLDs

Experimental design: Section D.6.3.

To establish that the judge functions correctly on "easy" cases with unambiguous causality, we evaluated it on four physics-based CLDs derived from well-established physical laws: (1) Thermostat Heating System based on Newton's Law of Cooling [106] and the First Law of Thermodynamics [107], (2) Predator-Prey dynamics from Lotka [108] and Volterra [109], (3) RC Circuit based on Ohm's Law [110] and Kirchhoff's Laws [111], and (4) Water Tank stock-flow dynamics from Pascal [112] and Torricelli [113]. These CLDs contain 31 edges total across 26 variables, with causal relationships directly derivable from foundational physics equations.

Table 7.8: Comparison of Correctness vs Citation-Based Judging on Physics CLDs

CLD	Correctness	Citation (All)	Citations Retrieved	Citation (Retr.)	Gap
Thermostat	100.0%	62.5%	75.0% (6/8)	83.3%	16.7%
Predator–Prey	100.0%	50.0%	60.0% (6/10)	83.3%	16.7%
RC Circuit	100.0%	14.3%	28.6% (2/7)	50.0%	50.0%
Water Tank	100.0%	50.0%	50.0% (3/6)	100.0%	0.0%
Total	100.0%	45.2%	54.8% (17/31)	82.4%	17.6%

Note. For the Correctness Judge, *approval* = edges with verdict `CORRECT` (score 1.0). For the Citation Judge, *approval* = edges with verdict `FULLY_SUPPORTED` or `PARTIALLY_SUPPORTED` (score ≥ 0.5); `NO_CITATION` = retrieval failure. "Citation (All)" counts `NO_CITATION` as not approved; "Citations Retrieved" = edges with $\geq$1 successfully retrieved citation; "Citation (Retr.)" = approval conditioned on retrieval. Gap = Correctness − Citation (Retr.).

Table 7.8 reports approval rates computed from discrete judge verdicts (see table caption). For the Correctness Judge, approval counts edges receiving `CORRECT` (score 1.0). For the Citation Judge, approval counts edges receiving `FULLY_SUPPORTED` or `PARTIALLY_SUPPORTED` (score $\geq$ 0.5), with `NO_CITATION` indicating retrieval failure (counted as not approved for *Citation (All)* and excluded when computing *Citation (Retrieved Only)*). Under these definitions, the correctness judge gives 100% approval on this small set of 31 physics edges, consistent with the expectation that these relationships are unambiguous and well grounded. We interpret this as a sanity check that the judge does not reject clearly valid edges in an "easy" domain. Citation-based judging shows lower overall approval (45.2%), but this is largely attributable to URL accessibility: 45% of edges received `NO_CITATION` verdicts because the web scraper could not retrieve content from reference URLs. When citations were successfully retrieved, the conditional approval rate rises to 82.4%. The remaining gap (correctness minus citation-retrieved approval) is consistent with cases where retrieved content was incomplete or insufficiently specific for the judge to credit as causal support.

7.4.9 Disentangling Retrieval Quality from Domain Dependence

Experimental design: Section D.6.4.

To test whether the weak discrimination (low scores on valid edges, hence poor separation from invalid edges) could plausibly be explained by retrieval coverage alone, we compare retrieval success and approval rates across domains with different citation sources: (1) expert-provided references (Physics CLDs and Uleman's Alzheimer's), and (2) automated web search (Depressive Symptoms, Social Norms, Emergency Department from RQ1a citation experiments).

Table 7.9: Retrieval Success Comparison

Domain	Citation Source	Retrieval
Physics CLDs	Expert-provided	54.8%
Uleman's (Alzheimer's)	Expert-provided	89.3%
Depressive Symptoms	Automated fetch	100.0%
Social Norms	Automated fetch	99.7%
Emergency Department	Automated fetch	97.9%

Note. Expert-provided references are curated by domain experts; automated fetch uses web search. Retrieval success = percentage of edges where citation content was successfully fetched.

Table 7.9 shows retrieval success rates across domains. The key comparison is between the two expert-provided datasets: Physics CLDs achieve only 54.8% retrieval (due to paywalled textbooks and academic sources), while Uleman's (Alzheimer's) achieves **89.3%** retrieval under standard web scraping (ContentScraper; see Table 7.7). Despite this retrieval advantage, Uleman's receives the lowest citation judge scores (see Table 7.7), while physics CLDs receive high citation judge scores when citations are retrieved (Table 7.8). This pattern—higher retrieval but lower scores on valid edges—supports the interpretation that domain dependence, not retrieval coverage alone, drives the weak discrimination observed in biomedical CLDs. Physics literature often contains more definitive causal statements, while biomedical literature frequently contains conditional, probabilistic, or correlational evidence that the judge may (appropriately) score as weaker causal support, even for valid edges. Moreover, this dependence operates not only at the domain level but also at the question level within domains: some causal claims have more direct experimental or meta-analytic support than others, regardless of the broader domain.

7.4.10 Synthesis of validation findings

The validation studies were designed to isolate the main explanations raised in Section 7.4.5.3. First, human validation confirms that LLM judges track expert judgment with substantial agreement ($\kappa = 0.618$, 80.6% agreement), making judge malfunction less likely as the primary explanation for weak discrimination. Second, cross-domain comparison shows that retrieval coverage alone does not explain weak discrimination: Uleman's achieves 89.3% retrieval yet receives the lowest citation-judge scores, while physics CLDs achieve only 54.8% retrieval but receive high scores when citations are successfully fetched. At the same time, retrieval quality remains a plausible constraint: direct causal evidence (RCTs, experimental manipulations, meta-analyses) may exist in the literature but remain unfound by the naive verbatim-query search that the Citation Judge employs, motivating the Deep Research (DR) approach explored in RQ3 (Section 7.7). Third, the binding constraint appears to be **evidence type**: biomedical literature often contains correlational, conditional, or hedged claims even for valid causal edges, while physics literature contains definitive causal statements directly derivable from established laws. Overall, these validation results indicate that weak citation-judge discrimination in biomedical CLDs reflects a combination of evidence-type constraints and retrieval-quality limits, rather than judge malfunction or retrieval coverage alone.

Sensitivity analysis (prompt robustness). As described in Methods §6.6, we include RM-ANOVA effect sizes as a *supplementary* variance-explained view; non-parametric blocked tests remain the primary evidence. Appendix A.1 consolidates the prompt-sensitivity analysis across the four RQ1a judge experiments using repeated-measures ANOVA (CLD×run blocks; $n = 9$). Prompt choice explains a **large** share of within-block variance for the correctness judge (Corruption Detection: $\eta_p^2 = 0.62$; GT Lit: $\eta_p^2 = 0.68$), and remains robust under Greenhouse–Geisser correction ($p_{\text{GG}} = 0.002$ for both). By contrast, citation-judge prompt effects are weaker and do not survive family-wise correction (e.g., Corruption Detection: $\eta_p^2 = 0.31$, $p_{\text{GG}} = 0.064$; GT Lit: $p_{\text{GG}} = 0.022 > \alpha_{\text{adj}} = 0.0125$).

7.4.11 RQ1a: Main Findings (Summary)

RQ1a asks whether LLM judges can detect incorrect causal edges (hallucinations) in CLDs. Across synthetic corruption tests, literature-ground-truth validation, external expert CLDs, and human agreement checks, we find:

1. **Correctness judges transfer only partially.** A correctness judge that performs strongly on a general hallucination benchmark (TruthfulQA) also detects *synthetic* CLD corruptions with high discrimination (F1 and AUC well above chance). However, performance drops substantially on *expert literature ground truth* CLDs: scores cluster closer to chance for some domains (notably Social Norms), and meaningful gains appear only in specific domains/prompts (e.g., Mechanistic in Depressive Symptoms and Emergency Department).
2. **Citation judges are unreliable as strict validators on expert CLDs.** In both corruption-detection and GT-lit settings, citation-based judging shows weak and domain-dependent discrimination: high recall can coincide with poor separation between correct vs. incorrect edges, and some prompt variants exhibit counterintuitive score–hallucination relationships (suggesting topical alignment rather than causal verification).
3. **Weak discrimination persists under external validation.** On the Uleman's expert CLD, citation-judge scores remain uniformly low on valid edges across multiple search providers and citation fetching methods, indicating the weak discrimination is not explained by a specific retrieval backend.
4. **Judges can match human raters, but with a systematic leniency bias.** Human validation shows substantial LLM–expert agreement overall ($\kappa = 0.618$, 80.6%; $r = 0.732$), comparable to a human baseline. Disagreements are consistently *lenient*: the LLM tends to over-accept evidence support, making it better suited for *screening/triage* than for definitive validation.
5. **Weak discrimination reflects domain and question dependence, not retrieval failure.** Cross-domain comparisons show high retrieval success in complex domains (89.3% on Uleman's), yet persistently low citation-judge scores on valid edges; by contrast, physics citations receive high scores when retrievable. The bottleneck is evidence type—hedged, correlational, or indirectly causal claims—which varies by domain and by specific causal question within domains.
6. **A retrieval quality gap motivates the RQ3 pivot to Deep Research.** The validation studies controlled for retrieval quality by providing expert-curated references; removing this control exposes a gap where direct causal evidence may exist but remain unfound by naive search. RQ3 explores whether causality-targeted, multi-query search can close this gap.

Takeaway: LLM judges provide useful, human-aligned *screening* signals and reliably catch obvious/synthetic corruptions, but they are not yet dependable standalone validators of causal edge correctness against expert ground truth across domains. Validation studies using multiple CLDs (human–LLM agreement, Uleman's expert CLD, and physics sanity checks) help disentangle judge failure from retrieval limitations and domain/question dependence. In particular, the Citation Judge appears reasonably aligned with expert evidence judgments when provided with relevant references (though systematically more lenient), suggesting that

poor discrimination in GT Lit settings is primarily driven by upstream factors—the type/strength of available evidence and whether shallow retrieval surfaces evidence that is sufficiently direct and specific for the causal claim. The strongest case is the Correctness Judge with a Mechanistic prompt (notably in Depressive Symptoms and Emergency Department), while citation-based judging is constrained by evidence type and retrieval quality, motivating the Deep Research approach explored in RQ3.

7.5 RQ1b: LLM-as-a-Corrector for Edge Quality Improvement

RQ1b evaluates whether an LLM-based corrector, conditioned on LLM-judge feedback, can improve CLD edge quality. The experimental design uses a repeated-measures structure with 9 blocks (3 CLDs × 3 runs), treating corrector prompt variant (Baseline, CoT, Mechanistic) as the within-block factor. The corrector receives judge-flagged edges and attempts to revise causal narratives or change edge types to address detected issues.

The objective is to determine (i) whether the corrector improves edge F1 relative to ground truth, (ii) whether improvements generalise from synthetic corruptions (GT Synth) to literature ground truth (GT Lit), and (iii) which corrector prompt variant performs best. We focus on the Correctness Judge type only, given its better RQ1a performance and the prohibitive costs of Citation Judge experiments. For GT Synth experiments, we use the Baseline Judge prompt; for GT Lit experiments, we use the Mechanistic Correctness Judge prompt, which showed better discrimination in RQ1a GT Lit experiments (see Figure 7.4).

Experimental design: Section D.7.

A consolidated corrector prompt sensitivity analysis (effect sizes and assumption checks) is provided in Appendix A.2.

7.5.1 GT Synth × Baseline Judge: Correction Results

Table 7.10: Corrector Performance on Synthetically Corrupted Data (Baseline Prompt)

CLD	F1 Δ	Precision Δ	Recall Δ	Judge Score Δ	Total Actions	Revise	Change Type
Social Norms	$+0.000 \pm 0.000$	$+0.000 \pm 0.000$	$+0.000 \pm 0.000$	$+0.028 \pm 0.108$	4.3 ± 8.0	3.3 ± 3.8	1.0 ± 4.3
Depressive	$+0.145 \pm 0.033$	$+0.164 \pm 0.120$	$+0.117 \pm 0.112$	$+0.186 \pm 0.030$	62.3 ± 14.6	31.7 ± 18.3	30.7 ± 3.8
Emergency Dept	$+0.082 \pm 0.045$	$+0.062 \pm 0.031$	$+0.121 \pm 0.085$	$+0.135 \pm 0.010$	267.7 ± 25.0	115.0 ± 17.9	152.7 ± 16.0
Macro Avg	$+0.075 \pm 0.180$	$+0.075 \pm 0.206$	$+0.079 \pm 0.171$	$+0.116 \pm 0.200$	111.4 ± 343.7	50.0 ± 144.2	61.4 ± 199.7

Note. Results show change in performance (Δ) from pre- to post-correction (3 runs per CLD). F1 Δ = change in F1 score (Post-Pre); Precision Δ, Recall Δ, Judge Score Δ = corresponding changes. Total Actions = Revise + Change (successful corrections only); Revise = motivation revisions; Change Type = edge type changes. Values presented as mean ± 95% CI (two-sided t, $df = 2$) across 3 runs per CLD. Macro average shows mean of CLD means ± 95% CI across CLDs.
Efficacy (blocked). One-sample Wilcoxon signed-rank test of median(F1 Δ)=0 on block-level values (blocks=(CLD, run), n=9): p=0.031*, $r_{rb} = +1.00$.

Table 7.10 presents corrector performance on synthetically corrupted CLDs (30% corruption rate, three runs per CLD). The corrector improves F1 scores for two of three CLDs, with the largest gains observed for Depressive Symptoms (+0.145) and Emergency Department (+0.082), while Social Norms shows no improvement. The aggregate improvement (+0.075) suggests modest recovery on average; however, the high variance indicates inconsistent performance across runs. Precision and recall similarly improve for the Depressive Symptoms and Emergency Department cases but not for Social Norms. Judge scores improve across all CLDs, with the largest gains for Depressive Symptoms, followed by Emergency Department and Social Norms.

The total actions, revise, and change type columns serve as validity checks, confirming that the corrector is functioning as intended. The action type distribution exhibits an approximately 45/55 split between revise and change type operations, deviating slightly from the expected 1/3 to 2/3 ratio. This expected ratio follows from the corruptor's equal probability selection of polarity flip, edge type change, and motivation corruption, where both polarity flip and edge type change are addressed through the change type operation. The observed distribution suggests that revision occurs more frequently than expected, with correspondingly fewer type changes. The macro-average of 111.4 total actions reflects the mean across three CLDs of varying size; Emergency Department dominates (267.7 actions) due to its larger edge count, while Social Norms contributes only 4.3 actions.

For the full corrector action breakdown (including edges where no correction was needed and API/parsing errors), see Table A.5 (Appendix A.2.1).

Overall, the results indicate that the corrector yields modest F1 improvements on average, with Social Norms as an exception. Under the blocked one-sample Wilcoxon signed-rank test on CLD×run blocks ($n = 9$; see Appendix E), the median F1 improvement is detectable under a nominal (unadjusted) $\alpha = 0.05$ criterion on Synthetic ($p = 0.031$). The improvement in judge scores with respect to uncorrected GT Synth is marginally larger but remains negligible. The following section examines whether alternative prompt variants yield different correction performance.

7.5.2 GT Synth × Baseline Judge: Prompt Ablation

To evaluate whether advanced prompting strategies improve correction performance, we tested three prompt variants on synthetically corrupted CLDs: Baseline (examples only), CoT (examples + step-by-step reasoning), and Mechanistic (examples + CoT + adapted Shimonovich's [68] refinement of Bradford Hill causal criteria).

Table 7.11: Corrector Prompt Ablation on Synthetic Corruptions: Performance Comparison

Prompt Variant	Overall F1 Δ (mean ± 95% CI)	Judge Δ (mean ± 95% CI)	Social Norms F1 Δ (mean ± 95% CI)	Depressive F1 Δ (mean ± 95% CI)	Emergency Dept F1 Δ (mean ± 95% CI)	Total Actions	Revise	Change Type
Baseline	$+0.075 \pm 0.180^{*}$	$+0.116 \pm 0.200$	$+0.000 \pm 0.000$	$+0.145 \pm 0.033$	$+0.082 \pm 0.045$	1003	450	553
CoT	$+0.069 \pm 0.153^{*}$	$+0.107 \pm 0.177$	$+0.000 \pm 0.000$	$+0.117 \pm 0.031$	$+0.091 \pm 0.052$	995	471	524
Mechanistic	$+0.026 \pm 0.091$	$+0.108 \pm 0.179$	$+0.000 \pm 0.000$	$+0.010 \pm 0.041$	$+0.068 \pm 0.050$	979	408	571
Aggregate	$+0.057 \pm 0.043$	$+0.111 \pm 0.050$	$+0.000 \pm 0.000$	$+0.090 \pm 0.177$	$+0.080 \pm 0.030$	2977	1329	1648

Note. Ablation study comparing three corrector prompt variants (3 runs per CLD per prompt, 27 total experiments). Aggregate row shows macro-averaged mean ± 95% CI across CLDs. F1 Δ = change in F1 score from pre- to post-correction; Judge Δ = change in LLM judge score. Total Actions = Revise + Change Type (successful corrections only); Revise = motivation revisions; Change Type = edge type changes. Edges with no correction needed or API errors excluded (see Appendix for full breakdown).
Blocked tests. Omnibus prompt effect tested with Friedman using (CLD, run) blocks (n=9); p=0.042*, W=0.35.
Post-hoc. Paired Wilcoxon prompt-pair tests with Bonferroni over 3 pairs ($\alpha_{adj} = 0.0167$): baseline=cot ($p_{adj} = 1.000$, $r_{rb} = +0.24$); baseline>mechanistic ($p_{adj} = 0.188$, $r_{rb} = +0.90$); cot>mechanistic ($p_{adj} = 0.094$, $r_{rb} = +1.00$).
Efficacy. One-sample Wilcoxon tests of median(F1 Δ)=0 on block-level values: baseline: p=0.031*, $r_{rb} = +1.00$, cot: p=0.031*, $r_{rb} = +1.00$, mechanistic: p=0.125, $r_{rb} = +1.00$.

In Table 7.11, "Total Actions" counts only *successful* corrections (Revise + Change Type), matching the action totals reported in the main RQ1b tables. The full action-count verification breakdown (including `None` decisions and API/parsing `Errors`) is provided in Appendix A.2.1 (Table A.5).

Table 7.11 indicates that the Baseline prompt (+0.075) and CoT (+0.069) yield larger mean improvements than the Mechanistic variant (+0.026). While Baseline and CoT show small but detectable improvements over doing nothing on synthetic corruptions under a nominal (unadjusted) $\alpha = 0.05$ criterion (one-sample Wilcoxon vs. $\Delta F1 = 0$: $p = 0.031$; see Appendix E), prompt-to-prompt differences under the same CLD×run blocks are weak and do not survive multiple-comparison correction (see Appendix A.2). Overall, the corrector remains

unreliable for ground-truth correction and its gains on synthetic data do not generalise. Detailed sensitivity analysis (Appendix A.2; Figure A.2) quantifies both efficacy and prompt effects under the blocked design.

Post-hoc paired Wilcoxon comparisons between prompts (Bonferroni over 3 pairs, $\alpha_{\text{adj}} = 0.0167$; see Appendix E) further support this conclusion: on Synthetic, CoT>Mechanistic is marginal at the raw level ($p = 0.031$) but not significant after correction ($p_{\text{adj}} = 0.094$), and Baseline vs. CoT shows no difference ($p = 0.688$). On Ground Truth, CoT>Mechanistic is again marginal raw ($p = 0.0195$) but does not survive correction ($p_{\text{adj}} = 0.0586$). Thus, any prompt ordering is weak compared to the setting shift (Synthetic vs. Ground Truth).

7.5.3 GT Lit × Mechanistic Judge: Correction Results

This section evaluates the corrector using the baseline corrector prompt across runs and CLDs, validated against GT Lit, with the **Mechanistic Correctness Judge prompt** providing the input verdicts. Results are aggregated over three runs per CLD, with mean ± standard deviation reported across CLDs, including a macro average.

Table 7.12: Corrector Performance on Ground Truth Data Using Correctness-Based Validation (Baseline Prompt)

CLD	F1 Δ	Precision Δ	Recall Δ	Judge Score Δ	Total Actions	Revise	Change Type
Social Norms	-0.044 ± 0.191	-0.114 ± 0.173	$+0.083 \pm 0.207$	$+0.083 \pm 0.052$	11.0 ± 7.5	2.3 ± 1.4	8.7 ± 7.6
Depressive	-0.036 ± 0.011	-0.043 ± 0.010	$+0.029 \pm 0.000$	$+0.100 \pm 0.012$	27.0 ± 4.3	14.7 ± 2.9	12.3 ± 3.8
Emergency Dept	$+0.030 \pm 0.017$	$+0.010 \pm 0.009$	$+0.131 \pm 0.057$	$+0.079 \pm 0.011$	99.7 ± 16.5	23.7 ± 10.0	76.0 ± 17.4
Macro Avg	-0.017 ± 0.102	-0.049 ± 0.154	$+0.081 \pm 0.127$	$+0.087 \pm 0.028$	45.9 ± 117.4	13.6 ± 26.6	32.3 ± 94.1

Note. Results show change in performance (Δ) from pre- to post-correction (3 runs per CLD). F1 Δ = change in F1 score (Post-Pre); Precision Δ, Recall Δ, Judge Score Δ = corresponding changes. Total Actions = Revise + Change (successful corrections only); Revise = motivation revisions; Change Type = edge type changes. Values presented as mean ± 95% CI (two-sided t, $df = 2$) across 3 runs per CLD. Macro average shows mean of CLD means ± 95% CI across CLDs.
Efficacy (blocked). One-sample Wilcoxon signed-rank test of median(F1 Δ)=0 on block-level values (blocks=(CLD, run), n=9): p=0.301, $r_{rb} = -0.42$.

Table 7.12 presents mixed results. The corrector improves Emergency Department F1 (+0.032) but degrades performance for Social Norms (−0.072) and Depressive Symptoms (−0.040), yielding a negative macro-average (−0.027). Precision decreases substantially for Social Norms (−0.129) and Depressive Symptoms (−0.047) while increasing marginally for Emergency Department (+0.008). Recall improves across all CLDs.

Judge scores improve substantially in all cases (macro Δ=+0.187), with the largest gains for Depressive Symptoms (+0.264). This pattern indicates that the corrector learns to satisfy the judge more than it improves agreement with expert ground truth. The total actions, revise, and change type columns serve as validity checks, confirming that the corrector functions as intended; the action distribution shows approximately 2/3 revise to 1/3 change type ratio.

For the full corrector action breakdown (including edges where no correction was needed and API/parsing errors), see Table A.6 (Appendix A.2.1).

Overall, the results indicate that even with the more discriminative Mechanistic Correctness Judge input, the corrector is ineffective at improving F1 scores of generated CLDs relative to GT Lit on average.

7.5.4 GT Lit × Mechanistic Judge: Prompt Ablation

We extend the ablation study to ground truth validation, comparing how different corrector prompts perform when evaluated against literature-derived expert CLDs (GT Lit) rather than GT Synth. All experiments use the **Mechanistic Correctness Judge prompt** as input for the corrector.

Table 7.13: Corrector Prompt Ablation on Ground Truth: Performance Comparison

Prompt Variant	Overall F1 Δ (mean ± 95% CI)	Judge Δ (mean ± 95% CI)	Social Norms F1 Δ (mean ± 95% CI)	Depressive F1 Δ (mean ± 95% CI)	Emergency Dept F1 Δ (mean ± 95% CI)	Total Actions	Revise	Change Type
Baseline	-0.017 ± 0.102	$+0.087 \pm 0.028$	-0.044 ± 0.191	-0.036 ± 0.011	$+0.030 \pm 0.017$	413	122	291
CoT	-0.009 ± 0.074	$+0.068 \pm 0.000$	-0.028 ± 0.169	-0.024 ± 0.009	$+0.026 \pm 0.009$	408	144	264
Mechanistic	-0.026 ± 0.107	$+0.090 \pm 0.031$	-0.054 ± 0.163	-0.047 ± 0.006	$+0.024 \pm 0.009$	409	80	329
Aggregate	-0.017 ± 0.026	$+0.086 \pm 0.011$	-0.042 ± 0.033	-0.036 ± 0.028	$+0.027 \pm 0.008$	1230	346	884

Note. Ablation study comparing three corrector prompt variants (3 runs per CLD per prompt, 27 total experiments). Aggregate row shows macro-averaged mean ± 95% CI across CLDs. F1 Δ = change in F1 score from pre- to post-correction; Judge Δ = change in LLM judge score. Total Actions = Revise + Change Type (successful corrections only); Revise = motivation revisions; Change Type = edge type changes. Edges with no correction needed or API errors excluded (see Appendix for full breakdown).
Blocked tests. Omnibus prompt effect tested with Friedman using (CLD, run) blocks (n=9); p=0.028*, W=0.40.
Post-hoc. Paired Wilcoxon prompt-pair tests with Bonferroni over 3 pairs ($\alpha_{adj} = 0.0167$): cot>baseline ($p_{adj} = 0.234$, $r_{rb} = -0.72$); baseline>mechanistic ($p_{adj} = 0.586$, $r_{rb} = +0.56$); cot>mechanistic ($p_{adj} = 0.059$, $r_{rb} = +0.87$).
Efficacy. One-sample Wilcoxon tests of median(F1 Δ)=0 on block-level values: baseline: p=0.301, $r_{rb} = -0.42$, cot: p=1.000, $r_{rb} = -0.02$, mechanistic: p=0.102, $r_{rb} = -0.67$.

In Table 7.13, "Total Actions" counts only *successful* corrections (Revise + Change Type). The full action-count verification breakdown (including `None` decisions and API/parsing `Errors`) is provided in Appendix A.2.1 (Table A.6).

Table 7.13 reveals three key findings: (1) all corrector prompt variants produce negative or near-zero F1 changes (Baseline: −0.027, CoT: −0.011, Mechanistic: −0.017); (2) Emergency Department is the only CLD showing consistent improvement across prompts (+0.032 to +0.034); and (3) no prompt variant achieves statistically significant efficacy (all $p > 0.16$). The Friedman omnibus test shows no significant prompt effect ($p = 0.439$, $W = 0.09$; see Appendix E).

The total number of corrective actions is similar across prompts (∼1100–1120), with the Mechanistic corrector variant revising causal narratives least frequently (663 revisions) while making more edge type changes (440). Judge scores improve uniformly across all prompts (+0.187 to +0.190), indicating that the corrector–judge loop optimises for judge satisfaction rather than ground truth alignment.

Sensitivity analysis (prompt robustness). As described in Methods §6.6, we include RM-ANOVA effect sizes as a *supplementary* variance-explained view; non-parametric blocked tests remain the primary evidence. Appendix A.2 indicates modest prompt sensitivity on GT Synth ($\eta_p^2 = 0.40$; Baseline improves most: +0.075 vs. Mechanistic +0.026), but under Greenhouse–Geisser correction this effect is borderline ($p_{GG} = 0.044$) and does not meet the Bonferroni-adjusted threshold for RQ1b ($\alpha_{adj} = 0.025$). On GT Lit, prompt effects remain non-significant ($p_{GG} = 0.073$), reinforcing that any prompt ordering is weak relative to the Synthetic→Ground Truth setting shift.

7.5.5 RQ1b: Main Findings (Summary)

RQ1b asks whether an LLM *corrector*, conditioned on LLM-judge feedback, can improve CLD edge quality. Across experiments, we find a clear setting dependence:

1. **Synthetic corruptions: modest, inconsistent gains.** On GT Synth (30% corruption), the corrector improves F1 on average (macro ΔF1 = +0.075; Table 7.10), driven by Depressive Symptoms and Emergency Department while Social Norms shows no F1 improvement. Under the blocked one-sample Wilcoxon signed-rank test on CLD×run blocks ($n = 9$; see Appendix E), the median F1 improvement is detectable under a nominal (unadjusted) $\alpha = 0.05$ criterion ($p = 0.031$), and variance across runs remains high.
2. **Prompt engineering has weak effects relative to the setting shift.** On synthetic corruptions, Baseline and CoT yield similar mean improvements and both exceed Mechanistic, but prompt-to-prompt differences are weak and do not survive multiple-comparison correction (Table 7.11). The same conclusion holds on ground truth (Table 7.13): the Friedman omnibus test shows no significant prompt effect ($p = 0.439$; see Appendix E), and any prompt ordering is marginal compared to the much larger drop in effectiveness from Synthetic→Ground Truth.
3. **Ground truth (GT Lit): no reliable improvement in edge recovery.** When validated against literature-derived expert CLDs using the **Mechanistic Correctness Judge** (which showed better discrimination in RQ1a), the corrector does *not* improve F1 on average (macro ΔF1 = −0.027): it slightly helps Emergency Department (+0.032) but degrades Depressive Symptoms (−0.040) and Social Norms (−0.072) under the baseline corrector prompt (Table 7.12). No corrector prompt variant achieves statistically significant efficacy (all $p > 0.16$; Table 7.13). Notably, **judge scores increase consistently** (+0.19 on average) after correction, indicating the corrector learns to satisfy the judge more than it improves agreement with expert ground truth.

Overall, RQ1b results do not support using an LLM corrector triggered by LLM-judge feedback as a reliable mechanism for improving CLDs against expert ground truth, regardless of corrector prompting strategy or judge variant used. Even with the more discriminative Mechanistic Correctness Judge input, the corrector fails to improve F1. Its limited success on synthetic corruption appears to reflect the ease and structure of the synthetic hallucinations correction rather than robust correction ability in the literature-grounded setting. Correctors however do improve judge scores, indicating they are learning to satisfy the judge more than they improve agreement with expert ground truth.

7.6 RQ2: Corpus-Independent Hallucination Detection via UQ Metrics

RQ2 evaluates whether generator-side uncertainty quantification (UQ) metrics can predict hallucinated edges *without* requiring external context (unlike the context-dependent judges in RQ1a). The experimental design uses a blocked aggregation structure with 9 blocks (3 CLDs × 3 runs), with inference performed on block-level means. Four UQ metrics are examined: perplexity (average token-level uncertainty), minimum probability (confidence floor), maximum window entropy (local uncertainty peaks)—all derived from generator logprobs—and cosine similarity (semantic alignment between narrative and retrieved citations, available for citation experiments only). Edges are labelled as *misaligned with GT Lit* (FP or FN under the edge-existence definition) versus *aligned with GT Lit* (TP or TN); this binary label is compared to the UQ scores.

The objective is to determine (i) whether individual UQ metrics discriminate hallucinations from correct edges better than chance, (ii) whether ensemble classifiers combining multiple UQ features improve detection, and (iii) whether UQ-based detection generalises across CLDs (leave-one-CLD-out validation). Synthetic corruption (GT Synth) experiments are excluded because post-hoc corruption cannot be reflected in generation-time logprobs; therefore, only GT Lit ground truth is used (hallucination = FP∨FN w.r.t. expert CLD).

Experimental design: Section D.8.

7.6.1 Single UQ Metric Performance

We evaluate each UQ metric's discriminative power for hallucination detection by treating the raw metric value as a classifier score and computing the area under the ROC curve (AUC-ROC). This measures the probability that a randomly chosen hallucination has a higher metric value than a randomly chosen correct prediction. Interpretation: AUC > 0.5 indicates higher metric values correlate with hallucinations; AUC < 0.5 indicates *lower* values correlate with hallucinations (the metric is still informative, but in the opposite direction—e.g., Gen Min Prob AUC $= 0.479$ means lower minimum probability $\rightarrow$ more likely hallucination, which aligns with the intuition that low-confidence generations are more error-prone). AUC $= 0.5$ indicates no discriminative power (random).

Class balance & metric choice. In this dataset, hallucinations are the minority class (15.2% of edges; Methods §D.8). We use AUC-ROC because it is threshold-free and summarises ranking/separation independent of any single operating point; however, under class imbalance a "moderate" AUC does *not* guarantee high precision at a practically useful recall. Above-chance claims in this section refer to **block-level** one-sample Wilcoxon tests on (AUC $- 0.5$) (Table 7.14); we do not treat edge-level tests as confirmatory due to dependence within files/blocks. We therefore complement AUC with distribution plots and effect-size diagnostics (Figure 7.10).

Table 7.14: Single UQ Metric Performance for Hallucination Detection (Meta-Analysis)

UQ Metric	N Edges	Mean AUC (95% CI)	Correlation r (95% CI)	Significant Files (%)	N Files
Gen Cosine Similarity	16,492	$0.672*$ $\pm$ 0.071	$+0.157^{ns}$ $\pm$ 0.107	65.7%	35
Gen Perplexity	31,704	0.563^{ns} $\pm$ 0.065	$+0.006^{ns}$ $\pm$ 0.082	44.4%	63
Gen Max Window Entropy	31,704	0.554^{ns} $\pm$ 0.052	$+0.070^{ns}$ $\pm$ 0.081	49.2%	63
Gen Min Prob	31,704	0.479^{ns} $\pm$ 0.049	-0.036^{ns} $\pm$ 0.075	14.3%	63

Note. Meta-analysis using three logprob-derived metrics and one retrieval-alignment metric (cosine similarity). N Edges = total causal edges analyzed; N Files = experiment files containing metric. Gen Cosine Similarity has fewer observations (citation-judging runs only).
Units of analysis: **Block-level** (CLD $\times$ Run, $N = 9$ blocks) used for inference on Mean AUC and Correlation r; 95% CIs via t-distribution ($df = 8$). **Edge-level** (n =16k–32k) used only for descriptive diagnostics. **Significant Files (%)** = percentage of files with edge-level Mann-Whitney U $p < 0.05$ (descriptive only).
Multiple comparisons: Bonferroni correction across 4 metrics ($\alpha_{\text{adj}} = 0.0125$). Significance: *** $p_{\text{adj}} < 0.001$, ** < 0.01, * < 0.05, ^{ns} = not significant. Hypothesis tests: one-sample Wilcoxon signed-rank on block-level (AUC $- 0.5$) or correlations.

Table 7.14 summarises the single-metric meta-analysis results. Statistical inference for RQ2 is performed at the block level (CLD$\times$run, $N = 9$) using one-sample Wilcoxon signed-rank tests on (AUC $- 0.5$) and t-based 95% CIs on block-mean AUCs (see Appendix E).

Gen Cosine Similarity achieves the highest mean AUC (0.671) and is the only single metric with above-chance discrimination that is significant under the block-level one-sample Wilcoxon tests after Bonferroni correction (Table 7.14). The three logprob-derived metrics (Gen Perplexity, Gen Max Window Entropy, Gen Min Prob)

are closer to chance and are not significant at the corrected threshold; in particular, Gen Min Prob has AUC slightly below 0.5, which implies the expected direction (lower minimum token probability → higher hallucination likelihood), but the effect is small and not distinguishable from chance at the block level.

Notably, cosine similarity also exhibits a counterintuitive direction in the correlation diagnostics: higher similarity to retrieved citations corresponds to higher hallucination likelihood (positive r), though these correlation estimates are non-significant after correction. A plausible explanation is that cosine similarity captures topical/lexical overlap between the model's motivation and whatever text is retrieved, rather than the presence of specific causal evidence for the claim; because retrieval is driven by narrative-derived queries, "high similarity" can co-occur with weak, correlational, or otherwise non-diagnostic evidence. Overall, single metrics provide at best weak screening signal, motivating the subsequent ensemble analyses (Table 7.16) and cross-domain generalisation tests.

7.6.2 Feature Distributions: Correct vs. Hallucinations

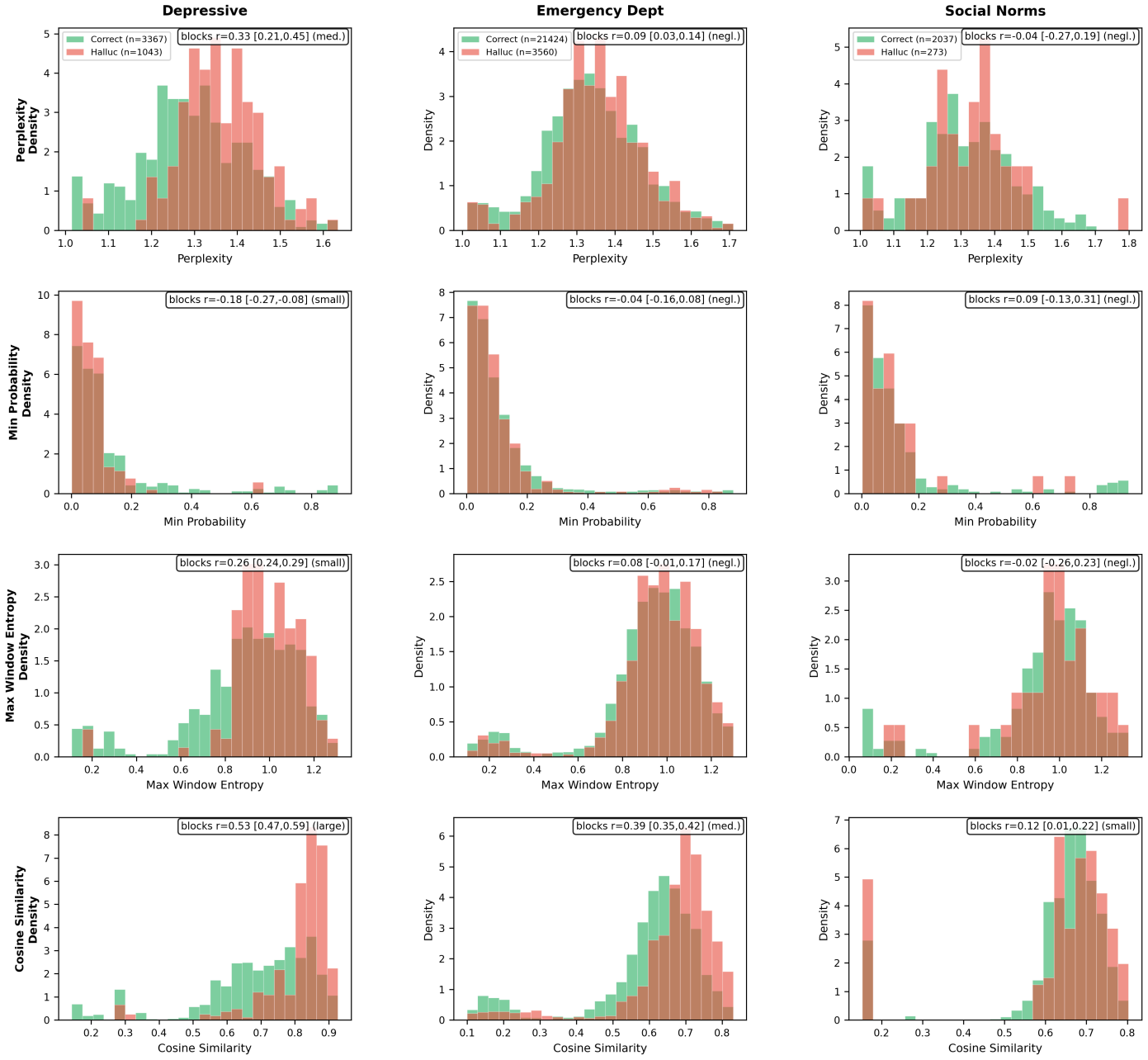


Figure 7.10: Generator UQ metric distributions comparing correct edges (green) vs. hallucinations (red) for each CLD (99th percentile). Rows show generator metrics; columns show CLDs. Subplot annotations report block-level rank-biserial effect sizes r aggregated across runs within each CLD (mean [95% CI] over the 3 (CLD×run) blocks; t-distribution, $df = 2$), rather than edge-level p-values.

To contextualize these distributions, we first assess whether the UQ metrics are approximately normal—a key assumption for many common parametric comparisons. Table 7.15 reports Shapiro–Wilk normality tests (overall, hallucinations only, and correct edges only) for each generator-based metric (see Appendix E). All tests reject normality at $\alpha = 0.05$, consistent with the pronounced skewness and heavy tails observed in Figure 7.10. Consequently, edge-level distribution comparisons are treated as descriptive/diagnostic.

Table 7.15: Shapiro-Wilk Normality Tests for UQ Metric Distributions

UQ Metric	Group	N	Skew	Kurt	W	p-value	Normal?
Perplexity	Overall	31,704†	+16.30	+263.7	0.037	$< 10^{-50}$	No
	Halluc	4,876	-0.07	+1.5	0.973	1.58×10^{-29}	No
	Correct	26,828†	+14.98	+222.4	0.035	$< 10^{-50}$	No
Min Prob	Overall	31,704†	+3.17	+10.5	0.608	$< 10^{-50}$	No
	Halluc	4,876	+3.71	+14.8	0.542	$< 10^{-50}$	No
	Correct	26,828†	+3.09	+10.0	0.598	$< 10^{-50}$	No
Max Window Entropy	Overall	31,704†	-1.60	+2.9	0.853	$< 10^{-50}$	No
	Halluc	4,876	-1.88	+4.9	0.831	$< 10^{-50}$	No
	Correct	26,828†	-1.56	+2.6	0.850	$< 10^{-50}$	No
Cosine Similarity	Overall	16,507†	-1.54	+2.4	0.842	$< 10^{-50}$	No
	Halluc	2,514	-1.87	+4.5	0.827	5.63×10^{-46}	No
	Correct	13,993†	-1.56	+2.4	0.836	$< 10^{-50}$	No

Note. Shapiro-Wilk test for normality (null hypothesis: data is normally distributed). †Subsampled to $n = 5{,}000$ due to Shapiro-Wilk sample size limit. Skew = skewness (0 = symmetric; positive = right-tailed); Kurt = excess kurtosis (0 = normal; positive = heavy-tailed). All metrics reject normality at $\alpha = 0.05$, justifying non-parametric methods (Mann-Whitney U, AUC-ROC).

The normality diagnostics confirm that the generator-based UQ metrics are strongly non-Gaussian at the edge level (both overall and within hallucination/correct strata). This cautions against over-interpreting visual separation alone and motivates non-parametric, distribution-robust summaries for within-CLD diagnostics. For inferential claims about RQ2 (above-chance discrimination across independent generation scenarios), we rely on block-level aggregation (CLD×run) rather than edge-level p-values to avoid pseudo-replication.

Figure 7.10 compares LLM generator uncertainty metric distributions for correct edges (green) versus hallucinations (red) across the three CLDs (99th percentile clipped; rank-biserial effect sizes shown per subplot). Columns show different CLDs; rows show different metrics. Overall, separation is visually subtle and domain-dependent: the block-level rank-biserial effects in the subplot annotations are often small and sometimes near zero, indicating that marginal distributions alone do not yield a robust, domain-invariant decision rule.

Perplexity shows the clearest directional effect in Depressive Symptoms (hallucinations tend to have higher perplexity), while effects in Emergency Department and Social Norms are weaker and closer to negligible by the block-level effect summaries.

Minimum token probability exhibits a consistent directional pattern across CLDs: hallucinations are concentrated toward lower minimum probabilities, while correct edges show a heavier tail of higher minimum-probability tokens. However, the block-level effect sizes indicate that this separation is modest and not uniformly strong across domains.

Maximum window entropy shows a domain-dependent pattern: hallucinations tend to exhibit higher local entropy peaks in Depressive Symptoms and Emergency Department, while Social Norms shows weaker separation and some apparent multimodality. Finally, cosine similarity to the retrieved context (available only for citation judging runs) often trends higher for hallucinations, reinforcing that semantic proximity to retrieved text does not guarantee causal correctness and may reflect topical alignment rather than faithful edge generation.

Overall, evidence suggests that these metrics carry some signal, but it appears domain-dependent, uneven, and subtle, and does not yield clear visual separation.

Even if no single metric yields a clear univariate threshold, the metrics may still be jointly informative: multivariate models can capture interactions and non-linear combinations that define a decision boundary not evident in marginal distributions. Therefore, we next train an ensemble classifier using all generator metrics jointly.

7.6.3 Recursive Feature Elimination Analysis

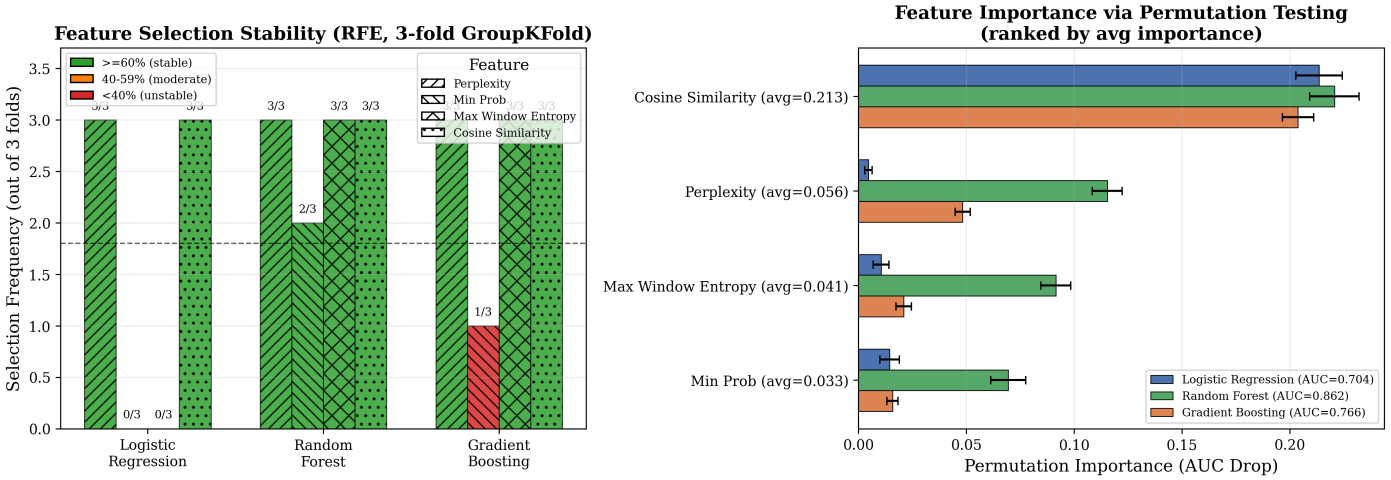


Figure 7.11: RFE analysis across classifiers with 3-fold block-level GroupKFold (blocks = CLD×run). **Left:** Feature selection stability with colour-coded thresholds (green ≥60% = stable, orange 40–59%, red <40% = unstable); labels show selection frequency. **Right:** Permutation importance (AUC drop when feature is shuffled) for each classifier; error bars show standard deviation across 30 permutations.

Figure 7.11 reveals substantial differences in feature selection and importance across classifiers. Gen Cosine Similarity is the most important feature for all classifiers (permutation importance 0.21–0.29), consistent with its stable selection across all models. For Logistic Regression (AUC = 0.704), Gen Min Prob, Gen Max Window Entropy, and Gen Perplexity show near-zero permutation importance, explaining why RFE never selects them (0/3 folds). In contrast, Random Forest (AUC = 0.862) shows high permutation importance for all 4 features (0.10–0.29), justifying its selection of all features. However, Random Forest's superior in-distribution performance comes at the cost of poor cross-domain generalisation (Table 7.16), suggesting that the additional features capture CLD-specific patterns rather than generalisable hallucination signals.

Having identified which features contribute to classification, we now evaluate ensemble classifiers using a cross-validation framework designed to detect overfitting and assess cross-CLD generalisation.

7.6.4 Ensemble Classification Results

Ensemble classifiers require a consistent feature vector per edge. Therefore, Phases 4–6 are computed on the subset of edges for which all four UQ metrics are available (i.e., citation-judging runs only), yielding 16,507 edges across 35 experiment files—a reduction from the 31,704 edges across 63 files used in the single-metric analysis.

Table 7.16 presents a cross-phase validation analysis designed to detect overfitting. Each phase uses a progressively stricter validation strategy: Phase 4 tests feature selection on all pooled data; Phase 5 evaluates on held-out samples from the same distribution (all CLDs); Phase 6 tests generalisation to entirely unseen CLDs

(leave-one-CLD-out cross-validation). Performance drops between phases indicate the degree to which models exploit CLD-specific patterns rather than learning generalizable hallucination signals.

Table 7.16: Ensemble Classifier Performance Across Validation Phases (Block-Level Cross-Validation)

Classifier	Phase 5 CV AUC (± 95% CI)	Phase 5 Test AUC	Phase 6: Cross-CLD AUC (± 95% CI)	Drop (5→6)	F1@t* (± 95% CI)
Logistic Regression	0.695 ± 0.067	0.702	0.664 ± 0.239	0.039	0.310 ± 0.289
Random Forest	0.832 ± 0.304	0.809	0.629 ± 0.179	0.179	0.215 ± 0.387
Gradient Boosting	0.743 ± 0.059	0.765	0.645 ± 0.296	0.120	0.317 ± 0.358
Neural Network	0.709 ± 0.032	0.739	0.648 ± 0.211	0.092	0.317 ± 0.347

Note. Block-level evaluation (blocks = CLD×run, $N = 9$); entire blocks stay together in train/test splits. **Phase 5:** 3-fold GroupKFold CV ($n = 3$); Test AUC on held-out ∼33% of blocks (point estimate). **Phase 6:** Leave-one-CLD-out ($n = 3$ folds); F1@t^* uses threshold selected on Phase 5 training only, then frozen. 95% CIs via t-distribution ($df = n - 1$). Drop = Phase 5 Test AUC − Phase 6 AUC.
Selected t^*: Logistic Regression: $t^* = 0.470$; Random Forest: $t^* = 0.467$; Gradient Boosting: $t^* = 0.162$; Neural Network: $t^* = 0.141$.

Table 7.16 shows cross-phase validation results using block-level GroupKFold (blocks = CLD×run, $N = 9$). Random Forest achieves the highest in-distribution performance (Phase 5 Test AUC = 0.809) but exhibits substantial degradation under leave-one-CLD-out evaluation (Phase 6 mean AUC = 0.629, drop = 0.179). Gradient Boosting shows similar overfitting (Phase 5 = 0.765, Phase 6 = 0.645, drop = 0.120). In contrast, Logistic Regression and the Neural Network show smaller cross-domain degradation (drops of 0.039 and 0.092) and very similar Phase 6 mean AUC (0.664 vs 0.648). The block-level CV prevents pseudo-replication from correlated edges within the same generation scenario, yielding more conservative but trustworthy performance estimates. Overall, these results suggest that simpler models generalise more robustly across CLDs, while higher-capacity tree ensembles are more prone to CLD-specific overfitting.

Practical operating-point comparison (F1 at a fixed threshold). For deployment, a detector must make binary decisions at a fixed operating point. Unlike RQ1a judges (whose scores have intrinsic semantics and are thresholded at 0.5), UQ-based classifiers do not have a natural default threshold under domain shift. To make the comparison principled, we select a fixed threshold t^* for each classifier on Phase 5 training blocks only by maximizing out-of-fold F1, and then freeze t^* for Phase 6 (cross-CLD) evaluation. Under this policy, the Phase 6 cross-CLD ensembles still achieve only modest fixed-threshold performance (mean F1@t* ≈ 0.22–0.32; Table 7.16). By contrast, the Mechanistic Correctness Judge in RQ1a GT Lit—using the fixed ≤ 0.5 decision rule for "hallucination"—achieves substantially higher F1 (overall F1 ≈ 0.45, with CLD-level F1 ≈ 0.56 Depressive, ≈ 0.48 Emergency Department, and ≈ 0.30 Social Norms; see Figure 7.4). This gap indicates that while UQ features carry some ranking signal (AUC ∼ 0.65), they are not reliable enough at a frozen operating point to serve as the primary hallucination flagger across domains. Practically, UQ metrics are better treated as auxiliary monitoring/triage signals, while context-sensitive judging provides a stronger decision signal for triggering downstream interventions, as optimal thresholds in the training set may prove brittle for CLDs in other domains while the Judge configuration proved robust in validation experiments on CLDs from other domains. Although AUC = 0.5 corresponds to random-ranking discrimination, we do not report a formal p-value comparison to 0.5 in Phase 6 because the leave-one-CLD-out procedure reuses the same CLDs across folds; blocks within a held-out CLD share the same trained model; and training sets overlap heavily across folds. These dependencies violate i.i.d. assumptions, so p-values would be easy to over-interpret.

7.6.5 RQ2: Main Findings (Summary)

RQ2 asks whether generator-time uncertainty (UQ) metrics can predict hallucinated edges (FP/FN w.r.t. GT Lit) in CLD generation. Across the single-metric and ensemble analyses, we find:

1. **Single metrics carry only weak-to-moderate signal.** Gen Cosine Similarity is the strongest single metric (AUC = 0.671; Table 7.14) and the only one that is Bonferroni-significant above chance at the block level; the remaining generator-time logprob metrics are not significant after correction and cluster near chance. Cosine similarity also shows a counterintuitive direction in the correlation diagnostics (higher similarity associated with hallucinations), consistent with topical overlap rather than causal faithfulness; correlation tests are non-significant after correction.
2. **High-capacity ensembles overfit across domains.** Random Forest achieves the highest in-distribution performance (Phase 5 Test AUC = 0.809) but degrades under leave-one-CLD-out evaluation (Phase 6 mean AUC = 0.629), indicating CLD-specific overfitting (Table 7.16).
3. **Simpler models generalise better, but remain limited.** Logistic Regression and the Neural Network show the smallest cross-domain performance drops (0.039 and 0.092) and comparable Phase 6 mean AUC (∼0.65–0.66; Table 7.16), implying some generalizable signal, but not strong enough to support reliable cross-domain hallucination detection in isolation.

Takeaway: Single generator UQ metrics provide a weak signal for hallucination detection. When used in an ensemble as input features for classifiers, they perform well in-distribution for some models (Gradient Boosting, Random Forest) but show limited cross-CLD generalisation, reducing practical applicability. In this study, UQ metrics are better viewed as auxiliary features for triage/monitoring rather than as a standalone, domain-robust hallucination detector.

RQ3 Pivot Rationale. The original RQ3 design—using generator UQ metrics to selectively deploy the corrector—is not supported by our findings. In RQ2, UQ-based ensembles retain above-chance ranking signal under cross-CLD evaluation (Phase 6 AUC ≈ 0.63–0.67), but their fixed-threshold decision performance remains modest even when the threshold is selected in-distribution and frozen out-of-domain (Phase 6 mean F1@t* ≈ 0.22–0.32; Table 7.16), making them a weak trigger for downstream compute across domains. In contrast, RQ1a shows that the Mechanistic Correctness Judge yields substantially higher GT Lit hallucination-flagging performance under its fixed thresholding rule (F1 ≈ 0.45 overall; Figure 7.4), providing a stronger and more stable uncertainty signal. Separately, RQ1b shows the corrector does not reliably improve GT Lit quality (macro ΔF1 ≈ −0.02), limiting the value of triggering correction even if we had a strong trigger. Additionally, the RQ1a validation studies suggest that weak GT Lit citation-judge discrimination is driven primarily by evidence-type constraints and a *retrieval quality gap* (direct causal evidence may exist but remain unfound by shallow search), rather than judge malfunction. Finally, RQ3 is motivated by a construct-validity question raised throughout GT Lit evaluation: some edges classified as false positives (FPs) may reflect literature-supported relationships that experts omitted for scope/theoretical reasons or that do not match the expert CLD due to construct/variable mapping differences (i.e., "false positives" may include overlooked-but-valid edges). We therefore pivot to an alternative form of smart test-time compute: using the Mechanistic Correctness Judge score as the uncertainty signal, and Deep Research (DR)—automated multi-agent literature search—as the form of deploying more (expensive) test-time compute that can measurably validate causal claims.

This preserves the spirit of RQ3: a computationally cheap uncertainty signal triggers selective deployment of more expensive but more effective test-time compute. The RQ1a physics CLD validation (Section 7.4.8) provides empirical motivation for DR: simple citation retrieval achieves 89.5% retrieval coverage on Uleman's CLD but yields low judge scores, while physics CLDs with lower retrieval (55%) achieve high approval when retrieved (82.4%). The bottleneck is not retrieval success but evidence type—standard search finds correlational

literature in complex domains while physics literature contains definitive causal statements. DR addresses this by specifically targeting direct causal evidence (RCTs, experimental manipulations, meta-analyses) through iterative, multi-agent search that filters for study types establishing causation rather than correlation. Due to time constraints from the late pivot, we did not explore the DR design space as systematically as in RQ1a/RQ1b with a component-wise ablation study. The DR system uses multiple search queries in parallel, iterative evidence gathering, and a multi-agent pipeline for scoring, selection, and judgement (see Methods, Section 4.7, for full system architecture).

Limitations of the exploratory design. The DR system has many degrees of freedom (query count, iteration depth, scoring thresholds, agent prompts, and components) that would ideally require a full ablation study with local and global sensitivity analysis. Additionally, searching for evidence *for* a claim introduces confirmation bias risk, and the system inherits publication bias from the underlying literature. These limitations are acknowledged; the following results should be interpreted as an exploratory pilot rather than a definitive evaluation.

7.7 RQ3: Deep Research Validation (Exploratory Pilot)

RQ3 evaluates whether automated multi-agent literature search (Deep Research) can validate LLM-generated causal claims by retrieving direct causal evidence (RCTs, experimental manipulations, meta-analyses) from the scientific literature. The experimental design is an exploratory pilot study (single-run, no replication) due to computational cost (∼$248 for 285 edges). Edges are stratified by GT Lit validation class (TP/FP/FN), and the primary metric is class-wise *detection rate*—the fraction of edges for which Deep Research retrieves direct causal evidence.

The objective is to determine (i) whether Deep Research can discriminate between valid (TP) and hallucinated (FP) edges better than the simple citation retrieval used in RQ1a, (ii) whether detection rates vary by CLD domain, (iii) whether DR-validated edges could improve generation performance under a ground-truth enrichment scenario, and (iv) whether adaptive test-time compute allocation (smart-trigger deployment of DR) can improve accuracy-per-dollar relative to always-on DR. Following the main analysis, we report computational costs, per-CLD detection breakdowns, an exploratory enrichment analysis, and human validation of DR evidence applicability.

Methodological pivot. The original RQ3 (smart test-time compute using UQ metrics to selectively deploy correctors) was contingent on RQ2 generalisation and RQ1b corrector efficacy—conditions not supported by our findings. We therefore repurposed RQ3 as an exploratory pilot of **Deep Research (DR)**: using automated literature search to *validate* (rather than correct) LLM-generated causal claims. This pilot is single-run and non-ablated; see Discussion (RQ3 pivot section) for rationale and limitations.

Experimental design: Section D.9.

We deployed DR on 285 edges (TP + FP + FN from the RQ1a GT Lit labels across all three CLDs; TN omitted to reduce cost) and asked whether automated literature search can validate the LLM's causal *narrative* for each edge. The key metric is **detection rate**: the fraction of edges for which DR retrieves *direct causal evidence* (RCTs, experimental manipulations, or meta-analyses; not purely correlational studies). We additionally report DR's self-assessed confidence (1–10).

Table 7.17: Deep Research Detection Rate and Discrimination Analysis (n=285 edges)

Classification	n	Detection Rate	Confidence	TP–X Gap	Cohen's h	p-value
TP (Expert-accepted)	44	70.5%	6.59	—	—	—
FP (LLM hallucinations)	178	62.4%	6.64	8.1 pp	0.17	.38
FN (Expert-rejected)	63	42.9%	6.78	27.6 pp	0.56	.006**

Note. Detection rate = % edges with direct causal evidence found. Confidence = self-assessed (1–10 scale). TP/FP/FN from RQ1a GT Lit validation across all 3 CLDs.

TP–FP test. Fisher's exact $p = .38$; OR = 1.44 [95% CI: 0.70, 2.94]; Cohen's $h = 0.17$ [95% CI: -0.16, 0.50] (negligible); power = 17%.

TP–FN test. Fisher's exact $p = .006**$; Cohen's $h = 0.56$ [95% CI: 0.18, 0.95] (medium); power = 82%.

FP–FN test. Fisher's exact $p = .008**$; Cohen's $h = 0.39$ [95% CI: 0.11, 0.68] (small); power = 76%.

Multiple comparisons. Bonferroni correction for 3 pairwise tests. TP–FP $p_{adj} = 1.00$; TP–FN $p_{adj} = .02*$; FP–FN $p_{adj} = .02*$.

Table 7.17 shows that TP edges have a higher detection rate than FP edges, but the TP–FP gap (8.1 pp) is not statistically significant ($p = .38$; low power, effect-size CI includes zero). In contrast, detection differs substantially between TP and FN (27.6 pp; significant after correction; medium effect) and between FP and FN (19.5 pp; significant after correction; small effect).

Given these aggregate results, we next report per-CLD detection rates to assess whether DR's discriminative behaviour is domain-dependent.

7.7.1 Computational Cost

Experimental design: Section D.9.0.2.

Overall computational costs are summarised in Table 7.25. Under our pricing assumptions, DR costs ∼ $248 for 285 edges ($0.87/edge), making it ∼119× more expensive than citation judging and ∼512× more expensive than correctness judging. True Negatives (1,109 edges) were not analysed due to cost (extrapolating DR to all TNs would cost ∼ $965 at $0.87/edge).

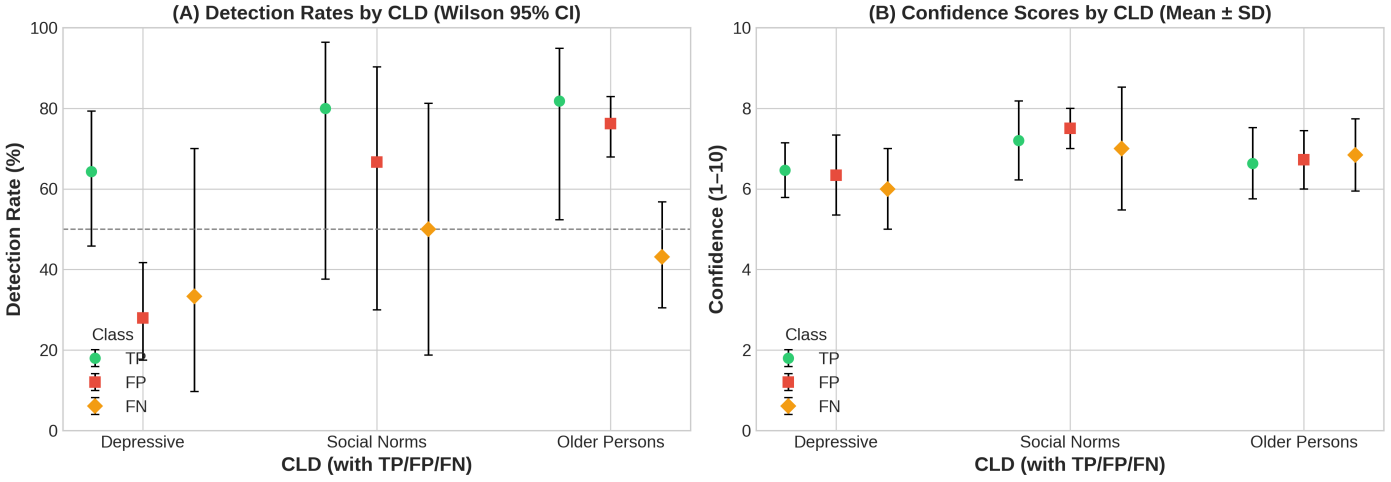


Figure 7.12: Deep Research results. **(A)** Detection rates by CLD and classification (TP/FP/FN) with Wilson 95% CIs. **(B)** Confidence scores by CLD and classification (mean ± SD).

Figure 7.12A shows substantial between-CLD variation in detection gaps between edge classes: Depressive Symptoms exhibits meaningful discrimination (TP–FP gap = +36 pp), while Social Norms (+13 pp) and Emergency Department (+6 pp) show weak discrimination; uncertainty is substantial for Social Norms due to small n. Figure 7.12B shows DR's self-reported confidence provides little discrimination between edge classes: means cluster tightly (range < 1 point on a 10-point scale) with substantial overlap across TP/FP/FN within each domain.

In aggregate, DR does not significantly discriminate between valid and hallucinated (as defined as FP in GT Lit) causal claims (TP–FP gap = 8.1 pp; $p = .38$; Cohen's $h = 0.17$, negligible). Nonetheless, DR retrieves direct causal evidence for 62.4% of FP edges and 42.9% of FN edges. This is consistent with multiple explanations: (i) some "false positives" may reflect literature-supported relationships that experts excluded for scope/theoretical reasons, and/or (ii) DR may retrieve evidence that is not sufficiently specific or causal for the exact claim—a motivation for the human validation causal evidence applicability analysis below. The FP–FN gap (+19.5 pp, $p = .008^{**}$, $p_{\text{adj}} = .024^{*}$) survives correction: FP edges have higher detection than FN edges.

7.7.2 Per-CLD Detection Rate Analysis

Experimental design: Section D.9.0.1.

Table 7.18: Per-CLD Pairwise Discrimination Tests (Fisher's Exact, Bonferroni $m = 3$)

CLD	Comparison	Δpp	Cohen's h	Effect size	p	p_{adj}
Depressive	TP–FP	+36.3	0.75	medium	.004**	.01*
	TP–FN	+31.0	0.63	medium	.20	.61
	FP–FN	-5.3	-0.12	negligible	1.00	1.00
Social Norms	TP–FP	+13.3	0.30	small	1.00	1.00
	TP–FN	+30.0	0.64	medium	.55	1.00
	FP–FN	+16.7	0.34	small	1.00	1.00
Older Persons	TP–FP	+5.6	0.14	negligible	1.00	1.00
	TP–FN	+38.7	0.83	large	.04*	.13
	FP–FN	+33.1	0.69	medium	<.001***	<.001***

Note. All pairwise comparisons (TP–FP, TP–FN, FP–FN) within each CLD using Fisher's exact test; Bonferroni correction applied within each CLD ($m = 3$, $\alpha_{\text{adj}} = 0.017$). Effect size interpretation: $|h| < 0.2$ negligible, 0.2–0.5 small, 0.5–0.8 medium, > 0.8 large. Detection rates and confidence distributions shown in Figure 7.12.

Table 7.18 reveals substantial domain variation in DR's TP–FP discriminative ability, relatively smaller variation in TP–FN, and substantial variation in the FP–FN gap. In the Depressive Symptoms CLD, DR clearly distinguishes valid from hallucinated edges (TP–FP: +36.3 pp, $h = 0.75$, $p_{\text{adj}} = .01$). The Social Norms CLD shows a smaller, non-significant gap (TP–FP: +13.3 pp, $h = 0.30$). In contrast, the Emergency Department CLD shows negligible TP–FP discrimination (+5.6 pp, $h = 0.14$), but a significant FP–FN gap (+33.1 pp, $h = 0.69$, $p_{\text{adj}} < .001$), indicating DR validates most edges—including false positives—while consistently missing edges the LLM failed to generate.

7.7.3 Exploratory: Ground Truth Enrichment Analysis

Experimental design: Section D.9.0.4.

Given that 62.4% of FP edges have `direct_causal_found=True` under DR, we explore an optimistic upper bound: *if* DR-validated FPs were treated as true causal relationships, how would generation performance

change? This assumption is intentionally strong and is used only to bound the potential impact; the human-validation analysis below provides a more conservative calibration.

Table 7.19: LLM Generation Metrics: Two Enrichment Scenarios (Aggregate and Per-CLD)

	Original	Scenario 1: Enriched GT		Scenario 2: LLM+DR System	
Dataset	**F1**	**F1**	**FPs → TP**	**F1**	**FNs found**
Aggregate	0.267	0.705 (+163%)	111/178 (62%)	0.779 (+191%)	27/63 (43%)
Depressive	0.500	0.667 (+33%)	14/50 (28%)	0.687 (+37%)	2/6 (33%)
Social Norms	0.455	0.692 (+52%)	4/6 (67%)	0.828 (+82%)	3/6 (50%)
Older Persons	0.113	0.722 (+540%)	93/122 (76%)	0.813 (+621%)	22/51 (43%)

Scenario 1 (Enriched GT): Promotes FP edges with DR evidence to TP. Tests: "What if experts missed valid relationships?"
Scenario 2 (LLM+DR System): Scenario 1 + DR discovers FN edges with literature support. Tests: "What could a complete LLM+DR pipeline achieve?"
Aggregate confusion matrices: Original: TP=44, FP=178, FN=63. Scenario 1: TP=155, FP=67, FN=63. Scenario 2: TP=182, FP=67, FN=36.

Of the 63 FN edges (expert-included, LLM-missed), 27 (42.9%) have DR causal evidence. Under Scenario 2, a complete LLM+DR pipeline would recover these edges as positive relationships, reducing FN from 63 to 36. The remaining 36 FN edges (57.1%) lack searchable evidence under our DR settings, consistent with tacit domain knowledge or evidence types not captured by the retrieved literature. Scenario 1 quantifies the impact of promoting FP→TP (F1: 0.267 → 0.705), while Scenario 2 estimates an upper bound when additionally recovering FN edges with DR evidence (F1: 0.267 → 0.779). The 36 residual FNs represent expert knowledge that neither LLM generation nor literature search recovers in this setting.

Human validation sampling and labels. We stratified-sampled 50 edges from the Deep Research output restricted to `direct_causal_found=True` and manually labelled each edge with a categorical verdict (`human_verdict`∈{`causal`, `different_vars, causal`, `Not relevant / non-causal`}) and a human-assigned evidence applicability score `Fully applicable`∈ $[0, 1]$. The sampling protocol and applicability rubric are defined in Methods (Section D.9.0.5).

Human validation evidence verdicts. Table 7.20 summarises the distribution of human verdicts by edge class. All TP edges are supported by direct causal evidence, but a majority of FP edges also have causal evidence (either exact-match `causal` or `different_vars, causal`). This corroborates that the primary failure mode is *variable mismatch* (causal evidence for related constructs/measures), rather than outright non-causal evidence.

Table 7.20: Human Validation: Evidence Verdict Distribution by Edge Classification ($n = 50$)

Verdict	**TP** ($n = 9$)	**FP** ($n = 32$)	**FN** ($n = 9$)	**Total**
`causal`	9 (100.0%)	21 (65.6%)	5 (55.6%)	35 (70.0%)
`different_vars, causal`	0 (0.0%)	8 (25.0%)	2 (22.2%)	10 (20.0%)
`Not relevant / non-causal`	0 (0.0%)	3 (9.4%)	2 (22.2%)	5 (10.0%)
Any causal	9 (100.0%)	29 (90.6%)	7 (77.8%)	45 (90.0%)

Note. `causal` = exact match to edge claim; `different_vars, causal` = causal evidence for related but not identical variables/constructs; `Not relevant / non-causal` = evidence does not support a causal relationship. "Any causal" = `causal` ∪ `different_vars, causal`.

Applicability-based calibration (thesis-ready filtering). Because DR's binary `direct_causal_found=True` flag only indicates that *some* direct causal evidence was retrieved—and often fails via variable/operationalisation mismatch—we calibrated it using the human `Fully applicable`$\in [0, 1]$ score (see Methods Section D.9.0.6 for threshold selection methodology). For the applicability scoring rubric, see Appendix Table C.31. For thresholds $t \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$, we computed the pooled coverage–correctness tradeoff (Figure 7.13/Table 7.21).

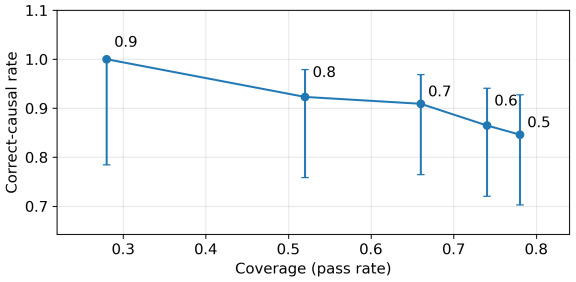


Figure 7.13: Human validation tradeoff curve (pooled; Wilson 95% CI). Each point corresponds to an applicability threshold $t \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$ applied to the validated sample ($n = 50$). Let $x(t)$ be *coverage* = $\Pr(\texttt{Fully\_applicable} \geq t)$ and $y(t)$ be *correct-causal rate* = $\Pr(\texttt{human\_verdict} = \texttt{causal} \mid \texttt{Fully\_applicable} \geq t)$. Points are annotated by t. Selected threshold: $t = 0.7$.

Table 7.21: Human validation tradeoff curve (pooled): coverage vs. correct-causal rate

Threshold	Coverage	Correct-causal rate	n pass
0.5	0.780	0.846	39
0.6	0.740	0.865	37
0.7	**0.660**	**0.909**	**33**
0.8	0.520	0.923	26
0.9	0.280	1.000	14

Coverage = fraction of validated edges with applicability $\geq t$. Correct-causal rate = fraction of passing edges labelled `causal`.
Bold row indicates selected threshold $t = 0.7$.

We then selected $t = 0.7$ via the elbow/gradient decision rule defined in Methods (§6.5.5) and used the human-validated pass rates to extrapolate expected FP promotions and FN discoveries to the full DR dataset (Table 7.22).

Table 7.22: LLM Generation Metrics with Human-Validated Ground Truth Enrichment (Aggregate, extrapolated)

Scenario	Precision	Recall	F1
Original (expert GT)	0.198	0.411	0.267
Scenario 1: Enriched GT (FP→TP)	0.459	0.618	0.527
Scenario 2: LLM+DR System (+FN found)	0.494	0.709	0.582
Conservative Scenario 1	0.396	0.583	0.472
Conservative Scenario 2	0.415	0.629	0.500

Threshold selected by elbow/gradient rule (Option B) on validation tradeoff curve: $t = 0.7$. FP promotions (point / conservative): 58 / 44. FN discoveries (point / conservative): 15 / 7. Counts are extrapolated from a stratified human-validation sample and should be interpreted as estimates.

Inter-rater reliability. On the 13 overlapping edges annotated by both raters, exact agreement on the four-category verdict was 46.2% (6/13). Chance-corrected agreement was $\kappa = 0.26$, indicating *fair* agreement [117]. For the applicability score, agreement was *poor*: ICC(2,1) = 0.03 [118], with a mean absolute difference of 0.34 on the 0–1 scale. These results suggest that rubric judgments—particularly the continuous applicability score—are sensitive to rater calibration, and downstream threshold calibration and enrichment extrapolations should be interpreted as exploratory. Future work should refine the rubric with clearer anchor examples and conduct a larger-scale reliability study before trusting Deep Research evidence applicability to the causal claim, and by extension also these optimistic ground truth enrichment results.

Critical Caveats: Both scenarios are **exploratory and preliminary**. Key limitations: (1) DR may validate edges as being directly causal irrespective of other CLD specific context such as mediators. (2) LLM-based DR validating LLM-generated edges risks circularity; (3) The review was primarily single-rater; a preliminary second-rater overlap ($n = 13$) indicates only fair agreement on categorical verdicts ($\kappa = 0.26$) and poor agreement on applicability scores (ICC=0.03), suggesting calibration-based extrapolations are potentially rater-sensitive; (4) **True Negatives not analysed**—DR covered only 20.4% of the edge space (285 of 1,394 possible edges); the 1,109 TN edges were not analysed due to cost ($0.87/edge, $\sim$ $965 for all TNs). These metrics represent **upper bounds** on achievable performance, pending domain expert validation. Dataset available for community verification (see Appendix C, especially §C.1).

7.7.4 Smart-Trigger DR Deployment Estimate

Experimental design: Section D.9.0.7.

The enrichment analysis above applies DR to *all* edges. In practice, DR is expensive ($0.87/edge), motivating a **smart-trigger** deployment that runs DR selectively. We estimate expected F1 gains if DR is triggered only for edges where the most reliable RQ1a Ground Truth Correctness judge (Mechanistic prompt) assigns a low aggregate score (≤ 0.5), indicating uncertainty about the edge (as opposed to the original RQ3 intention of using RQ2 UQ metrics as the trigger).

Procedure. We parsed per-edge Mechanistic judge scores from the 9 judged workbooks (3 CLDs $\times$ 3 runs) and computed the mean score per edge. We then joined these to the 285 DR edges and marked edges as *triggered* if their mean score ≤ 0.5. Of the triggered edges, we counted how many had `direct_causal_found=True` in the DR oracle. Finally, we applied the human-validated calibration rates at $t = 0.7$ (FP: 89.5%, FN: 83.3%) to estimate expected promotions/discoveries.

Results. Of 285 DR edges, 142 triggered (50%). Among triggered edges with DR evidence: 59 FP edges and 9 FN edges had `direct_causal_found=True`. Applying calibration:

Table 7.23: Smart-trigger Deep Research enrichment estimates. Trigger: Mechanistic GT correctness score ≤ 0.5. Calibration: human-validated applicability ≥ 0.7.

Scenario	Precision	Recall	F1 (95% CI)
Original	0.198	0.411	0.267
Enriched GT (+52.8 FP→TP)	0.436	0.606	0.507 [0.457, 0.524]
LLM+DR (+7.5 FN discovered)	0.454	0.653	0.536 [0.474, 0.557]

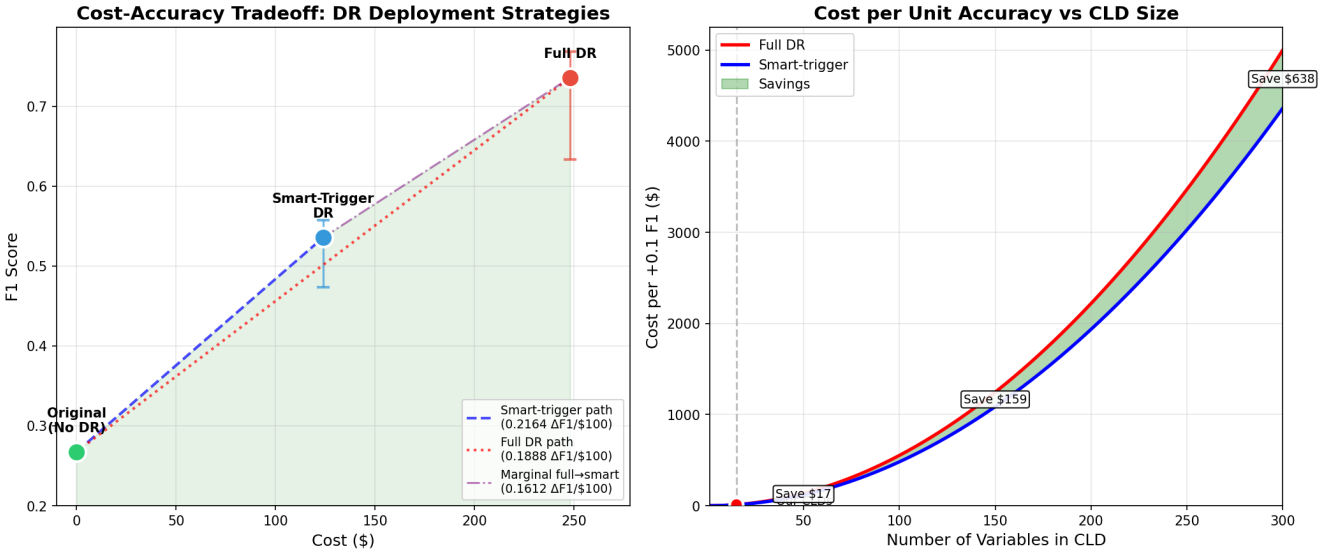


Figure 7.14: Cost–accuracy tradeoff and cost-per-unit-accuracy scaling for Deep Research (DR) deployment. Left: observed cost vs. expected F1 for always-on DR vs. smart-trigger DR (judge-triggered deployment). Right: extrapolated cost per +0.1 F1 improvement as CLD size increases under a sparse-edge growth assumption, highlighting the widening cost gap.

Note. Figure 7.14 shows two-sided uncertainty bands (Wilson 95% bounds) propagated from the human-calibrated promotion/discovery rates (see Methods, Sections D.9.0.5 and D.9.0.4). For interpretation, we prefer the conservative estimate obtained by using the Wilson 95% lower bound, because it reduces the risk of overstating gains due to sampling variability in the human validation rates. While this lower bound addresses sampling uncertainty in the calibration step, it does not account for other sources of uncertainty (e.g., generalisation of applicability to non-validated edges, edge dependence, domain shift, and not considering CLD mediators). Further research should assess how the estimated gains change when these assumptions are relaxed.

We evaluated whether a selective test-time DR deployment policy could improve cost-efficiency relative to always-on DR, using the RQ1a Ground Truth Correctness judge (Mechanistic prompt) as a trigger (mean aggregate score ≤ 0.5). On the 285-edge RQ3 dataset, 142 edges (50%) triggered; among these, DR reported `direct_causal_found=True` for 59 FP and 9 FN edges. Using the human-validated calibration at applicability ≥ 0.7 (FP: 17/19 causal; FN: 5/6 causal; Wilson 95% CIs reported above), the smart-trigger policy yields an estimated ΔF1 of 0.268 at $\sim$\$124 total compute (judge on all edges + DR on triggered edges), compared to ΔF1 of +0.468 at $\sim$\$248 for full DR.

Under these assumptions, smart-trigger achieves higher ΔF1 per dollar (0.216 vs. 0.189 ΔF1 per \$100; $\sim$15% relative improvement) while recovering $\sim$57% of the full-deployment F1 gain at $\sim$50% of the DR cost. The scaling panel should be interpreted as a cost extrapolation, not a validated performance claim: it assumes the observed trigger rate and per-edge applicability distribution generalise to other edges and other CLDs. Under this stylized model, the absolute dollar gap between full and smart-trigger deployment increases with CLD size due to quadratic scaling of the edge space, while the efficiency ratio remains approximately constant.

7.7.5 RQ3: Main Findings (Summary)

Our restated RQ3 asks whether automated literature search (Deep Research; DR) can validate LLM-generated causal edges. The original RQ3 formulation—"Can adaptive test-time compute allocation improve CLD accuracy efficiency?"—is addressed through this lens.

As an exploratory pilot (single-run, non-ablated), we find:

1. **DR detection is not a reliable FP filter in aggregate.** Across 285 TP/FP/FN edges, DR retrieves `direct_causal_found=True` for many FP edges (62.4%), yielding a small, non-significant TP–FP detection gap (8.1 pp; $p = .38$; Cohen's $h = 0.17$; Table 7.17). DR does, however, distinguish FN from both TP and FP (TP–FN and FP–FN gaps significant after correction; Figure 7.12).
2. **Utility is domain-dependent and DR confidence is weakly informative.** Per-CLD results show strong TP–FP discrimination in Depressive but weak discrimination in Social Norms and Emergency Department (Table 7.18; Figure 7.12). DR's self-reported confidence shows substantial overlap across TP/FP/FN classes within domains (Figure 7.12B), limiting its value as a standalone accept/reject criterion.
3. **Enrichment effects can be large if gains extrapolated from the human validation sample hold.** A naive upper bound (treat all DR-positive edges with `direct_causal_found=True` as correct) yields very large gains (Scenario 1/2; Table 7.19). However, DR sometimes finds causal evidence for highly related, but not directly equivalent, concepts or using different measures. Human validation shows that the dominant failure mode is *variable mismatch*—causal evidence for different constructs than the edge's stated variables. To form a more realistic estimate based on the human validation sample, an optimal trade-off between evidence quality and coverage can be found by applying a threshold on the pooled tradeoff curve ($t = 0.7$; Figure 7.13), which yields more calibrated enrichment estimates than the naive "any causal evidence found by DR is valid" assumption: F1 improves from 0.267 (expert GT) to 0.527 (Scenario 1) and 0.582 (Scenario 2), with conservative estimates of 0.472 and 0.500 (Table 7.22). Based on the quality of causal evidence observed in the human validation sample, the human-calibrated enrichment extrapolation should be treated as *tentative*: the second-rater overlap indicates only fair agreement on categorical verdicts ($\kappa = 0.26$) and poor agreement on applicability scores (ICC(2,1)=0.03), suggesting that the calibration curve and selected threshold are rater-sensitive. Accordingly, any extrapolated gains should be interpreted cautiously and prioritise conservative lower-bound scenarios. This still relies on strong assumptions: (i) the within-CLD edge-class applicability distribution in the validated sample generalises to the remaining (out-of-sample) edges for that CLD, and (ii) edges are evaluated largely in isolation rather than in the full CLD causal context (e.g., mediators/confounders) that could disqualify a relationship from being directly causal.
4. **Smart-trigger DR slightly improves cost-efficiency (exploratory).** Triggering DR only on edges flagged as uncertain by the strongest correctness judge (Mechanistic; score ≤ 0.5) recovers a notable fraction of the full-deployment DR enrichment gains at roughly half the DR cost, with an estimated ΔF1 of 0.268 after human-calibrated applicability filtering (Figure 7.14). This provides a proof of concept that DR efficiency can be improved by deploying DR selectively using lower-compute uncertainty estimates.

Takeaway: In this pilot, DR is best interpreted as a high-cost, domain-dependent *enrichment signal* that, based on this sample, can provide applicable causal evidence for a subset of claims although defined as hallucinations (FP & FN w.r.t. GT Lit). However, applicability of the evidence differs, and due to low inter-rater reliability observed, we recommend further validation studies before forming strong conclusions. Consequently, the evidence does not support DR being a drop-in validator that cleanly separates expert-true edges from LLM false positives across domains. It can, however, yield direct causal evidence for a majority of edges in our CLD ground truth sample (under strong assumptions) that, if validated, could be used to improve CLD quality, with smart triggering used to partially offset DR's computational cost and achieve higher accuracy per dollar spent.

7.8 Effect Size Summary

Table 7.24 summarises the standardised effect sizes observed across all research questions, linking each finding to its stated hypothesis, quantitative formulation, statistical test, and conclusion. This provides a unified view of which null hypotheses were rejected and the magnitude of effects detected.

Table 7.24: Effect Size Summary: Hypothesis Operationalization, Statistical Tests, and Conclusions

RQ	Hypothesis	H_0 / H_1	Test	Effect	Value	p	Conclusion (decision; effect)
RQ1a: LLM-as-a-Judge Validation							
1a	LLM-human agreement	H_0: $\kappa = 0$ vs H_1: $\kappa > 0$	Cohen's κ (n=72)	κ	0.618	—	H_0 rejected; substantial
1a	Prompt affects F1 (Corr. Citation)	H_0: equal medians H_1: $\geq$1 differs	Friedman (9 blocks)	W	0.28	.058	H_0 not rejected; weak
	Prompt affects F1 (Corr. Correctness)			W	0.70	.001**	H_0 rejected; strong
	Prompt affects F1 (GT Citation)			W	0.30	.042*	H_0 rejected; moderate
	Prompt affects F1 (GT Correctness)			W	0.48	.013*	H_0 rejected; moderate
RQ1b: LLM-as-a-Corrector Evaluation							
1b	Corrector efficacy (Synthetic)	H_0: med(ΔF1)=0 H_1: med(ΔF1)$\neq$0	Wilcoxon signed-rank (9 blocks)	d	+1.18	.031*	H_0 rejected; large
	Corrector efficacy (Ground Truth)	H_0: med(ΔF1)=0 H_1: med(ΔF1)$\neq$0	Wilcoxon signed-rank (9 blocks)	d	−0.32	.301	H_0 not rejected; medium
RQ2: Context-Insensitive Hallucination Detection							
2	UQ discriminates hallucinations	H_0: median(AUC−0.5) = 0 H_1: median(AUC−0.5) $\neq$ 0	Wilcoxon signed-rank (9 blocks)	AUC	0.672	.015*	H_0 rejected; moderate
	Cross-CLD generalization	—	Descriptive (Phase 5→6)	ΔAUC	0.18	—	Generalization gap
*RQ3: Deep Research Literature Validation**							
3	DR discriminates (TP–FP)	H_0: $p_{TP} = p_{FP}$ H_1: $p_{TP} \neq p_{FP}$	Fisher's exact (Bonf. m=3)	h	0.17	1.000	H_0 not rejected; negl.
	DR discriminates (TP–FN)	H_0: $p_{TP} = p_{FN}$ H_1: $p_{TP} \neq p_{FN}$	Fisher's exact (Bonf. m=3)	h	0.56	0.018	H_0 rejected; medium
	DR discriminates (FP–FN)	H_0: $p_{FP} = p_{FN}$ H_1: $p_{FP} \neq p_{FN}$	Fisher's exact (Bonf. m=3)	h	0.39	0.024	H_0 rejected; small
	Inter-rater reliability	—	Cohen's κ (n=13)	κ	0.26	—	Descriptive; fair
	(Applicability score)	—	ICC(2,1) (n=13)	ICC	0.03	—	Descriptive; poor
*RQ3: Enrichment Analysis (Exploratory)**							
	Enrichment baseline	—	Descriptive	F1	0.267	—	Baseline
	Scenario 1: Enriched GT	—	Descriptive	ΔF1	+0.438	—	Exploratory
	Scenario 2: LLM+DR	—	Descriptive	ΔF1	+0.512	—	Exploratory
	Human-val. Sc. 1 (cons.)	—	Extrapolation	ΔF1	+0.205	—	Exploratory
	Human-val. Sc. 2 (cons.)	—	Extrapolation	ΔF1	+0.233	—	Exploratory
	Smart compute	—	Descriptive	ΔF1/\$100	+0.027	—	Exploratory

Effect size interpretation. κ: Landis-Koch (0.61–0.80 = substantial). ICC: <0.5 poor, 0.5–0.75 moderate, 0.75–0.9 good, >0.9 excellent. W: <0.3 weak, 0.3–0.5 moderate, >0.5 strong. d: <0.2 small, 0.2–0.8 medium, >0.8 large. r: <0.2 weak, 0.2–0.5 moderate, >0.5 strong. h: <0.2 negligible, 0.2–0.5 small, 0.5–0.8 medium, >0.8 large.
Significance. * $p < .05$, ** $p < .01$, *** $p < .001$. RQ3 p-values are Bonferroni-adjusted ($\alpha_{adj} = 0.0167$). RQ2 single-metric p uses Bonferroni adjustment across 4 metrics ($\alpha_{adj} = 0.0125$).
* *RQ3 exploratory.* Enrichment estimates are extrapolated; human-validated scenarios use Wilson CI lower bound (conservative). Not independently validated ground truth.
RQ1a prompt tests. Friedman with (CLD×run) blocking; W = Kendall's concordance. *RQ1b efficacy.* One-sample Wilcoxon signed-rank on 9 blocks.
RQ2 discrimination. AUC = Area Under Curve; one-sample Wilcoxon signed-rank test on (AUC − 0.5) over 9 blocks (CLD×run). Cross-CLD generalization is reported descriptively (Phase 5→6).

In summary, Table 7.24 shows: **RQ1a**—LLM-human agreement is substantial (κ=0.618, H_0 rejected); prompt effects on citation judging are weak-to-moderate (W=0.28–0.30) while correctness judging shows strong prompt sensitivity (W=0.70, H_0 rejected). **RQ1b**—corrector efficacy is confirmed for synthetic corruptions (H_0 rejected, d=+1.18, large effect) but not for ground truth (H_0 not rejected, d=−0.32, medium effect). **RQ2**—UQ metrics show modest discrimination (AUC=0.672; Wilcoxon $p_{\text{adj}} = .015$) with a cross-CLD generalisation gap (ΔAUC=0.18). **RQ3**—no TP–FP discrimination (H_0 not rejected, h=0.17, negligible), but DR distinguishes TP and FP from FN (TP–FN: H_0 rejected, h=0.56, medium; FP–FN: H_0 rejected, h=0.39, small; both survive Bonferroni correction). Inter-rater reliability on the second-rater overlap is limited (categorical verdicts: $\kappa = 0.26$; applicability scores: ICC=0.03), so extrapolated enrichment gains should be treated as exploratory and rater-sensitive. While exploratory enrichment analyses indicate large aggregate improvements (ΔF1=+0.438 to +0.512), we recommend proceeding with caution and therefore to instead go with conservative lower bound scenarios (Human-val. Sc. 1 (cons.): Extrapolation ΔF1=+0.205; Human-val. Sc. 2 (cons.): Extrapolation ΔF1=+0.233) and even then interpret these with caution until further validation work is done.

7.9 Computational Scaling Analysis

Experimental design: Section 6.6.

Full scaling methodology, scripts, and supplementary scaling results are provided in Appendix C.4.14.

7.9.1 Time and Cost Scaling

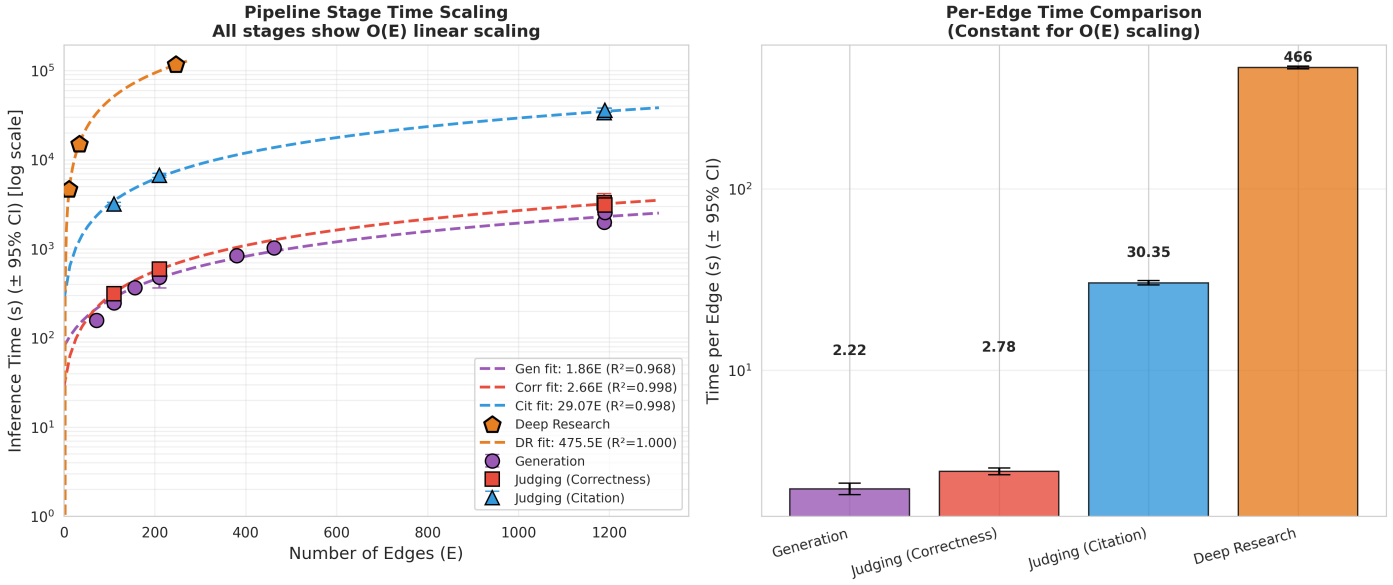


Figure 7.15: Generation vs. judging scaling comparison (mean ± SD). **Left:** Time scaling vs. number of edges. **Right:** Per-edge time comparison across pipeline stages. Error bars show SD.

Figure 7.15 shows that both generation and correctness judging exhibit $O(E)$ linear scaling with similar slopes (∼2 s/edge). Generation (2.2 ± 0.5 s/edge) and correctness judging (2.8 ± 0.2 s/edge) have comparable costs, while citation judging (30.1 ± 1.4 s/edge) dominates the pipeline.

7.9.2 Token Cost Analysis

Table 7.25: Computational Efficiency Summary by Experiment

Experiment	Model	Files	Edges	Tok/Edge	¢/Edge	Total ($)
RQ1a Judge Experiments (actual)						
Corruption Citation	5m	36	18,120	42,992	1.37	247.67
Ground Truth Citation	4.1	36	18,117	43,933	9.22	1669.69
Corruption Correctness	4.1	27	13,590	480	0.16	22.29
Ground Truth Correctness	4.1	27	13,587	470	0.16	21.34
RQ1b Corrector Experiments (estimated)						
Synth Baseline	4.1	9	4,527	980	0.33	14.74
Synth CoT	4.1	8	4,318	980	0.33	14.06
Synth Mechanistic	4.1	9	4,527	980	0.33	14.74
GT Baseline	4.1	9	4,529	980	0.33	14.75
GT CoT	4.1	9	4,529	980	0.33	14.75
GT Mechanistic	4.1	9	4,529	980	0.33	14.75
RQ3 Deep Research (actual)						
Deep Research	5/5m	3	285	246,893	87.00	247.95
Total						**2296.72**

Note. RQ1a costs from actual LLM Usage Stats. RQ1b estimated: correction + rejudging = 2× correctness tokens/edge. RQ3 from Deep Research logs (mix of GPT-5 + GPT-5-mini). Models: 4.1 = GPT-4.1 ($2.00/$8.00 per 1M tokens), 5 = GPT-5 ($1.25/$10.00 per 1M tokens), 5m = GPT-5-mini ($0.25/$2.00 per 1M tokens).
Reproduction: `python3 final_runs/supp_cost_analysis/analysis_scripts/calculate_all_costs.py --data-dir final_runs`

Table 7.25 reveals that Deep Research dominates per-edge cost (247K tokens/edge, $0.87/edge) due to multi-agent orchestration and iterative search. Additional cost-analysis scripts and breakdowns are provided in Appendix C.4.16.

For a 66-edge CLD (Emergency Department), generation plus correctness judging costs approximately $0.29 and takes 6 minutes cumulative API time. Citation validation adds significant overhead (33 minutes, $3.68 for GPT-4.1 or $0.13 for GPT-5-mini). Linear $O(E)$ scaling enables predictable cost estimation for arbitrary CLD sizes.

7.9.3 Parallelization Speedup

We benchmarked parallelization using the configuration described in Methods: 34 edges (Depressive Symptoms CLD), GPT-4.1 (temperature=0.3), worker counts of 1, 3, 5, and 10, with 3 runs per configuration for both correctness judging (short prompts) and citation judging (long prompts with retrieved citations).

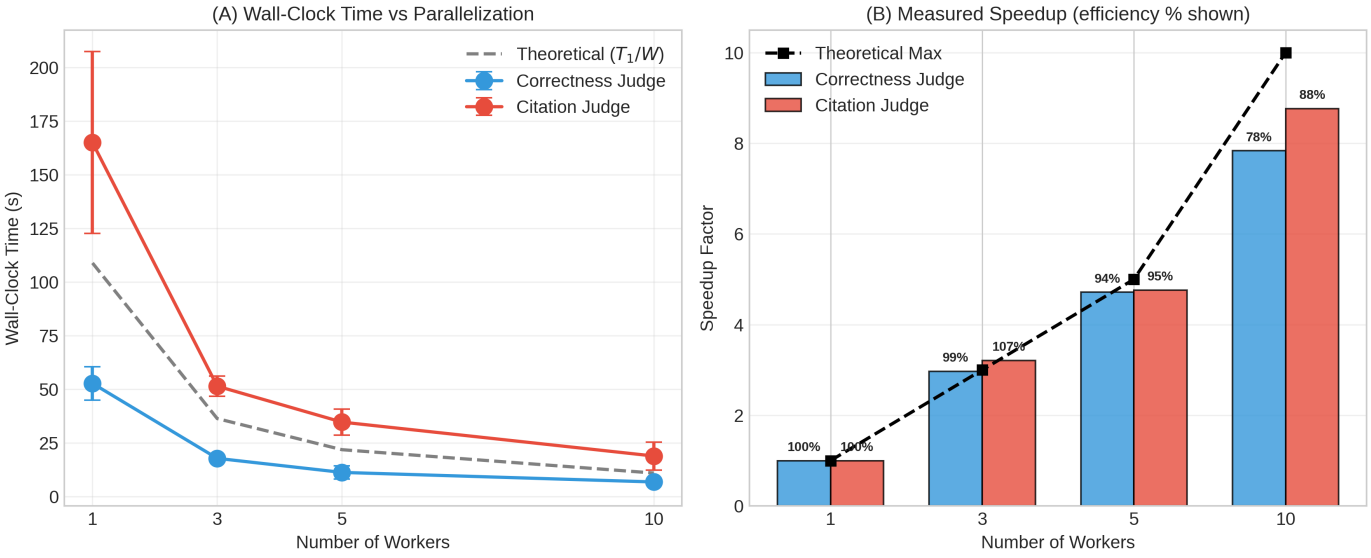


Figure 7.16: Parallelization benchmark (34 edges, GPT-4.1, 3 runs per config, mean ± SD). **(A)** Wall-clock time vs. worker count; dashed line shows theoretical perfect scaling (T_1/W). **(B)** Measured speedup with efficiency percentages.

Table 7.26: Parallelization Benchmark: Wall-Clock Time and Speedup by Judge Type

Workers	Correctness Judge			Citation Judge		
	Time (s)	Speedup	Eff.	Time (s)	Speedup	Eff.
1	52.7 ± 7.8	1.00×	100%	165.0 ± 42.4	1.00×	100%
3	17.7 ± 1.3	2.97×	99%	51.4 ± 4.6	3.21×	107%
5	11.2 ± 3.1	4.71×	94%	34.6 ± 6.1	4.76×	95%
10	6.7 ± 1.1	7.84×	78%	18.8 ± 6.5	8.77×	88%

Note. Benchmark on Depressive CLD (34 edges, GPT-4.1, 3 runs per config). Time = wall-clock time (mean ± 95% CI). Speedup = T_1/T_W. Efficiency = Speedup/W × 100%.
Citation judging uses longer prompts with retrieved citations, resulting in higher per-call latency.

Figure 7.16A shows citation judging (165s sequential) takes 3.1× longer than correctness (52.7s) due to longer prompts with retrieved citations. At 10 workers (Figure 7.16B), correctness achieves 7.8× speedup (78% efficiency) while citation achieves 8.8× speedup (88% efficiency). Citation judging shows slightly better parallel efficiency due to higher per-call latency dominating overhead. Table 7.26 details the wall-clock times with confidence intervals, showing that efficiency remains above 94% up to 5 workers before degrading at higher parallelism due to API rate-limiting and scheduling overhead.

Chapter 8

Discussion

This chapter interprets the experimental findings and situates them within the broader context of LLM-assisted CLD generation. Specifically, we examine *why* performance shifts across evaluation settings and what those shifts imply for the construct validity of "hallucination" in CLD generation. We reconcile strong performance in the controlled GT Synth setting with weaker, domain-dependent behaviour under GT Lit by distinguishing three sources of disagreement: (i) genuine model errors, (ii) retrieval and evidence-type constraints that bound what can be supported by accessible literature, and (iii) construct and scope mismatches between expert CLDs and edge-level claims. This framing is then used to analyse failure modes in judge–corrector loops (including proxy optimisation), to clarify the role and limits of uncertainty signals for triage, and to situate literature-oriented search (Deep Research) as a form of evidence-grounded test-time compute that can inform the contested edge space. The objective is to articulate an empirically grounded interpretation of the results and to derive methodological implications for hybrid workflows in which LLMs support scalable candidate generation and evidence gathering while expert judgement remains central to construct definition and applicability assessment.

8.1 Direct Answers to the Research Questions

For reader convenience, we provide a direct, concise answer to each research question before the deeper interpretive discussion that follows.

RQ1: Can LLM-as-a-Judge methods detect hallucinations in LLM-generated CLDs?

Answer: Partially. LLM judges reliably detect *synthetic* CLD corruptions (GT Synth) with strong discrimination, but their ability to detect deviations from *expert literature ground truth* (GT Lit) is substantially weaker and domain-dependent. Overall, LLM-as-a-Judge is best interpreted as a *screening/triage* signal rather than a dependable standalone validator across domains.

RQ1a: How effectively can correctness-based and citation-based judges distinguish valid from hallucinated causal relationships?

Answer: Correctness-based judging outperforms citation-based judging, but both have important failure modes; prompt selection matters and interacts with judge type.

- **Correctness judge:** Strong on GT Synth, but performance drops sharply on GT Lit; on GT Lit, block-level analyses in Results indicate a significant prompt effect on discrimination, with Mechanistic providing the most robust ordering overall, though performance remains domain dependent and Social Norms remains difficult across prompts.
- **Citation judge:** Unreliable as a strict validator on expert CLDs; high retrieval coverage does not imply strong causal evidence, and the judge can over-accept or mis-rank edges due to retrieval confounds. Notably, mechanistic prompting *underperforms* simpler prompts (Baseline, CoT) on citation judging, with counterintuitive positive score–hallucination correlations suggesting mechanistic framing biases retrieval toward topically related but causally non-specific evidence.

RQ1b: Can LLM-as-a-Corrector improve CLD accuracy when guided by judge feedback?

Answer: Not reliably on literature ground truth. The corrector yields modest, inconsistent improvements on synthetic corruptions, but does not improve (and can degrade) alignment with expert GT Lit on average. Judge scores increase consistently after correction, suggesting a Goodhart-like dynamic where the corrector learns to satisfy the judge more than it improves expert-grounded accuracy.

RQ2: Can uncertainty quantification (UQ) metrics predict hallucinations in generated CLDs?

Answer: Only weakly, and not in a domain-robust way. Single generator-time UQ metrics provide at most modest discrimination. While ensembles can perform well in-distribution, cross-CLD generalisation remains limited, and fixed-threshold decision quality remains modest even under a principled threshold-freeze policy. In practice, these metrics are better suited to *flagging/triage* than to standalone hallucination detection across domains.

RQ3: Can adaptive test-time compute allocation improve CLD accuracy efficiency?

Answer: The original UQ-triggered judge+corrector design is not supported; a modified judge-triggered Deep Research strategy shows proof-of-concept efficiency gains. Because (i) UQ metrics do not generalise reliably across CLDs (RQ2) and (ii) the corrector does not reliably improve GT Lit accuracy (RQ1b), the original "adaptive triggering" pathway is untenable as specified. However, selectively deploying higher-cost test-time compute for *literature validation* (Deep Research) when a lower-cost signal flags uncertainty (e.g., the Mechanistic correctness judge score) recovers a substantial fraction of the full-deployment gains at roughly half the Deep Research cost in our exploratory analysis, albeit with strong assumptions and clear domain dependence.

8.2 Central Interpretive Claim: Error vs. Scope Disagreement

The core interpretive challenge in this thesis is distinguishing *genuine model error* from *expert scope choices*. Treating expert CLDs as absolute ground truth is a practical operationalisation, but it conflates at least two sources of disagreement: (i) the generator produces a factually incorrect or unsupported causal claim (a true hallucination), and (ii) the generator produces a defensible causal claim that experts omitted due to scope, construct-mapping, or literature-coverage decisions (a contested edge). An edge absent from an expert CLD may reflect genuine non-causality, but it may also be excluded for representation reasons unrelated to factual correctness.

This ambiguity is not merely methodological: it shapes the interpretation of every downstream metric. Under GT Lit, "false positive" labels include both true hallucinations and contested edges; under GT Synth, corruptions are known but may not reflect the distribution of naturally occurring errors. The performance gap between GT Synth and GT Lit observed throughout this thesis is therefore partly a *construct shift*, not only a detection-difficulty increase.

The RQ3 Deep Research analysis provides direct evidence for this claim: a majority of edges labelled "false positive" under GT Lit admit retrievable causal evidence from the scientific literature (see Section 8.8 for the full analysis), suggesting that a substantial fraction of FP mass represents contested edges rather than straightforward confabulations. This reframes the practical objective from "detecting hallucinations" to "triaging contested edges for expert review."

For definitions of GT Lit and GT Synth, see Results Section 7.4.1.

8.3 CLD Generation Paradigm

Although corpus-independent CLD generation was not a primary research contribution of this thesis—the focus being hallucination *detection* and mitigation—the generator results provide important context. The LLM generator exceeded random baselines by 2–3× across all three CLDs (Results Section 7.2), demonstrating that LLMs capture non-trivial causal structure from prompts alone.

This thesis evaluates a *generate-then-validate* paradigm: the model proposes candidate causal structure without being anchored to a specific source document, and downstream components (judges, corrector, and Deep Research) attempt to detect and mitigate unsupported edges. This differs from *extraction* paradigms (text-to-CLD) where grounding is provided by the source text, shifting the central challenge from evidence attribution to model faithfulness. As a consequence, "hallucination" in corpus-independent generation is tightly coupled to the ground-truth construct (GT Synth vs. GT Lit) and to expert scope choices (Section 8.2); disagreements can reflect genuine errors, but also defensible edges absent from expert CLDs.

These generator results are encouraging but domain-limited: all primary evaluations used health-related CLDs. Future work should validate generator performance on CLDs from other scientific domains (economics, ecology, engineering) to establish whether the observed baseline-exceeding behaviour generalises or is artefactual to health-domain training coverage.

8.4 Mechanisms of Judge Failure

Investigation into edge-level reasoning reveals three distinct failure modes explaining poor performance of both Citation and Correctness judges. These mechanisms—Evidence of Absence, Plausibility Bias, and Partial Evidence Acceptance—create structural blind spots that quantitative metrics alone fail to capture. All qualitative examples in this section are drawn directly from the per-edge judged spreadsheets and run logs saved under `final_runs/` (see footnotes for exact paths).

8.4.1 The "Evidence of Absence" Paradox (Citation Judge)

The Citation Judge exhibits a perverse incentive structure when evaluating negative claims, and the same asymmetry can also surface in the Correctness Judge. For the Citation Judge, when retrieval fails to find relevant literature—often due to keyword mismatches, variable specificity, or simply the rarity of *direct* causal

evidence—the judge can treat *absence of evidence* as *evidence of absence*, validating `NONE` with non-trivial confidence. For the Correctness Judge, an analogous failure mode arises because `NONE` claims are hard to falsify without external evidence: a coherent "no clear mechanism" rationale often contains no internal contradiction for the judge to seize on, so it can assign partial (or even high) support despite the edge being present in the expert graph.

For example, in the Emergency Department CLD evaluation (CoT prompt, run 2), the judge evaluated an expert edge that the generator missed (a GT Lit **false negative** under the generator-vs-GT definition) where the generator asserted a NONE relationship between "Functioning of healthcare professionals" and "Availability of alternatives for ED".[1]

- **Motivation:** "Functioning of healthcare professionals does not directly cause availability of alternatives for ED. There is no clear causal mechanism..."
- **Judge reasoning:** "...it does not report that changes in how healthcare professionals function directly create or make available alternative care options..."
- **Outcome:** Aggregate verdict **Partially supported**, score **0.5**

This edge exists in the validation graph (hence FN), yet the judge assigned a non-trivial score by treating "lack of direct evidence in retrieved snippets" as support for the negative claim.

This pattern is visible in the judge-score distributions: across multiple settings, **TN and FN outcomes receive substantially higher mean scores than TP and FP**, and critically, the FN bar is often close to TN (e.g., in citation-based corruption detection, FN ≈ TN in the score-by-outcome panel; Figure 7.7). In other words, many missed edges are scored as "clean" rather than being pushed below the hallucination threshold, which is exactly what the evidence-of-absence asymmetry predicts.

8.4.2 Plausibility Bias (Correctness Judge)

The Correctness Judge, lacking external retrieval, relies entirely on internal logical consistency. This makes it highly susceptible to "Plausible Hallucinations"—spurious edges that are factually unverified but logically coherent.

For example, in the Emergency Department CLD (mechanistic prompt, run 2), a corrupted edge between "Healthy lifestyle" and "Acute care demand" received the maximum score.[2]

- **Spurious Motivation:** "Healthy lifestyle... promotes greater health awareness... reduces likelihood of escalating... direct negative effect."
- **Judge Reasoning:** "The explanation clearly articulates a plausible causal mechanism... Reasoning is internally consistent... CORRECT."
- **Outcome:** Score **1.0**, despite being corrupted

[1] Source artefact (GT Lit, citation judge, Emergency Department, run 2, CoT prompt): `final_runs/RQ1a_gt_lit_citation/emergency_department/run_2/judged_older_persons_emergency_department_visits_and_interactionstitled_spreadsheet_cot_20251118_100404_20251118_105236.xlsx`.

[2] Source artefacts (GT Synth, correctness judge, Emergency Department, run 2, mechanistic prompt): judged spreadsheet `final_runs/RQ1a_gt_synth_correctness/emergency_department/run_2/judged_older_persons_emergency_department_visits_and_interactionstitled_spreadsheet_mechanistic_20251116_020717_20251116_021236.xlsx`; run log shows repeated max-score assignments for `Healthy lifestyle → Acute care demand leading to an ED visit`, e.g. `final_runs/RQ1a_gt_synth_correctness/emergency_department/run_2/judging_20251116_015224.log`.

The judge validated the *logic* of the corruption, not its factuality. In this evaluation file, 204/257 corrupted edges (79%) received aggregate scores $\geq$0.5. Under synthetic "spurious motivation" corruptions designed to sound plausible, correctness-only judging collapses into a coherence check.

This failure mode directly instantiates the factuality–faithfulness distinction in hallucination taxonomies [9]: the Correctness Judge evaluates *faithfulness* (internal logical consistency) but cannot verify *factuality* (correspondence with external facts) without retrieval. Plausible hallucinations pass the faithfulness test—their reasoning is internally coherent—while failing the factuality test. This distinction explains why correctness-only judging is not expected to reliably detect well-motivated but factually incorrect causal claims in the absence of external evidence, and why prompting alone is unlikely to fully resolve this limitation.

8.4.3 Partial Credit and Non-Scientific Evidence Acceptance (Citation Judge)

Even when edges are labelled corrupted, the Citation Judge can assign high scores because: (i) it grants partial credit for plausibly related mechanisms, and (ii) retrieval may return non-peer-reviewed sources containing supportive statements. Although whitelisting of scientific domains and BHC prompting partially mitigates this, the degree to which this was mitigated was not systematically evaluated. Future work should systematically quantify this effect by stratifying retrieved sources by evidence tier (e.g., systematic reviews/RCTs vs. observational studies vs. expert opinion) and measuring judge discrimination and calibration under controlled retrieval settings, with and without domain whitelisting and BHC prompting.

In the Emergency Department evaluation, we found high-scoring corrupted edges.[3]

- **Non-scientific source:** An edge between "Proactive attitude" and "Language and cultural barriers" received score 1.0 (Fully supported) based on a non-peer-reviewed blog providing prescriptive "strategies" rather than empirical evidence.
- **Partial credit on intermediate mechanisms:** An edge between "Accessibility of information" and "Knowledge" received score 0.625 by validating the intermediate concept of "information overload," though direct causal evidence for the final claim was absent.

These cases demonstrate that citation judging is sensitive to retrieval artefacts (including non-scientific sources) and partial-credit aggregation on plausible intermediate mechanisms.

8.4.4 Domain-Dependent Failure: The Social Norms Anomaly

In the Social Norms CLD, all prompts failed to reliably discriminate hallucinations from valid edges. This could reflect the complexity of the Social Norms domain, where causal mechanisms are behavioural and context-dependent rather than physiological, or it could indicate limitations in the literature ground truth quality for this domain. Future research should investigate whether domain-adapted prompts or alternative causal frameworks would improve performance in behavioural domains.

[3]Source artefacts (GT Synth, citation judge, Emergency Department, run 2): judged spreadsheet `final_runs/RQ1a_gt_synth_citation/emergency_department/run_2/judged_citation_older_persons_emergency_department_visits_and_interactionstitled_spreadsheet_mechanistic_20251204_092254_20251204_102947.xlsx`; run log contains per-edge updates including `(Proactive attitude)-[:POSITIVE]->(Language and cultural barriers)` with `aggregate_score=1.0` and `(Accessibility of information)-[:NEGATIVE]->(Knowledge)` with `aggregate_score=0.625`: `final_runs/RQ1a_gt_synth_citation/emergency_department/run_2/judging_20251204_065138.log`.

8.5 RQ1 Interpretation: Judges as Screening Signals

The failure mechanisms in Section 8.4 explain why LLM-based judges should be interpreted as *screening signals* rather than definitive validators. This section synthesises the practical implications.

8.5.1 Correctness Judge

Correctness judging is informative but unreliable as a standalone validator. Its performance depends heavily on the ground-truth construct (GT Synth vs. GT Lit) and the domain. Under plausible corruptions, correctness-only judging collapses into a coherence check—validating the *logic* of a claim rather than its factuality. This failure intensifies in behavioural domains (e.g., Social Norms) where causal mechanisms are context-dependent and harder to falsify through internal consistency alone.

Prompt engineering significantly influences correctness-judge performance, particularly on GT Lit. The Mechanistic prompt—which explicitly requests causal mechanism articulation—achieved the strongest discrimination in Depressive Symptoms and Emergency Department, and the block-level analysis in Results indicates a significant prompt effect on GT Lit AUC that is primarily driven by Mechanistic outperforming the other prompts (Results Section 7.4.3). However, the effect is domain dependent: in Social Norms, discrimination remains weak and no stable prompt ordering emerges.

The parametric sensitivity analysis (Appendix A) confirms these non-parametric findings with large effect sizes: correctness-judge prompt effects explain the majority of within-subject variance, with both GT Synth and GT Lit effects significant after Bonferroni correction. On GT Lit, the Mechanistic prompt more than doubled F1 relative to baseline; on GT Synth, the pattern reverses, with Mechanistic underperforming simpler prompts. These effect sizes indicate that prompt selection is a first-order design decision for correctness judging, not a marginal tuning parameter.

8.5.2 Citation Judge

Citation judging reframes edge validation as an *evidence assessment* problem, but its practical utility is constrained by retrieval bias, evidence specificity, and aggregation choices. The generate-then-search design introduces confirmation bias (queries derived from the model's own narrative) and an "evidence of absence" asymmetry for negative claims. Aggregation by simple mean is sensitive to citation count, search ranking, and fetch failures. These limitations are reflected in the failure mechanisms and motivate treating citation-based scores as soft signals requiring expert review rather than automated accept/reject thresholds.

Future work should therefore treat aggregation and selection as first-class design choices (not merely implementation details), e.g., using median or top-k/max rules, or adding an explicit evidence-selection step (reranker/verifier) that chooses the single most informative citation for a more expensive causal judgement rather than averaging over mixed-quality evidence.

Taken together, the RQ1 validation studies (human agreement, Uleman's external CLD, and physics CLDs; Results Sections 7.4.6, 7.4.7, 7.4.8) suggest that the Citation Judge is usable as a permissive evidence-screening signal, but has weak discrimination in GT Lit settings because performance is bounded by upstream factors: whether retrieval succeeds, and whether retrieved excerpts contain sufficiently direct and specific causal evidence for the edge claim (as opposed to topically related or correlational discussion).

Notably, prompt effects on citation judging exhibit an *opposite* pattern from correctness judging: the Baseline and CoT prompts significantly outperformed Mechanistic variants on GT Lit AUC (Results Section 7.4.5).

Mechanistic prompting yielded counterintuitive *positive* correlations between judge scores and hallucination labels—higher citation scores associated with higher hallucination probability—consistent with a failure mode where mechanistic framing leads the judge to over-accept topically relevant but causally non-specific evidence. The parametric sensitivity analysis (Appendix A) confirms this pattern: on GT Lit, mechanistic variants reduced citation-judge F1 relative to baseline with a moderate-to-large effect size, though this effect is marginal after Greenhouse–Geisser adjustment for sphericity violations. On GT Synth, citation-judge prompt effects were non-significant, confirming that prompt engineering matters more for literature ground truth than for synthetic corruptions.

This prompt×judge-type interaction implies that prompt selection should be tailored to the validation strategy: mechanistic prompting suits correctness judging (where explicit causal reasoning aids discrimination), whereas simpler prompting suits citation judging (where mechanistic framing may lead the judge to over-accept insufficiently specific evidence—for example, by overvaluing whether a retrieved paper uses the *right experimental design* for establishing causality (e.g., an RCT or quasi-experimental identification) relative to whether the evidence is actually specific to the particular variables/edge claim). However, this interpretation carries a model confound caveat: due to budget constraints, the citation-judge GT Synth experiments used GPT-5-mini rather than GPT-4.1 (Results Section 7.4.4), so observed prompt effects in citation judging may be partially attributable to model differences rather than prompt–task interactions alone.

8.5.3 Human Validation: Systematic Leniency

Human validation (72 edges, two judge models) yielded substantial LLM–human agreement ($\kappa = 0.618$, 80.6% exact match), meeting prior GMB inter-rater baselines [117] and aligning with the ∼80% LLM–human agreement rates reported for LLM-as-a-Judge evaluations [21]. However, 100% of disagreements showed the LLM judge being *more permissive* than the human validator—an asymmetric error profile suggesting LLM judges are useful for flagging low-confidence edges but insufficient for strict validation without expert oversight. Agreement also varied by generating model: Claude achieved almost perfect agreement ($\kappa = 0.923$) on the same CLD where GPT-4.1 showed only moderate agreement ($\kappa = 0.435$), suggesting generating model choice matters for alignment with human judgement.

8.5.4 Implication: Active Learning for Judge Calibration

The systematic leniency of LLM judges suggests an active learning paradigm: edges flagged as low-confidence by the judge could be prioritised for human expert review, with expert feedback used to calibrate judge thresholds and potentially fine-tune judge models. This approach would leverage the judge's screening capability while compensating for its permissive bias through targeted human oversight. With the dataset obtained from such prioritised review, the judging LLM could then be fine-tuned to reduce its permissive bias. Evidence from causal graph validation supports this direction: supervised fine-tuning can substantially outperform prompt-only LLM baselines on causal-relation classification [59], suggesting that even modest human-labelled calibration sets could improve judge discrimination.

8.6 Mechanisms of Corrector Failure

The key implication from RQ1b is that judge-guided correction can exhibit setting dependence and can improve judge scores without reliably improving expert-grounded alignment. This motivates conservative, human-in-the-loop deployment for correction: treat the corrector as an assistive drafting tool rather than an autonomous repair mechanism.

Section 8.4 analyzes why judges can fail. The corrector adds a further layer: it *acts* on judge feedback. To understand why judge-guided correction can increase judge scores while failing to improve (or degrading) GT Lit F1, we inspected edge-level correction artefacts saved under `final_runs/RQ1b_corrector_ablation/Data/` (notably `correction_outcomes_*.json` and the paired "before/after correction" spreadsheets).

8.6.1 Judge-Satisficing Revisions (Goodhart-Style Metric Gaming)

Many corrections operate by rewriting motivations to better match the judge's preferences (e.g., acknowledging confounders, bidirectionality, mediators, and adding plausible mechanisms) rather than changing the underlying edge set toward expert GT Lit. This directly explains the characteristic RQ1b pattern—**judge scores improve consistently while GT Lit F1 does not** (Results Table 7.12). For example, multiple "revise" actions in a Depressive Symptoms GT Lit correction session primarily reframe motivations to be more cautious and mechanistically explicit without changing the edge type.[4]

This pattern is a direct manifestation of Goodhart's Law: "when a measure becomes a target, it ceases to be a good measure" [123]. The judge score serves as a proxy for expert-grounded quality; when the corrector optimises for judge approval, it games the proxy rather than improving the true objective. This dynamic is not a failure of implementation but an inherent risk when using LLM judges to supervise LLM correctors: the corrector can learn surface features that satisfy the judge (e.g., more explicit mechanism articulation, acknowledgement of confounders) without changing the underlying causal claim toward expert ground truth.

8.6.2 Precision Loss via Over-Addition Bias

The corrector is incentivised to "fix" low-quality `NONE` edges whose motivations are missing or vacuous (e.g., "I don't know") by converting them into motivated causal claims. This tends to increase recall (consistent with Results Table 7.12, where recall increases across CLDs) but can reduce precision when these added edges are not present in expert GT Lit. In the same Depressive Symptoms session, many `NONE` edges are changed to `POSITIVE/NEGATIVE` with plausible rationales (e.g., `Perceived stress (PSS)→Loneliness (UCLS)` changed `NONE→POSITIVE`).[5]This behavior is helpful under GT Synth corruption (where the "missing motivation" label is often truly erroneous), but it can be harmful under GT Lit where absence from the expert CLD may reflect scope or construct-mapping choices rather than "nonsense" (see RQ3 discussion).

8.6.3 Polarity and Type Inconsistencies (Schema Slips)

Even when the corrector's narrative sounds plausible, it can produce mechanically inconsistent edits: the *new type* can contradict the stated mechanism. For instance, in one Depressive Symptoms correction outcome, the corrector changes `Mindfulness (FFMQ)→Body mass index (BMI)` from `NONE` to `POSITIVE` while the accompanying motivation argues mindfulness can *reduce* BMI via healthier eating and self-control—a sign mismatch that would increase false positives and directly depress precision.[6]

[4]Example correction outcomes (GT Lit, Depressive Symptoms, run 2): `final_runs/RQ1b_corrector_ablation/Data/RQ1b_corrector_experiment_ground_truth_correctness_mechanistic/depressive/run_2/correction_outcomes_ca7a8e72.json`.

[5]Example `NONE`→causal changes (GT Lit, Depressive Symptoms, run 2): `final_runs/RQ1b_corrector_ablation/Data/RQ1b_corrector_experiment_ground_truth_correctness_mechanistic/depressive/run_2/correction_outcomes_ca7a8e72.json`.

[6]Example polarity inconsistency (GT Lit, Depressive Symptoms, run 2): `final_runs/RQ1b_corrector_ablation/Data/RQ1b_corrector_experiment_ground_truth_correctness_mechanistic/depressive/run_2/correction_outcomes_b3ef9444.json`.

8.6.4 Noisy-Label Amplification Under Weak Judge Signal

Because the corrector is supervised by judge feedback, it inherits the judge's blind spots. When the judge provides weak or biased discrimination (notably Social Norms under GT Lit in our experiments), correction becomes a learning-from-noisy-labels problem: the corrector can confidently apply edits that optimise judge score yet reduce GT Lit F1. This mechanism is the correction analogue of judge failure: if the judge cannot reliably separate hallucination from valid-but-scope-excluded edges, the corrector cannot reliably learn which edges to prune versus retain.

8.6.5 Instability Across Correction Sessions

We also observe inconsistent decisions for the same edge across correction sessions, indicating high variance and limited robustness. For example, within Depressive Symptoms, `Mindfulness (FFMQ)`→`Physical inactivity (SBQ)` is changed from `NONE` to `NEGATIVE` in one correction outcome, while in another it is retained as `NONE` with an explicit "no strong evidence" justification.[7]This instability helps explain why prompt variants do not yield reliable differences in correction performance and why improvements do not consistently generalise from GT Synth to GT Lit. This high variance suggests that aggregation strategies from the test-time compute literature may improve robustness. Self-consistency [66] addresses analogous instability in reasoning tasks by sampling multiple reasoning paths and selecting by majority vote; multi-agent debate [67] further reduces variance through adversarial deliberation. Applying these strategies to correction—sampling multiple correction proposals per edge and selecting by consensus or debate—could reduce session-to-session variance and provide calibrated confidence estimates for which corrections are reliable.

8.6.6 Per-CLD Variation in Corrector Effectiveness

Corrector effectiveness varied substantially across CLDs: Emergency Department showed consistent F1 improvement across all prompts, while Social Norms exhibited F1 degradation (ΔF1 = -0.072) and Depressive Symptoms showed F1 degradation across all prompts (ΔF1 ≈ -0.040). This heterogeneity is expected under our pipeline design: the corrector is supervised by judge feedback, so when the judge provides informative signals, correction can repair real errors; when the judge is weak or systematically biased, correction becomes a "learning-from-noisy-labels" problem and can amplify the judge's failure modes rather than improve expert-grounded alignment.

This interpretation is consistent with the RQ1a results: Social Norms is a boundary case where even the Mechanistic correctness judge fails to discriminate on GT Lit, while Emergency Department retains meaningful discrimination. In such a setting, the corrector is more likely to perform harmful edits (e.g., over-pruning plausible edges or rewriting relationships toward generic-but-judge-pleasing formulations), reducing GT Lit F1 even as judge scores improve—a manifestation of Goodhart's law [123], where optimising for the proxy (judge approval) degrades the true target (expert-grounded quality). Depressive Symptoms sits between these extremes: it contains a mixture of more "biomedical/mechanistic" relationships (where judge feedback may be useful) and more context-dependent psychosocial relationships (where the judge may be less reliable), making net effects prompt- and edge-mix-dependent.

[7]Example disagreement across sessions (GT Lit, Depressive Symptoms, run 2): compare `final_runs/RQ1b_corrector_ablation/Data/RQ1b_corrector_experiment_ground_truth_correctness_mechanistic/depressive/run_2/correction_outcomes_ca7a8e72.json` vs. `final_runs/RQ1b_corrector_ablation/Data/RQ1b_corrector_experiment_ground_truth_correctness_mechanistic/depressive/run_2/correction_outcomes_b3ef9444.json`.

Taken together, these mechanisms suggest that correction success is primarily limited by judge signal quality and by the domain's causal ambiguity; we therefore focus next on how to diagnose and measure judge–corrector coupling in a way that can support more robust policies.

8.6.7 Diagnosing Judge–Corrector Coupling

Two concrete analyses could test this explanation by explicitly probing the *judge→corrector dependence*: whether correction success is primarily limited by judge signal quality. First, test whether per-CLD ΔF1 correlates with judge quality (e.g., AUC on GT Lit) and whether the correlation is mediated by which component of F1 moves (precision vs. recall). Second, characterise correction edits by outcome class: for each CLD, measure how often edits convert FP→TN or FN→TP versus causing regressions (TP→FN, TN→FP), and relate these transitions to judge confidence and error type (polarity, spurious edge, motivation/revision). Together, these analyses would directly validate whether correction success is primarily limited by judge signal quality and domain-specific causal ambiguity.

8.7 RQ2: UQ Metrics and Cross-Domain Transfer

The key implication from RQ2 is that generator-side UQ metrics provide limited, domain-dependent signal and do not robustly generalise across CLDs, undermining threshold-based triggering as a universal policy. In practice, these metrics are better treated as *triage/monitoring signals* than as standalone hallucination detectors.

8.7.1 Counterintuitive Cosine Similarity Direction

Cosine similarity was intended as a baseline metric, with the hypothesis that higher similarity would correlate with lower hallucination probability. However, the observed correlation was positive—higher similarity associated with *higher* hallucination probability (Results Table 7.14). Several explanations are plausible: (i) semantic similarity captures topical alignment without establishing causal evidence; (ii) true positives may use different terminology than searchable literature; (iii) false positives may actually have literature support—as the RQ3 Deep Research analysis confirms, a majority of FPs had retrievable causal evidence (Section 8.8). Additionally, domain-specific embedding models (SPECTER, SciBERT, PubMedBERT) may better capture scientific terminology alignment than general-purpose encoders and should be evaluated in future work.

8.7.2 Cross-Domain Generalisation Failure

Leave-one-CLD-out cross-validation (Results Table 7.16) revealed a clear overfitting–generalisation trade-off: high-capacity models (Random Forest, Gradient Boosting) achieved strong in-distribution performance but exhibited the largest cross-domain degradation, while simpler models (Logistic Regression, Neural Network) showed more modest drops and comparable or better out-of-CLD AUC. This pattern suggests that tree ensembles exploit CLD-specific patterns that do not transfer, whereas linear or low-capacity models learn more generalisable—but weaker—hallucination signals. Even the best cross-domain AUC (Logistic Regression) remains in the 0.6–0.7 range, providing only marginal screening value and insufficient discrimination for reliable threshold-based detection.

8.7.3 Alternative UQ Metric Designs

One design limitation of our logprob-based metrics is that they aggregate uncertainty over a variable-length completion that includes both the relationship type and a free-form narrative, which can introduce noise from benign wording variation. An alternative approach—used by Theoraizer [34]—forces single-token outputs for relationship judgements (e.g., "C/N", "yes/no") and extracts log probabilities directly from the relationship-type token. Future work should compare single-token versus full-narrative UQ strategies for hallucination detection, especially under cross-domain evaluation.

For CLD generation, where the full causal narrative is required for explanation and downstream citation search, a hybrid approach may be more appropriate: extracting logprobs from specific tokens known to be semantically important—for example, the probability of the word "causes" in "A causes B," the probability of the target variable name, the gap between that probability and the next-best token, or how that gap changes when retrieved evidence is included versus excluded from the prompt. Additionally, our current implementation captured output probabilities over the entire generated sequence (including preamble and formatting tokens), not just the causal narrative itself; isolating the narrative portion would reduce noise in future UQ designs.

A further limitation is that our logprob-based metrics conflate epistemic uncertainty (genuine model ignorance) with aleatoric and lexical uncertainty (legitimate ambiguity or stylistic variation), as discussed in Section 3.7. This conflation is particularly pronounced in causal extraction: when generating relationship types or variable names, multiple tokens may be contextually appropriate (semantically equivalent phrasings), causing probability-based methods to flag outputs as unreliable even when the model has robust knowledge of the underlying causal structure. Semantic entropy [44] directly addresses this conflation: by sampling multiple generations and clustering semantically equivalent outputs, it isolates the epistemic component (genuine model ignorance about facts) from aleatoric and lexical variation (multiple valid phrasings of the same knowledge). Farquhar et al. show that this isolation substantially improves hallucination detection in question-answering tasks—precisely because it separates "the model doesn't know" from "the model knows but expresses it differently." For CLD generation, this would require defining semantic equivalence over causal claims (e.g., treating "A causes B" and "A promotes B" as equivalent when clustering), a domain-specific adaptation that our current single-generation logprob metrics lack. This multi-sample requirement increases computational cost but may yield cleaner uncertainty signals.

8.7.4 Temperature–Uncertainty Trade-off

UQ metric values are sensitive to the temperature parameter used during generation. As discussed in Section 2.4.0.1 and illustrated in Figure 2.8, temperature scaling reshapes the next-token distribution: lower temperatures concentrate probability mass on high-probability tokens (more deterministic output), while higher temperatures flatten the distribution (more diverse output). This creates a fundamental trade-off for logprob-based UQ: lowering temperature would produce less noisy probability signals—making metrics like min-probability and entropy more interpretable—but would simultaneously reduce the generator's exploratory capacity. In CLD generation, exploratory sampling is desirable because it allows the model to propose novel or less obvious causal relationships that might otherwise be suppressed.

More critically, highly deterministic (low-temperature) generation may trigger earlier model refusal behaviour (e.g., "I don't know" or declining to posit a relationship), particularly for edges where the model has moderate uncertainty. While such refusals may avoid hallucinations in a narrow sense, they also eliminate causal hypotheses from the pipeline entirely—leaving no edge narrative to validate via citation search or Deep Research. In a generate-then-validate workflow, overly conservative generation can be counterproductive: while false positives may be filtered downstream, missed hypotheses may require additional generation passes or expert elicitation

to recover. Future work should systematically probe how temperature affects both UQ metric reliability and the yield of valid causal hypotheses.

8.8 RQ3: Deep Research Validation

The poor cross-CLD generalisation of UQ metrics undermined the original RQ3 design of using them as early hallucination indicators. Poor UQ metric F1 scores observed even at optimal thresholds, together with expected domain-dependent threshold brittleness, pointed toward an alternative: the Mechanistic Correctness Judge—which achieved better F1 performance and more interpretable outputs—could serve as a compute-cheap hallucination flagger to selectively deploy more expensive validation. However, ineffective corrector results from RQ1b—where LLM-as-a-Corrector failed to reliably improve CLD quality and often degraded it—disqualified correction as a viable form of verification test-time compute. We instead designed a more expensive form of test-time compute: a Deep Research solution based on principles from literature (Section 4.7) aimed to maximise retrieval chances.

An added benefit of this approach is that it could also address a question arising from the RQ1 validation study, which showed that citation judges found sufficient evidence for ground-truth positives when controlling for retrieval factors. That question is: if we remove the control of providing a set, causally unambiguous reference for a causal narrative in an edge, can we design a system that finds causal evidence for the narrative even in causally ambiguous domains where our naive search judge fails? And can this system discriminate FP and TP with respect to the ground truth, or yield supporting evidence suggesting that some FPs and FNs should perhaps be included in the ground truth?

8.8.1 Why Literature Validation Has Weak Discriminative Power

The moderate TP–FP discrimination gap (Results Section 7.7) reflects a fundamental limitation of literature-based validation: retrievable causal evidence is a necessary but not sufficient condition for expert-grounded correctness. Deep Research finds supporting literature for a majority of both true positives *and* false positives because the biomedical and social-science literatures are vast, contradictory, and often report correlational or context-specific relationships that do not transfer to the expert's CLD scope. The implication is that literature retrieval should not be treated as an oracle for hallucination detection in our setting—it more plausibly shifts the burden from "does this claim sound plausible?" to "does this claim have *some* published support?" and, crucially, whether that support is applicable to the specific edge claim.

Domain variation in the TP–FP gap reinforces this interpretation. Depressive Symptoms showed the largest gap, likely because its causal claims span both well-evidenced biomedical mechanisms (where TP validation is high) and contested psychosocial relationships (where FP validation is lower). Emergency Department, by contrast, showed the smallest gap—consistent with a domain where most causal claims, whether expert-included or generator-added, can find *some* healthcare-operations literature support.

8.8.2 The Variable-Mismatch Problem

Human validation revealed that the dominant Deep Research failure mode was construct mismatch: retrieved evidence often supported a *related* causal relationship rather than the specific variable pair in question (Results Section 7.7). This failure mode is not a retrieval bug but a structural consequence of LLM-based search: the system queries for semantic neighbours, and the literature rarely uses identical variable operationalisations.

The practical implication is that high-confidence DR validation still requires expert review to confirm construct alignment—a bottleneck that limits full automation.

Applying an applicability threshold ($\geq$0.7) trades recall for precision: fewer edges pass, but those that do achieve high correctness rates. This calibration approach offers a pragmatic path forward—treating DR as a *triage* tool that elevates high-confidence candidates for streamlined expert review rather than as a standalone validator. A direction for future research could investigate whether an *adversarial* judge can succeed in assessing applicability—i.e., a second model tasked with challenging the evidence–claim alignment rather than confirming it. However, calibration would be essential given the systematic leniency bias: the fact that an LLM flagged the evidence as applicable in the first place weakens the case that another LLM will correctly reject it, unless variables that increase discrimination beyond the first LLM are introduced (e.g., different prompting strategies, a model with different training data, a fine-tuned/calibrated verifier, or an entirely different model architecture that does not share transformer-specific biases).

8.8.3 What Ground Truth Enrichment Reveals

The enrichment analysis (Results Tables 7.19 and 7.22) should be interpreted as an *upper bound on potential value* rather than a validated performance gain. Under the strong assumption that DR-validated FPs represent genuine causal relationships that experts omitted, F1 gains are substantial—but this assumption conflates "has literature support" with "should be in the expert CLD." Expert exclusion may reflect deliberate scoping, parsimony, or domain-specific validity judgements that literature retrieval cannot recapitulate. The more conservative human-calibrated estimates are therefore more realistic, though still tentative given limited inter-rater reliability on the calibration sample.

8.8.4 Residual False Negatives: Partial Recovery and Tacit Expert Knowledge

Deep Research focuses on validating generator output (FPs and TPs), but also provides insight into false negatives—expert-included edges that the generator missed. DR retrieved direct causal evidence for 42.9% of FN edges (Results Section 7.7), demonstrating that literature-based validation can partially recover missed expert relationships. Under an enrichment scenario, these DR-validated FNs could be promoted to positive relationships, reducing false negatives from 63 to 36.

However, the remaining 57.1% of FN edges lack searchable evidence under our DR settings. While this could reflect tacit expert knowledge not captured in retrievable literature—domain experts may encode relationships based on clinical intuition, unpublished observations, or context-specific experience that leaves limited searchable trace—alternative explanations include limitations in our query formulation, search depth, or source coverage. These results suggest that literature-based validation may reach diminishing returns for certain edge types, though the precise ceiling and its causes require further investigation with varied retrieval strategies. Targeted elicitation methods (e.g., prompting with related concepts, multi-turn dialogue, or explicit "what's missing?" queries) may help surface these implicit relationships, but fully recovering tacit expert knowledge remains an open challenge for automated CLD generation. Structural limitations of scientific literature that may further explain unavailable causal evidence are discussed in Section 8.11.8.

8.8.5 Judge-Triggered Deployment: Efficiency and Risks

Selective deployment using judge scores as a trigger demonstrates that not all edges require expensive multi-agent validation. The finding that half the edges can be skipped while retaining most of the validation benefit

suggests a practical cost-saving strategy: reserve Deep Research for edges where the judge is uncertain or negative, and accept generator output directly for high-confidence edges. This efficiency profile aligns with findings on compute-optimal test-time allocation [43]: Snell et al. demonstrate that adaptive allocation of inference-time compute—deploying expensive verification selectively based on difficulty signals—can yield up to 4× efficiency gains over uniform application. Our judge-triggered DR instantiates this principle: the Mechanistic Correctness Judge serves as a cheap difficulty signal, and Deep Research serves as the expensive verification reserved for uncertain cases. The observed ∼15% accuracy-per-dollar improvement represents a first estimate of compute-optimal gains in CLD validation, though the efficiency frontier likely extends further with threshold tuning and multi-signal triggering.

However, this efficiency gain introduces a critical dependency: the entire adaptive-compute strategy inherits the judge's failure modes. If the judge systematically passes hallucinations (as in Social Norms), triggering logic will skip Deep Research precisely for edges that most need validation. This creates a compound failure: domains where the judge is weakest are also domains where expensive validation is least likely to be deployed. Mitigating this requires either domain-aware triggering thresholds (which reintroduces the calibration problem) or ensemble triggering that combines judge scores with complementary signals (e.g., UQ metrics, retrieval confidence). The practical implication is that judge-triggered Deep Research is most reliable in domains where the judge already provides meaningful discrimination—precisely the domains where Deep Research is least necessary. One limitation of using the Mechanistic Correctness Judge as a hallucination flagger is that judges introduce their own biases—systematic leniency, prompt sensitivity, and domain-dependent discrimination (Section 8.4)—so this trade-off exchanges one set of limitations for another rather than eliminating them. In practice, judge-triggered deployment may still miss edges that the judge incorrectly approves or unnecessarily validate edges that were already correct.

8.8.6 Limitations of the Deep Research Pivot

The RQ3 pivot occurred late in the research timeline, constraining investigation depth: single-run design prevents uncertainty quantification, search parameters were not systematically ablated, and Deep Research uses GPT-5 while generation used GPT-4.1 (model confound). Moreover, the ground-truth enrichment estimates rely heavily on extrapolating from a limited human-validation sample; the preliminary second-rater overlap indicated low inter-rater reliability on categorical verdicts and poor agreement on applicability scores, so enrichment-derived F1 gains should be interpreted as exploratory rather than confirmatory. A consequence of the recall-oriented design (and of DR's edge-wise evaluation) is that DR evaluates each relationship largely *in isolation*—without conditioning on the full CLD context (mediators, confounders, feedback structure) or expert scope choices—so retrieved evidence may support a relationship that is plausible in general but not appropriate as a *direct* edge or not in-scope for the expert CLD. This can inflate apparent validation rates for edges that would be revised or excluded once CLD context and expert boundary decisions are considered.

In the spirit of exploration and to maximise chances of causal evidence retrieval, we deployed a relatively non-parsimonious multi-agent system with search parameters configured for both broad and deep search, aiming to establish an upper bound in line with the objectives set out in the motivation (Section 8.8). While the system's components were each conceived with literature support (Section 4.7), by starting off with high complexity we forgo Occam's razor—the principle of choosing the simplest explanation consistent with the data—and its modelling analogue, the Minimum Description Length (MDL) principle: choosing the model that minimises summed model and data complexity. The expected upside of this was maximising causal evidence retrieval probability, as was (barring limited applicability in some cases) supported by our results. The downside is a massive parameter and component space in which combinatorial explosion from high degrees of freedom makes finding an optimum infeasible. For future studies, we therefore recommend starting with a primitive of this system; component-wise, this could be an iterative search-refining agent instead of a single naïve search, as

this allows incorporation of retrieved evidence into search refinement and permits multiple steps to steer the search toward more relevant sources.

8.9 Synthesis: Compounding Dependencies and the Division of Labour

The individual RQ findings, when considered together, reveal a *cascading dependency structure* that explains why no single automated component can reliably replace expert judgement in CLD construction.

8.9.1 Compounding Failure Modes Across Components

Each component in the pipeline depends on upstream signal quality, creating compound failure modes:

1. **Judge → Corrector dependency:** The corrector is supervised by judge feedback. When the judge provides weak discrimination (as in Social Norms under GT Lit), the corrector inherits this weakness and can amplify rather than repair errors—the Goodhart-like dynamic observed in RQ1b.
2. **UQ → Triggering dependency:** The original RQ3 design assumed UQ metrics could trigger selective correction. When UQ metrics failed to generalise across CLDs (RQ2), this triggering pathway became untenable.
3. **Judge → Deep Research dependency:** The pivoted RQ3 design uses judge scores to trigger Deep Research. This inherits the judge's domain-dependent discrimination: in domains where the judge is weakest, expensive validation is least likely to be deployed precisely where it is most needed.

This dependency chain means that upstream failures propagate downstream. A weak judge produces noisy corrector supervision, unreliable triggering decisions, and ultimately degraded end-to-end quality—even when individual components perform well in isolation under controlled conditions.

8.9.2 The GT Synth to GT Lit Transfer Gap

A consistent pattern across RQ1, RQ2, and RQ3 is the *construct shift* between synthetic and literature ground truth. Components that perform well on GT Synth (where corruptions are known and unambiguous) often fail on GT Lit (where "errors" conflate genuine hallucinations with contested expert scope choices). This gap is not merely a detection-difficulty increase—it reflects a fundamental ambiguity in what constitutes a "hallucination" when the ground truth itself encodes subjective boundary decisions.

The Deep Research finding that a majority of GT Lit "false positives" admit retrievable causal evidence provides direct evidence for this interpretation: the FP label does not reliably distinguish factual errors from defensible-but-excluded edges. Any automated system optimised on GT Lit will therefore be optimising partly for expert scope reproduction rather than purely for causal correctness.

8.9.3 Emergent Division of Labour

These findings point toward a specific division of labour between LLM and human:

- **LLMs excel at:** Scalable candidate generation (2–3× random baseline), literature retrieval and evidence gathering, and triage/flagging of uncertain edges.

- **Humans remain essential for:** Construct-validity judgements (what *should* be in the CLD), scope and boundary decisions, applicability assessment (does retrieved evidence match the specific claim?), and final validation of LLM-surfaced candidates.

This division can be interpreted as a principled allocation given current model and evaluation constraints: LLMs provide *breadth* (surfacing candidates that experts might overlook) while humans provide *depth* (contextual judgement that LLMs cannot reliably replicate). Future work may shift this boundary, but we do not assume full automation is achievable from our evidence. The practical workflow emerging from this thesis is one of *LLM-assisted candidate generation and evidence gathering, followed by expert-mediated boundary and validity decisions.*

As compute becomes cheaper, finding the optimal allocation of human and LLM effort becomes a dynamic process. Where preference alignment between system output and expert judgement is high and can be reliably indicated, the candidate contributions of the system warrant less human review; in cases where this alignment diverges, more human review is warranted. The contribution this thesis has made is in mapping where these divergences and convergences occur:

- **Convergence zones (less review needed):** GT Synth corruption detection achieved F1 of 0.65–0.82; the Emergency Department CLD showed consistent corrector F1 improvement across all prompts; the Mechanistic Correctness Judge was the only prompt variant exceeding chance-level AUC on GT Lit. Deep Research reduces literature review burden by surfacing evidence meeting causal criteria for contested edges (FPs and FNs vs. GT Lit), shifting expert time from *finding* causal literature to *scope and applicability* decisions—enabling faster edge-inclusion judgements and providing causal-evidence annotations even for true positives.
- **Divergence zones (more review needed):** Social Norms showed corrector F1 *degradation* (ΔF1 $= -0.072$) and near-chance judge discrimination; UQ metrics failed to generalise across CLDs; the GT Synth to GT Lit transition produced substantial performance drops across all components.

8.9.4 Implications for Group Model Building

Traditional expert-driven CLD construction involves iterative boundary negotiation and literature review—processes that LLM-assisted generation could accelerate but not replace. The high rate of DR-validated "false positives" suggests that LLM generators may surface causally defensible relationships that experts excluded for scope reasons, or that experts do not make explicit because they rely on tacit domain knowledge (e.g., context-specific operational experience or clinical intuition) that leaves limited retrievable trace in the published literature, making automated generation a potential *enrichment* tool for Group Model Building (GMB) workshops rather than a replacement for facilitated expert deliberation. This enrichment interpretation aligns with the annotated CLD (aCLD) framework of Crielaard et al. [5], which formalises a three-way distinction for candidate links: *present with consensus*, *uncertain*, and *known-to-be-absent*. DR-validated false positives likely represent the "uncertain" category—relationships with retrievable causal evidence that experts excluded for scope or construct-mapping reasons, not because they are factually incorrect. Systematic GMB practice [3] emphasises that such boundary decisions are inherently context-dependent and resource-intensive; LLM-assisted generation can reduce the resource burden of candidate enumeration while preserving the context-dependent deliberation that determines final inclusion. The practical workflow emerging from this thesis—LLM-generated candidates with DR-provided evidence annotations, followed by expert boundary negotiation—maps directly onto the GMB facilitation structure, with LLMs automating the "breadth" phase while humans retain authority over the "depth" phase.

However, the domain-dependent performance and compounding failure modes reinforce that LLM components should augment, not substitute, expert facilitation. In domains where causal mechanisms are well-established and literature is abundant (e.g., physics), LLM-assisted workflows may approach semi-autonomous operation with periodic expert review—although it should be noted that when causality is clearly established, the need for exploratory CLD modelling itself is reduced. In domains where mechanisms are contested, context-dependent, or poorly documented (e.g., behavioural social science), LLMs can still generate initial CLD drafts, but heavier human oversight of the generated candidates remains necessary.

8.10 Practical Recommendations

Based on our findings, we offer the following recommendations for practitioners using LLMs for CLD generation:

1. **Avoid relying solely on automated correction.** In our GT Lit experiments, LLM correctors showed setting dependence and Goodhart-like effects: judge-score gains did not reliably translate into expert-grounded quality improvements. Human review remains important for quality assurance.
2. **Treat UQ metrics as flagging tools, not filters.** Logprob-based and embedding-based metrics can help prioritise edges for human review but should not be used for automated accept/reject decisions without target-domain calibration, especially across domains.
3. **Match prompt strategy to judge type.** In our experiments, mechanistic prompting (e.g., Mechanistic) improved correctness judging by eliciting explicit causal reasoning, whereas for citation judging, simpler prompts (Baseline, CoT) tended to avoid retrieval bias toward topically related but causally non-specific evidence. This interaction should be replicated without judge-model confounds before being treated as a general guideline.
4. **Expect domain-dependent performance.** Judge and metric effectiveness varies substantially by CLD domain. Validate on held-out data from your target domain before deployment.
5. **Consider literature retrieval for enrichment and evidence gathering, not filtering.** Deep Research can surface potentially valid relationships that experts may have omitted and provide evidence for triage, but evidence often varies in applicability/specificity to the exact edge claim, requiring calibration and expert review.
6. **Maintain human domain expertise.** The gap between synthetic and literature ground truth performance, combined with the high rate of DR-supported edges among GT Lit "hallucinations," underscores that human expert judgement remains essential for CLD validation.

8.11 Limitations and Validity Threats

This section consolidates threats to internal, construct, external, and conclusion validity. Several issues were anticipated and mitigated in Methods (Section 6.10); here we discuss their implications for interpreting results.

8.11.1 Model Dependence

All experiments used GPT-4.1 for generation and GPT-4.1/GPT-5-mini for judging. Performance may differ for other foundation models (Claude, Gemini, open-source alternatives), and conclusions may not transfer across LLM families with different training data, architectures, or alignment procedures.

8.11.2 Domain Dependence

All three primary CLDs focus on health-related domains. Generalisation to other scientific domains (physics, economics, ecology) cannot be assumed without validation on domain-diverse CLD corpora. Auxiliary validation on physics CLDs, Uleman's external CLD, and the Lotka–Volterra CLDs (Results Sections 7.4.8, 7.4.7) indicated that both judge types can work, but these studies used small samples, had other limitations (Section 8.11.3), and did not systematically test prompt variants.

8.11.3 Validation Studies

Human validation showed Claude 4 Sonnet-generated edges achieved near-perfect judge–human agreement ($\kappa = 0.923$) versus moderate agreement for GPT-4.1 ($\kappa = 0.435$; Table 7.5), though sampling may have influenced this contrast. We recommend GPT-4.1 human–LLM-judge agreement figures to be interpreted cautiously until validated on a larger dataset of edges using CLDs from different domains. RQ1 human validation covered Depressive Symptoms and Social Norms but not Emergency Department, so LLM–human agreement may not transfer uniformly. Physics and Uleman's CLD validation studies used Claude 4.5 Opus to define CLD structure and generate narratives (Appendix C.4.15), whereas main experiments used GPT-4.1 and GT Lit input nodes. High judge scores on physics CLDs may thus reflect Opus's superior narrative generation rather than solely domain ease; future work should explore this model confound more systematically. RQ3 Deep Research validation used a small sample ($N = 50$ edges), limiting power and making estimates sensitive to sampling variance. Preliminary second-rater overlap ($N = 13$) indicated low inter-rater reliability on categorical verdicts and poor applicability-score agreement, though the sample was small. Primary validators for RQ1 and RQ3 had limited domain expertise; the RQ3 primary rater also served as RQ1 secondary rater and is the author of this thesis (Section 10). This may have introduced noise and bias, reducing inter-rater reliability and limiting generalisability to expert adjudication.

8.11.4 Construct Validity

GT Synth and GT Lit represent different constructs. GT Synth provides high internal validity but may not capture the full spectrum of natural hallucination types. GT Lit provides ecological validity but conflates genuine errors with scope decisions—the central interpretive issue discussed in Section 8.2. Measurement choices also influence construct validity: the fixed threshold (judge score ≤ 0.5) trades precision vs. recall, and citation judging evaluates *retrieved evidence support* rather than the causal relationship in the abstract. The fixed synthetic corruption rate (30%) limits interpretability; future work should ablate across rates to calibrate transfer to natural error distributions.

Additionally, comparability to the broader literature on LLM hallucinations [8] is limited by operationalisation differences. General hallucination taxonomies focus primarily on false positives—claims the model fabricates—whereas our operationalisation also treats false negatives (missed expert relationships) as a hallucination type. Furthermore, the CLD-specific hallucination types adopted from the SD-AI benchmark [54] (spurious edges, polarity errors, missing edges, wrong causal narratives) are directly comparable within the CLD context but do not capture the full range of hallucination types LLMs can produce in open-ended generation (e.g., citation fabrication). This scope restriction is appropriate for CLD evaluation but limits generalisability claims about hallucination detection beyond structured causal graph tasks.

A further construct-validity threat is evaluator circularity: evaluating LLM outputs with LLM judges risks shared biases and "model-on-model" error amplification. We mitigate this by: (1) evaluating against GT Synth/GT Lit rather than judge preference alone; (2) human validation revealing substantial agreement ($\kappa = 0.618$)

but systematic leniency asymmetry, making circularity bias observable (Results Section 7.4.6). Nonetheless, LLM judges should be interpreted as screening signals requiring human calibration for deployment.

8.11.5 Edge-Isolation Design Constraint

A cross-cutting design limitation affects all downstream components: the generator is prompted with the full variable set and incorporates CLD-level constraints (including a mediation check) to only propose an edge when the relationship is intended to be *directly causal*. However, after generation, the judge, corrector, and Deep Research evaluate edges largely *in isolation*—they do not receive the current CLD structure, candidate mediators, or feedback context as explicit input. This creates a potential mismatch: the generator's mediation-aware *directness* constraint produces narratives that reflect CLD context (e.g., downgrading a seemingly direct effect to `NONE` when a mediating pathway exists), but Deep Research decomposes this narrative into keyword queries and may update the claim based on supporting evidence found *in isolation*, without conditioning on the CLD's candidate mediators. This can introduce an omitted-variable bias: Deep Research may converge on evidence for a direct effect that would be judged indirect once the CLD context is considered. A practical mitigation is to re-apply a mediation/directness check after Deep Research proposes an edge update, ensuring that evidence-driven revisions remain consistent with CLD-level constraints. More broadly, future system iterations should propagate the generator's mediation check to downstream components so that judging, correction, and Deep Research explicitly condition on candidate mediators and the evolving CLD context.

8.11.6 Node-Set Conditioning (Variable Extraction Omitted)

To make evaluation against GT Lit feasible and to isolate edge-level hallucination behaviour, our primary experiments conditioned generation and downstream components on a fixed node set derived from the expert CLDs. This design choice likely makes the task easier than fully corpus-independent CLD generation: it removes variable extraction and naming errors, reduces construct-mapping ambiguity, and ensures the evaluated edge space is defined over the same variables as the ground truth. In practical deployments, users may provide only a domain description or a partial variable set; if the node set is inferred from context alone, additional divergence is expected because downstream "false positives" and "false negatives" can arise from variable-set mismatch (missing or merged constructs, synonymy, overly broad labels) rather than edge-level causal error. Future work should therefore evaluate end-to-end pipelines under weaker conditioning regimes (context-only prompts and/or noisy user-provided variable lists), and explicitly measure variable-level failure modes alongside edge-level performance.

8.11.7 Retrieval and Temporal Dependence

Citation-based judging depends on the retrieval stack (search index, query formulation, paywalls, HTML structure). Search results and website content can drift over time, limiting strict reproducibility. Both Citation Judge and Deep Research inherently search for *supporting* rather than disconfirming evidence, introducing confirmation bias that may inflate apparent validation rates for both true and false positives. We primarily used the Brave Search API for citation retrieval; different search providers (PubMed API, Semantic Scholar, Google Scholar) can return different result sets, potentially affecting judge scores and detection rates. Our Uleman's provider×fetcher validation uses fixed query sets and controlled time snapshots, but two limitations weaken its generalisability: (i) it does not include counterfactual evidence retrieval (searching for disconfirming evidence), and (ii) the reference citations are expert-provided—meaning that the supporting citation is already matched to the edge—representing an idealised upper bound; in the corpus-independent generation approach used throughout the rest of this thesis, such curated references are unavailable, so real-world retrieval quality

is likely lower. The fetcher used to convert URLs to judge-readable text affects both cost and coverage: our custom fetcher achieved 89–100% coverage on health CLDs but only 54.8% on physics CLDs, illustrating how retrieval infrastructure constraints (e.g., paywalls, non-standard HTML, domain-specific publication venues) can substantially affect downstream judge and validation outcomes.

8.11.8 Structural Limitations of Scientific Literature

Beyond retrieval infrastructure, several structural features of the scientific literature itself may explain why causal evidence is unavailable for certain relationships. Publication bias systematically under-represents null results and confirmatory findings for established relationships [124, 125], meaning that causal relationships that failed to replicate or showed context-dependent effects may leave sparse searchable traces. Some relationships are considered so self-evident within a domain that they are rarely investigated or explicitly stated—experts may encode foundational knowledge that is taught but not empirically tested (e.g., basic physiological mechanisms in medicine, first-principles causal chains in physics). Ethical and practical constraints further limit available evidence: certain causal hypotheses cannot be experimentally tested due to restrictions on randomising harmful exposures, leaving only observational studies that may not use direct causal language. Additionally, relationships spanning disciplinary boundaries may be documented in specialised literatures using different terminologies or indexed in databases not covered by general-purpose search APIs. Finally, temporal factors create coverage gaps: recent findings may exist only in preprints, conference proceedings, or institutional reports not indexed by standard search engines, while older foundational findings may predate digital indexing entirely. These structural limitations suggest that literature-based validation will systematically underperform for certain relationship types regardless of retrieval sophistication, reinforcing the need for expert elicitation to complement automated evidence gathering.

8.11.9 Statistical Power and Sample Size

The experimental design uses block-level (CLD×run) aggregation as the unit of analysis for statistical inference ($N = 9$ blocks: 3 CLDs × 3 runs), following the repeated-measures framework specified in Methods Section 6.4. This design controls for CLD-level and run-to-run variance but limits statistical power: with only 9 independent observations, the framework can detect moderate-to-large treatment effects but may fail to detect smaller effects. At the edge level, the three CLDs contribute asymmetrically to the 1,394-edge space (Emergency Department: 1,122 of 1,394 possible directed pairs, or 80%; Depressive Symptoms: 182 pairs; Social Norms: 90 pairs), so edge-level diagnostics are dominated by Emergency Department patterns while block-level tests treat each CLD equally. Human validation samples (72 edges for RQ1a human–LLM agreement; 50 edges for RQ3 Deep Research applicability) were sufficient for effect-size estimation but would benefit from larger samples for subgroup analyses and more robust inter-rater reliability estimation. Additionally, class imbalance—we observed 15.2% hallucinations in our dataset—can make ROC-AUC appear optimistic; we therefore also report thresholded precision/recall/F1, which provides a practically relevant indication of discrimination quality even when class prevalence is skewed.

8.11.10 Sensitivity Analysis Constraints

Systematic sensitivity analyses were not conducted due to computational costs: a full grid over temperature, top-p, prompt variants, and CLD domains would require hundreds of full CLD generations, each triggering downstream judging and correction. Our generator temperature ANOVA (Results Section 7.2.2) showed no significant effect within the tested range, but this represents a limited probe. Key LLM parameters were selected through theoretical arguments and limited empirical validation rather than exhaustive search [21, 77, 100];

ensemble classifier hyperparameters used fixed values without tuning. The synthetic corruption rate (30%) was also fixed; future work should ablate across rates (e.g., 10%/30%/50%) to calibrate transfer to natural error distributions. Temperature sensitivity analysis may become less relevant as frontier models evolve, since several recent models (e.g., OpenAI's o1/o3) do not expose temperature controls. The Deep Research system has many tuneable parameters (query count, iteration depth, scoring thresholds, agent prompts, evidence-type filters) that were not systematically ablated due to the late pivot to this approach and per-edge costs ($0.87/edge); future work should conduct component-wise ablation to identify which DR design choices most affect detection-rate discrimination and cost-efficiency.

8.11.11 Budget and Design Constraints

Cost limitations affected multiple design decisions: top-5 search results for citation retrieval, single-run Deep Research per edge, use of GPT-5-mini where GPT-4.1 would have maintained consistency, and exclusion of true negative edges from Deep Research evaluation (1,109 of 1,394 possible).

In particular, the switch from GPT-4.1 to GPT-5-mini for citation judging creates a model confound: weaker prompt sensitivity in citation judging cannot be cleanly attributed to judge type versus model-specific behaviour. Budget constraints also precluded systematic sensitivity analyses of search parameters, retrieval depth, and agent configurations in Deep Research.

8.12 Future Work

The directions for future work are those directly implied by the boundary conditions and failure mechanisms established in this thesis (Sections 8.4–8.9).

8.12.1 General Directions

Several limitations cut across all pipeline components and motivate overarching future work:

1. **Model generalisation across LLM families.** All experiments used OpenAI models (GPT-4.1, GPT-5-mini, GPT-5). Model and domain dependence were strong throughout (Section 8.11); future work should replicate the full judge/corrector/UQ/DR pipeline across LLM families (Claude, Gemini, open-source models such as Llama and Mistral) to distinguish architecture-specific behaviours from general LLM limitations.
2. **Domain generalisation beyond health CLDs.** Primary experiments used three health-related CLDs and exhibited substantial domain dependence and GT Synth to GT Lit transfer gaps (Sections 8.7, 8.11). Future work should validate the pipeline on additional scientific domains (e.g., economics, ecology, engineering, social policy) to characterise which failure modes persist outside health and which are domain-specific.
3. **Replication and sample size for uncertainty quantification.** Several components (notably Deep Research) used single runs, limiting statistical inference; human validation samples were informative but limited in size (Section 8.5). Future work should run all components with $n \geq 3$ replicates to estimate uncertainty, report confidence intervals, and scale expert evaluation across domains and edge types to enable robust agreement estimates and subgroup analyses. Larger-scale validation would also test how results change when extrapolation assumptions are relaxed—particularly for Deep Research, where enrichment estimates currently depend on extrapolating from limited human-validated samples.

4. **CLD-context integration to address edge-isolation.** All downstream components currently evaluate edges largely in isolation, despite CLD-level directness and mediation constraints (Section 8.11). Future work should provide judges, correctors, and DR with local CLD context (candidate mediators, neighbourhood edges, feedback structure) and integrate mediation/directness checks after component proposals to reduce directness-related errors, polarity confusions, and CLD-incoherent validations.
5. **Adversarial evidence search to counter confirmation bias.** Citation Judge and Deep Research search only for supporting evidence. Future work should add parallel disconfirming-evidence queries and evaluate whether the support/disconfirm ratio improves hallucination discrimination.

8.12.2 LLM-as-a-Judge: From Screening Signal to Calibrated Triage

1. **Targeted judge calibration from disagreement cases.** Human validation showed substantial agreement but systematic leniency (Section 8.5). Future work should therefore build a small expert-labelled calibration set enriched for judge–human disagreement cases and evaluate whether calibration reduces permissiveness while preserving recall (e.g., PR-AUC at fixed review workloads). This direction is consistent with evidence that supervised fine-tuning can substantially outperform prompt-only LLM baselines on causal-relation validation [59].
2. **Reject-first prompting to counter plausibility bias.** Given the plausibility bias and "evidence of absence" asymmetries we observed (Section 8.4), future work should compare confirmatory prompts against adversarial/disconfirmatory prompts (e.g., "argue against the edge") and quantify changes in false-positive acceptance and calibration under GT Lit.
3. **Citation-judge failure taxonomy.** Future work should report a standardised per-edge taxonomy (e.g., no retrieval/no fetch vs. weak evidence vs. related-but-not-identical vs. causal/correlational mismatch) to separate infrastructure failures from evidence-quality failures and to guide escalation policies (deeper retrieval, query reformulation, or human review).

8.12.3 LLM-as-a-Corrector: Making Edits Translate to Expert-Grounded Gains

1. **Action-factorised correction to reduce schema slips.** We observed polarity/type inconsistencies and instability across correction sessions (Section 8.6). Future work should separate prompts (or models) for revise vs. delete vs. type/polarity changes and test whether this reduces mechanically inconsistent edits and improves GT Lit quality.
2. **Evaluate correction by transition outcomes, not judge scores.** Because judge-score gains did not reliably translate into GT Lit gains (Goodhart-style metric gaming; Section 8.6), future work should treat per-edge transitions (FP→TN, FN→TP, and regressions) as the primary outcome and analyse how these transitions depend on judge confidence and error type.
3. **Evidence-grounded correction with post-hoc directness checks.** Over-addition bias and scope disagreement can harm precision on GT Lit (Sections 8.2, 8.6). Future work should integrate retrieval so correctors can propose evidence-backed revisions, and then re-apply mediation/directness constraints after evidence integration to prevent evidence-driven edits from violating CLD-level assumptions.
4. **Iterative refinement with stopping criteria.** Single-pass correction exhibited high variance (Section 8.6). Allowing multiple correction rounds with explicit stopping rules (e.g., no improvement in transition metrics or calibrated confidence) could improve robustness while controlling cost.
5. **Improved correctors via training and tooling.** Since the corrector can satisfy the judge without improving GT Lit (Section 8.6), future work should investigate per-edge action mismatch and train against expert-grounded objectives (e.g., RLHF on correction quality, or tool-assisted evidence grounding) and evaluate whether this shifts behaviour toward expert-aligned edits rather than judge-satisficing rewrites.

8.12.4 UQ Metrics: From Universal Thresholds to Robust Ranking Signals

1. **Token-level LUQ for the relationship decision.** Narrative-level aggregation likely adds noise and may hinder transfer (Section 8.7). Future work should extract logprobs for the relationship decision itself (e.g., via single-token relationship outputs or relationship-type tokens) and re-run leave-one-CLD-out evaluation to test whether cross-domain generalisation improves.
2. **Use UQ for prioritisation rather than accept/reject filtering.** Because fixed-threshold policies were brittle across CLDs (Sections 8.7, 8.8), a more realistic use of UQ is to rank edges for review. Future work should report how many errors are captured when reviewing the top $k\%$ most-uncertain edges and the resulting workload reduction.
3. **Domain-specific embeddings for similarity baselines.** Cosine similarity behaved counterintuitively and may be capturing topical rather than causal alignment (Section 8.7.1). Future work should evaluate scientific embeddings (SPECTER, SciBERT, PubMedBERT) and test whether directionality and transfer improve.
4. **Multi-sample uncertainty for cleaner epistemic signals.** Since single-generation LUQ conflates epistemic and lexical uncertainty (Section 8.7), future work should test multi-sample approaches (e.g., semantic entropy) by sampling multiple edge narratives per pair, clustering semantically equivalent outputs, and evaluating the cost–benefit trade-off under cross-CLD validation.

8.12.5 Deep Research: Optimising Applicability, Not Just Retrieval

1. **Explicit evidence–claim alignment verification (applicability-first).** The dominant failure mode was construct mismatch (Section 8.8). Future work should add a dedicated applicability checker (potentially adversarial) and validate it against the human rubric, including calibration and inter-rater reliability.
2. **Negative controls and TN coverage to estimate over-validation.** Because literature is vast and recall-oriented search can "support" many claims (Section 8.8), future work should evaluate sampled TN edges and synthetic negative-control claims to estimate false validation rates.
3. **Component-wise ablation toward a minimal DR primitive.** The pivot favoured a non-parsimonious system with a large parameter space (Section 8.8). Future work should compare a minimal iterative query-refinement agent against the full multi-agent stack to identify which components drive applicability improvements per dollar.
4. **Parameter ablation for robust, cost-efficient configurations.** DR has many tunable parameters and the pilot did not systematically explore them (Section 8.8). Future work should systematically vary search iterations (1–5), evidence source limits (3–10), and scoring/quality thresholds to identify stable settings under cost constraints.

8.12.6 Adaptive Test-Time Compute: Policies that Avoid Compounding Failure

1. **Ensemble triggering to mitigate compounding failures.** Judge-triggered DR inherits judge failure modes (Section 8.8). Future work should trigger DR using a combination of judge confidence, retrieval diagnostics, and UQ ranking and compare cost–accuracy trade-offs under leave-one-CLD-out policy selection.
2. **Shift-aware policies without labelled calibration.** Universal thresholds are brittle and labelled data may be unavailable early (Section 8.8). Future work should use unsupervised drift indicators (e.g., judge-score distribution shift, retrieval/fetch failure rates) to adaptively increase validation intensity and test robustness across domains.

If one were to prioritise a single high-value next step, it would be to conduct a systematic Deep Research ablation (single-agent vs. multi-agent, search depth, evidence filters, and trigger policies), replicated and human-validated, on a larger, domain-diverse CLD corpus (beyond health), to quantify generalisation and isolate which components drive gains.

Chapter 9

Conclusion

This thesis investigated whether methods independent of researcher-provided corpora can detect and mitigate hallucinations in LLM-generated Causal Loop Diagrams (CLDs), motivated by the reliability requirements of using LLMs for structured scientific knowledge generation. Across three research questions, we evaluated LLM-as-a-Judge configurations, LLM-as-a-correctors, uncertainty quantification (UQ) signals, automated literature-oriented validation via Deep Research, and adaptive deployment of Deep Research based on judge uncertainty signals.

Overall, the results support a nuanced conclusion: corpus-independent detection and correction methods can work in controlled settings, but their effectiveness degrades when the target is expert-grounded causal structure. In our dataset of three health-domain CLDs, correctness-judge performance is high in a controlled synthetic hallucination setting (GT Synth), but it drops in the literature ground-truth setting (GT Lit), with only the causal mechanistic prompt beating random baselines. In expert domains, disagreement with an expert CLD should not automatically be treated as fabrication: it can also reflect scope choices, tacit knowledge, or construct mismatch between an edge claim and what is explicitly evidenced in accessible sources. As a consequence, LLM-generated CLDs are best used as candidate drafts that focus expert attention on the contested edge space, rather than as self-validating outputs.

For corpus-independent hallucination signals, generator-side UQ metrics (perplexity, minimum probability, maximum window entropy, and "corpus-dependent" cosine similarity) do not effectively flag hallucinations independently. They can contribute as features to supervised ensemble classifiers in-distribution, but they fail to generalise out-of-CLD, reinforcing their role as auxiliary signals for monitoring and triage rather than as standalone detectors.

For judge-guided correction, LLM-as-a-corrector consistently improves judge scores and slightly improves CLD quality in the synthetic setting, but not versus literature ground truth. This highlights a Goodhart-like risk [123] in judge–corrector feedback loops: optimizing for proxy judge satisfaction without improving expert-grounded alignment.

Finally, for literature-oriented validation, we piloted a multi-agent Deep Research (DR) system as adaptive test-time compute. DR retrieves direct causal evidence for 62.4% of false positive edges and 42.9% of false negative edges. Single-rater human validation ($n = 50$) suggests that at applicability threshold $t = 0.7$, 66% of DR-evidenced edges pass the threshold and, among passing edges, 90.9% are correct-causal; however, a preliminary second-rater overlap ($n = 13$) indicates only fair agreement on categorical verdicts and poor agreement on applicability scores, so calibration-based extrapolations should be interpreted as exploratory and potentially rater-sensitive. Under this calibration, extrapolated ground-truth enrichment suggests substantial gains in aggregate edge F1: $0.267 \rightarrow 0.527$ (Scenario 1: FP$\rightarrow$TP) and $0.267 \rightarrow 0.582$ (Scenario 2: LLM+DR),

with conservative estimates of 0.472 and 0.500. Using a mechanistic correctness judge to flag uncertain edges for adaptive DR deployment suggests an estimated 15% accuracy-per-dollar increase.

9.1 Key Contributions

- We provide an end-to-end experimental evaluation of corpus-independent CLD generation and post-hoc validation, spanning generation, judging, correction, uncertainty signals, and evidence-oriented verification, with methods established in literature but not tested systematically in CLD context.
- We show that the *Correctness Judge* performs strongly in controlled synthetic hallucination settings (GT Synth) but does not reliably transfer to literature ground truth (GT Lit), where only the Mechanistic prompt exceeds random baselines.
- We find that the *Citation Judge* does not reliably discriminate valid from hallucinated edges under either GT Synth or GT Lit: while it shows substantial alignment with a human rater when provided with relevant references (with a systematic leniency bias), weak discrimination appears to stem primarily from upstream constraints—evidence type/strength and retrieval quality. Retrieval and evidence asymmetries (notably, *absence of evidence* being treated as *evidence of absence* for negative edges) further make citation judging better suited to permissive screening than automated filtering.
- We benchmark generator-side uncertainty quantification (UQ) metrics as corpus-independent hallucination signals and show that they are weak as standalone detectors and do not generalise reliably across CLDs, but can serve as auxiliary features for monitoring and triage.
- We identify and characterise failure modes that matter specifically for CLDs (e.g., scope-vs-factual disagreement, plausibility bias, evidence-of-absence effects, and construct/applicability mismatch).
- We demonstrate that judge-guided correction can optimise for judge satisfaction without improving expert-grounded alignment, highlighting a Goodhart-like risk [123] in judge–corrector feedback loops.
- We provide evidence that literature-oriented validation can enrich the disagreement space: calibrated with human validation and under strong assumptions (relationship isolation, a single human validator, and extrapolation), Deep Research yields substantial but bounded F1 gains; future work should test enrichment performance as these assumptions are relaxed. We also show a proof-of-concept judge-triggered deployment policy that improves cost-efficiency by selectively invoking Deep Research on uncertain edges.

9.2 Concluding Remarks

Taken together, these findings position LLMs as scalable assistants for proposing and stress-testing causal structures, not as autonomous validators. The most promising workflow is hybrid: use LLMs to expand the edge space, use judges and UQ signals to triage uncertainty, and selectively apply evidence-oriented validation (e.g., Deep Research with an explicit applicability threshold) to focus expert attention on the contested edges that most affect the final model.

A key finding is that "hallucination" in this setting may not be equivalent to fabrication. In our experiments, a substantial proportion of LLM-generated edges classified as false positives against expert ground truth nonetheless had retrievable direct causal evidence under Deep Research (62.4%). This indicates that the expert–LLM disagreement space can mix multiple phenomena: genuine fabrications, expert scoping choices, and construct or operationalisation mismatch where evidence supports a related—but not identical—claim. Accordingly, disagreement with an expert CLD should not automatically be treated as error; it can also be a signal for targeted evidence review and model refinement.

This perspective positions LLMs not merely as reproducers of expert consensus, but as tools for systematically surfacing candidate relationships for expert adjudication—an increasingly valuable function as system

complexity grows and the edge space ($O(|V|^2)$) exceeds the capacity for exhaustive expert deliberation. As LLM capabilities continue to advance, the methods evaluated here—particularly uncertainty-guided selective deployment combined with literature-grounded validation—provide a foundation for integrating LLM-assisted causal synthesis into rigorous research workflows. The path forward is not to replace expert judgement, but to augment it: use LLMs to handle scalable generation and evidence gathering, while reserving expert attention for construct definition, applicability assessment, and the causal modelling decisions that require domain knowledge.

The most valuable next step would be a systematic Deep Research ablation (single-agent vs. multi-agent, search depth, evidence filters, and trigger policies) replicated and (human) validated, across a larger, domain-diverse CLD corpus (beyond health) to quantify generalisation and isolate which components drive gains.

Chapter 10

Ethics and Data Management

This thesis adheres to the ethical code (https://student.uva.nl/en/topics/ethics-in-research) and research data management policies (https://rdm.uva.nl/en) of UvA and IvI.

10.1 Ethical Considerations

This research involves the use of large language models (LLMs) for automated generation and validation of causal loop diagrams in health domains. Several ethical considerations are relevant:

- **No human subjects:** This research does not involve human participants except in human validation of LLM outputs. All experiments use publicly available datasets and published causal loop diagrams from peer-reviewed literature.
- **AI transparency:** LLM-generated outputs are clearly labelled as such. We emphasise that LLM-generated CLDs should serve as starting points for expert review rather than definitive scientific conclusions.
- **Potential for misuse:** While this research aims to assist domain experts, there is potential for misuse if LLM-generated causal structures are treated as ground truth without expert validation. We explicitly recommend human oversight for any practical applications.
- **Environmental impact:** The computational experiments involve substantial API calls to commercial LLM providers. We have optimised experimental designs to minimise unnecessary compute while maintaining scientific rigor, for example by capturing generator UQ metrics during generation in RQ1 runs such that these needn't be recomputed for RQ2.

10.2 Data Management

The full software environment used to generate experimental artefacts for this thesis is contained within a private GitHub repository (https://github.com/CausalixAI/platform, branch: `thesis-nitai`). The repository is private because the code pertains to a startup (Causalix) that Professor Quax and the author are launching. As detailed in the Declaration of Authorship, this thesis was conducted within a shared software

environment, with clear delineation of individual contributions. Access to the full platform repository can be granted to examiners upon request.

For computational reproducibility of the analyses and the Results chapter figures/tables, a public reproduction snapshot is provided at https://github.com/NitaiNijholt/cld-hallucination-detection (branch: `thesis-reproduction`). This public repository includes the analysis scripts and result datasets required to regenerate the thesis outputs; it does not include the full simulation/data-generation pipeline or proprietary platform infrastructure.

Description	**Availability**	**License**
Full platform source code and simulation/data-generation pipeline	Private GitHub repository (access on request)	Proprietary (Causalix)
Ground truth CLDs: Depressive Symptoms, Social Norms, Emergency Department, Alzheimer's, Physics	Extracted from cited publications	Fair use for research
Result datasets for analysis reproduction (RQ1–RQ3, SUPP, PRELIM)	Public GitHub reproduction snapshot (https://github.com/NitaiNijholt/cld-hallucination-detection)	Mixed (see repository)
RQ1 human validation data (n=72)	Private GitHub repository	Proprietary (Causalix)
RQ3 human validation data (n=50)	Public GitHub reproduction snapshot (https://github.com/NitaiNijholt/cld-hallucination-detection)	CC-BY
Analysis scripts (reproduction snapshot)	Public GitHub reproduction snapshot (https://github.com/NitaiNijholt/cld-hallucination-detection)	MIT (code)

Table 10.1: Data and code availability for thesis reproducibility.

Human Validation Raters. Human validation labels for the datasets listed in Table 10.1 were provided by the following individuals:

- **RQ1a human validation data ($n = 72$ edges):** Primary rater: Mick Scheide (Data Manager, Antoni van Leeuwenhoek Hospital) validated File 1 (Depressive Symptoms, GPT-4.1, $n = 30$) and File 2 (Depressive Symptoms, Claude 4 Sonnet, $n = 30$). For File 3 (Social Norms, GPT-4.1, $n = 12$), both Mick Scheide and Nitai Nijholt (MSc Student, UvA FNWI) provided ratings to establish inter-rater reliability.
- **RQ3 human validation data ($n = 50$ edges):** Primary rater: Nitai Nijholt validated all 50 sampled Deep Research edges from the Depressive Symptoms CLD (`final_runs/RQ3_deep_research_validation/valida`
  Secondary rater: Cillian Hourican (PhD Candidate, UvA FNWI) validated a 13-edge overlap subset for inter-rater reliability assessment.

All experimental artefacts, including generated CLDs, judge outputs, corrector outputs, and Deep Research evidence, are preserved in structured formats (JSON, CSV) with full provenance tracking (run IDs, timestamps, model versions, prompts). Reproducibility procedures are described in Section C.4.

Appendix A

Supplementary Sensitivity Analysis

This appendix provides extended sensitivity analyses for both Judge (RQ1a) and Corrector (RQ1b) prompts, quantifying performance stability across prompt variations and reporting the corresponding assumption checks and effect sizes.

A.1 Judge Prompt Sensitivity

To quantify how prompt engineering choices affect judge performance, we perform a sensitivity analysis measuring the relationship between prompt variation (input) and F1 change (output). For judge prompts (RQ1a), prompt distances are computed as cosine distance from the baseline prompt using MPNet embeddings. We use **repeated-measures ANOVA** with CLD×run as subjects and prompt as the within-subject factor, reporting **partial** η^2 as the effect size (proportion of within-subject variance explained by prompt). Bonferroni correction is applied across the four RQ1a tests ($\alpha_{\text{adj}} = 0.0125$).

A.1.1 Derivation: First-Order Sobol Index for a Categorical Factor

For a single categorical factor $A \in \{1, \ldots, K\}$ with $p_a = \Pr(A = a)$, define $\mu_a := \mathbb{E}[Y \mid A = a]$ and $\mu := \mathbb{E}[Y] = \sum_{a=1}^{K} p_a \mu_a$. The first-order Sobol index for A is:

$$S_A := \frac{\operatorname{Var}(\mathbb{E}[Y \mid A])}{\operatorname{Var}(Y)}. \tag{A.1}$$

Since $\mathbb{E}[Y \mid A]$ takes value μ_a when $A = a$, we have

$$\operatorname{Var}(\mathbb{E}[Y \mid A]) = \sum_{a=1}^{K} p_a (\mu_a - \mu)^2. \tag{A.2}$$

By the law of total variance,

$$\operatorname{Var}(Y) = \mathbb{E}[\operatorname{Var}(Y \mid A)] + \operatorname{Var}(\mathbb{E}[Y \mid A]) = \sum_{a=1}^{K} p_a \operatorname{Var}(Y \mid A = a) + \sum_{a=1}^{K} p_a (\mu_a - \mu)^2. \tag{A.3}$$

Hence,

$$S_A = \frac{\sum_{a=1}^{K} p_a(\mu_a - \mu)^2}{\mathrm{Var}(Y)} = \eta^2, \tag{A.4}$$

i.e., the population one-way ANOVA effect size (correlation ratio) for a categorical factor [126]. In the main experiments, we report η_p^2 (partial eta-squared) because our design is blocked by CLD×run (repeated measures), so the sensitivity measure is computed after controlling for these subject-level differences.

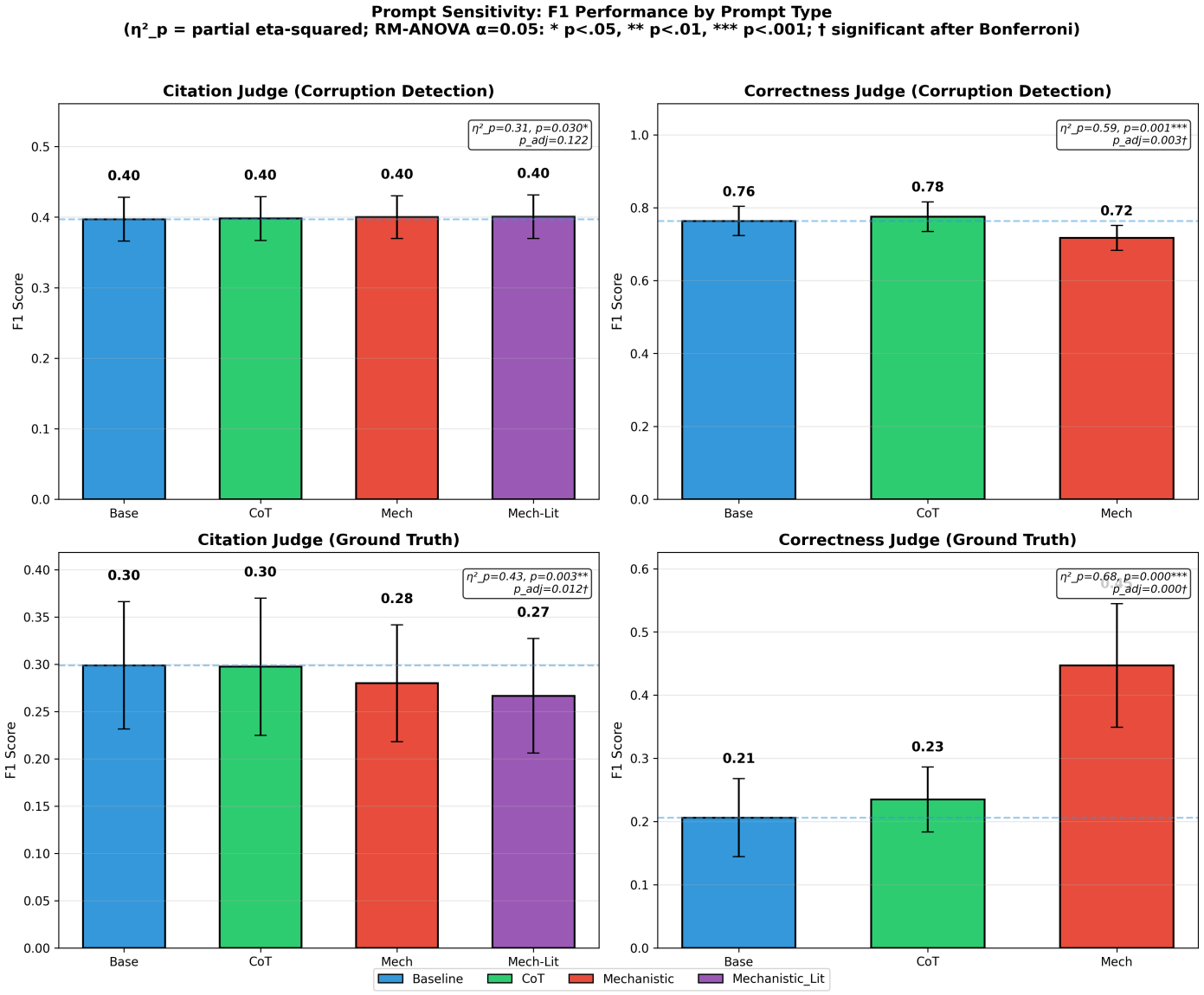


Figure A.1: Prompt sensitivity analysis across RQ1a experiments. Each panel shows F1 performance (mean ± 95% CI across CLDs and runs) for prompt variants. Statistical test: repeated-measures ANOVA with partial η^2; significance indicated by * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; † significant after Bonferroni correction. Dashed lines indicate baseline.

Figure A.1 shows that prompt effects are strongest for the correctness judge (Corruption Detection: $\eta_p^2 = 0.62$, $p < 0.001$; Ground Truth: $\eta_p^2 = 0.68$, $p < 0.001$), with large effect sizes indicating that 62–68% of within-subject variance is explained by prompt choice. Citation judge effects are weaker (Corruption Detection: $\eta_p^2 = 0.31$, $p = 0.030$) and do not survive Bonferroni correction.

Table A.1: Judge Prompt Sensitivity Analysis

Experiment	Prompt	Dist.	F1	Std	95% CI	ΔF1	η_p^2	p
Corr. Cit.	Baseline	0.00	0.397	0.038	[0.366, 0.428]	+0.000	0.31	0.122
	CoT	0.02	0.398	0.038	[0.367, 0.429]	+0.001		
	Mechanistic	0.32	0.400	0.037	[0.370, 0.430]	+0.003		
	Mech-Lit	0.31	0.401	0.038	[0.370, 0.431]	+0.004		
Corr. Corr.	Baseline	0.00	0.764	0.049	[0.724, 0.804]	+0.000	0.59	0.003**
	CoT	0.01	0.776	0.050	[0.735, 0.816]	+0.012		
	Mechanistic	0.18	0.717	0.042	[0.683, 0.751]	-0.047		
GT Cit.	Baseline	0.00	0.299	0.083	[0.231, 0.366]	+0.000	0.43	0.012*
	CoT	0.02	0.297	0.089	[0.225, 0.370]	-0.001		
	Mechanistic	0.32	0.280	0.076	[0.218, 0.342]	-0.019		
	Mech-Lit	0.31	0.267	0.074	[0.206, 0.327]	-0.032		
GT Corr.	Baseline	0.00	0.206	0.076	[0.144, 0.268]	+0.000	0.68	<.001***
	CoT	0.01	0.235	0.063	[0.183, 0.286]	+0.029		
	Mechanistic	0.18	0.447	0.120	[0.349, 0.545]	+0.241		

Note. Dist. = MPNet cosine distance from baseline. η_p^2 = partial eta-squared (RM-ANOVA). N=9 per prompt.
95% CI computed using t-distribution ($df = 8$) for block-level meta-inference per uncertainty reporting rule.
p-values are Bonferroni-corrected (4 tests, α_{adj}=0.0125). * $p < .05$, ** $p < .01$, *** $p < .001$.

Table A.2: RM-ANOVA Assumption Checks for Judge Prompt Sensitivity (Appendix K)

Exp.	n	k	Shapiro p	Mauchly p	ϵ_{GG}	p (unc.)	p (GG)
Corr. Cit.	9	4	0.002	<.001	0.57	0.030	0.064
Corr. Corr.	9	3	0.352	<.001	0.74	<.001	0.002
GT Cit.	9	4	0.684	<.001	0.48	0.003	0.022
GT Corr.	9	3	0.509	<.001	0.57	<.001	0.002

Note. Shapiro-Wilk is applied to RM-ANOVA residuals (subject + prompt additive model). Mauchly's test assesses sphericity of the within-subject covariance. ϵ_{GG} is Greenhouse–Geisser epsilon; p (GG) uses ϵ_{GG}-adjusted degrees of freedom. For $k = 2$, sphericity is not defined and Mauchly/GG are omitted.

Interpretation. Sphericity is violated in all four experiments (Mauchly $p < .001$), so the Greenhouse–Geisser correction is appropriate. Under p (GG), prompt effects remain significant for Corr. Corr., GT Cit., and GT Corr., but Corr. Cit. no longer reaches $p < .05$ (p (GG)= 0.064). Residual normality is violated only for Corr. Cit. (Shapiro $p = 0.002$); the other three analyses are consistent with approximately normal residuals.

Interpreting the assumption checks (Table A.2), Mauchly's test indicates **sphericity violations** in all judge prompt analyses ($p < .001$), so inferential conclusions should be based on **Greenhouse–Geisser corrected** p-values. Under GG correction, the corruption-detection citation judge prompt effect is no longer significant ($p_{\text{GG}} = 0.064$), while both correctness judge prompt effects remain robust ($p_{\text{GG}} = 0.002$). The ground-truth citation judge effect is marginal under GG ($p_{\text{GG}} = 0.022$) and still does not survive Bonferroni correction ($\alpha_{\text{adj}} = 0.0125$). Shapiro–Wilk rejects residual normality only for Corr. Cit.; given $n = 9$, we treat p-values as approximate and emphasise effect sizes alongside corrected inference.

Table A.1 shows that prompt choice significantly affects the correctness judge (large effects, $\eta_p^2 > 0.60$), while citation-judge prompt effects are moderate ($\eta_p^2 \approx 0.30$–0.43) with the corruption detection effect not surviving family-wise correction.

A.2 Corrector Prompt Sensitivity

Extending the sensitivity analysis to the LLM-as-a-Corrector (RQ1b), we evaluate how corrector prompt complexity affects correction performance. Unlike judge prompts which had distinct semantic content, corrector prompts follow a cumulative design: Baseline provides examples only, CoT adds step-by-step reasoning (1.45× length), and Mechanistic adds Bradford Hill criteria (1.91× length). Because these prompts are long and share a common prefix, we compute corrector prompt distances using long-context Jina embeddings v2 (to avoid truncation artefacts in short-context encoders).

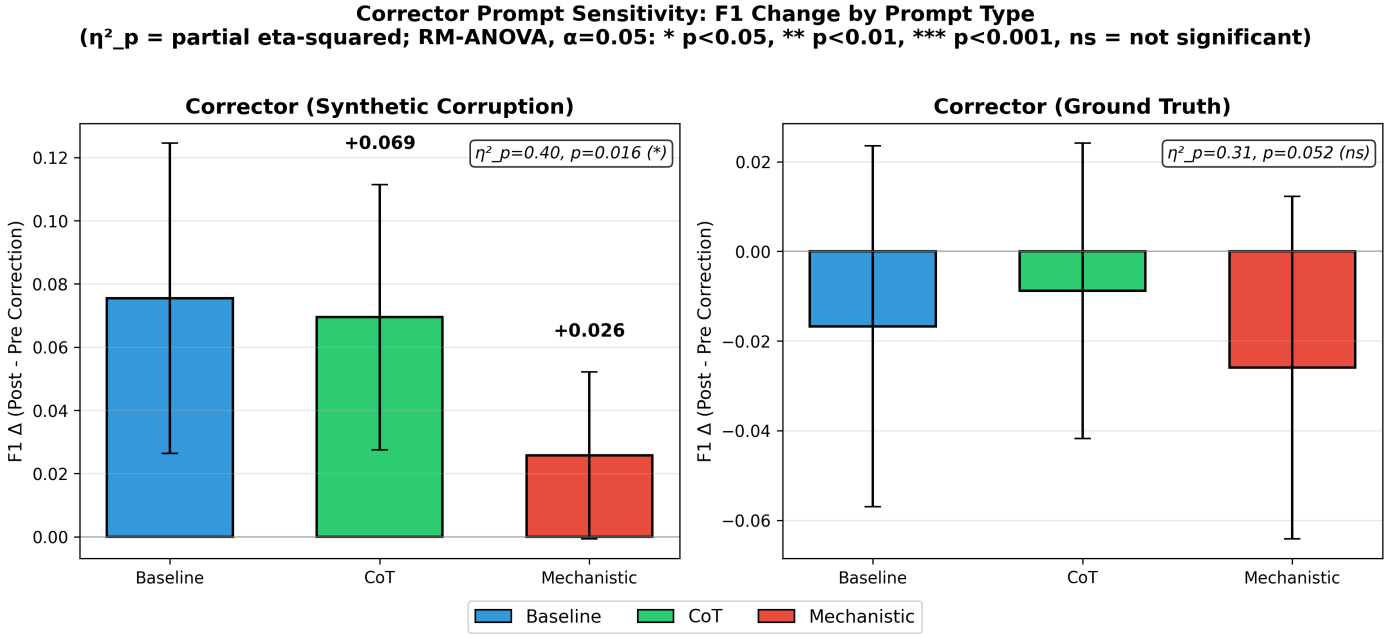


Figure A.2: Corrector prompt sensitivity analysis. F1 Δ (post-correction minus pre-correction) by prompt type. Statistical test: repeated-measures ANOVA with partial η^2; significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Error bars show 95% CI across CLDs and runs.

Figure A.2 shows a significant prompt effect for synthetic corruptions ($\eta_p^2 = 0.40$, $F(2, 16) = 5.43$, $p = 0.016$), indicating that simpler prompts (Baseline: +0.075) outperform complex prompts (Mechanistic: +0.026). Ground truth correction shows a marginally significant effect ($\eta_p^2 = 0.31$, $p = 0.052$).

Table A.3: Corrector Prompt Sensitivity Analysis

Experiment	Prompt	Dist.	F1Δ	Std	95% CI	Actions	η_p^2	p
Synth.	Baseline	0.00	+0.08	0.06	[+0.03, +0.12]	1006		
	CoT	0.01	+0.07	0.05	[+0.03, +0.11]	1006	0.40	0.032*
	Mechanistic	0.03	+0.03	0.03	[-0.00, +0.05]	1006		
GT	Baseline	0.00	-0.02	0.05	[-0.06, +0.02]	418		
	CoT	0.01	-0.01	0.04	[-0.04, +0.02]	408	0.31	0.104
	Mechanistic	0.03	-0.03	0.05	[-0.06, +0.01]	418		

Note. Dist. = Jina v2 cosine distance from baseline. η_p^2 = partial eta-squared (RM-ANOVA). N=9 per prompt.
95% CI computed using t-distribution ($df = 8$) for block-level meta-inference per uncertainty reporting rule.
p-values are Bonferroni-corrected (2 tests, α_{adj}=0.025). * $p < .05$, ** $p < .01$, *** $p < .001$.

Table A.4: RM-ANOVA Assumption Checks for Corrector Prompt Sensitivity (Appendix K)

Exp.	n	k	Shapiro p	Mauchly p	ϵ_{GG}	p (unc.)	p (GG)
Synth.	9	3	0.516	<.001	0.54	0.016	0.044
GT Lit	9	3	0.119	<.001	0.72	0.052	0.073

Note. Shapiro-Wilk is applied to RM-ANOVA residuals (subject + prompt additive model). Mauchly's test assesses sphericity of the within-subject covariance. ϵ_{GG} is Greenhouse–Geisser epsilon; p (GG) uses ϵ_{GG}-adjusted degrees of freedom. For $k = 2$, sphericity is not defined and Mauchly/GG are omitted.

Interpreting the assumption checks (Table A.4), Mauchly's test again indicates **sphericity violations** ($p < .001$), so GG-corrected p-values provide the most appropriate inference. Under GG correction, the synthetic prompt effect is borderline ($p_{\text{GG}} = 0.044$) and would not meet the Bonferroni-adjusted threshold for the two RQ1b tests ($\alpha_{\text{adj}} = 0.025$), while the GT Lit prompt effect remains non-significant ($p_{\text{GG}} = 0.073$). Shapiro–Wilk does not reject residual normality for either experiment.

The table above summarises the repeated-measures ANOVA results for corrector prompt sensitivity.

A.2.1 Corrector Action Counts

Tables A.5 and A.6 provide the full breakdown of corrector actions for verification purposes.

Table A.5: RQ1b Corrector Action Counts - Synthetic Corruptions (Full Breakdown)

Prompt Variant	Edges Processed	Successful Actions	Revise	Change Type	None	Error
Baseline	1003	1003	450	553	0	0
CoT	995	995	471	524	0	0
Mechanistic	979	979	408	571	0	0
Aggregate	2977	2977	1329	1648	0	0

Note. Full breakdown of corrector actions for verification. Edges Processed = total candidate edges for correction. Successful Actions = Revise + Change (used in main text tables). None = edges where corrector determined no change needed. Error = API/parsing errors during correction. Verification: Edges Processed = Revise + Change + None + Error.

Table A.6: RQ1b Corrector Action Counts - Ground Truth (Full Breakdown)

Prompt Variant	Edges Processed	Successful Actions	Revise	Change Type	None	Error
Baseline	1122	1111	728	383	0	11
CoT	1122	1105	724	381	4	13
Mechanistic	1122	1103	663	440	11	8
Aggregate	3366	3319	2115	1204	15	32

Note. Full breakdown of corrector actions for verification. Edges Processed = total candidate edges for correction. Successful Actions = Revise + Change (used in main text tables). None = edges where corrector determined no change needed. Error = API/parsing errors during correction. Verification: Edges Processed = Revise + Change + None + Error.

Appendix B

Function-to-Algorithm Mapping

Table B.1 provides the complete tracing from result directories to simulation scripts to core function implementations, enabling verification of the algorithmic descriptions in Appendix I.

Table B.1: Complete Function-to-Algorithm Mapping

Algo	Result Directory	Script	Core Function (Line)
1	RQ1a_ground_truth_correctness/ RQ1a_corruption_det_correctness/	run_rq1a_ground_truth_judge.py :179 → run_discovery_experiment	judge_all_edges_serial (modules.py:4421)
2	RQ1a_ground_truth_citation/ RQ1a_corruption_det_citation/	run_rq1a_ground_truth_judge.py :179 → run_discovery_experiment	judge_all_edges_with_citations_serial (modules.py:5688)
3	RQ1a_corruption_detection_*/	run_rq1_phase1_single_corrupt.py:74	_corrupt_relationships (modules.py:1748)
4	RQ1b_correction_*/	run_rq1b_correction_single.py:121	correct_edges_serial (modules.py:7050)
5	(computed during generation)	via run_discovery_experiment	compute_avg_nll (logit_metrics.py:177) compute_min_prob (:251) compute_max_window_entropy (:259) get_embedding_local (:98)
6	RQ2_uq_hallucination_detection/	rq2_ensemble_classifier.py	train_classifiers (:32)
7	(see Algorithm I.9)	modules.py	fetch_citations_with_jina (:265)
8	RQ3_deep_research_validation/	test_deep_research_ground_truth.py	DeepResearchManager (pydantic_ai_deepresearch.py:642)

Internal Dispatch: `run_discovery_experiment()` (modules.py:11264) dispatches to:

- Line 11731: `_corrupt_relationships()` when `corruption_rate > 0`
- Line 11792: `judge_all_edges_with_citations_serial()` when `judge_edges=True`
- Line 11844: `correct_edges_serial()` when `run_correction=True`

Appendix C

Reproducibility

This appendix provides comprehensive information for replicating all experiments, including data availability, code locations, software environment specifications, and exact execution commands.

Reproduction Scope. A public reproduction snapshot is available at https://github.com/NitaiNijholt/cld-hallucination-detection (branch: `thesis-reproduction`). This repository supports **analysis reproduction**—regenerating all thesis figures and tables from the included experimental result datasets using the unified analysis pipelines. **Simulation/data-generation** (i.e., re-running the original experiments to produce new result datasets) requires the private platform repository (`CausalixAI/platform`, branch: `thesis-nitai`), which contains proprietary infrastructure and external API dependencies not included in the public snapshot.
Paths in this appendix reference the public reproduction repository structure unless otherwise noted. Execution scripts for simulation (Section C.4, Tables C.5) are documented for completeness but require access to the private platform.

C.1 Data Availability

C.1.1 Ground Truth CLD Datasets

The three ground truth CLD datasets used in all experiments are available in JSON format within the public reproduction repository:

`final_runs/analysis_lib/data/ground_truth_clds/`

These JSON files contain edge definitions, variable sets, and context data extracted from the original literature-derived CLDs. The analysis scripts use these files to compute ground-truth labels (TP/FP/FN) for edge evaluation.

Table C.1: Ground Truth CLD Dataset Files

Domain	Nodes	Edges	Source	Filename
Depressive Symptoms	14	34	Uleman et al. [114]	`Depressive_symptoms_..._healthy_adults.xlsx`
Social Norms	10	12	Crielaard et al. [115]	`Social_norms_and_obesity_prevalence.xlsx`
Emergency Department	34	66	Smeekes et al. [116]	`older_persons_ED_..._spreadsheet.xlsx`

Each Excel file contains: edge definitions (source/target), relationship polarity, causal motivations, scientific citations, and generation-time UQ metrics.

C.1.2 Validation and Auxiliary Datasets

Table C.2: Validation and Auxiliary Datasets

Dataset	Used In	Size	Location
TruthfulQA [127]	RQ1 Verification	1,634 Q&A pairs	`final_runs/RQ1_verification_Truth_QA/`
Physics CLDs (see below)	RQ1a Validation	31 edges, 4 CLDs	`final_runs/validation_RQ1a_physics_clds/cld_data_files/`
Uleman's Alzheimer CLD	RQ1a Validation	150 edges	`final_runs/validation_RQ1a_ulemans_external/`
Human Validation Set	RQ1a Validation	72 edges	`final_runs/validation_RQ1a_human_agreement/`
RQ3 DR Validation	RQ3 Human Val.	50 edges	`final_runs/RQ3_deep_research_validation/validation/`

C.1.2.1 Physics CLD Details.

Four physics-based CLDs were constructed for sanity-check validation, derived from well-established physical laws:

Table C.3: Physics CLD Validation Dataset

CLD	Nodes	Edges	Physical Laws / Sources
Thermostat Heating	7	8	Newton's Law of Cooling [106]; First Law of Thermodynamics [107]
Predator-Prey	6	8	Lotka-Volterra equations [108, 109]
RC Circuit	7	8	Ohm's Law [110]; Kirchhoff's Laws [111]
Water Tank	6	7	Pascal's Law [112]; Torricelli's Law [113]
Total	26	31	

Construction process. The physics CLDs were constructed as follows:

1. **Causal structure definition:** Variables, directed edges, and polarities were proposed in an interactive session with Claude 4.5 Opus [128] by translating established physical laws into CLD form (Newton's Law of Cooling [106], First Law of Thermodynamics [107], Lotka–Volterra equations [108, 109], Ohm's and Kirchhoff's Laws [110, 111], Pascal's and Torricelli's Laws [112, 113]). The resulting edge directions and polarities (`POSITIVE`/`NEGATIVE`) were then verified against these sources.
2. **Motivation generation:** For each edge, causal explanations ("motivations") were authored in the same interactive Opus session and verified against the same source texts. The prompt pattern was: "*Write a motivation for the edge [Source] → [Target] with polarity [POSITIVE/NEGATIVE] based on [Physical Law]. Include the relevant equation and cite sources.*"
3. **Reference mapping:** Each edge was annotated with bibliographic references to seminal papers and standard textbooks. Post-processing scripts (`add_urls_to_references.py`, `add_supporting_quotes.py`) enriched the references with URLs (where publicly accessible) and supporting quotes extracted from source texts.
4. **Verification:** All motivations and supporting quotes were verified against source texts to ensure scientific accuracy. URLs were browser-verified for accessibility (`verify_and_fix_urls.py`).
5. **Data format:** Final JSON structure: {`source, target, polarity, motivation, references`[{`short_citat` `full_citation, url, supporting_quote`}]}. Converted to Excel format via `create_thermostat_excel.py`.

Reproducibility note. The motivation text was authored interactively (not via automated script) to ensure scientific precision. To reproduce similar motivations, one can prompt any capable LLM with the edge structure and underlying physical law, then verify against authoritative sources. The complete authored motivations, references, and all post-processing scripts are provided for full transparency.

Data locations. JSON files: `final_runs/validation_RQ1a_physics_clds/Data/physics_clds_with_references/*` post-processing scripts: same directory; Excel export: `create_thermostat_excel.py`; pipeline documentation: `PIPELINE_EXPLANATION.md`.

C.1.3 Experiment Result Datasets by Research Question

Table C.4: Experiment Result Datasets by Research Question

RQ	Experiment	Files	Location
RQ1a	GT Lit Correctness	27	`final_runs/RQ1a_gt_lit_correctness/`
	GT Lit Citation	36	`final_runs/RQ1a_gt_lit_citation/`
	GT Synth Correctness	27	`final_runs/RQ1a_gt_synth_correctness/`
	GT Synth Citation	36	`final_runs/RQ1a_gt_synth_citation/`
RQ1b	Correction GT Synth	27	`final_runs/RQ1b_corrector_experiment_corruption_*/`
	Correction GT Lit	27	`final_runs/RQ1b_corrector_experiment_ground_truth_*/`
RQ2	UQ Metrics Analysis	63	`final_runs/RQ2_uq_hallucination_detection/`
RQ3	Deep Research	285 edges	`final_runs/RQ3_deep_research_validation/`
Prelim.	Generator Comparison	9	`final_runs/prelim_generator_model_comparison/`
	Temperature Sensitivity	45	`final_runs/prelim_temperature_sensitivity/`
	Random Baseline	3,000	`final_runs/prelim_random_baseline_generator/`
Sens.	Prompt Sensitivity	4 exp.	`final_runs/supp_prompt_sensitivity/`

C.1.4 Final Experiment Results

All experimental results are stored in the `final_runs/` directory:

```
final_runs/
+-- reproduce_all_thesis_assets.py    [Master reproduction script]
+-- RQ1_unified_analysis.py           [RQ1 analysis pipeline]
+-- RQ2_unified_analysis.py           [RQ2 analysis pipeline]
+-- RQ3_unified_analysis.py           [RQ3 analysis pipeline]
+-- SUPP_unified_analysis.py          [Supplementary analyses]
+-- PRELIM_unified_analysis.py        [Preliminary analyses]
+-- create_thesis_figure_structure.py [Maps outputs to thesis paths]
+-- analysis_lib/                     [Self-contained analysis code]
|   +-- analyze_rq1a_*.py             [RQ1a analysis scripts]
|   +-- logit_metrics.py              [Embedding utilities]
|   +-- data/ground_truth_clds/       [Ground truth CLD JSON files]
+-- RQ1a_gt_lit_correctness/          [GT Lit - Correctness Judge]
|   +-- depressive/run_{1,2,3}/       [Per-CLD result files]
|   +-- emergency_department/run_{1,2,3}/
|   +-- social_norms/run_{1,2,3}/
+-- RQ1a_gt_lit_citation/             [GT Lit - Citation Judge]
```

```
+-- RQ1a_gt_synth_correctness/        [GT Synth - Correctness Judge]
+-- RQ1a_gt_synth_citation/           [GT Synth - Citation Judge]
+-- RQ1b_corrector_experiment_*/      [Corrector experiments (by prompt)]
+-- RQ2_uq_hallucination_detection/   [UQ metrics analysis]
+-- RQ3_deep_research_validation/     [Deep Research validation]
+-- validation_RQ1_truthfulqa_judge/  [TruthfulQA benchmark]
+-- validation_RQ1a_physics_clds/     [Physics CLD validation]
+-- validation_RQ1a_ulemans_external/ [Uleman's CLD validation]
+-- prelim_generator_model_comparison/    [Generator model selection]
+-- prelim_random_baseline_generator/     [Random graph baseline]
+-- prelim_temperature_sensitivity/       [Temperature sensitivity]
+-- supp_parallelization_benchmark/       [Parallelization speedup]
+-- supp_prompt_sensitivity/              [Prompt sensitivity]
+-- supp_time_cost_scaling/               [Time/cost scaling]
```

C.2 Code Availability

C.2.1 Experiment Execution Scripts

Table C.5: Primary Experiment Execution Scripts

Experiment	Script	Dependencies
RQ1a Ground Truth	`run_rq1a_ground_truth_multirun.py`	`run_rq1a_ground_truth_judge.py`
RQ1a Corruption (Corr.)	`run_rq1a_corruption_multirun.py`	`run_corruption_detection_pipeline.sh`
RQ1a Corruption (Cit.)	`run_rq1a_corruption_citation_multirun.py`	`run_rq1_judge_citations_all_3_prompts.py`
RQ1b Corrector	`run_rq1b_correction_multirun.py`	`run_rq1b_correction_single.py`
Temperature Sensitivity	`run_temperature_sensitivity.py`	`modules.py` (run_discovery_experiment)

All execution scripts located in: `data_science/parameter_tuning_experiments/`

See also: Table C.6 (analysis scripts) and Table C.7 (complete script inventory).

C.2.2 Analysis Scripts

Table C.6: Analysis and Evaluation Scripts

Analysis	Script Path
RQ1a: LLM-as-Judge	
RQ1a Aggregate Analysis	`final_runs/analysis_lib/analyze_rq1a_aggregate_enhanced.py`
RQ1a Ground Truth Analysis	`final_runs/analysis_lib/analyze_rq1a_ground_truth_enhanced.py`
Physics CLD Validation	`final_runs/validation_RQ1a_physics_clds/analyse_physics_cld_results.py`
Physics CLD Test Runner	`final_runs/validation_RQ1a_physics_clds/run_physics_cld_tests.py`
Uleman's Provider Comparison	`final_runs/validation_RQ1a_ulemans_external/judge_citation_providers.py`
TruthfulQA Verification	`final_runs/RQ1_verification_Truth_QA/runner/truthqa_judge_baseline_correctness.py`
RQ1b: Corrector	
RQ1b Results Analysis	`final_runs/RQ1b_corrector_ablation/analysis_scripts/analyze_corrector_ablation.py`
RQ2: UQ Metrics	
RQ2 Master Report	`final_runs/RQ2_uq_hallucination_detection/rq2_master_report_v2_enhanced.py`
RQ2 Batch Analyser	`final_runs/RQ2_uq_hallucination_detection/rq2_simple_batch_analyser.py`
RQ3: Deep Research	
RQ3 Unified Analysis	`final_runs/RQ3_deep_research_validation/rq3_unified_analysis.py`
RQ3 Statistical Tests	`final_runs/RQ3_deep_research_validation/rq3_statistical_test.py`
RQ3 Per-CLD Detection Rates	`final_runs/RQ3_deep_research_validation/rq3_per_cld_detection_rates.py`
RQ3 Enrichment Metrics	`final_runs/RQ3_deep_research_validation/compute_enriched_gt_metrics.py`
RQ3 Human Validation	`final_runs/RQ3_deep_research_validation/scripts/enrichment_extrapolation.py`
RQ3 Smart Trigger Analysis	`final_runs/RQ3_deep_research_validation/scripts/estimate_smart_trigger_dr_enrichment.py`
RQ3 Cost Efficiency Plot	`final_runs/RQ3_deep_research_validation/scripts/plot_dr_cost_efficiency.py`
RQ3 Figure Generation	`final_runs/RQ3_deep_research_validation/generate_figures.py`
RQ3 Edge Sampling	`final_runs/RQ3_deep_research_validation/scripts/sample_edges_for_validation.py`
RQ3 Validation Tables	`final_runs/RQ3_deep_research_validation/scripts/generate_validation_distribution_tables.py`
Preliminary & Validation	
Generator Model Comparison	`final_runs/prelim_generator_model_comparison/analysis_scripts/analyze_generator_models.py`
Random Baseline Analysis	`final_runs/random_baseline_generator/compute_random_baseline.py`
Embedding Model Benchmark	`final_runs/supp_embedding_benchmark/embedding_benchmark_real_cld.py`
Cost & Scaling Analysis	
Complete Cost Analysis	`final_runs/supp_cost_analysis/calculate_all_costs.py`
Per-Experiment Costs	`final_runs/supp_cost_analysis/calculate_costs_by_experiment.py`
Deep Research Costs	`final_runs/supp_cost_analysis/compute_dr_costs.py`
Token Analysis	`final_runs/supp_token_analysis_rq1a/analyse_output_tokens_tiktoken.py`
Time/Cost Scaling	`final_runs/supp_time_cost_scaling/aggregate_time_cost_scaling.py`
Parallelization Benchmark	`final_runs/supp_parallelization_benchmark/analysis_scripts/analyze_parallelization_results.`
Sensitivity Analysis	
Judge Prompt Sensitivity	`final_runs/supp_prompt_sensitivity/analysis_scripts/sensitivity_analysis_simple.py`
Corrector Prompt Sensitivity	`final_runs/supp_prompt_sensitivity/analysis_scripts/sensitivity_analysis_corrector.py`
Temperature Assumption Tests	`final_runs/temperature_sensitivity_assumption_tests.py`

See also: Table C.5 (execution scripts) and Table C.7 (complete script inventory).

C.2.3 Complete `final_runs/` Script and Command Inventory

To make the artefact set auditable end-to-end, Table C.7 enumerates the key runnable Python scripts present under `final_runs/` in the public reproduction repository, together with canonical invocations from the repository root.

Note: Some scripts listed (particularly those requiring external API calls or database connections) may require additional configuration not included in the public snapshot. The unified analysis entry points are the recommended entry points for reproduction.

Table C.7: Complete `final_runs/` Script and Command Inventory

Script	Canonical command (from repo root)
Unified analysis entry points	
final_runs/reproduce_all_thesis_assets.py	uv run python final_runs/reproduce_all_thesis_assets.py
final_runs/RQ1_unified_analysis.py	uv run python final_runs/RQ1_unified_analysis.py
final_runs/RQ2_unified_analysis.py	uv run python final_runs/RQ2_unified_analysis.py
final_runs/RQ3_unified_analysis.py	uv run python final_runs/RQ3_unified_analysis.py
final_runs/SUPP_unified_analysis.py	uv run python final_runs/SUPP_unified_analysis.py
final_runs/PRELIM_unified_analysis.py	uv run python final_runs/PRELIM_unified_analysis.py
final_runs/VALIDATION_unified_analysis.py	uv run python final_runs/VALIDATION_unified_analysis.py
Top-level utilities	
final_runs/aggregate_time_cost_scaling.py	python final_runs/aggregate_time_cost_scaling.py
final_runs/retrieve_effect_sizes.py	python final_runs/retrieve_effect_sizes.py
final_runs/temperature_sensitivity_assumption_tests.py	python final_runs/temperature_sensitivity_assumption_tests.py
RQ1: TruthfulQA verification	
final_runs/RQ1_verification_Truth_QA/runner/truthqa_judge_baseline_correctness.py	python final_runs/RQ1_verification_Truth_QA/runner/truthqa_judge_baseline_correctness.py
final_runs/RQ1_verification_Truth_QA/analysis_script/recalculate_metrics.py	python final_runs/RQ1_verification_Truth_QA/analysis_script/recalculate_metrics.py
RQ1a validation: physics CLDs	
final_runs/validation_RQ1a_physics_clds/run_physics_cld_tests.py	python final_runs/validation_RQ1a_physics_clds/run_physics_cld_tests.py
final_runs/validation_RQ1a_physics_clds/analyse_physics_cld_results.py	python final_runs/validation_RQ1a_physics_clds/analyse_physics_cld_results.py
final_runs/validation_RQ1a_physics_clds/generate_comparison_table.py	python final_runs/validation_RQ1a_physics_clds/generate_comparison_table.py
final_runs/validation_RQ1a_physics_clds/generate_retrieval_comparison_table.py	python final_runs/validation_RQ1a_physics_clds/generate_retrieval_comparison_table.py
final_runs/validation_RQ1a_physics_clds/get_avg_scores.py	python final_runs/validation_RQ1a_physics_clds/get_avg_scores.py
final_runs/validation_RQ1a_physics_clds/cld_data_files/create_thermostat_excel.py	python final_runs/validation_RQ1a_physics_clds/cld_data_files/create_thermostat_excel.py
final_runs/validation_RQ1a_physics_clds/physics_clds_with_references/add_supporting_quotes.py	python final_runs/validation_RQ1a_physics_clds/physics_clds_with_references/add_supporting_quotes.py
final_runs/validation_RQ1a_physics_clds/physics_clds_with_references/add_urls_to_references.py	python final_runs/validation_RQ1a_physics_clds/physics_clds_with_references/add_urls_to_references.py
final_runs/validation_RQ1a_physics_clds/physics_clds_with_references/show_example.py	python final_runs/validation_RQ1a_physics_clds/physics_clds_with_references/show_example.py
final_runs/validation_RQ1a_physics_clds/physics_clds_with_references/verify_and_fix_urls.py	python final_runs/validation_RQ1a_physics_clds/physics_clds_with_references/verify_and_fix_urls.py
final_runs/validation_RQ1a_physics_clds/physics_clds_with_references/verify_urls.py	python final_runs/validation_RQ1a_physics_clds/physics_clds_with_references/verify_urls.py
RQ1a validation: Uleman's citation provider comparison	
final_runs/validation_RQ1a_ulemans_external/judge_citation_providers.py	python final_runs/validation_RQ1a_ulemans_external/judge_citation_providers.py
final_runs/validation_RQ1a_ulemans_external/compare_jina_vs_scraper.py	python final_runs/validation_RQ1a_ulemans_external/compare_jina_vs_scraper.py
RQ2: hallucination detection (UQ metrics)	
final_runs/RQ2_uq_hallucination_detection/rq2_simple_batch_analyser.py	python final_runs/RQ2_uq_hallucination_detection/rq2_simple_batch_analyser.py
final_runs/RQ2_uq_hallucination_detection/rq2_master_report_v2_enhanced.py	python final_runs/RQ2_uq_hallucination_detection/rq2_master_report_v2_enhanced.py
RQ3: Deep Research analysis	
final_runs/RQ3_deep_research_validation/analysis_scripts/rq3_unified_analysis.py	python ...rq3_unified_analysis.py
final_runs/RQ3_deep_research_validation/analysis_scripts/analyze_deep_research_results.py	python ...analyze_deep_research_results.py
final_runs/RQ3_deep_research_validation/analysis_scripts/rq3_statistical_test.py	python ...rq3_statistical_test.py
final_runs/RQ3_deep_research_validation/analysis_scripts/generate_figures.py	python ...generate_figures.py
final_runs/RQ3_deep_research_validation/analysis_scripts/enrichment_extrapolation.py	python ...enrichment_extrapolation.py
final_runs/RQ3_deep_research_validation/analysis_scripts/plot_dr_cost_efficiency.py	python ...plot_dr_cost_efficiency.py
Cost, scaling, and diagnostics	
final_runs/supp_cost_analysis/calculate_all_costs.py	python final_runs/supp_cost_analysis/calculate_all_costs.py
final_runs/supp_cost_analysis/calculate_costs_by_experiment.py	python final_runs/supp_cost_analysis/calculate_costs_by_experiment.py
final_runs/supp_cost_analysis/calculate_experiment_costs.py	python final_runs/supp_cost_analysis/calculate_experiment_costs.py
final_runs/supp_cost_analysis/calculate_costs_max_edges.py	python final_runs/supp_cost_analysis/calculate_costs_max_edges.py
final_runs/supp_cost_analysis/analyse_token_and_call_stats.py	python final_runs/supp_cost_analysis/analyse_token_and_call_stats.py
final_runs/supp_cost_analysis/analyse_output_tokens_tiktoken.py	python final_runs/supp_cost_analysis/analyse_output_tokens_tiktoken.py
final_runs/supp_cost_analysis/compute_dr_costs.py	python final_runs/supp_cost_analysis/compute_dr_costs.py
final_runs/supp_cost_analysis/generate_table21_scaling_summary.py	python final_runs/supp_cost_analysis/generate_table21_scaling_summary.py
final_runs/supp_token_analysis_rq1a/analyse_output_tokens_tiktoken.py	python final_runs/supp_token_analysis_rq1a/analyse_output_tokens_tiktoken.py
final_runs/supp_time_cost_scaling/aggregate_time_cost_scaling.py	python final_runs/supp_time_cost_scaling/aggregate_time_cost_scaling.py
final_runs/supp_time_cost_scaling/collect_corrector_timing.py	python final_runs/supp_time_cost_scaling/collect_corrector_timing.py
Sensitivity, benchmarks, and baselines	
final_runs/supp_prompt_sensitivity/analysis_scripts/sensitivity_analysis_simple.py	python final_runs/supp_prompt_sensitivity/analysis_scripts/sensitivity_analysis_simple.py
final_runs/supp_prompt_sensitivity/analysis_scripts/sensitivity_analysis_corrector.py	python final_runs/supp_prompt_sensitivity/analysis_scripts/sensitivity_analysis_corrector.py
final_runs/supp_prompt_sensitivity/analysis_scripts/sensitivity_analysis_simple_rm_anova.py	python final_runs/supp_prompt_sensitivity/analysis_scripts/sensitivity_analysis_simple_rm_anova.py
final_runs/supp_prompt_sensitivity/analysis_scripts/sensitivity_analysis_corrector_rm_anova.py	python final_runs/supp_prompt_sensitivity/analysis_scripts/sensitivity_analysis_corrector_rm_anova.py
final_runs/supp_parallelization_benchmark/analysis_scripts/analyze_parallelization_results.py	python ...analyze_parallelization_results.py
final_runs/power_analysis/compute_power_analysis.py	python final_runs/power_analysis/compute_power_analysis.py
final_runs/prelim_random_baseline_generator/analysis_scripts/compute_random_baseline.py	python ...compute_random_baseline.py
final_runs/supp_embedding_benchmark/embedding_benchmark_real_cld.py	python ...embedding_benchmark_real_cld.py
final_runs/prelim_generator_model_comparison/analysis_scripts/analyze_generator_models.py	python ...analyze_generator_models.py

See also: Table C.5 (execution scripts) and Table C.6 (analysis scripts) for organised by-experiment views.

C.2.4 Prompt Configuration Files

All prompts use three core variants (display names): *Baseline* (direct instruction), *Mechanistic* (Bradford Hill criteria [68]), and *CoT* (chain-of-thought reasoning [60]). In prompt configuration files, these correspond to the variant IDs `baseline`, `mechanistic`, and `cot`. Citation judges additionally include a fourth variant *Mechanistic-Lit* (variant ID `mech-lit`), which extends Mechanistic with explicit literature grounding requirements.

Table C.8: Prompt Configuration Files (13 prompts: 2 judge tasks × 3 variants + 1 mech-lit + 1 corrector × 3 variants + 3 corruption)

Task	Variants	File(s)
Judge-Correctness	baseline, mech, cot	`alternative_prompts/prompts_correctness_{baseline,mechanistic,cot}.yaml`
Judge-Citation	baseline, mech, cot, mech-lit	`alternative_prompts/prompts_citation_{baseline,mechanistic,cot,mech_lit}.ya`
Corrector	baseline, mech, cot	`corrector_prompts.yaml` (variants within single file)
Corruption	`prompts_Nitai_C.yaml` (prompt IDs below):	
– motivation	(single)	`corruptor`: generates spurious but plausible motivation
– spurious edge	(single)	`generateSpuriousEdge`: NONE → false positive causal edge
– polarity flip	(single)	`generateFlippedPolarity`: POS ↔ NEG with new motivation

C.3 Software Environment

Table C.9: System and Runtime Environment

Component	Details
Hardware	ASUSTeK ROG Zephyrus G16 GU605CX_GU605CX
CPU	Intel(R) Core(TM) Ultra 9 285H (16 cores, 16 logical processors)
GPU	NVIDIA GeForce RTX 5090 Laptop GPU (24463 MiB VRAM; driver 576.88)
System RAM	64 GB DDR5 (8×8 GB; 7467 MT/s configured)
Operating System	Ubuntu 24.04.2 LTS (WSL2; kernel 6.6.87.2-microsoft-standard-WSL2; Windows build 26100.7462)
Python Version	3.10.18 (`.python-version`=3.10)
Package Manager	uv 0.8.3
CUDA Version	12.9

C.3.1 Environment Installation

All dependencies are managed using `uv`, a fast Python package manager with lockfile support for reproducible environments. To replicate the computational environment:

```
# Install uv (if not already installed)
curl -LsSf https://astral.sh/uv/install.sh | sh

# Clone public reproduction repository
git clone https://github.com/NitaiNijholt/cld-hallucination-detection.git
cd cld-hallucination-detection

# Sync exact versions from lockfile
uv sync

# Verify installation
uv run python -c "import pandas; import sklearn; print('OK')"
```

The `uv.lock` file ensures deterministic dependency resolution, pinning all 104 packages to exact versions used during experiments. This dependency management approach is informed by principles from the Workflow for Open Reproducible Code in Science [129].

C.3.2 Key Software Dependencies

Full package dependencies with pinned versions are specified in `pyproject.toml`. Table C.10 lists key software components with citations.

Table C.10: Key Software Dependencies and Citations

Package	Version	Citation
LLM APIs		
OpenAI API	1.107.3	OpenAI (2023). GPT-4 Technical Report. arXiv:2303.08774
Anthropic API	0.61.0	Anthropic (2024). Claude 3 Model Card
Data Analysis		
NumPy	2.2.3	[130]
pandas	2.2.3	[131]
SciPy	1.15.3	[132]
Machine Learning		
scikit-learn	1.7.2	[133]
imbalanced-learn	0.14.0	[134]
Visualisation		
Matplotlib	3.10.1	[135]
seaborn	0.13.2	[136]
Graph & Database		
NetworkX	3.4.2	[137]
Neo4j	5.27.0	[138]
LLM Integration		
Pydantic	2.11.7	[139]
Interactive Computing		
Jupyter	1.1.1	[140]

Complete BibTeX citations for all software are available in the repository at `thesis/final_thesis/software.bib`.

C.4 Execution Commands

C.4.1 Unified Analysis Entry Points (Single-command pipelines)

The directory `final_runs/` contains three unified analysis entry points intended to reproduce the thesis results tables/figures with minimal manual orchestration:

```
cd <repo_root>

# RQ1 (TruthfulQA verification) + RQ1a (Judge) + RQ1b (Corrector)
uv run python final_runs/RQ1_unified_analysis.py \
  --run-dir final_runs/_runs/RQ1_<timestamp>

# RQ2 (UQ metrics + classifier pipeline + reporting)
uv run python final_runs/RQ2_unified_analysis.py \
  --run-dir final_runs/_runs/RQ2_<timestamp>
```

```
# RQ3 (Deep Research analysis + figures + validation artefacts)
uv run python final_runs/RQ3_unified_analysis.py \
  --run-dir final_runs/_runs/RQ3_<timestamp>
```

For a complete directory-level overview, see `final_runs/README_DIRECTORY_MAP.md`. The master reproduction script `reproduce_all_thesis_assets.py` orchestrates all unified analyses and maps outputs to thesis figure paths.

Note on Execution Scripts. The subsections below (RQ1a–RQ3 execution commands) document the original simulation scripts used to generate experimental data. These scripts reside in the private platform repository (`data_science/parameter_tuning_experiments/`) and are **not included** in the public reproduction snapshot. They are documented here for completeness and to enable full replication by examiners with platform access. For analysis reproduction using pre-computed result datasets, use the unified entry points above.

C.4.2 RQ1 Judge Verification (TruthfulQA)

Before evaluating judges on CLD-specific tasks, we verify baseline hallucination detection capability using the TruthfulQA benchmark. This sanity check helps ensure the evaluation pipeline behaves as expected before moving to the more domain-specific CLD setting; it does not, by itself, imply that any CLD-specific failures are purely domain-driven.

```
cd <repo_root>/final_runs/validation_RQ1_truthfulqa_judge/

# Run the TruthfulQA judge verification (baseline correctness prompt structure)
uv run python runner/truthqa_judge_baseline_correctness.py

# Optional: recompute summary metrics from a saved run JSON
# uv run python analysis_script/recalculate_metrics.py --input <path_to_results_json>
```

Script Location: `final_runs/RQ1_verification_Truth_QA/runner/ truthqa_judge_baseline_correctness.py`

Results Location: `final_runs/RQ1_verification_Truth_QA/` (including `Data_runs/`)

Table C.11: TruthfulQA Verification Configuration

Parameter	Value
Dataset	TruthfulQA (817 questions)
Samples	1,634 (817 truthful + 817 hallucinated answers)
Judge Model	GPT-4.1
Temperature	0.0
Seed	42
Prompt	Baseline correctness
Max Tokens	250

Output Files:

- `truthfulqa_results.json`: Per-sample verdicts
- `truthfulqa_metrics.json`: Accuracy, precision, recall, F1
- `confusion_matrix.png`: Classification confusion matrix

C.4.3 RQ1a Ground Truth Experiments

[Requires private platform repository]

```
# Private platform: data_science/parameter_tuning_experiments/
cd data_science/parameter_tuning_experiments/

# Correctness-based ground truth
python run_rq1a_ground_truth_multirun.py \
    --approach correctness \
    --output-dir ../../final_runs/RQ1a_gt_lit_correctness

# Citation-based ground truth
python run_rq1a_ground_truth_multirun.py \
    --approach citation \
    --output-dir ../../final_runs/RQ1a_gt_lit_citation
```

C.4.4 RQ1a Corruption Detection Experiments

```
# Correctness-based corruption detection
python run_rq1a_corruption_multirun.py \
    --output-dir ../../final_runs/RQ1a_corrupted_correctness

# Citation-based corruption detection
python run_rq1a_corruption_citation_multirun.py \
    --output-dir ../../final_runs/RQ1a_corrupted_citation
```

Table C.12: GT Synth Citation Judge Configuration (GPT-5-mini)

Parameter	**Value**
Judge Model	GPT-5-mini
Temperature	0.0
Top-p	1.0
Approach	Citation-based judging
Parallelization	10 workers
Corruption	
Corruption Rate	30%
Corruption Types	Motivation (spurious explanations), Structural (new edges, polarity flips)
Dataset	
CLDs	Depressive Symptoms, Social Norms, Emergency Department
Runs per CLD	3 (seeds 1, 2, 3)
Prompt Variants	4 (Baseline, CoT, Mechanistic, Mechanistic-Lit)
Total Experiments	36 (3 CLDs $\times$ 3 runs $\times$ 4 prompts)

Table C.13: RQ1a GT Lit Correctness Judge Configuration

Parameter	Value
Judge Model	GPT-4.1
Temperature	0.0
Seed	42
Approach	Correctness-based judging
Parallelization	10 workers
Web Search	Disabled
Dataset	
CLDs	Depressive Symptoms, Social Norms, Emergency Department
Runs per CLD	3 (seeds 10, 20, 30)
Prompt Variants	3 (Baseline, CoT, Mechanistic)
Total Experiments	27 (3 CLDs × 3 runs × 3 prompts)

Table C.14: RQ1a GT Synth Correctness Judge Configuration

Parameter	Value
Judge Model	GPT-4.1
Temperature	0.0
Seed	42
Approach	Correctness-based judging
Parallelization	10 workers
Web Search	Disabled
Corruption	
Corruption Rate	30%
Corruption Types	Motivation (spurious), Structural (new edges, polarity flips)
Dataset	
CLDs	Depressive Symptoms, Social Norms, Emergency Department
Runs per CLD	3 (seeds 1, 2, 3)
Prompt Variants	3 (Baseline, CoT, Mechanistic)
Total Experiments	27 (3 CLDs × 3 runs × 3 prompts)

Table C.15: RQ1a GT Lit Citation Judge Configuration

Parameter	Value
Judge Model	GPT-4.1
Temperature	0.0
Seed	42
Approach	Citation-based judging
Parallelization	10 workers
Web Search	Enabled
Citation Fetcher	ContentScraper (Jina AI Reader disabled)
Dataset	
CLDs	Depressive Symptoms, Social Norms, Emergency Department
Runs per CLD	3 (seeds 10, 20, 30)
Prompt Variants	4 (Baseline, CoT, Mechanistic, Mechanistic-Lit)
Total Experiments	36 (3 CLDs × 3 runs × 4 prompts)

C.4.5 RQ1b Corrector Experiments

```
python run_rq1b_correction_multirun.py \
```

```
--input-dir ../../final_runs/RQ1a_gt_lit_correctness \
--output-dir ../../final_runs/RQ1b_corrector/ground_truth
```

Table C.16: RQ1b Corrector (GT Synth) Configuration

Parameter	Value
Corrector	
Corrector Model	GPT-4.1
Temperature	0.0
Corrector Prompts	3 (Baseline, CoT, Mechanistic)
Evaluation Judge	
Judge Model	GPT-4.1
Judge Prompt	Baseline Correctness
Temperature	0.0
Dataset	
Input	RQ1a GT Synth flagged edges
CLDs	Depressive Symptoms, Social Norms, Emergency Department
Runs per CLD	3 (seeds 1, 2, 3)
Total Experiments	27 (3 CLDs × 3 runs × 3 prompts)

Table C.17: RQ1b Corrector (GT Lit) Configuration

Parameter	Value
Corrector	
Corrector Model	GPT-4.1
Temperature	0.0
Corrector Prompts	3 (Baseline, CoT, Mechanistic)
Evaluation Judge	
Judge Model	GPT-4.1
Judge Prompt	Baseline Correctness
Temperature	0.0
Dataset	
Input	RQ1a GT Lit flagged edges
CLDs	Depressive Symptoms, Social Norms, Emergency Department
Runs per CLD	3 (seeds 10, 20, 30)
Total Experiments	27 (3 CLDs × 3 runs × 3 prompts)

C.4.6 RQ1a Validation Studies

Two validation studies assess judge reliability: external validation on an independently expert-constructed CLD, and human-LLM agreement analysis.

External Validation: Uleman's CLD Citation Provider Comparison

This study validates the citation-based judge on the Uleman's Alzheimer's disease CLD (150 edges) constructed through Group Model Building with 17 domain experts.

```
cd final_runs/validation_RQ1a_ulemans_external/

# Run citation-based judging across provider sessions and write JSON/XLSX outputs
```

```
python judge_citation_providers.py

# Optional: compare two saved runs (e.g., Jina vs ContentScraper) and export XLSX
# python compare_jina_vs_scraper.py
```

Script Location: `final_runs/validation_RQ1a_ulemans_external/ judge_citation_providers.py`

Table C.18: Uleman's Citation Provider Comparison Configuration

Parameter	Value
Dataset	Uleman's Alzheimer's CLD (150 edges)
Search Providers	OpenAlex, Brave Search (with/without whitelist), Perplexity, PubMed
Citation Fetchers	Jina AI Reader, ContentScraper (BeautifulSoup-based, PDF-capable)
Configurations	4 providers $\times$ 2 fetchers = 8
Total Judgements	1,200 (8 configs $\times$ 150 edges)
Judge Settings	
Judge Model	GPT-4.1
Temperature	0.0
Top-p	1.0
Prompt	Baseline citation prompt

Human-LLM Agreement Validation

This study measures inter-rater reliability between LLM citation judge outputs and human expert annotations.

Inputs Location: `final_runs/validation_RQ1a_human_agreement/` (validation spreadsheets).

Analysis Script (provenance): The one-off analysis used to compute agreement metrics is preserved in the repository snapshot:

```
python final_runs/validation_RQ1a_human_agreement/combined_human_validation_analysis.py
```

Table C.19: Human-LLM Agreement Validation Configuration

Parameter	Value
Sample Size	72 edges across 3 validation files
CLDs	Depressive Symptoms, Social Norms
Generators	GPT-4.1, Claude 4 Sonnet
Human Validator	Domain expert (categorical verdicts)
LLM Judge	Citation-based (GPT-4.1, T=0.0, baseline)
Metrics	
Agreement	Cohen's κ, Agreement Rate, Pearson r
Interpretation	Landis & Koch (1977) guidelines

Table C.20: Human validation rubric for evaluating causal evidence quality in Citation Judge validation (RQ1). Original rubric in Dutch with English translation.

Stappen Causal Narrative Check / *Steps for Causal Narrative Check*

1. Worden de variabelen of synoniemen genoemd en in de zelfde richting als in motivatie (A→B en niet B→A)?
Are the variables or synonyms named and in the same direction as in motivation (A→B and not B→A)?

2. Is er een relatie statistisch significante ($p < 0{,}05$) relatie?
Is there a statistically significant ($p < 0.05$) relationship?

3. Is de relatie causaal (A veroorzaakt B)?
Is the relationship causal (A causes B)?

4. Als A→C veroorzaakt via mediator B is het indirect, benoem in dat geval de mediator in annotatie. Wij zijn opzoek naar directe causale relaties.
If A→C is caused via mediator B it is indirect; in that case name the mediator in annotation. We are looking for direct causal relationships.

5. Kijken of de populatie uit de bron overeenkomt met de populatie van de stelling. Als dit anders of beperkt is moet dat benoemd worden in een aparte kolom.
Check if the population from the source matches the population of the statement. If different or limited, this must be noted in a separate column.

6. Algemene check van kwaliteit bewijs: is het een longitudinale studie, wetenschappelijke studie of blogpost?
General quality check of evidence: is it a longitudinal study, scientific study, or blog post?

Classificatie / *Classification*

Kleur/Colour	**Score**	**Notitie / Note**
Dark Green	1.0	Hoogste kwaliteit bewijs, causaal + wetenschappelijk, longitudinaal, RCT. Wanneer het duidelijk causaal bewijs hoeft de rest van de bronnen niet bekeken te worden. *Highest quality evidence, causal + scientific, longitudinal, RCT. When clear causal evidence exists, the rest of the sources need not be reviewed.*
Light Green	0.8	Wel causaal verband maar geen wetenschappelijke studie, bijv. blog medische professionals. *Causal relationship but no scientific study, e.g., blog by medical professionals.*
Yellow	0.5	Relatie maar geen causaal verband, cross-sectioneel onderzoek. *Relationship but no causal link, cross-sectional study.*
Red	0.0	Geen toegang tot link, onjuiste variabelen, geen verband. *No access to link, incorrect variables, no relationship.*
Gray	0.0	Als de bron echt niet duidelijk is of er iets vreemds aan de hand is. Goed beschrijven wat er niet duidelijk is. *If the source is really unclear or something strange is going on. Describe well what is not clear.*

Output Files:

- `provider_comparison_results.json`: Per-provider metrics
- `human_llm_agreement.json`: Cohen's κ, agreement rates
- `confusion_matrices/`: Per-file agreement visualisations

C.4.7 RQ2 Hallucination Detection Analysis

[Requires private platform repository]

```
# Private platform: data_science/
cd data_science/

python rq2_master_report.py \
    --input-dir ../final_runs \
    --output-dir ../final_runs/RQ2_uq_hallucination_detection

python rq2_ensemble_classifier.py \
    --input-dir ../final_runs/RQ2_uq_hallucination_detection
```

Table C.21: RQ2 UQ Logprobs Configuration

Parameter	Value
UQ Metrics	
Perplexity (PPL)	Capped NLL averaging (token cap 20.0, avg cap 10.0)
$P_{\min}$	Minimum token probability
H_{win}	Max sliding-window entropy (window 5, top-k=5)
Dataset	
Source	RQ1a GT Lit experiments (27 correctness files)
Total Edges	31,704 (from 63 deduplicated files)
Hallucination Rate	15.2%
Analysis	
Block Definition	CLD $\times$ run ($N = 9$ blocks)
Statistical Test	One-sample t-test vs. 0.5

Table C.22: RQ2 Cosine Similarity (Embeddings) Configuration

Parameter	Value
Embedding Model	
Model	all-mpnet-base-v2
Source	sentence-transformers (Hugging Face)
Similarity	Max narrative–citation-chunk cosine similarity
Dataset	
Source	RQ1a GT Lit Citation experiments only (36 files)
Ensemble Subset	16,507 edges (35 files with all 4 metrics)
Hallucination Rate	15.2%
Analysis	
Block Definition	CLD $\times$ run ($N = 9$ blocks)
Statistical Test	One-sample t-test vs. 0.5

Table C.23: RQ2 Shared Experimental Configuration

Parameter	Value
CI Metrics	Gen Perplexity, Gen Min Prob, Gen Max Window Entropy, Gen Cosine Similarity
Block Definition	CLD × run ($N = 9$: 3 CLDs × 3 runs)
Hallucination Label	FP or FN w.r.t. GT Lit
Scaler	StandardScaler
Random State	42

Table C.24: RQ2 Phases 1–3 (Single-Metric Analysis) Configuration

Parameter	Value
Data Loading	
Input Files	63 deduplicated experiment files (36 citation, 27 correctness)
Filter	Exclude corruption experiments (`exp_type` ∈ {`citation, correctness`})
Hallucination Labeling	
Label	`is_hallucination` = True if Classification ∈ {FP, FN}
Ground Truth	GT Lit (literature-based correctness)
Data Cleaning	
Missing Values	Remove rows with NaN in CI metric columns
Infinite Values	Replace $\pm\infty$ with NaN, then drop
Min Edges/File	$\geq$10 edges required
Min Class Size	$\geq$2 per class (hallucination / correct) required
Block Aggregation	
Block Key	(CLD, run) $\Rightarrow N = 9$ blocks
Aggregation	Mean AUC per block
Statistical Testing	
Test	One-sample Wilcoxon signed-rank on (AUC − 0.5)
Alternative	Two-sided
Bonferroni Correction	$m = 4$ tests, $\alpha_{\text{adj}} = 0.0125$

Table C.25: RQ2 Analysis Pipeline Configuration

Phase	Parameter	Value
Phase 4 (RFE)	Classifiers	LogReg, RandomForest, GradientBoosting
	Cross-validation	GroupKFold ($n\_splits = 3$)
	Feature range	$k \in \{1, 2, 3, 4\}$
Phase 5 (Ensemble)	Classifiers	LogReg, RF, GB, MLP
	Train/Test Split	67%/33% (GroupShuffleSplit)
	CV on training	3-fold GroupKFold
	Class weighting	balanced (LogReg, RF)
Phase 6 (LOCO)	Protocol	Leave-one-CLD-out
	Scaling	StandardScaler (fit on training CLDs)
Classifier Hyperparameters		
Logistic Regression		max_iter=1000, class_weight=balanced
Random Forest		n_estimators=100, class_weight=balanced
Gradient Boosting		n_estimators=100
Neural Network (MLP)		hidden_layers=(32,16), max_iter=2000

C.4.8 RQ3 Deep Research Experiments

The RQ3 Deep Research pipeline validates causal claims through automated literature search using a multi-agent system. Execution proceeds in three stages: (1) running Deep Research on edges, (2) running the unified analysis pipeline, and (3) human validation with extrapolation.

Stage 1: Deep Research Execution

```
cd final_runs/RQ3_deep_research_validation/

# Run Deep Research on all TP/FP/FN edges from GT Lit experiments
python test_deep_research_ground_truth.py \
    --input-dir ../RQ1a_gt_lit_correctness \
    --output-dir data/
```

Stage 2: Unified Analysis Pipeline

The unified analysis script orchestrates all RQ3 analysis scripts in sequence:

```
cd <repo_root>

# Run complete analysis pipeline (calls all scripts below) and write outputs
# under a single run folder:
uv run python final_runs/RQ3_unified_analysis.py \
  --run-dir final_runs/_runs/RQ3_<timestamp>

# Individual scripts (called by unified analysis):
# (When invoked through the unified runner, these scripts write to RQ3_OUTPUT_DIR.)
```

Stage 3: Human Validation and Extrapolation

```
cd final_runs/RQ3_deep_research_validation/

# Sample edges for human validation (stratified by CLD x classification)
uv run python scripts/sample_edges_for_validation.py

# After manual annotation: run extrapolation analysis
uv run python scripts/enrichment_extrapolation.py
```

Table C.26: RQ3 Deep Research Configuration

Parameter	Value
Heavy Model	GPT-5 (hypothesis evaluation, judgement, verdict)
Light Model	GPT-5-mini (iterative search, evidence scoring)
Temperature	1.0 (both models)
Top-p	1.0 (both models)
Search API	Brave Search with academic domain filtering
Max Iterations	1
Max Sources per Query	5
Max Searches per Edge	5
Time Limit	900 seconds (15 minutes) per edge
Batch Size	5 edges processed concurrently
Dataset	
Total Edges	285 (44 TP, 178 FP, 63 FN)
CLDs	Depressive Symptoms (84), Social Norms (17), Emergency Department (184)
Human Validation	
Sample Size	50 edges (29.6% of edges with direct_causal_found=True)
Sampling	Stratified by CLD × classification (seed=42)
Thresholds Tested	{0.5, 0.6, 0.7, 0.8, 0.9}

Table C.27: GT Enrichment Experiment Configuration

Parameter	Value
Deep Research Settings	
Heavy Model	GPT-5 (T=1.0, top_p=1.0)
Light Model	GPT-5-mini (T=1.0, top_p=1.0)
Max Iterations	1
Max Sources per Query	5
Max Searches per Edge	5
Enrichment Scenarios	
Scenario 1 (Enriched GT)	Promote DR-validated FPs to TP in ground truth
Scenario 2 (LLM+DR)	Scenario 1 + add DR-validated FNs to system output
Dataset	
Source Data	RQ3 Deep Research outputs (285 edges)
Classifications	TP (44), FP (178), FN (63)
Eligible for Promotion	Edges with direct_causal_found=True
Human Calibration (for extrapolation)	
Applicability Threshold	0.7 (selected via elbow/gradient rule)
Sample Size	50 edges (stratified by CLD × classification)

Table C.28: RQ3 Human Validation Configuration

Parameter	Value
Sample Size	50 edges (29.6% of 169 eligible)
Random Seed	42
Sampling Method	Proportional stratified (CLD × classification)
Filter	direct_causal_found = True
Verdict Categories	causal, different_vars, invalid
Applicability Score	Continuous $\in [0, 1]$

Table C.29: RQ3 Threshold Selection Configuration

Parameter	**Value**
Thresholds Tested	$t \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$
CI Method	Wilson score interval ($\alpha = 0.05$)
Elbow Rule	score $= \Delta x / \lvert\Delta y\rvert$
Selected Threshold	$t^* = 0.7$

Table C.30: RQ3 Smart Trigger Configuration

Parameter	**Value**
Trigger Threshold	mean(Aggregate Score) ≤ 0.5
Judge Used	Mechanistic Correctness (GT Lit)
Applicability Threshold	0.7
Aggregation	Mean score across 9 runs

Table C.31: Human validation rubric for evaluating Deep Research evidence applicability (RQ3). This rubric guides human validators in assessing whether retrieved evidence supports the claimed causal relationship.

Validation Process

For each edge (row), the human validator performs the following steps:

Step 1: Check the link in the `strongest_causal_evidence_url` column.

Step 2: Assess whether the source contains evidence for (or against) the causal claim: "*source* causes *target*".

Step 3: Assign a categorical verdict in the `human_verdict` column (see below).

Step 4: Assign an applicability score (0.0–1.0) in the `Fully_applicable` column (see scoring rubric below).

Step 5: Add clarifying notes in the `note` column if needed.

Categorical Verdicts (`human_verdict`)

Verdict	Description
`causal`	Evidence directly supports a causal relationship between the exact source and target variables.
`different_vars, causal`	Evidence supports a causal relationship, but for related but not identical variables (e.g., similar constructs, different operationalisations).
`non-causal`	Evidence shows only correlation, association, or co-occurrence—no causal mechanism or temporal precedence established.
`not relevant`	Evidence does not address the relationship between source and target variables, or the link is inaccessible/broken.

Applicability Score Rubric (`Fully_applicable`: 0.0–1.0, in 0.1 increments)

Colour	Score	Criteria
	1.0	**Exact match:** Correct variables, correct constructs/measures, causal-level evidence (RCT, experimental manipulation, longitudinal with controls).
	0.9	Correct variables, causal evidence, but **one variable has wrong/different measurement** (e.g., self-report vs. objective measure).
	0.8	Correct variables, causal evidence, but **two variables have wrong/different measurements**.
	0.7	**One variable doesn't quite match** but is still similar enough (e.g., "knowledge access" vs. "internet access"), other variable matches exactly. **OR:** Evidence states "A causes B and C" but only evidence for "A causes B" is relevant.
	0.6	**Correlational evidence only** (no causal mechanism established), but variables match correctly.
	0.5	**Two variables slightly mismatching** but still plausible, with causal evidence.
	0.4	Weak match: variables are conceptually related but operationally different; evidence is correlational.
	0.3	Poor match: only one variable loosely related; evidence type unclear or weak.
	0.2	Very poor match: tangential relevance only.
	0.1	Minimal relevance: only keyword overlap; no substantive connection to claimed relationship.
	0.0	**Not applicable:** Link broken, content inaccessible, completely irrelevant, or no evidence found.

Output Files:

- `data/deep_research_results_*.json`: Per-edge DR verdicts with evidence
- `rq3_comprehensive_stats.json`: Complete statistical analysis
- `rq3_statistical_test_results_latest.json`: Fisher's exact tests, power analysis
- `rq3_per_cld_detection_rates_latest.json`: Per-CLD breakdown
- `enriched_gt_analysis_latest.json`: Enrichment scenario metrics
- `Figures/rq3_combined_figure.png`: Publication-ready figure
- `rq3_main_table.tex`: Main detection rate table
- `rq3_per_cld_table.tex`: Per-CLD pairwise discrimination table
- `rq3_enriched_gt_table.tex`: Enrichment scenarios comparison
- `validation/enrichment_estimates_*.csv`: Threshold sensitivity results
- `validation/threshold_selection_*.json`: Selected threshold with rationale

Table C.32: RQ3 Human Validation: Evidence Applicability Score Distribution by Edge Classification ($n = 50$)

Applicability Range	**TP** ($n = 9$)	**FP** ($n = 32$)	**FN** ($n = 9$)	**Total**
$[0.0, 0.5)$	0	8	3	11 (22.0%)
$[0.5, 0.6)$	1	1	0	2 (4.0%)
$[0.6, 0.7)$	0	4	0	4 (8.0%)
$[0.7, 0.8)$	2	4	1	7 (14.0%)
$[0.8, 0.9)$	4	7	1	12 (24.0%)
$[0.9, 1.0]$	2	8	4	14 (28.0%)
Mean ± SD	0.77 ± 0.12	0.64 ± 0.27	0.67 ± 0.39	0.67 ± 0.27

Note. Applicability score (`Fully applicable`) measures how well retrieved evidence matches the specific (source, target, mechanism) claim. Higher scores indicate more applicable evidence. TP edges have the highest mean applicability and lowest variance, consistent with literature-grounded relationships.

C.4.9 Generator Temperature Sensitivity Analysis

This experiment validates the generator temperature choice (T=0.7) by testing whether temperature significantly affects edge recovery performance against literature ground truth.

[Requires private platform repository for execution; pre-computed results available in public snapshot]

```
# Private platform: data_science/parameter_tuning_experiments/
cd data_science/parameter_tuning_experiments/
python run_temperature_sensitivity.py
```

Script Location (private): `data_science/parameter_tuning_experiments/run_temperature_sensitivity.py`

Results Location (public): `final_runs/prelim_temperature_sensitivity/`

Table C.33: Temperature Sensitivity Experiment Configuration

Parameter	Value
CLDs	3 validation CLDs (Depressive Symptoms, Social Norms, Emergency Department)
Generator Model	GPT-4.1
Generator Prompt	Nitai_C (`prompts_Nitai_C.yaml`)
Temperature Values	{0.0, 0.3, 0.5, 0.7, 1.0}
Runs per Temperature	3
Seeds	{10, 20, 30}
Total Runs	45 (5 temperatures × 3 seeds × 3 CLDs)
Metric & Analysis	
Metric	Edge F1 vs. literature ground truth
Statistical Test	One-way ANOVA
Null Hypothesis	Mean F1 equal across temperatures

Output Files:

- `temperature_sensitivity_results.json`: Per-run results with F1, precision, recall
- `temperature_sensitivity_analysis.json`: Statistical analysis (ANOVA, per-temperature stats)
- `temperature_sensitivity_analysis.txt`: Human-readable summary report
- `result_excel_path_*.xlsx`: Per-run Excel files with edge classifications (45 files)

Key Code Sections:

```
# Temperature values and runs configuration (lines 48-50)
TEMPERATURE_VALUES = [0.0, 0.3, 0.5, 0.7, 1.0]
RUNS_PER_TEMP = {0.0: 3, 0.3: 3, 0.5: 3, 0.7: 3, 1.0: 3}
SEEDS = [10, 20, 30]

# Generator config with varying temperature (lines 85-91)
generator_config = {
    "provider": "openai",
    "model": "gpt-4.1",
    "temperature": temp,  # varies per run
    "seed": seed,
    "logprobs": True,
    "top_logprobs": 5
}
```

Edge F1 Computation: Generated edges are compared against the literature-derived ground truth CLD using exact matching (same source, target, and polarity). The `run_discovery_experiment()` function computes edge-level TP/FP/FN counts (via `edge_comparison`), from which $\text{F1} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN}$ is derived.

ANOVA Computation: One-way ANOVA via `scipy.stats.f_oneway()` tests whether the mean F1 differs significantly across temperature groups. Groups with $n \geq 2$ are included (all 5 temperatures qualify with $n = 3$ each).

C.4.10 Generator Model Comparison

This one-shot comparison selects the generator model for all subsequent experiments by evaluating GPT-4.1, Sonar-Pro, and Claude Sonnet 4 on edge recovery against literature ground truth.

```
cd final_runs/prelim_generator_model_comparison/

# Run generation experiments (one per model per CLD)
python run_generator_model_comparison.py

# Analyse results and generate thesis table
python analyse_generator_models.py
```

Script Location: `final_runs/generator_model_comparison_analysis/`

Results Location: `final_runs/generator_model_comparison_analysis/`

Table C.34: Generator Model Comparison Configuration

Parameter	Value
Models	GPT-4.1, Sonar-Pro, Claude Sonnet 4
CLDs	3 validation CLDs
Generator Prompt	Nitai_C
Temperature	0.7
Corruption Rate	0.0 (generation only, no corruption)
Judge Edges	False (no judging, compare directly to GT)
Metric	Micro-averaged Edge F1 across all CLDs
Selection Rule	Highest aggregate F1 with logprobs availability

Output Files:

- `generator_model_comparison.tex`: LaTeX table for thesis
- `*_comparison_results.json`: Per-model metrics

C.4.11 Random Baseline Generator

This analysis establishes a structural random baseline by computing expected edge F1 for Erdős-Rényi random graphs matching each validation CLD's node and edge counts.

```
cd final_runs/prelim_random_baseline_generator/

python compute_random_baseline.py
```

Script Location: `final_runs/random_baseline_generator/compute_random_baseline.py`

Results Location: `final_runs/random_baseline_generator/`

Table C.35: Random Baseline Configuration

Parameter	Value
Random Graphs per CLD	1000
Graph Model	Erdős-Rényi G(n,m)
Node Sets	Same as validation CLDs
Edge Counts	Matched to validation CLDs
Metric	Edge F1 vs. literature ground truth
CI Method	z-distribution (n=1000)

Output Files:

- `random_baseline_comparison.png`: Comparison figure
- `random_baseline_summary.txt`: Summary statistics

C.4.12 Prompt Sensitivity Analysis

This analysis quantifies how prompt engineering choices affect judge and corrector performance by measuring the input-output relationship between semantic prompt distance and F1 change.

Script Location (Judge): `final_runs/supp_prompt_sensitivity/analysis_scripts/sensitivity_analysis_simple.py`

Script Location (Corrector): `final_runs/supp_prompt_sensitivity/analysis_scripts/sensitivity_analysis_corrector.py`

Results Location: `final_runs/supp_prompt_sensitivity/`

Table C.36: Prompt Sensitivity Analysis Configuration

Parameter	Value
Embedding Model	MPNet (all-mpnet-base-v2, 768-dim)
Distance Metric	Cosine distance from baseline prompt
Prompt Variants	Judge: Baseline, CoT, Mechanistic (+ Mechanistic-Lit for citation); Corrector: Baseline, CoT, Mechanistic
Judge Experiments	4 (Corruption Correctness/Citation, GT Correctness/Citation)
Corrector Experiments	2 (Synthetic Corruption, Ground Truth)
Statistical Test	Friedman (blocked by CLD×run), post-hoc paired Wilcoxon (Bonferroni)

Output Files:

- `prompt_sensitivity_figure.png`: Judge sensitivity visualisation
- `corrector_sensitivity_figure.png`: Corrector sensitivity visualisation
- `prompt_sensitivity_table.tex`: LaTeX summary table for judge prompt effects
- `corrector_sensitivity_table.tex`: LaTeX summary table for corrector prompt effects

C.4.13 Parallelization Benchmark

This benchmark measures wall-clock time speedup from parallelization by comparing sequential (1 worker) vs. parallel (3, 5, 10 workers) execution.

Script Location: `final_runs/parallelization_benchmark/scripts/`

Results Location: `final_runs/parallelization_benchmark/`

```
cd final_runs/supp_parallelization_benchmark/scripts/

python run_parallelization_benchmark.py
python analyse_parallelization_results.py
```

Table C.37: Parallelization Benchmark Configuration

Parameter	Value
Task	Correctness judging (34 API calls)
Simulated CLD	Depressive Symptoms (34 edges)
Model	GPT-4.1
Temperature	0.3
Max Tokens	200
Worker Configurations	1 (sequential), 3, 5, 10
Runs per Configuration	3
Total Benchmark Runs	12

Output Files:

- `data/benchmark_results_*.xlsx`: Raw timing data (wall-clock, cumulative API time)
- `Figures/speedup_comparison.png`: Speedup analysis figure
- `Figures/efficiency_analysis.png`: Parallel efficiency visualisation
- `parallelization_summary.txt`: Results summary with speedup/efficiency
- `BENCHMARK_METADATA.txt`: Full configuration JSON

C.4.14 Time and Cost Scaling Analysis

This analysis aggregates inference time and cost metrics across all RQ1 experiments to characterise computational scaling.

[Pre-computed results available in public snapshot; script available for re-analysis]

Script Location (public): `final_runs/supp_time_cost_scaling/analysis_scripts/aggregate_time_cost_scali`

Results Location (public): `final_runs/supp_time_cost_scaling/`

```
# From repository root:
uv run python final_runs/supp_time_cost_scaling/analysis_scripts/\
    aggregate_time_cost_scaling.py \
    --base_dir final_runs \
    --output_dir final_runs/supp_time_cost_scaling
```

Table C.38: Time and Cost Scaling Analysis Configuration

Parameter	Value
Judging Data	121 Excel files (correctness + citation experiments)
Generation Data	33 Excel files (parameter tuning results)
CLDs	Depressive Symptoms (34 edges), Social Norms (12 edges), Emergency Department (66 edges)
Judge Types	Correctness, Citation
Prompt Variants	Baseline, CoT, Mechanistic
Judge Models	GPT-4.1, GPT-5-mini
Pricing (OpenAI, December 2024)	
GPT-4.1	$2.00/1M input, $8.00/1M output tokens
GPT-5-mini	$0.15/1M input, $0.60/1M output tokens

Output Files:

- `time_cost_scaling_*.xlsx`: Raw aggregated data with per-file metrics
- `Figures/figure1_scaling_analysis.png`: Time vs. edges/nodes scaling
- `Figures/figure2_configuration_comparison.png`: Cost/time/tokens by config
- `Figures/figure3_generation_vs_judging.png`: Generation vs. judging comparison
- `latex/*.tex`: LaTeX tables for thesis
- `ANALYSIS_METADATA.txt`: Complete list of input file paths
- `aggregate_time_cost_scaling.py`: Copy of analysis script

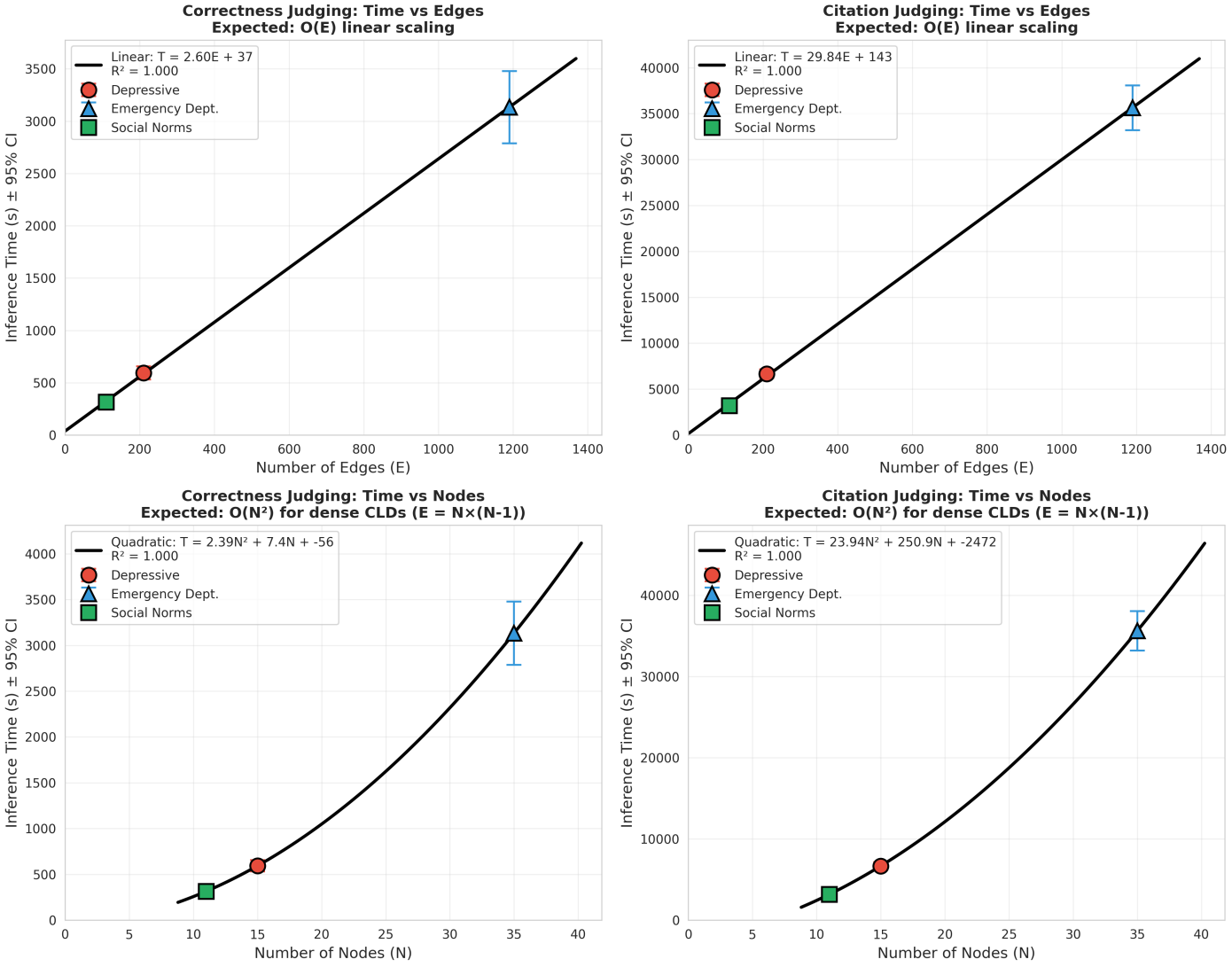


Figure C.1: Time scaling analysis for CLD judging. **Top row:** Time vs. number of edges shows linear $O(E)$ scaling. **Bottom row:** Time vs. number of nodes shows quadratic $O(N^2)$ scaling, consistent with dense CLD structure where $E \approx N(N-1)$. Citation judging (right) is $\sim$11$\times$ slower than correctness (left) due to web search and document retrieval. Points represent per-CLD means $\pm$ 95% CI (error bars); regression lines show fitted models with R^2 values. All three CLDs shown: Depressive Symptoms (D), Social Norms (S), Emergency Department (E).

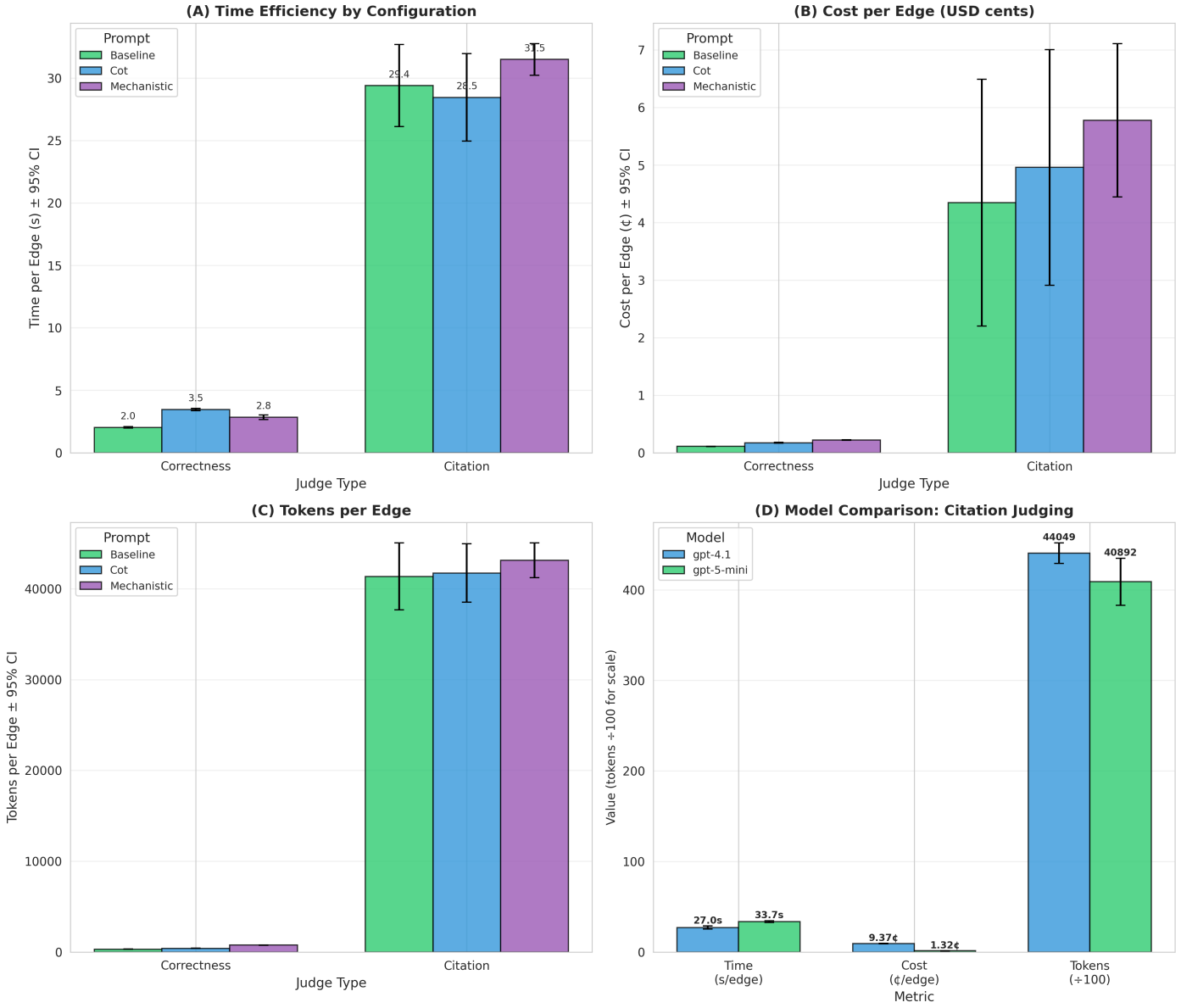


Figure C.2: Computational efficiency by configuration (mean ± 95% CI). **(A)** Time per edge: Citation judging (30.1 ± 1.4 s/edge) vs. correctness (2.8 ± 0.2 s/edge). **(B)** Cost per edge: Citation (5.6¢/edge) vs. correctness (0.17¢/edge). **(C)** Token usage: Citation requires ∼11× more tokens due to retrieved document context. **(D)** Model comparison for citation judging: GPT-5-mini achieves comparable time/tokens at 97% lower cost than GPT-4.1. All error bars show 95% CI.

C.4.15 Physics CLD Validation

This experiment validates judge functionality using physics-based CLDs grounded in well-established scientific principles (Newton's laws, conservation laws, Lotka-Volterra equations), where 100% approval is expected.

```
cd final_runs/validation_RQ1a_physics_clds/

# Run correctness-based judging on physics CLDs
python run_physics_cld_tests.py --approaches correctness
```

```
# Run citation-based judging on physics CLDs
python run_physics_cld_tests.py --approaches citation

# Analyse results and generate LaTeX tables
python analyse_physics_cld_results.py
```

Table C.39: Physics CLD Validation Configuration

Parameter	Value
CLDs	Thermostat (8 edges), Predator-Prey (10), RC Circuit (7), Water Tank (6)
Total Edges	31
Judge Model	GPT-4.1
Temperature	0.0
Structure + motivations authored with	Claude 4.5 Opus [128] (verified against sources)
Physics Principles	
Thermostat	Newton's Law of Cooling [106], First Law of Thermodynamics [107]
Predator-Prey	Lotka–Volterra population dynamics [108, 109]
RC Circuit	Ohm's Law [110], Kirchhoff's Laws [111]
Water Tank	Pascal's Law [112], Torricelli's Law [113]

Output Files:

- `results/physics_cld_raw_results_*.json`: Per-edge judge verdicts
- `results/physics_cld_tables_*.tex`: LaTeX tables for thesis

C.4.16 Cost Analysis

Computes API costs from token usage in experiment Excel files.

```
cd final_runs/supp_cost_analysis/

# Complete cost analysis (RQ1a + RQ1b estimates)
python calculate_all_costs.py

# Per-experiment breakdown
python calculate_costs_by_experiment.py --output-dir .

# Deep Research costs
python compute_dr_costs.py
```

Table C.40: Cost Analysis Configuration

Parameter	Value
Data Source	`LLM Usage Stats` sheet in experiment Excel files
RQ1a Files	121 judging experiment files
RQ1b Estimation	2× correctness tokens/edge (correction + rejudging)
Pricing (OpenAI, December 2024)	
GPT-4.1	\$2.00/1M input, \$8.00/1M output tokens
GPT-5-mini	\$0.15/1M input, \$0.60/1M output tokens

Output Files:

- `cost_complete_report_*.txt`: Human-readable cost report
- `cost_complete_table.tex`: LaTeX table for thesis
- `cost_by_experiment_results.json`: Detailed per-experiment breakdown

C.4.17 Embedding Model Benchmark

Compares embedding models for cosine similarity computation in RQ2 hallucination detection.

```
cd final_runs/supp_embedding_benchmark/

python embedding_benchmark_real_cld.py
```

Table C.41: Embedding Model Benchmark Configuration

Parameter	Value
Test Documents	30 causal narratives from CLD experiments
Average Length	451 characters
Models Tested	
all-mpnet-base-v2	110M params, 768-dim, via sentence-transformers
Qwen3-Embedding-8B	8B params Q4_K_M, 4096-dim, via Ollama

Key Result: MPNet is 23× faster (3.0 ms/doc vs. 67.3 ms/doc), justifying its selection for batch workloads; published MTEB results for both models are reported in Appendix H for context.

Output Files:

- `benchmark_results.txt`: Full output with LaTeX table

C.5 Output Artefacts

C.5.1 Metadata Files

Each experiment run generates:

- `session_info.txt`: Base CLD session identifier and generation parameters
- `corruption_metadata.json`: Corruption seed, temperature, model, corrupted edge indices
- `judging_metadata_*.json`: Per-prompt judging parameters (prompt version, temperature, model)

C.5.2 Result Files

Table C.42: Experiment Output File Types

File Type	Contents
`judged_*.xlsx`	Per-edge validation results with scores and classifications
`*_metrics.json`	Aggregate metrics (precision, recall, F1, AUC)
`*_confusion_matrix.png`	Visualisation of classification performance
`*_score_distribution.png`	Distribution of judge scores by ground truth label

C.5.3 Naming Convention

Output files follow: `judged_{CLD_NAME}_{PROMPT_TYPE}_{TIMESTAMP}.xlsx`

Where `PROMPT_TYPE` ∈ {`baseline`, `mechanistic`, `cot`} and `TIMESTAMP` = `YYYYMMDD_HHMMSS`.

C.6 Output Token Verification

To verify no judge outputs were truncated due to `max_tokens` limits, we analysed all judge messages using the GPT-4 tokenizer (`tiktoken`).

Table C.43: Output Token and API Call Analysis by Prompt Variant

Variant	Description	Max Tok	Limit	Messages	Calls	Success	Rate
baseline	Direct evaluation	1,499	1,500	18,548	34,660	34,641	99.9%
mechanistic	Bradford Hill criteria	255	1,500	9,265	9,262	9,262	100.0%
mech_lit	Mechanistic (citation, lit.)	1,487	1,500	13,466	64,993	57,863	89.0%
mech_orig	Mechanistic (citation, orig.)	1,110	1,500	4,736	21,026	21,008	99.9%
cot	Chain-of-thought reasoning	1,534	10,000	24,266	59,795	59,304	99.2%
Total		—	—	**70,281**	**189,736**	**182,078**	**96.0%**

Note. Max Tok = maximum observed output tokens via GPT-4 tokenizer (tiktoken). Limit = `max_tokens` API parameter. Calls = total API calls. Success = HTTP 200 completions. Rate = success percentage.
Reproduction: `python final_runs/supp_token_analysis_rq1a/analyse_output_tokens_tiktoken.py`

Conclusion: All outputs remained below their respective token limits. The closest approach was baseline at 1,499/1,500 tokens. No truncation artefacts (incomplete JSON, mid-sentence cutoffs) were detected. The mech_lit variant shows lower success rate (89.0%) due to web search failures in citation retrieval.

Appendix D

Detailed Experimental Specifications

Structure of Experimental Details

This appendix expands each row of Table 6.2 into a reproducible specification. To make experiments easy to scan and compare, each experiment section follows a consistent structure:

- **RQ & Goal / Hypotheses:** What the experiment is testing and the expected direction of effects.
- **Source Data:** Which datasets/edge tables are used and how ground truth labels are defined (GT Lit vs. GT Synth, TP/FP/FN).
- **Simulation Parameters:** What varies vs. what is fixed (models, prompts, seeds, corruption rate, judge temperature), aligned with the **Varies/Fixed** columns in Table 6.2.
- **Simulation/Analysis Procedure:** The operational steps (generation → judging → correction / Deep Research) and any filtering/deduplication rules.
- **Outputs:** The artefacts produced (Excel/JSON tables, scores, plots) and where they enter downstream analyses.
- **Metrics / Statistical Analysis:** The reported metrics and inference procedure (unit of analysis, tests, and multiple-comparison family).
- **Reproducibility:** The exact scripts/entry points and appendix pointers needed to rerun the experiment end-to-end.

When an RQ contains multiple experimental cells (e.g., RQ1a's judge×GT combinations), we list the shared setup first, then provide a short **Experiment Details** breakdown for each cell.

D.1 Preliminary: Generator Selection

Overview.
Before running the main experiments, we conducted preliminary studies to select the generator model, prompt, and temperature. These configuration choices are fixed across all subsequent experiments.

D.1.0.1 Generator Model Selection

RQ & Goal.
Select a single frontier LLM for all CLD generation experiments, prioritizing edge recovery performance and availability of token-level log probabilities for downstream uncertainty quantification.

Source Data.

Three validation CLDs (Depressive Symptoms, Social Norms, Emergency Department) with literature-derived ground truth. Node sets are fixed to ground truth variables. Ground-truth CLD files in Appendix C.1 (Table C.1).

Simulation Parameters.

- **Models compared:** GPT-4.1 (OpenAI), Sonar-Pro (Perplexity), Claude Sonnet 4 (Anthropic)
- **Prompt:** Nitai_C (fixed across models)
- **Temperature:** T=0.7
- **Corruption/judging:** Disabled (pure edge recovery comparison)

Simulation Procedure.

For each model–CLD combination, generate a complete `All Edges` table using the same node set as the literature ground truth. Models are compared purely on edge recovery against GT Lit. Generation procedure in Appendix I (Algorithm I.1).

Simulation Outputs.

Per-model `All Edges` tables with relationship types and causal narratives. Result directory and scripts in Appendix C.4.10 (Table C.34).

Unit of Analysis.

Run-level: each (model, CLD, seed) combination constitutes one observation ($N = 27$ total: 3 models $\times$ 3 CLDs $\times$ 3 runs).

Metrics.

Confusion-matrix counts (TP/FP/FN) computed from `All Edges` sheets, yielding precision, recall, and F1. Aggregate micro-averaged F1 computed by pooling TP/FP/FN across all three CLDs.

Selection Rule.

Select the generator with highest aggregate (micro-averaged) F1, prioritizing balanced precision/recall. Secondary constraint: token-level logprobs availability for RQ2; among compared models, this is available for GPT-4.1.

Reproducibility.

Full configuration and script locations in Appendix C.4.10 (Table C.34).

Results.

Table 7.1 (Chapter 7).

D.1.0.2 Generator Prompt Selection (Nitai_C)

RQ & Goal.

Select a fixed generator prompt variant that maximises edge recovery performance.

Source Data.

Three validation CLDs (Depressive Symptoms, Social Norms, Emergency Department) with literature-derived ground truth (edge ablation), plus the node-generation ablation dataset described in Appendix G. Ground-truth CLD files in Appendix C.1 (Table C.1).

Simulation Procedure.

Two ablation studies were conducted:

- **Edge Generation Ablation:** Nine prompt variants evaluated across three CLDs, testing CoT, task decomposition, RAG/citation requirements, and relationship taxonomy complexity.
- **Node Generation Ablation:** Seven prompt variants evaluated in a 7×3×5 factorial design (105 runs total).

Generation procedure in Appendix I (Algorithm I.1).

Simulation Outputs.

Per-variant F1 scores for edge and node generation tasks. Edge ablation appendix in Appendix F.6; node ablation appendix in Appendix G.

Unit of Analysis.

Prompt variant-level: Edge ablation uses 9 prompts × 3 CLDs = 27 observations; node ablation uses 7 prompts × 3 CLDs × 5 runs = 105 observations.

Key Findings.

- **Edge generation:** Nitai_C (dual-level CoT, three relationship types) achieved highest F1 (0.460), outperforming baseline (0.371) by 24%. CoT improved progressively (Nitai_A→B→C); mandatory citations reduced performance by 25%; overly restrictive grounding caused complete failure (Cillian_C: F1=0.00).
- **Node generation:** ANOVA revealed no significant differences ($F = 0.81$, $p = 0.57$; see Appendix E), with F1 ranging 0.617–0.677. Node generation is robust to prompt engineering choices.

Reproducibility.

Exact Nitai_C prompt in Appendix F.8 (Listing F.1). Edge ablation: Appendix F.6. Node ablation: Appendix G.

Results.

Edge ablation: Appendix F.6. Node ablation: Appendix G.

D.1.0.3 Generator Temperature Selection

RQ & Goal.

Validate generator temperature T=0.7 through sensitivity analysis measuring edge recovery performance.

Source Data.

Three validation CLDs: Social Norms (10 nodes, 12 edges), Depressive Symptoms (14 nodes, 34 edges), Emergency Department (34 nodes, 66 edges). Ground-truth CLD files in Appendix C.1 (Table C.1).

Simulation Parameters.

- **Generator:** GPT-4.1 with Nitai_C prompt
- **Ground Truth:** Literature-derived CLDs (GT Lit)
- **Temperatures:** T∈{0.0, 0.3, 0.5, 0.7, 1.0} (5 levels)
- **Runs:** 3 per temperature per CLD (seeds 10, 20, 30), totaling 45 experiments

Simulation Procedure.

For each temperature–CLD–seed combination, generate edges and compute F1 against GT Lit. Generation procedure in Appendix I (Algorithm I.1).

Simulation Outputs.

Per-run edge F1 scores (45 total). Temperature sensitivity configuration in Appendix Table C.33.

Unit of Analysis.

Run-level: each (temperature, CLD, seed) combination constitutes one observation ($N = 45$ total: 5 temperatures × 3 CLDs × 3 runs).

Metrics.

Edge F1 per run; aggregated by temperature level.

Statistical Analysis.

One-way ANOVA tested whether temperature significantly affects edge F1 (see Appendix E). Results in Table 7.3.

Selection Rule.

Temperature T=0.7 retained as it shows no significant performance difference from other values.

Results.

Table 7.3 (Chapter 7).

Reproducibility.

Full configuration in Appendix Table C.33.

D.2 Preliminaries: Generator Performance

RQ & Goal:

Contextualize generator performance via a structural random baseline—the expected edge-recovery for random graphs with the same (n, m) as each CLD.

Source Data:

Three validation CLDs with known node and edge counts (Appendix C.1, Table C.1).

Parameters:

Random baseline: 1000 Erdős–Rényi directed graphs $G(n, m)$ per CLD; LLM generator: 3 runs per CLD (seeds 10, 20, 30). Full configuration in Appendix C.4.11 (Table C.35).

Unit of Analysis:

Graph-level for random baseline ($N = 1000$ per CLD); run-level for LLM ($N = 3$ per CLD).

Simulation Procedure:

For each random graph, compute edge precision/recall/F1 vs. the ground truth; for the LLM runs, compute edge F1 likewise (Appendix C.4.11, Table C.35).

Outputs:

Random-baseline per-graph F1 distributions (1000 samples per CLD) and per-run LLM F1 values; directory mapping in Appendix C.1 (Table C.4).

Metrics/CI:

Precision, recall, F1; 95% CI uses z for random (n=1000) and t for LLM (n=3).

Results:

Table 7.2 and Figure 7.1 (Chapter 7).

Reproducibility:

Full configuration in Appendix C.4.11 (Table C.35).

D.3 Preliminary: TruthfulQA Verification

RQ & Goal:

Sanity-check the correctness-based judge on a general-domain hallucination benchmark before CLD-specific evaluation; strong TruthfulQA performance with weak CLD performance would suggest CLD hallucinations require domain grounding or citation validation beyond correctness judging.

Hypothesis:

H_0: Accuracy ≤ 0.80; H_1: Accuracy > 0.80.

Source Data:

TruthfulQA [127] (817 questions). We construct a balanced set of 1,634 QA pairs: 817 truthful answers and 817 imitative falsehoods (Appendix C.1, Table C.2).

Parameters:

GPT-4.1, temperature 0.0, seed 42, baseline correctness prompt (same structure as RQ1a), max tokens 250. Full configuration in Appendix C.4.2 (Table C.11).

Unit of Analysis:

Q&A pair-level ($N = 1634$ total: 817 truthful + 817 hallucinated).

Simulation Procedure:

The judge labels each answer as `CORRECT` vs. `INCORRECT` based solely on factual accuracy (Algorithm I.2).
Outputs:
Per-answer verdicts with reasoning (artefact schema: Appendix C.5, Table C.42).
Metrics:
Accuracy; Precision/Recall/F1 treating `Hallucinated` as the positive class.
Reproducibility:
Data in `final_runs/RQ1_verification_Truth_QA/`; execution/configuration in Appendix C.4.2 (Table C.11).
Results:
Table 7.4 (Chapter 7).

D.4 RQ1a: LLM-as-a-Judge Hallucination Detection

RQ & Goal:
Test whether LLM judges can detect hallucinated CLD edges: Correctness Judge (Section 4.4.2) and Citation Judge (Section 4.4.3) under two ground-truth regimes (GT Lit and GT Synth), yielding four experiment cells.
Hypotheses:
H1 (Correctness): lower correctness score implies hallucination (FP/FN w.r.t. GT).
H2 (Citation): lower citation score implies hallucination (FP/FN w.r.t. GT).
Source Data:
Three validation CLDs (Depressive Symptoms, Social Norms, Emergency Department) with literature-derived ground truth; node sets fixed to GT variables (edge-only evaluation; node generation ablation in Appendix G). Ground-truth files and output directories: Appendix C.1 (Tables C.1, C.4).
Parameters:
Correctness judge: 3 prompt variants (Baseline, CoT, Mechanistic) $\times$ 3 CLDs $\times$ 3 runs = 27.
Citation judge: 4 prompt variants (Baseline, CoT, Mechanistic, Mechanistic-Lit) $\times$ 3 CLDs $\times$ 3 runs = 36.
GT Lit seeds 10/20/30 (generation variation); GT Synth seeds 1/2/3 (corruption variation) with 30% corruption.
Citation retrieval in ContentScraper-only mode (Jina AI Reader disabled; Appendix I.9). Full config in Table 6.2.
Simulation Procedure:
For each CLD–seed–prompt: generate edges (GPT-4.1 with Nitai_C); for GT Synth apply Corruptor; run the judge to score each edge; compare scores to GT labels (Algorithms I.1, I.2, I.3, and for GT Synth Algorithm I.4).
Outputs:
Per-edge scores (0.0/0.5/1.0) with verdicts and reasoning; schemas in Appendix C.5 (Table C.42).
Metrics:
AUC-ROC (baseline 0.5), point-biserial correlation r, and thresholded classification (score ≤ 0.5 = hallucination): Precision/Recall/F1.
Unit of Analysis:
Block-level (CLD$\times$run, $N = 9$); edge-level ($N \approx$ 10,000–31,000 depending on experiment cell).
Stats:
Prompt effects under blocked repeated-measures (block = CLD$\times$run, n=9): Friedman omnibus + paired Wilcoxon with Bonferroni correction (Section 6.6; Appendix E); RM-ANOVA sensitivity decomposition (η_p^2) in Appendix A.
Reproducibility:
Data directories: `final_runs/RQ1a_ground_truth_*` and `final_runs/RQ1a_corruption_*`; see Appendix C.1 (Table C.4) and Appendix C.2 (Tables C.5, C.6).
Results:

GT Lit Correctness: Figures 7.4, 7.5. GT Synth Correctness: Figures 7.2, 7.3. GT Lit Citation: Figures 7.8, 7.9. GT Synth Citation: Figures 7.6, 7.7 (Chapter 7).

Experiment Details

D.4.0.1 RQ1a: Correctness Judge (Lit)

RQ & Goal:
Evaluate Correctness Judge on GT Lit edges (literature-derived ground truth). See shared RQ1a setup (Section D.4).
Source Data:
GT Lit edges from 3 CLDs; ground truth = FP/FN w.r.t. expert CLDs. GT files: Appendix Table C.1.
Simulation Parameters:
Appendix Table C.13. Fixed: GPT-4.1, T=0.0. Varies: 3 prompts × 3 CLDs × 3 runs = 27.
Simulation Procedure:
Algorithm I.1 (generation) → Algorithm I.2 (judging).
Outputs:
Directory: `final_runs/RQ1a_gt_lit_correctness/`. Schemas: Appendix Table C.42.
Unit of Analysis:
CLD×run×prompt ($N = 27$: 3 prompts × 3 CLDs × 3 runs).
Metrics / Stats:
See shared RQ1a setup; Friedman + Wilcoxon.
Reproducibility:
Scripts: Appendix Tables C.5, C.6.
Results:
Figures 7.4, 7.5 (Chapter 7).

D.4.0.2 RQ1a: Correctness Judge (Synth)

RQ & Goal:
Evaluate Correctness Judge on GT Synth edges (30% corruption). See shared RQ1a setup (Section D.4).
Source Data:
GT Synth edges from 3 CLDs; ground truth = corruption status (corrupted = hallucination).
Simulation Parameters:
Appendix Table C.14. Fixed: GPT-4.1, T=0.0, 30% corruption. Varies: 3 prompts × 3 CLDs × 3 runs = 27.
Simulation Procedure:
Algorithm I.1 (generation) → Algorithm I.4 (corruption) → Algorithm I.2 (judging).
Outputs:
Directory: `final_runs/RQ1a_gt_synth_correctness/`. Schemas: Appendix Table C.42.
Unit of Analysis:
CLD×run×prompt ($N = 27$: 3 prompts × 3 CLDs × 3 runs).
Metrics / Stats:
See shared RQ1a setup; Friedman + Wilcoxon.
Reproducibility:
Scripts: Appendix Tables C.5, C.6.
Results:
Figures 7.2, 7.3 (Chapter 7).

D.4.0.3 RQ1a: Citation Judge (Lit)

RQ & Goal:
Evaluate Citation Judge on GT Lit edges (literature retrieval). See shared RQ1a setup (Section D.4).
Source Data:
GT Lit edges from 3 CLDs; ground truth = FP/FN w.r.t. expert CLDs. GT files: Appendix Table C.1.
Simulation Parameters:
Appendix Table C.15. Fixed: GPT-4.1, T=0.0, ContentScraper. Varies: 4 prompts × 3 CLDs × 3 runs = 36.
Simulation Procedure:
Algorithm I.1 (generation) → Algorithm I.9 (retrieval) → Algorithm I.3 (judging).
Outputs:
Directory: `final_runs/RQ1a_gt_lit_citation/`. Schemas: Appendix Table C.42.
Unit of Analysis:
CLD×run×prompt ($N = 36$: 4 prompts × 3 CLDs × 3 runs).
Metrics / Stats:
See shared RQ1a setup; Friedman + Wilcoxon.
Reproducibility:
Scripts: Appendix Tables C.5, C.6.
Results:
Figures 7.8, 7.9 (Chapter 7).

D.4.0.4 RQ1a: Citation Judge (Synth)

RQ & Goal:
Evaluate Citation Judge on GT Synth edges (30% corruption, GPT-5-mini). See shared RQ1a setup (Section D.4).
Source Data:
GT Synth edges from 3 CLDs; ground truth = corruption status (corrupted = hallucination).
Simulation Parameters:
Appendix Table C.12. Fixed: GPT-5-mini, T=0.0, 30% corruption. Varies: 4 prompts × 3 CLDs × 3 runs = 36.
Simulation Procedure:
Algorithm I.1 (generation) → Algorithm I.4 (corruption) → Algorithm I.9 (retrieval) → Algorithm I.3 (judging).
Outputs:
Directory: `final_runs/RQ1a_gt_synth_citation/`. Schemas: Appendix Table C.42.
Unit of Analysis:
CLD×run×prompt ($N = 36$: 4 prompts × 3 CLDs × 3 runs).
Metrics / Stats:
See shared RQ1a setup; Friedman + Wilcoxon.
Reproducibility:
Scripts: Appendix Tables C.5, C.6.
Results:
Figures 7.6, 7.7 (Chapter 7).

D.5 Sensitivity: Prompt Sensitivity Analysis

RQ & Goal.

Quantify prompt engineering impact on judge performance using variance-based sensitivity analysis inspired by Sobol indices [141].

Source Data.

Prompt-level and block-level results from the four RQ1a judge experiments (summarised in Table 6.2). Sensitivity output directory in Appendix C.1 (Table C.4).

Simulation Parameters.

- **Design:** Blocked with CLD×run as subjects, prompt as within-subject factor
- **Prompt distance:** MPNet [142] embeddings (768-dim); cosine distance from baseline: $d(p_i) = 1 - \cos(\mathbf{e}_{\text{baseline}}, \mathbf{e}_i)$
- **Scope:** 4 experiments × 3–4 prompt variants

Full configuration in Appendix Table C.36.

Simulation Procedure.

Repeated-measures ANOVA with CLD×run as subjects and prompt as within-subject factor. Full RM-ANOVA tables and diagnostics in Appendix A.

Simulation Outputs.

RM-ANOVA effect sizes (η_p^2), assumption diagnostics, and sensitivity figures for each judge×GT setting. Output artefacts in Appendix C.5 (Table C.42); sensitivity output directory in Appendix C.1 (Table C.4).

Metrics.

Following Sobol's variance-based sensitivity framework, for a categorical factor $A \in \{1, \ldots, K\}$ (here: prompt choice) the first-order index is:

$$S_A := \frac{\text{Var}(\mathbb{E}[Y \mid A])}{\text{Var}(Y)}. \tag{D.1}$$

For categorical A, this reduces to $S_A = \eta^2$, i.e., the population one-way ANOVA effect size (correlation ratio) for a categorical factor [126]. In the main experiments, we report η_p^2 (partial eta-squared) because our design is blocked by CLD×run (repeated measures), so the sensitivity measure is computed after controlling for these subject-level differences. Partial eta-squared: $\eta_p^2 = \frac{SS_{\text{prompt}}}{SS_{\text{prompt}} + SS_{\text{error}}}$. Measures proportion of within-subject variance in F1 explained by prompt choice. Interpretation (Cohen): <0.01 negligible, 0.01–0.06 small, 0.06–0.14 medium, ≥0.14 large.

Unit of Analysis.

Block-level (CLD×run, $N = 9$) for RM-ANOVA variance decomposition; prompt as within-subject factor.

Statistical Analysis.

RM-ANOVA assumption diagnostics: Shapiro–Wilk (residual normality), Mauchly (sphericity), Greenhouse–Geisser correction when violated, and Bonferroni correction (see Appendix E). Full tables in Appendix A.

Reproducibility.

Data: `final_runs/supp_prompt_sensitivity/`. Scripts: `sensitivity_analysis_simple_rm_anova.py`. Dataset mapping in Appendix C.1 (Table C.4).

Results.

Appendix A (Figure A.1).

D.6 Validation: Uleman's Provider Study

Overview.

Four validation studies assess judge reliability: (1) search provider comparison, (2) human-LLM agreement, (3) physics CLD sanity check, and (4) retrieval success analysis.

Source Data.

Validation datasets (Uleman's CLD, physics CLDs, human validation set) and their output directories. Dataset locations in Appendix C.1 (Table C.2).

Simulation Procedure.

Apply the relevant judge pipeline(s) (correctness/citation) to validation edges; citation judging includes retrieval and fetching. Algorithms in Appendix I (Algorithms I.2, I.3, I.9).

Simulation Outputs.

Per-edge verdicts/scores (and retrieval diagnostics for citation judging) stored in the corresponding validation output directories. Output artefacts in Appendix C.5 (Table C.42); validation directories in Appendix C.1 (Table C.2).

Reproducibility.

Dataset specifications: Appendix C.1 (Tables C.2, C.3). Scripts and configurations: Appendix C.4.6 (Tables C.18, C.19).

D.6.1 Search Provider Comparison (Uleman's CLD)

RQ & Goal.

Validate citation-based judge on an independently expert-validated dataset. Determine whether low GT Lit scores are due to search infrastructure limitations.

Hypothesis.

If low scores are infrastructure-driven, we expect significant variation across providers/fetchers. Consistently low scores would confirm a fundamental validation gap between expert reasoning and literature-based verification.

Source Data.

Uleman's CLD: 150-edge causal loop diagram for Alzheimer's disease, constructed through Group Model Building (GMB) with 17 domain experts [114]. Unlike primary CLDs (extracted from literature), represents direct expert consensus with bibliographic references.

Simulation Parameters.

- **Search Providers (4):** OpenAlex, Brave Search (with/without domain whitelist), Perplexity, PubMed
- **Citation Fetcher:** ContentScraper (BeautifulSoup-based, PDF-capable); Jina AI Reader disabled
- **Design:** 4 providers × 2 fetchers × 150 edges = 1,200 judgements
- **Judge:** Citation-based (GPT-4.1, baseline prompt)

Full configuration in Appendix C.4.6 (Table C.18).

Outputs.

Per-edge scores and provider comparison summaries; output artefacts in Appendix C.5 (Table C.42).

Metrics.

Average score (mean 0–1), coverage (successful fetch %), score difference (ContentScraper − Jina AI).

Unit of Analysis.

Edge-level ($N = 150$); crossed with 4 providers × 2 fetchers = 1,200 total judgements.

Reproducibility.

Data: `final_runs/validation_RQ1a_ulemans_external/`. Configuration: Appendix Table C.18.

Results.

Table 7.7 (Chapter 7).

D.6.2 Human–LLM Agreement

RQ & Goal.

Validate that LLM citation judges produce reliable verdicts by measuring inter-rater agreement with human experts.

Hypothesis.

Expect substantial agreement ($\kappa > 0.60$), performance comparable to human baseline, and consistency across CLDs/generators.

Source Data.

72 causal edges across 3 validation files (2 CLDs, 2 generators): 30 edges Depressive Symptoms/GPT-4.1, 30 edges Depressive Symptoms/Claude 4 Sonnet, 12 edges Social Norms/GPT-4.1.

Simulation Parameters.

Human validator (domain expert) and LLM judge (GPT-4.1, T=0, baseline prompt) each provide categorical verdicts: Not supported (0), Partially supported (0.5), Fully supported (1). Judge settings follow RQ1a Citation Judge configuration. Full configuration in Appendix C.4.6 (Table C.19).

Outputs.

Per-edge human and LLM verdicts with agreement statistics; output artefacts in Appendix C.5 (Table C.42).

Metrics.

- **Cohen's κ:** Chance-corrected agreement. Interpretation (Landis & Koch): <0.20 slight, 0.21–0.40 fair, 0.41–0.60 moderate, 0.61–0.80 substantial, >0.80 almost perfect.
- **Agreement Rate:** Exact match percentage
- **Pearson r:** Score correlation

Baseline: Human inter-rater reliability from prior GMB studies ($\kappa \approx 0.60$, 80% agreement).

Unit of Analysis.

Edge-level ($N = 72$ total across 3 validation files: 30 + 30 + 12 edges).

Reproducibility.

Data: `final_runs/RQ1_human_validation_citation_judge/`. Configuration: Appendix Table C.19.

Results.

Tables 7.5, 7.6 (Chapter 7).

D.6.3 Physics Sanity Check

RQ & Goal.

Verify that judges correctly identify valid causal relationships when unambiguous (sanity check under curated inputs).

Source Data.

Four physics-based CLDs derived from fundamental laws (31 edges, 26 variables):

1. Thermostat Heating System: Newton's Law of Cooling [106], First Law of Thermodynamics [107]
2. Predator-Prey Dynamics: Lotka-Volterra equations [108, 109]
3. RC Circuit: Ohm's Law [110], Kirchhoff's Laws [111]
4. Water Tank: Pascal's Law [112], Torricelli's Law [113]

CLD structure (variables, edges, polarities) and edge rationales were authored in an interactive session with Claude 4.5 Opus [128] by translating these laws into CLD form, then verified against source texts. Per-CLD breakdown: Appendix Table C.3.

Metrics.

We report *approval* rates computed from discrete judge verdicts. For the Correctness Judge, approval is the fraction of edges with verdict `CORRECT` (score 1.0). For the Citation Judge, approval is the fraction of edges with aggregate verdict `FULLY_SUPPORTED` or `PARTIALLY_SUPPORTED` (score ≥ 0.5); `NO_CITATION` indicates retrieval failure. We report approval across all edges and conditional approval restricted to edges with at least one retrieved citation ($C_e \geq 1$).

Simulation Parameters.

Full configuration in Appendix C.4.15 (Table C.39). Judge settings follow RQ1a configuration (GPT-4.1, T=0.0).

Simulation Procedure.

Both Correctness Judge and Citation Judge applied to curated physics edges (Algorithms I.2, I.3).

Outputs.

Per-edge verdicts and scores; output artefacts in Appendix C.5 (Table C.42).

Unit of Analysis.

Edge-level ($N = 31$ edges across 4 physics CLDs).

Reproducibility.

Data: `final_runs/validation_RQ1a_physics_clds/`. Configuration: Appendix Table C.39.

Results.

Table 7.8 (Chapter 7).

D.6.4 Retrieval Success Analysis

RQ & Goal.

Investigate whether low citation scores are driven by retrieval failures (missing documents) or evidence characteristics (weak/hedged claims).

Source Data.

1. Expert-provided citations: Physics CLDs, Uleman's CLD (150 edges)
2. Automated web search: Three main validation CLDs via generate-then-search pipeline

For Uleman's and Physics CLDs, CLD structure and causal narratives were authored with Claude 4.5 Opus [128].

Metrics.

Retrieval success rate = percentage of edges with successfully fetched content. Discrepancies between retrieval success and judge approval distinguish infrastructure failures from validation failures. Fetch success criteria: Appendix I.9.

Unit of Analysis.

Edge-level (diagnostic analysis comparing retrieval success vs. approval rates across datasets).

Reproducibility.

Analysis integrated with Uleman's and Physics CLD validation studies; see Appendix C.4.6 for scripts and configurations.

Results.

Table 7.9 (Chapter 7).

D.7 RQ1b: LLM-as-a-Corrector Evaluation

RQ & Goal:

Evaluate whether the Corrector (Section 4.6) can repair judge-flagged hallucinations.

Hypothesis:
H3: correction improves edge quality for flagged edges ($\Delta F1 > 0$).
Source Data:
Judge-flagged edges from RQ1a (GT Synth and GT Lit); output directories in Appendix C.1 (Table C.4).
Parameters:
3 corrector prompts (Baseline, CoT, Mechanistic); design = 3 prompts × 3 CLDs × 3 runs = 27 runs per GT type; evaluation judge = baseline Correctness Judge; scope = correctness-based correction only (citation-corrector omitted due to retrieval cost). Full configuration in Table 6.2.
Simulation Procedure:
For each flagged edge, apply the Corrector (revise relationship type and/or narrative), re-judge the corrected edge, and compare pre/post metrics against ground truth (Algorithms I.5 and I.2).
Outputs:
Corrected edges plus re-judge scores; artefact schema in Appendix C.5 (Table C.42).
Metrics:
$\Delta F1/\Delta Precision/\Delta Recall$ (pre→post vs. GT) and $\Delta JudgeScore$ = mean(corrected) − mean(original).
Unit of Analysis:
Block-level (CLD×run, $N = 9$) for statistical inference; edge-level for individual correction evaluation.
Stats:
Efficacy via one-sample Wilcoxon signed-rank on block-level $\Delta F1$ (blocks = CLD×run, n=9) testing median($\Delta F1$) = 0; prompt effects via Friedman omnibus + paired Wilcoxon with Bonferroni correction (Section 6.6; Appendix E); RM-ANOVA sensitivity in Appendix A.
Reproducibility:
Data in `final_runs/RQ1b_corrector_experiment_*/`; scripts in Appendix C.2 (Table C.5); analysis script `analyze_corrector_ablation.py`.
Results:
GT Synth: Tables 7.10, 7.11. GT Lit: Tables 7.12, 7.13 (Chapter 7).
Experiment Details

D.7.0.1 RQ1b: Correctness Corrector (Synth)

RQ & Goal:
Evaluate Corrector on GT Synth flagged edges. See shared RQ1b setup (Section D.7).
Source Data:
Judge-flagged edges from RQ1a GT Synth experiments (corrupted CLDs).
Simulation Parameters:
Appendix Table C.16. Fixed: GPT-4.1, T=0.0, baseline judge. Varies: 3 prompts × 3 CLDs × 3 runs = 27.
Simulation Procedure:
Algorithm I.5 (correction) → Algorithm I.2 (re-judging).
Outputs:
Directory: `final_runs/RQ1b_corrector_experiment_corruption_*/`. Schemas: Appendix Table C.42.
Unit of Analysis:
Block-level (CLD×run, $N = 9$); see shared RQ1b setup (Section D.7).
Metrics / Stats:
$\Delta F1$ pre→post; Wilcoxon signed-rank.
Reproducibility:
Scripts: Appendix Tables C.5, C.6.
Results:
Tables 7.10, 7.11 (Chapter 7).

D.7.0.2 RQ1b: Correctness Corrector (Lit)

RQ & Goal:
Evaluate Corrector on GT Lit flagged edges. See shared RQ1b setup (Section D.7).
Source Data:
Judge-flagged edges from RQ1a GT Lit experiments. GT files: Appendix Table C.1.
Simulation Parameters:
Appendix Table C.17. Fixed: GPT-4.1, T=0.0, baseline judge. Varies: 3 prompts × 3 CLDs × 3 runs = 27.
Simulation Procedure:
Algorithm I.5 (correction) → Algorithm I.2 (re-judging).
Outputs:
Directory: `final_runs/RQ1b_corrector_experiment_ground_truth_*/`. Schemas: Appendix Table C.42.
Unit of Analysis:
Block-level (CLD×run, $N = 9$); see shared RQ1b setup (Section D.7).
Metrics / Stats:
ΔF1 pre→post; Wilcoxon signed-rank.
Reproducibility:
Scripts: Appendix Tables C.5, C.6.
Results:
Tables 7.12, 7.13 (Chapter 7).

D.8 RQ2: Corpus-Independent Hallucination Detection

RQ & Goal:
Test whether UQ metrics captured during generation (Section 4.2.1) can predict hallucinations without external context.
Hypotheses:
H4 (UQ): higher perplexity / lower $P_{\min}$ / higher window entropy ⇒ hallucination (block-level AUC > 0.5).
H5 (CosSim): lower narrative–citation cosine similarity ⇒ hallucination (block-level AUC > 0.5).
Source Data:
63 deduplicated GT Lit files (36 citation, 27 correctness), 31,704 edges, hallucination rate 15.2%. Hallucination label: (FP or FN) w.r.t. GT Lit. *Metric availability:* Logprob-derived metrics (perplexity, min prob, max window entropy) available for all 63 files; cosine similarity available only for citation-judging runs (35 files, 16,492 edges) because it requires retrieved citation text. Ensemble phases use a 16,507-edge subset (35 citation files) where all four metrics are available. Outputs and GT files: Appendix C.1 (Tables C.4, C.1).
UQ Metrics + Parameters:
Computed from generator logprobs (Algorithms I.6, I.7). Perplexity uses capped NLL averaging (token cap 20.0; average cap 10.0, bounding PPL to $\leq e^{10} \approx 22{,}026$); $P_{\min}$ is the minimum token probability; H_{win} is max sliding-window entropy (window size 5 over top-k entropies with k=5); CosSim is the max narrative–citation-chunk embedding similarity for citation experiments (embedding model all-mpnet-base-v2 from [101]). Full shared configuration in Appendix Table C.23.
Outputs:
Per-edge metric values, per-file AUCs, and classifier predictions; schemas in Appendix C.5 (Table C.42).
Metrics:
AUC-ROC, point-biserial correlation r, and classifier CV/test AUC. Because hallucinations are the minority class (15.2%), AUC is used as a threshold-free separation metric; thresholded Precision/Recall depend on the chosen operating point and class prevalence.
Unit of Analysis:
Block-level (CLD×run, $N = 9$) for statistical inference; edge-level ($N = 31{,}704$) for descriptive analysis.

Stats:
Block-level aggregation (CLD×run, N=9) as the unit of inference. *Single-metric analysis:* one-sample **Wilcoxon signed-rank** tests on (AUC−0.5) with Bonferroni correction across 4 metrics (m=4, α_{adj}=0.0125); correlation r aggregated via Fisher z-transform across blocks with separate Bonferroni family. *Edge-level:* Mann–Whitney U tests reported as **Significant Files (%)**—the percentage of files where edge-level $p < 0.05$; this is a *descriptive* consistency measure only (not used for confirmatory inference due to within-file dependence). *File inclusion:* files excluded if <10 edges or <2 per class (hallucination/correct). *Uncertainty:* 95% CIs via t-distribution over blocks (df=8). Full procedures: Appendix E.
Results:
Figures 7.10, 7.11 (Chapter 7).
Experiment Details

D.8.0.1 RQ2: UQ (Logprobs)

RQ & Goal:
Test whether logprob-derived UQ metrics can predict hallucinations. See shared RQ2 setup (Section D.8).
Source Data:
RQ1a GT Lit experiments (27 correctness + 36 citation files, 31,704 edges); hallucination = FP/FN.
Simulation Parameters:
Appendix Table C.21. Metrics: PPL, P_{min}, H_{win} (computed during generation).
Simulation Procedure:
Algorithm I.6 (metric extraction from logprobs).
Outputs:
Directory: `final_runs/RQ2_uq_hallucination_detection/`. Schemas: Appendix Table C.42.
Unit of Analysis:
Block-level (CLD×run, $N = 9$); see shared RQ2 setup (Section D.8).
Metrics / Stats:
Block-level AUC; one-sample **Wilcoxon signed-rank** test on (AUC − 0.5).
Reproducibility:
Scripts: Appendix Tables C.5, C.6.
Results:
Figure 7.10 (Chapter 7).

D.8.0.2 RQ2: CosSim (Embeddings)

RQ & Goal:
Test whether narrative–citation cosine similarity can predict hallucinations. See shared RQ2 setup (Section D.8).
Source Data:
RQ1a GT Lit Citation experiments only (36 files); ensemble subset = 16,507 edges.
Simulation Parameters:
Appendix Table C.22. Model: all-mpnet-base-v2; max narrative–citation-chunk similarity.
Simulation Procedure:
Embedding computation + cosine similarity calculation. Benchmark: Appendix H.
Outputs:
Directory: `final_runs/RQ2_uq_hallucination_detection/`. Schemas: Appendix Table C.42.
Unit of Analysis:
Block-level (CLD×run, $N = 9$); see shared RQ2 setup (Section D.8).
Metrics / Stats:
Block-level AUC; one-sample **Wilcoxon signed-rank** test on (AUC − 0.5).

Reproducibility:
Scripts: Appendix Tables C.5, C.6.
Results:
Figure 7.10 (Chapter 7).

D.8.0.3 RQ2: Analysis Pipeline (6 Phases)

Simulation Procedure (6 phases):
Phases 1–3 (single-metric analysis):
Data loading: Load 63 deduplicated experiment files from inventory (36 citation, 27 correctness); exclude corruption experiments.
Hallucination labeling: `is_hallucination` = True if Classification ∈ {FP, FN} w.r.t. GT Lit.
Data cleaning: Remove rows with missing or infinite CI metric values (dropna on feature columns + label).
Per-file analysis: Compute AUC-ROC for each file (hallucination vs. correct); require ≥10 edges and ≥2 per class.
Block aggregation: Group files by (CLD × run) to form N=9 blocks; compute mean AUC per block.
Statistical test: One-sample Wilcoxon signed-rank test on (AUC − 0.5) across blocks (Bonferroni: m=4, α_{adj}=0.0125).
Full configuration in Appendix Table C.24.
Phase 4 (RFE): Recursive Feature Elimination [133] with `GroupKFold(n_splits=3)` over blocks; classifiers: LogReg, RF, GradBoost.
Phase 5 (ensemble): block-level train/test split (67%/33%) via `GroupShuffleSplit` and model selection on training blocks via `GroupKFold`; four classifiers including a neural network.
Classifier families: brief algorithm notes for LR/RF/GB/MLP appear in Appendix E.0.0.4.
Phase 6 (LOCO): leave-one-CLD-out evaluation for cross-domain generalisation. Full formal specification in Appendix I.7.
Simulation Parameters:
Full pipeline configuration in Appendix Table C.25.
Reproducible splitting protocol (Phases 4–6):
Grouping key is `block_id=cld_run` (3 CLDs × 3 runs ⇒ N=9); rows are deterministically sorted by (`block_id`, `source_file`) prior to any group split; scaling uses `StandardScaler` fit on training blocks only.
Phase 4 holds out 3 blocks and trains on 6; evaluates $k \in \{1, 2, 3, 4\}$ and selects the k maximizing validation AUC per fold.
Phase 5 holds out 3 blocks for testing and uses the remaining 6 for training; CV on training uses 3 folds (2 blocks held out per fold). For F1@t^* reporting, the threshold t^* is selected by maximizing out-of-fold F1 on training blocks only; this fixed t^* is then frozen and applied to both the Phase 5 test set and all Phase 6 evaluations.
Phase 6 holds out an entire CLD as test and trains on the other two CLDs (scaling fit on training CLDs only). 95% CIs are computed via the t-distribution with $df = n - 1$ over folds ($n = 3$ for both phases).
Unit of Analysis:
Block-level (CLD×run, $N = 9$) for Phases 1–5; CLD-level ($N = 3$) for Phase 6 (LOCO).
Limitations:
UQ metrics are not applicable for GT Synth (corruption is post-generation); cosine similarity is only defined for Citation Judge runs.
Reproducibility:
Data in `final_runs/RQ2_uq_hallucination_detection/`; dataset mapping in Appendix C.1 (Table C.4). Analysis scripts in Appendix C.2 (Table C.6).
Results:
Figures 7.10, 7.11; Section 7.6.4 (Chapter 7).

D.9 RQ3: Deep Research Literature Validation

Methodological Note: The original RQ3 design proposed *smart test-time compute allocation*—using UQ metrics to selectively deploy the LLM-as-a-corrector. This was predicated on: (1) RQ2 UQ metrics generalizing across CLDs, and (2) RQ1b correctors effectively repairing hallucinations. Neither condition was met (see Discussion, Section 8.8). We therefore pivoted to **Deep Research validation** (system documented in Section 4.7).
This section presents Deep Research as an **exploratory pilot study**. Given time constraints from the late pivot, the experimental design is less comprehensive than RQ1–RQ2: single-run (no replication), limited parameter ablation, and no end-to-end pipeline integration.

D.9.0.1 Deep Research

RQ & Goal.
Evaluate whether Deep Research (Section 4.7) can distinguish valid from hallucinated causal claims by finding *direct causal* evidence in the scientific literature.

Hypothesis.
H6: high detection for TP, low detection for FP, and a TP–FP gap >20 percentage points.

Source Data.
285 edges from RQ1a GT Lit experiments (TP=44, FP=178, FN=63). Output directory: Appendix C.1 (Table C.4).

Parameters.
Single run per edge (∼ $0.87/edge) with `max_iterations=1`, `max_sources=5`, `MAX_SEARCHES=5`, `time_limit=900s`.

Simulation Procedure.
Generate search queries from the causal claim, run parallel Brave searches, score evidence, apply strict direct-causality criteria, and synthesize a verdict (Algorithm I.11).

Outputs.
Per-edge verdict, confidence (1–10), `direct_causal_found` flag, and source URLs; schemas in Appendix C.5 (Table C.42).

Metrics.
Detection rate of `direct_causal_found`; TP–FP gap $\Delta > 20$pp; mean confidence by class.

Stats.
Pairwise Fisher exact tests (Bonferroni: $m = 3$, $\alpha_{\text{adj}} = 0.017$); effect size Cohen's h (Appendix E).

Unit of Analysis.
Edge-level ($N = 285$ edges: TP=44, FP=178, FN=63).

Reproducibility.
Config in Appendix C.4.8 (Table C.26); script `run_rq3_analysis.py`.

Results.
Figure 7.12, Table 7.25 (Chapter 7).

D.9.0.2 Computational Cost

Cost summary (reported in Results).
See Results Section 7.7.1 (Table 7.25) for the Deep Research computational cost breakdown and the overall cost comparison ($0.87/edge; ∼119× citation judging; ∼512× correctness judging).

D.9.0.3 Limitations

- **No replication:** Single run per edge prevents uncertainty quantification
- **Evidence applicability:** Retrieved causal evidence may be only partially aligned to specific edge claims
- **Full-text access:** Paywalled articles limit evidence quality
- **Search API dependency:** Results depend on Brave Search indexing

D.9.0.4 Exploratory Analysis: Ground Truth Enrichment

RQ & Goal.

Explore whether DR results could enrich expert-constructed GT by validating "hallucinations" that may represent valid causal relationships experts omitted.

Extrapolation Procedure.

Two enrichment scenarios:

- **Scenario 1 (Enriched GT):** Promote DR-validated FPs to TP in ground truth; recompute Precision/Recall/F1.
- **Scenario 2 (LLM+DR):** Additionally add DR-validated FNs to system output.

Simulation Parameters.

Full configuration in Appendix Table C.27.

Caveats.

Domain variation (28% Depressive Symptoms vs 76% Emergency Department FP promotion), evidence specificity concerns, circularity risk (LLM validating LLM), no expert review. Results exploratory.

Results.

Section 7.7.3, Table 7.19 (Chapter 7).

Reproducibility.

```
python final_runs/RQ3_deep_research_validation/compute_enriched_gt_metrics.py
```

D.9.0.5 Human Validation

RQ & Goal.

Validate DR verdicts via human review to establish applicability thresholds and enable extrapolation.

Source Data.

50 edges (29.6% of 169 eligible) via proportional stratified sampling across CLD × classification. Seed 42 for reproducibility.

Annotation Procedure.

Human reviewer assigns categorical verdict (`causal`, `different_vars`, `invalid`) and continuous applicability score $\in [0, 1]$ using the applicability rubric in Appendix Table C.31: 1.0 = exact match; 0.8–0.9 = different operationalisations; ≈0.7 = partial claim; ≈0.6 = proxy constructs; ≤0.5 = weak/non-causal evidence.

Simulation Parameters.

Full configuration in Appendix Table C.28.

Results.

Figure 7.13, Table 7.21, Table 7.20 (Chapter 7).

Reproducibility.

```
python final_runs/RQ3_deep_research_validation/scripts/sample_edges_for_validation.py
```

Inter-rater reliability check.

To assess rubric reliability, a second rater independently annotated an overlapping subset of 13 edges using the same rubric. Agreement for the categorical verdict was quantified using Cohen's κ (chance-corrected; interpretation per Landis & Koch as defined in Section D.6.2). Agreement for the continuous applicability score was quantified using the Intraclass Correlation Coefficient, ICC(2,1) (two-way random effects, absolute agreement, single measures) [118]. The ICC is defined as:

$$\text{ICC} = \frac{\sigma^2_{\text{between}}}{\sigma^2_{\text{between}} + \sigma^2_{\text{within}}}$$

where $\sigma^2_{\text{between}}$ is the variance between subjects (edges) and σ^2_{within} is the variance within subjects (across raters). Interpretation thresholds [118]: <0.50 poor, 0.50–0.75 moderate, 0.75–0.90 good, >0.90 excellent.
Reproducibility: `python final_runs/RQ3_deep_research_validation/validation/calculate_inter_rater_r`

D.9.0.6 Threshold Selection and Extrapolation

RQ & Goal.

Determine applicability threshold for filtering DR verdicts and extrapolate enrichment potential to full dataset.

Extrapolation Procedure.

Sensitivity analysis across $t \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$. For each threshold, compute coverage (pass rate) and correct-causal rate with Wilson 95% CI. Select threshold via gradient-based elbow rule on coverage–correctness curve.

Simulation Parameters.

Full configuration in Appendix Table C.29.

Metrics.

- **Coverage:** $x(t) = \Pr(\text{applicability} \geq t)$
- **Correct-causal rate:** $y(t) = \Pr(\texttt{verdict} = \texttt{causal} \mid \text{applicability} \geq t)$

Elbow Rule (Threshold Selection).

For each pair of adjacent thresholds (t_i, t_{i+1}) moving from stricter to looser (i.e., descending t), compute:

$$\text{score}_i = \frac{\Delta x}{|\Delta y|} = \frac{x(t_{i+1}) - x(t_i)}{|y(t_{i+1}) - y(t_i)|}$$

where $\Delta x > 0$ is the coverage gain and $|\Delta y|$ is the absolute drop in correct-causal rate. Select $t^* = t_{i+1}$ for the segment with maximum score. Intuition: choose the threshold where coverage gain per unit correctness loss is maximised (the "elbow" or diminishing-returns point).

Extrapolation.

Two-stage: (1) compute pass rate and causal rate from validated sample at t^*; (2) project to full dataset ($n = 285$) with conservative lower CI bounds. Compute projected F1 under Scenarios 1 and 2.

Caveats.

Small strata (e.g., Social Norms $n = 3$) have high sampling variance. Results reported with point estimates and conservative bounds.

Results.

Figure 7.13, Table 7.21, Table 7.22 (Chapter 7).

Reproducibility.

`python final_runs/RQ3_deep_research_validation/scripts/enrichment_extrapolation.py`

D.9.0.7 Smart Trigger

Estimate a cost-efficient DR deployment policy by triggering DR only on edges flagged as uncertain by the strongest GT Lit Correctness Judge (Mechanistic prompt). We compute an aggregate per-edge judge score (mean over runs) and mark edges with score ≤ 0.5 as triggered. We then evaluate the triggered subset using DR outputs and extrapolate expected ΔF1/\$ under human-calibrated applicability thresholds (see Results Figure 7.14 and Methods Section D.9.0.6). Full configuration in Appendix Table C.30.
Reproducibility: Scripts in `final_runs/RQ3_deep_research_validation/scripts/` (see Appendix C.2).

Appendix E

Statistical Procedures Reference

For transparency and reproducibility, we summarise the statistical hypotheses and assumption diagnostics associated with the procedures used throughout the thesis:

E.0.0.1 Omnibus and Post-Hoc Tests.

- **Friedman test** [143] **(blocked prompt effect; RQ1a/RQ1b):** omnibus test of prompt effects in a repeated-measures (blocked) design using ranks within blocks (CLD×run). H_0: prompt conditions have equal rank distributions within blocks. H_1: at least one prompt differs. Effect size: Kendall's W [144].
- **Wilcoxon signed-rank test** [145] **(paired post-hoc; RQ1a/RQ1b):** pairwise test on matched blocks. H_0: median$(\Delta) = 0$ for the paired prompt difference across blocks. H_1: median$(\Delta) \neq 0$.
- **Mann–Whitney U test** [146] **(RQ2 diagnostic):** compares two independent samples by rank. Used descriptively to characterise separation between hallucinated vs. correct edge distributions within CLDs. Effect size: rank-biserial r.
- **One-sample t-test (RQ2):** tests whether the mean of block-level AUCs differs from chance (0.5). H_0: $\mu_{\text{AUC}} = 0.5$. H_1: $\mu_{\text{AUC}} \neq 0.5$. Valid for $n = 9$ blocks given approximate normality of block means (Shapiro–Wilk $p > 0.05$).
- **Fisher's exact test** [147] **(RQ3):** tests independence in 2×2 contingency tables (e.g., detection rate TP vs. FP). H_0: rates are equal. H_1: rates differ. Effect size: Cohen's h.
- **Kruskal–Wallis test** [148] **(robustness check):** non-parametric one-way ANOVA on ranks. Used to verify robustness when parametric assumptions are violated. H_0: all groups have the same distribution. H_1: at least one group differs.

E.0.0.2 Parametric Sensitivity Analysis (Appendix A).

- **Repeated-measures ANOVA (RM-ANOVA):** parametric repeated-measures omnibus test used for sensitivity-style variance decomposition. H_0: equal within-subject means across prompts. H_1: at least one prompt mean differs. Effect size: η_p^2.
- **Shapiro–Wilk test** [121] **(residual normality):** diagnostic for approximate normality of RM-ANOVA residuals. H_0: residuals are normally distributed.
- **Mauchly's test** [149] **(sphericity, $k > 2$):** diagnostic for sphericity (equal variance of all pairwise condition differences). H_0: sphericity holds.
- **Greenhouse–Geisser correction** [150]**:** degrees-of-freedom adjustment using $\epsilon_{GG} \in (0, 1]$ when sphericity is violated; yields a more conservative p-value. This is a correction procedure (not a separate hypothesis test).

- **Levene's test** [122] **(homogeneity of variance):** diagnostic for equal variances across groups. H_0: variances are equal.

E.0.0.3 Multiple-Comparison and Interval Methods.

- **Bonferroni correction** [151, 152]**:** family-wise error control across a predefined family of tests, using $\alpha_{\text{adj}} = \alpha/m$ (or equivalently reporting adjusted p values).
- **Wilson score interval** [120] **(RQ3):** confidence interval for proportions that avoids overshoot near 0 or 1. Used for detection rates and human-validation accuracy.
- **Fisher z-transformation** [153] **(RQ2):** variance-stabilizing transform for Pearson r; used when aggregating correlations across blocks.

E.0.0.4 RQ2 Classifier Algorithms (Ensemble Models, Hyperparameters).

- **Shared implementation (RQ2 Phases 4–6):** classifiers are implemented in scikit-learn [133] and trained on standardised CI metrics using `StandardScaler` (fit on training blocks only). Splits are *group-aware* with `GroupShuffleSplit(test_size=0.33, random_state=42)` for the held-out test blocks and `GroupKFold(n_splits=3)` for block-level CV, with groups defined as `block_id = CLD×run` (9 blocks total).
- **Random Forest (RF)** [154] **(as implemented):** `RandomForestClassifier(n_estimators=100, class_weight='b random_state=42)`.
- **Gradient Boosting (GB)** [155] **(as implemented):** `GradientBoostingClassifier(n_estimators=100, random_state=42)`.
- **Logistic Regression (LR)** [156] **(as implemented):** `LogisticRegression(max_iter=1000, class_weight='bala random_state=42)`.
- **Neural Network (MLP)** [157] **(as implemented):** `MLPClassifier(hidden_layer_sizes=(32,16), max_iter=200 random_state=42)` trained via backpropagation.

Appendix F

Experimental Prompts

This section documents the prompts used in the final experiments. Full prompt YAML files are archived in `final_runs/prompts/` (see `final_runs/prompts/README.md`). Below we reproduce the key prompt texts verbatim.

F.1 Prompt File Locations

Correctness Judge prompts:

- `final_runs/prompts/RQ1a_judge/judge_correctness/prompts_correctness_baseline.yaml`
- `final_runs/prompts/RQ1a_judge/judge_correctness/prompts_correctness_cot.yaml`
- `final_runs/prompts/RQ1a_judge/judge_correctness/prompts_correctness_mechanistic.yaml`

Citation Judge prompts:

- `final_runs/prompts/RQ1a_judge/judge_citation/prompts_citation_baseline.yaml`
- `final_runs/prompts/RQ1a_judge/judge_citation/prompts_citation_cot.yaml`
- `final_runs/prompts/RQ1a_judge/judge_citation/prompts_citation_mechanistic.yaml`
- `final_runs/prompts/RQ1a_judge/judge_citation/prompts_citation_mechanistic_lit.yaml`

Corrector prompts:

- `final_runs/prompts/RQ1b_corrector/corrector/corrector_prompts.yaml`

Generator prompt:

- `final_runs/prompts/PRELIM_generator/final_generator/prompts_Nitai_C.yaml`

F.2 Full Prompt Text: Generator (Nitai_C)

The generator uses two prompts: `generateNodes` (node discovery) and `generateEdges` (edge discovery). Both employ chain-of-thought reasoning and few-shot examples.

F.2.1 generateNodes Prompt

System Prompt:

```
You are a systems dynamics modeling expert. Your role is to identify a single,
clearly defined, variable that causally influences a target variable within a
specific context, time, and space. The variable must be measurable, not
composite, and must not overlap with the target or any excluded variables.

Let's think step-by-step:
1. First, understand the target variable, its definition, and the context
2. Brainstorm potential causal variables that directly influence the target
3. Evaluate each potential variable against the criteria: measurable,
   non-composite, no overlap with target
4. Select the most appropriate variable with the clearest causal link
5. Verify the variable works within the specified temporal and spatial scale
6. Define the variable concisely and specify appropriate units

Once you've completed this reasoning process, respond only with a JSON including
the variable name and its concise definition, formatted as:
{"Variable":"Concise name", "Definition":"Definition", "Units":"Units"}
```

User Prompt Template:

```
Given the following constraints:
Target Variable: {{ target.title() }} (measured in {{ target_units }})
Temporal Scale: {{ temporal.title() }}
Spatial Scale: {{ spatial.title() }}
Context: {{ context | join(', ') }}

I need you to think carefully through this problem:

1. First, understand what {{ target.lower() }} means: "{{ target_definition }}"
2. Consider what directly causes changes in {{ target.lower() }} within the
   {{ temporal.lower() }} timeframe and {{ spatial.lower() }} spatial scale
3. Evaluate potential variables against these criteria:
   - Must be directly measurable (not abstract)
   - Must not be composite (not combining multiple factors)
   - Must have a clear causal effect on {{ target.lower() }}
   - Must work within the specified temporal and spatial scales

After this analysis, name a variable (2-3 words) with a direct causal effect.
Output format: {"Variable":"Name", "Definition":"Definition", "Units":"Units"}
```

F.2.2 generateEdges Prompt

System Prompt:

```
You are a scientific system dynamics model expert specializing in causal loop
diagrams. Given two variables and their definitions, your task is to determine
the precise causal relationship between them.

Approach this systematically:
```

```
1. First, clearly understand what each variable means and how it's measured
2. Consider the mechanisms by which the first variable might directly affect
   the second
3. Evaluate whether this mechanism works within the given spatial and temporal
   scales
4. Check for potential mediating variables that might explain the relationship
5. Distinguish between correlation and causation - focus only on true causal
   relationships
6. Determine if the relationship is positive, negative, or non-existent
7. Verify your reasoning by considering hypothetical increases and decreases
8. Provide a concise explanation that justifies your conclusion

Select the most appropriate relationship type based on causal logic, not mere
correlation. Respond with: {"Relationship":"TYPE","Explanation":"Brief text."}
```

User Prompt Template:

```
I need to determine if there is a causal relationship between "{{ var1 }}" and
"{{ var2 }}" for a Causal Loop Diagram (CLD). Let me think step by step:

Step 1: Understand the variables
- Variable 1: "{{ var1 }}" defined as "{{ def1 }}"
- Variable 2: "{{ var2 }}" defined as "{{ def2 }}"
- Scale: Spatial = "{{ spatial }}", Temporal = "{{ temporal }}"

Step 2: Consider potential causal mechanisms
- Does a change in "{{ var1 }}" directly cause a change in "{{ var2 }}"?
- Does this mechanism operate within the given spatial and temporal scales?

Step 3: Check for mediating variables
- Are there variables in the CLD that mediate this relationship?
- If a mediator exists, the direct relationship should not be included

Step 4: Determine relationship type
1. POSITIVE - increase in var1 causes increase in var2 (or decrease->decrease)
2. NEGATIVE - increase in var1 causes decrease in var2 (or decrease->increase)
3. NONE - No causal relationship exists

Example Responses:
1. {"Relationship":"POSITIVE","Explanation":"Force causes acceleration..."}
2. {"Relationship":"NEGATIVE","Explanation":"Resistance causes current to..."}
3. {"Relationship":"NONE","Explanation":"No direct causal link exists..."}

IMPORTANT: In case you don't know, respond with
{"Relationship":"NONE","Explanation":"I don't know"}
```

F.3 Full Prompt Text: Correctness Judge

The correctness judge evaluates causal explanations for logical soundness. Three variants were tested.

F.3.1 Baseline Variant

System Prompt:

```
You are a precise and fair judge evaluating the quality of causal reasoning.
Your job is to carefully read the causal explanation and determine if it is
logically sound and scientifically coherent. Base your judgment only on the
internal logic of the explanation, not on external knowledge or citations.
```

User Prompt Template:

```
Evaluate the quality of this causal explanation:

SOURCE VARIABLE: {{ source }}
TARGET VARIABLE: {{ target }}
RELATIONSHIP TYPE: {{ relationship }}
EXPLANATION: {{ motivation }}

Based on the explanation, assess whether the causal reasoning is sound.

Respond in this format:
REASON: [brief explanation of your judgment]
VERDICT: [CORRECT|PARTIALLY_CORRECT|INCORRECT]
SCORE: [1.0 for CORRECT, 0.5 for PARTIALLY_CORRECT, 0.0 for INCORRECT]
```

F.3.2 Chain-of-Thought (CoT) Variant

Based on Kojima et al. (2022) [60]. Same system prompt as Baseline. User prompt adds the trigger phrase:

```
Let's think step by step to assess whether the causal reasoning is sound.

Respond in this format:
REASON: [Start with your step-by-step chain of thought, then provide your
        final judgment. Make your reasoning process explicit.]
VERDICT: [CORRECT|PARTIALLY_CORRECT|INCORRECT]
SCORE: [1.0 for CORRECT, 0.5 for PARTIALLY_CORRECT, 0.0 for INCORRECT]
```

F.3.3 Mechanistic Variant (Bradford Hill Criteria)

Based on Shimonovich et al. (2020) [68], adapting four Bradford Hill criteria for qualitative CLDs: (1) Plausibility, (2) Temporality, (3) Strength, (4) Coherence.

System Prompt:

```
You are a precise and fair judge evaluating causal explanations using core
principles from the Bradford Hill causal criteria framework.

You will assess explanations on FOUR key dimensions adapted for qualitative
causal models:

1. PLAUSIBILITY - Does it describe a plausible causal mechanism?
2. TEMPORALITY - Is the causal direction justified and temporally coherent?
3. STRENGTH - Does it convey that this is an important (not trivial) causal
   relationship?
4. COHERENCE - Is the reasoning internally consistent and non-circular?
```

```
These criteria align with Bradford Hill's framework (Shimonovich et al., 2020)
while being practical for conceptual models. Base your judgment only on the
internal logic of the explanation, not on external knowledge.
```

User Prompt Template:

```
Evaluate this causal explanation using four core Bradford Hill criteria:

CLAIMED CAUSAL PATHWAY:
{{ source }} -> [{{ relationship }}] -> {{ target }}

PROPOSED EXPLANATION:
{{ motivation }}

Assess the explanation on these four dimensions:

1. PLAUSIBILITY (Causal Mechanism):
  - Does the explanation describe HOW {{ source }} affects {{ target }}?
  - Is a specific mechanism or pathway described (not just assertion)?
  - Is the mechanism scientifically plausible and non-circular?

2. TEMPORALITY (Causal Direction):
  - Does the explanation support the direction {{ source }} -> {{ target }}?
  - Is temporal ordering appropriate (cause precedes effect)?
  - Does it avoid reverse causation?

3. STRENGTH (Relative Importance):
  - Does the explanation convey that {{ source }} is an IMPORTANT influence?
  - Does it suggest this is more than trivial or merely correlational?
  - Is the causal claim stated with appropriate confidence?

4. COHERENCE (Internal Consistency):
  - Is the reasoning logically consistent and non-contradictory?
  - Does it avoid tautological or circular arguments?
  - Are the claims coherent with each other?

Respond in this format:
VERDICT: [CORRECT|PARTIALLY_CORRECT|INCORRECT]
SCORE: [1.0 for CORRECT, 0.5 for PARTIALLY_CORRECT, 0.0 for INCORRECT]
REASON: [2-3 sentence explanation referencing which criteria were met/failed]

Guidelines:
- CORRECT: Addresses all four criteria reasonably well
- PARTIALLY_CORRECT: Addresses 2-3 criteria well but has notable gaps
- INCORRECT: Fails most criteria (vague, circular, wrong direction, etc.)
```

F.4 Full Prompt Text: Citation Judge

The citation judge evaluates whether causal claims are supported by their cited sources. Four variants were tested.

F.4.1 Baseline Variant

System Prompt:

```
You are a precise and fair judge evaluating whether scientific claims are
supported by their cited sources. Your job is to carefully read the
relationship motivation and the cited sources, and determine if the motivation
is supported. Base your judgment only on the content of the citations provided,
not on general knowledge.
```

User Prompt Template:

```
Verify if this relationship is supported by its citations:

SOURCE VARIABLE: {{ source }}
TARGET VARIABLE: {{ target }}
RELATIONSHIP TYPE: {{ relationship }}
MOTIVATION: {{ motivation }}

Here are the contents of the cited sources:
{{ citations }}

Based on the cited sources, evaluate if the relationship motivation is
supported by the citations.

Respond in this format:
VERDICT: [SUPPORTED|PARTIALLY_SUPPORTED|CONTRADICTED|UNADDRESSED]
REASON: [brief explanation of your judgment]
```

F.4.2 Chain-of-Thought (CoT) Variant

Based on Kojima et al. (2022) [60]. Same system prompt as Baseline. User prompt adds the CoT trigger:

```
Evaluate whether the citations support this relationship claim:

SOURCE VARIABLE: {{ source }}
TARGET VARIABLE: {{ target }}
RELATIONSHIP TYPE: {{ relationship }}
EXPLANATION: {{ motivation }}

CITED SOURCES:
{{ citations }}

Let's think step by step to assess whether the explanation is adequately
supported by the citations.

Respond in this format:
REASON: [brief explanation of your judgment]
VERDICT: [SUPPORTED|PARTIALLY_SUPPORTED|NOT_SUPPORTED]
SCORE: [1.0 for SUPPORTED, 0.5 for PARTIALLY_SUPPORTED, 0.0 for NOT_SUPPORTED]
```

F.4.3 Mechanistic Variant (Bradford Hill)

Based on Shimonovich et al. (2020) [68]. Evaluates citations against Bradford Hill criteria.

System Prompt:

```
You are a precise and fair judge evaluating cited evidence using the Bradford
Hill framework.

You will assess whether citations provide evidence for four key causality
criteria identified as having strong support across modern causal inference:

1. STRENGTH OF ASSOCIATION - including confounding analysis
2. TEMPORALITY - temporal precedence and ruling out reverse causation
3. PLAUSIBILITY - causal mechanisms and mediation pathways
4. EXPERIMENT/DESIGN - study design implications for exchangeability

Base your judgment only on what the citations actually say, not on general
knowledge.
```

User Prompt Template:

```
Evaluate the cited evidence for this causal relationship using the Bradford
Hill framework:

CLAIMED CAUSAL PATHWAY:
{{ source }} -> [{{ relationship }}] -> {{ target }}

PROPOSED MECHANISM:
{{ motivation }}

CITED SOURCES:
{{ citations }}

Let's think step by step. Assess whether the citations provide evidence for:

1. PLAUSIBILITY (Causal Mechanism):
   - Does the explanation describe HOW {{ source }} affects {{ target }}?
   - Is a specific mechanism or pathway described (not just assertion)?

2. TEMPORALITY (Causal Direction):
   - Does the explanation support the direction {{ source }} -> {{ target }}?
   - Does it avoid reverse causation?

3. STRENGTH (Relative Importance):
   - Does the explanation convey that {{ source }} is an IMPORTANT influence?
   - Is this more than trivial or merely correlational?

4. COHERENCE (Internal Consistency):
   - Is the reasoning logically consistent and non-contradictory?
   - Does it avoid tautological or circular arguments?

Respond in this format:
REASON: [brief explanation of your judgment]
VERDICT: [SUPPORTED|PARTIALLY_SUPPORTED|NOT_SUPPORTED]
SCORE: [1.0 for SUPPORTED, 0.5 for PARTIALLY_SUPPORTED, 0.0 for NOT_SUPPORTED]
```

F.4.4 Mechanistic-Literature Variant

Extends Mechanistic variant with explicit literature grounding requirements. Adds experimental evidence criterion:

```
4. EXPERIMENT/DESIGN (Implications for Exchangeability):
   - Do citations include experimental or quasi-experimental studies?
   - Do they discuss study designs that enhance causal inference (RCTs,
     natural experiments)?
   - Do they address comparability between exposed and unexposed groups?
   - Do they consider or control for factors affecting exchangeability?
```

This variant requires citations to provide evidence of experimental design considerations beyond mere mechanism description, emphasizing literature-grounded causal inference.

F.5 Full Prompt Text: Corrector

The corrector uses two tools: `revise_motivation` (fix explanation, keep edge type) and `change_edge_type` (change polarity or remove edge). Three variants were tested.

F.5.1 Baseline Variant

```
You are an expert causal reasoning corrector. You have 2 tools:

1. revise_motivation: Fix incorrect motivation text (keeps same edge type)
2. change_edge_type: Change edge type (POSITIVE/NEGATIVE/NONE) with new
   motivation

IMPORTANT: NONE represents "no causal relationship" - it's a valid type, not
deletion.

TOOL SELECTION EXAMPLES:

EXAMPLE 1: Bad motivation, correct type -> revise_motivation
Edge: "Exercise" -> "Mental Health" (POSITIVE)
Motivation: "Exercise makes people happy"
Judge: "Overly simplistic; doesn't explain mechanism"
ACTION: revise_motivation(new_motivation="Exercise increases endorphin
production and reduces cortisol, improving mental health outcomes")

EXAMPLE 2: Wrong polarity -> change_edge_type
Edge: "Vaccination Rate" -> "Disease Spread" (POSITIVE)
Motivation: "More vaccinations lead to more disease spread"
Judge: "Wrong polarity - vaccinations reduce disease spread"
ACTION: change_edge_type(old_type='POSITIVE', new_type='NEGATIVE',
new_motivation="Higher vaccination rates create herd immunity, reducing
disease transmission")

EXAMPLE 3: Should be NONE -> change_edge_type
Edge: "Shoe Size" -> "Reading Ability" (POSITIVE)
```

```
Motivation: "Larger shoe size indicates better reading ability"
Judge: "No causal relationship - confounded by age"
ACTION: change_edge_type(old_type='POSITIVE', new_type='NONE',
new_motivation="No direct causal link between shoe size and reading. The
correlation is due to age as a confounding variable")

DECISION RULES:
revise_motivation: Edge type is correct, but explanation is flawed
change_edge_type: Edge type itself is wrong (polarity, existence)

Analyze the judge feedback and call the appropriate tool.
```

F.5.2 CoT Variant

Adds a 5-step reasoning framework after the examples:

```
Let's think step by step to determine the best correction action.

STEP 1: Analyze the Judge Feedback
- What specific problem did the judge identify?
- Is it about: (a) polarity/sign, (b) mechanism/reasoning, (c) type, or
  (d) existence?

STEP 2: Assess the True Causal Relationship
- What is the CORRECT causal relationship between these variables?
- Should it be: POSITIVE, NEGATIVE, or NONE (no causal relationship)?

STEP 3: Identify the Minimal Correction
- If motivation is flawed but type is correct -> Use revise_motivation
- If edge type itself is wrong -> Use change_edge_type

STEP 4: Verify Your Decision
- Does this action address the specific judge feedback?
- Could revise_motivation work instead of change_edge_type?
- If changing to NONE, can you explain WHY no causal relationship exists?

STEP 5: Execute the Tool Call
Call the appropriate tool with your validated reasoning.
```

F.5.3 Mechanistic Variant (Bradford Hill)

Extends CoT with Bradford Hill criteria:

```
Let's think step by step, applying Bradford Hill criteria for causal assessment.

STEP 1: Analyze Judge Feedback + Assess PLAUSIBILITY (Causal Mechanism)
- What specific problem did the judge identify?
- Does a plausible causal MECHANISM exist between source and target?
- Is the current motivation describing the wrong mechanism, or is there NO
  mechanism?

STEP 2: Evaluate TEMPORALITY (Causal Direction)
```

```
- What is the CORRECT causal relationship: POSITIVE, NEGATIVE, or NONE?
- Does the causal direction make sense (source must precede target)?

STEP 3: Assess STRENGTH (Relative Importance)
- Is there a SUBSTANTIAL causal relationship, or is it trivial/negligible?
- If motivation is flawed but strong causal relationship exists
  -> revise_motivation

STEP 4: Evaluate COHERENCE (Internal Consistency) + Verify Decision
- Is the reasoning logically consistent and non-contradictory?
- Does judge feedback suggest type is wrong, or just explanation needs fixing?

STEP 5: Execute Tool Call
Based on Bradford Hill assessment:
- IF Plausibility+Temporality+Strength satisfied, but Coherence fails
  -> revise_motivation
- IF Temporality shows wrong direction -> change_edge_type (flip polarity)
- IF Plausibility or Strength fail -> change_edge_type (to NONE)
```

F.6 Prompt Engineering Ablation Study

Prompt engineering has emerged as a critical technique for optimizing large language model (LLM) performance on domain-specific tasks. To systematically evaluate the effectiveness of various prompting strategies for automated causal loop diagram generation, we conducted an ablation study comparing nine prompt variants across three causal loop diagrams.

F.6.1 Experimental Design

We evaluated nine prompt variants across three ground-truth causal loop diagrams:

- **CLD 1**: Depressive Symptoms in response to stressors (34 edges)
- **CLD 2**: Social Norms and obesity prevalence (12 edges)
- **CLD 3**: Emergency Department (66 edges)

All experiments used GPT-4.1 as the generator model with consistent hyperparameters (temperature=0.7, top_p=1.0). Performance was measured using the F1 score, which balances precision and recall in edge discovery.

F.6.2 Prompt Engineering Techniques

Table F.1 presents a classification matrix of prompt engineering techniques employed in each variant, based on established literature.

Table F.1: Prompt Engineering Techniques Classification Matrix

Prompt	CoT[1]	Few-Shot[2]	Role[3]	Decomp[4]	RAG[5]
Nitai_A		✓	✓		
Nitai_B	✓	✓	✓	✓	
Nitai_C	✓✓	✓✓	✓	✓	
Nitai_E	✓✓	✓	✓	✓	✓✓
Rick_A		✓	✓✓		
Rick_B	✓	✓✓	✓✓	✓✓	
Cillian_B		✓	✓	✓	
Cillian_C		✓	✓	✓✓	
Current		✓			

[1]Chain-of-Thought [15]; [2]Few-Shot Learning [16]; [3]Role Prompting; [4]Task Decomposition; [5]RAG with mandatory citations.
Legend: ✓= basic; ✓✓= strong implementation

Key Prompt Distinctions:

- **Nitai variants**: Progressive complexity from baseline (A) to Chain-of-Thought reasoning (B, C), culminating in RAG-augmented prompting (E). In the final experiments, Nitai_C uses three relationship types (`POSITIVE`, `NEGATIVE`, `NONE`).
- **Rick variants**: Emphasis on academic rigor and system dynamics best practices. Rick_B features comprehensive systematic decomposition with explicit confounding analysis.
- **Cillian variants**: Cillian_B employs intervention-based framing. Cillian_C uses hypothesis testing with definition-strict evaluation.
- **Current**: Minimal baseline control with basic prompting.

F.6.3 Results

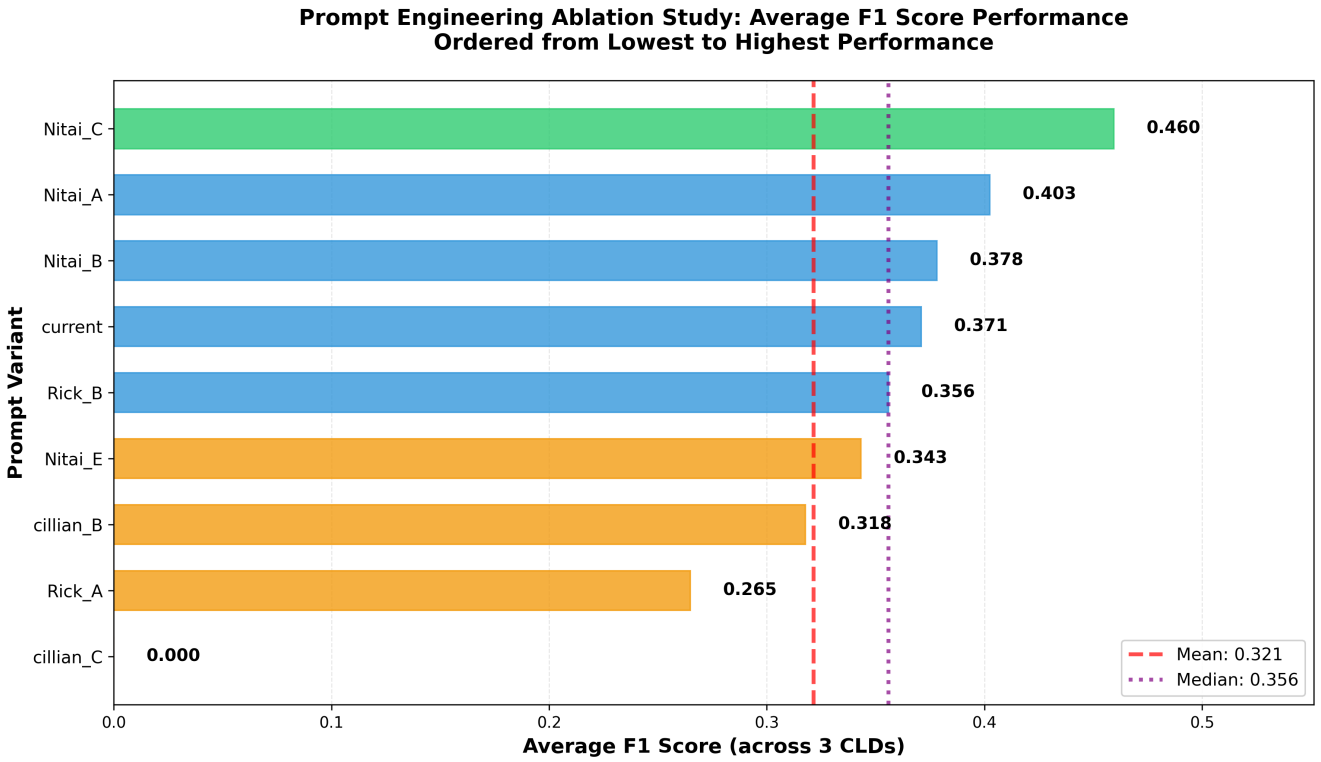


Figure F.1: Average F1 scores for prompt variants across three CLDs, ordered from lowest to highest. Nitai_C achieved highest performance (F1=0.460). Colour coding: green (F1 $\geq$ 0.45), blue (0.35–0.45), orange (0.25–0.35), red (<0.25).

Table F.2: F1 Scores by Prompt Variant and Causal Loop Diagram

Prompt	CLD 1	CLD 2	CLD 3	Average
Nitai_C	0.591	0.204	0.583	**0.460**
Nitai_A	0.590	0.218	0.400	0.403
Nitai_B	0.588	0.092	0.455	0.378
Current	0.588	0.239	0.286	0.371
Rick_B	0.444	0.186	0.438	0.356
Nitai_E	0.438	0.207	0.385	0.343
Cillian_B	0.515	0.152	0.286	0.318
Rick_A	0.380	0.145	0.269	0.265
Cillian_C	0.000	0.000	0.000	0.000

F.6.4 Key Findings

Overall Performance:

- **Best Performance**: Nitai_C achieved the highest average F1 score (0.460), differing from others in multiple dimensions (dual-level CoT, relationship taxonomy, examples).

- **Performance Range**: Range of 0.195 between best (0.460) and second-worst (0.265) performers. Mean F1 = 0.321, SD = 0.132.

CLD-Specific Patterns:

- **CLD 2 Challenge**: All variants performed worst on CLD 2 (Social Norms), with F1 scores 0.00–0.24.
- **High Variance**: Within-prompt SD across CLDs ranged 0.18–0.25, indicating sensitivity to problem characteristics.

Technique-Specific Observations:

- **RAG Trade-off**: Nitai_E (mandatory citations) showed 25% lower performance than Nitai_C, suggesting strict citation requirements limit valid relationship discovery.
- **Complete Failure**: Cillian_C's zero performance indicates overly restrictive definition-grounded evaluation can be counterproductive.

F.6.5 Interpretation

The progression Nitai_A → Nitai_B → Nitai_C demonstrates incremental performance gains with Chain-of-Thought, achieving 14% improvement over baseline. This supports Wei et al.'s [15] findings that explicit reasoning steps enhance complex task performance.

Rick_B (F1=0.356), employing systematic decomposition, underperformed Nitai_C's CoT approach by 23%, suggesting decomposition strategies may be less effective when tasks require holistic pattern recognition rather than sequential problem solving.

F.6.6 Limitations

- **Sample Size**: Only three CLDs limits generalizability.
- **Single Run**: Each variant tested once per CLD, preventing statistical significance testing.
- **Model Specificity**: Results apply to GPT-4.1; patterns may differ for other LLMs.
- **Prompt Authorship**: Different variants designed by different researchers, introducing potential confounding.

F.7 Generator Design Rationale

This section motivates the generator's decomposition into edge-level decisions and the accompanying safeguards used in our experiments.

In our CLD generation system, given temporal and spatial scales and domain context, we generate a set of nodes (or, in the main experiments, insert validated nodes by fixing the node set to the ground-truth validation variables V), and then generate relationships pairwise as directed queries over ordered variable pairs (A, B) and (B, A) (i.e., $|V|(|V|-1)$ candidate edges). For each directed pair, the LLM is asked whether A *directly* causes B and to return a relationship type and a short causal narrative; in this study, relationship types are constrained to {`POSITIVE`, `NEGATIVE`, `NONE`} (Section F.8, Listing F.1). To address the limitation of reviewing links in isolation, we perform a mediation check by passing the current node set V to the edge prompt as candidate

mediators. When supported by the LLM API, we also capture token-level logprobs for the edge-generation response (relationship label + narrative) and derive logprob-based uncertainty features (LUQ metrics) used in downstream hallucination detection. Algorithmic details are traced from the implementation in Section I (Algorithm I.1). Relative to Crielaard et al.'s CLD-to-SDM pipeline (Section 2.3; Crielaard et al. [5]), this operationalises the *causal link identification* step and approximates *intermediary-variable documentation* via mediation checks over the current node set, while deferring richer aCLD elements such as functional forms and interaction terms to future work (i.e., these are not addressed in this thesis). Our downstream phases address evidence-quality annotation through post-hoc edge validation and remediation (LLM-as-a-Judge and LLM-as-a-Corrector) and, when allocating additional test-time compute, external evidence acquisition via Deep Research (RQ3).

F.8 Generator Prompt (Nitai_C)

The final experiments use the **Nitai_C** generator prompt configuration for both node generation (`generateNodes`) and edge generation (`generateEdges`). The prompt is stored at:

`data_science/parameter_tuning_experiments/prompts_Nitai_C.yaml`

and is archived (as a copied file) at:

`final_runs/prompts/PRELIM_generator/final_generator/prompts_Nitai_C.yaml`

(see `final_runs/prompts/README.md` for IDs and checksums).

```
PROMPT ARCHIVE NOTE:
- Full prompt text is archived as a copied YAML at:
  final_runs/prompts/PRELIM_generator/final_generator/prompts_Nitai_C.yaml
- See: final_runs/prompts/README.md
```

Listing F.1: Generator Prompt Configuration: Nitai_C (`generateNodes` + `generateEdges`)

Three prompting strategies were evaluated for each component:

1. **Baseline**: Direct instruction without reasoning scaffolding (control condition)
2. **Chain-of-Thought (CoT)**: Zero-shot CoT via "Let's think step by step" [60]
3. **Mechanistic**: Domain-specific criteria based on the Bradford Hill causal framework [68]

For citation judges, an additional **Mechanistic-Lit** variant combines mechanistic evaluation with explicit literature grounding requirements.

F.9 Edge Generation Ablation Prompts

The full text of all nine edge-generation prompt variants used in the ablation study (Section F.6) is archived (as copies) under:

`final_runs/prompts/PRELIM_generator/edge_ablation/`

See `final_runs/prompts/README.md` for stable IDs and file checksums.

(Prompts are not reproduced verbatim here to minimize appendix length.)

Appendix G

Node Generation Prompt Ablation

Seven prompt variants were evaluated in a full factorial design: 7 prompts × 3 CLDs × 5 runs = 105 total experiments.

Table G.1: Node Generation Ablation: Hybrid F1 Scores by Prompt Variant

Prompt	Mean F1	Std	95% CI	Cohen's d (vs. baseline)	N
Node_V6_Andrew	0.677	0.079	[0.638, 0.717]	+0.44	15
Node_V3_CoT_System	0.657	0.123	[0.595, 0.719]	+0.16	15
Node_V0_Minimal (baseline)	0.639	0.094	[0.592, 0.687]	—	15
Node_V5_Nitai_C	0.636	0.088	[0.591, 0.680]	-0.04	15
Node_V1_Role_Only	0.628	0.080	[0.588, 0.669]	-0.13	15
Node_V4_CoT_User	0.619	0.104	[0.567, 0.672]	-0.20	15
Node_V2_Constraints	0.617	0.077	[0.578, 0.656]	-0.26	15

Statistical Analysis:

- Node ablation analysis: No significant differences detected

Interpretation: Unlike edge generation, node generation is robust to prompt engineering variations. The minimal baseline performed comparably to sophisticated Chain-of-Thought variants, suggesting that node identification is a simpler task for LLMs that does not benefit substantially from advanced prompting strategies.

G.1 Node Generation Prompts

The full text of all seven node-generation prompt variants used in the node ablation study (Section G) is archived (as copies) under:

`final_runs/prompts/PRELIM_generator/node_ablation/`

See `final_runs/prompts/README.md` for stable IDs and file checksums.

(Prompts are not reproduced verbatim here to minimize appendix length.)

Appendix H

Embedding Model Benchmark

To justify the choice of embedding model for cosine similarity computation (Section 2.5), we benchmarked `all-mpnet-base-v2` against the MTEB-leading Qwen3-Embedding-8B using 30 real causal narratives from our CLD experiments.

Table H.1: Embedding Model Benchmark ($n = 30$ causal narratives from CLD)

Model	Params	Embed dim	ms/doc	docs/s
all-mpnet-base-v2	110M	768	3.0	336
Qwen3-Embedding-8B (Q4)	8B	4096	67.3	14.9

Compute setup: Hardware details are reported in Appendix C.3 (Table C.9). Qwen3 was run via Ollama with 4-bit quantization (Q4_K_M).

Test data: Causal narratives averaged 451 characters (∼100 tokens). In production, fetched citation documents are chunked up to 4,000 characters (∼1,000 tokens) before embedding—longer inputs may increase per-document latency.

MTEB Benchmark Comparison:

Table H.2: MTEB Scores: MPNet vs. Qwen3-Embedding-8B

Task	MPNet (110M)[a]	Qwen3-8B[b]
STS (Spearman ρ)	80.28	**88.58**
Reranking (MAP)	**59.36**	51.56
Pair Classification (AP)	83.04	**87.52**
Classification (Acc)	65.07	**90.43**
Clustering (V-measure)	43.69	**58.57**
Retrieval (nDCG@10)	43.81	**69.44**
Summarization	27.49	**34.83**
Average	57.78	**75.22**

[a]Muennighoff et al. [101], Table 1 (MTEB v1, 2022). [b]Zhang et al. [102], Table 7 (MTEB eng v2, 2025).

Result: MPNet achieves 22× faster inference (3.0 vs. 67.3 ms/doc; Table H.1). Since cosine-similarity scoring is used as a throughput-bound heuristic for max-similarity chunk selection, we prioritise latency; MPNet is a strong general-purpose embedding model and is widely used in practice.

Note: The MTEB scores are taken from different benchmark versions (v1 vs. v2) and evaluation setups and are therefore **not directly comparable**. We include them only to illustrate that both models are competitive across common embedding tasks.

Script: `final_runs/supp_embedding_benchmark/embedding_benchmark_real_cld.py`

Appendix I

Algorithmic Descriptions

This section provides formal algorithmic descriptions of the core system components, traced directly from the implementation. For experimental design rationale, see the corresponding Methods sections.

I.1 Algorithm 0: CLD Generation

The CLD generator constructs causal loop diagrams through a two-phase process: node generation followed by edge generation. Both phases capture UQ metrics via logprobs for downstream hallucination detection.

Table I.1: CLD Generation (`generate_variables` + `discover_relationships`)

Input: target_variable, temporal_scale, spatial_scale, num_variables, context
Output: Neo4j session with nodes (variables) and edges (relationships) + UQ metrics

// Phase 1: Node Generation (generateNodes prompt)
session_id ← uuid.generate()
Neo4j.create_node(target_variable, is_target=True, session_id)
variables ← [target_variable]
while len(variables) < num_variables:
  vars ← {target, target_units, target_definition, temporal, spatial, context}
  (sys_prompt, usr_prompt) ← tree.build_prompt("generateNodes", vars)
  usr_prompt ← usr_prompt + "Exclude: " + join(variables)
  response ← generator_llm.send_message(prompts, logprobs=True, top_logprobs=5)
  (name, definition, units, citations) ← tree.read_variable_response(response)
  log_metrics ← compute_uq_metrics(response.logprobs)
  Neo4j.create_node(name, definition, citations, log_metrics, session_id)
  variables.append(name)

// Phase 2: Edge Generation (generateEdges prompt)
pairs ← generate_ordered_pairs(variables) *// N^2 pairs, excludes self-loops*
for each (source, target) ∈ pairs:
  (source_def, target_def) ← Neo4j.fetch_definitions(source, target)
  vars ← {var1: source, var2: target, def1, def2, temporal, spatial, variables}
  (sys_prompt, usr_prompt) ← tree.build_prompt("generateEdges", vars)
  response ← generator_llm.send_message(prompts, logprobs=True, top_logprobs=5)
  (rel_type, motivation, citations) ← tree.read_relationships(response)
  log_metrics ← compute_ci_metrics(response.logprobs)
  Neo4j.create_relationship(source, target, rel_type, motivation, citations, log_metrics)
return session_id, variables, relationships

Implementation: `modules.py:698` (`generate_variables`), `:1147` (`discover_relationships`), `:908` (`process_variable_pair`)

Acknowledgement: The CLD generation algorithm was conceived by Rick Quax with feedback from Cillian Hourican and implemented by Andrew Crossley. The author (Nitai Nijholt) contributed the mediation check, multi-provider LLM infrastructure with rate limiting and fallbacks, and prompt testing framework.

I.2 Algorithm 1: Judge-Correctness

The correctness-based judge evaluates causal reasoning quality without external evidence.

Table I.2: Judge-Correctness (`judge_all_edges_serial`, approach="correctness")

Input: session_id, judge_models[], prompt_config
Output: Per-edge verdicts with scores

```
edges ← Neo4j.query("MATCH (s)-[r]-¿(t) WHERE s.session_id = $sid")
judges[] ← create_llm_clients(judge_models)
for each edge ∈ edges:
    text ← edge.spurious_motivation ∨ edge.motivation
    if text = ∅: verdict ← "NO_MOTIVATION"; continue
    variables ← {source, target, relationship, motivation: text}
    (sys_prompt, usr_prompt) ← tree.build_prompt("judgeCorrectness", vars)
    scores[], verdicts[] ← [], []
    for each judge ∈ judges:
        response ← judge.send_message(sys_prompt, usr_prompt, logprobs=True)
        (verdict, score, reason) ← parse_regex(response, "VERDICT: SCORE:")
        verdicts.append(verdict); scores.append(score)
    aggregate_score ← mean(scores)
    aggregate_verdict ← majority_vote(verdicts)
    update_edge(edge, aggregate_verdict, aggregate_score)
return judged_edges
```

Implementation: `modules.py:4421` → `:4740` (correctness branch)

I.3 Algorithm 2: Judge-Citation

The citation-based judge validates causal claims against fetched scientific literature.

Table I.3: Judge-Citation (`judge_all_edges_with_citations_serial`)

Input: session_id, judge_models[], citation_fetcher
Output: Per-edge aggregate verdicts with per-citation scores

```
edges ← Neo4j.query("MATCH (s)-[r]-¿(t) WHERE s.session_id = $sid")
judges[] ← create_llm_clients(judge_models)
for each edge ∈ edges:
  citations ← edge.citations
  if citations = ∅: verdict ← "NO_CITATION"; continue
  citation_contents ← ContentScraper.process_citations(citations)
  per_citation_results ← []
  for each (url, content) ∈ citation_contents:
    prompt ← build_citation_prompt(edge.motivation, content)
    judge_verdicts ← []
    for each judge ∈ judges:
      response ← judge.send_message(prompt)
      verdict ← parse(response) // SUPPORTED/PARTIAL/CONTRADICTED
      judge_verdicts.append(verdict)
    citation_verdict ← majority_vote(judge_verdicts)
    citation_score ← verdict_to_score(citation_verdict)        // 1.0, 0.5, 0.0
    per_citation_results.append({url, citation_verdict, citation_score})
  aggregate_score ← mean(per_citation_results.scores)
  aggregate_verdict ← aggregator_llm(per_citation_results)
  update_edge(edge, aggregate_verdict, aggregate_score)
return judged_edges
```

Implementation: `modules.py:5688` (`judge_all_edges_with_citations_serial`) → `modules.py:4988`

I.4 Algorithm 3: Corruption

The corruptor introduces controlled errors using stratified sampling for balanced corruption types.

Parameters used in this thesis (GT Synth). Unless otherwise stated, corruption experiments use `corruption_rate=0.3` with a balanced 1/3 split across corruption types (motivation corruption, spurious edges, polarity flips). The corruptor LLM configuration is `model=gpt-4.1`, `temperature=0.7`, `top_p=1.0`; corruption is seeded for reproducibility (see `corruption_metadata.json` in the corresponding `final_runs/` output directory).

Table I.4: Corruption (`_corrupt_relationships`)

Input: session_id, corruption_rate (default 0.3), seed, corruptor_llm
Output: Number of corrupted edges, corruption metadata

```
random.seed(seed)                                   // Reproducibility
edges ← Neo4j.query("MATCH (s)-[r]-¿(t)")
none_edges ← {e ∈ edges : e.type = "NONE"}
causal_edges ← {e ∈ edges : e.type ∈ {"POSITIVE", "NEGATIVE"}}
n_corrupt ← ⌊len(edges) × corruption_rate⌋
target_per_type ← n_corrupt ÷ 3                     // Balanced: 1/3 each
spurious_sample ← shuffle(none_edges)[:target_per_type]
causal_sample ← shuffle(causal_edges)[:target_per_type × 2]
for each edge ∈ spurious_sample ∪ causal_sample:
  if edge.type = "NONE": corruption_type ← "spurious_edge"
  else: corruption_type ← argmin(assigned["polarity_flip"], assigned["motivation"])
  assigned[corruption_type] += 1
  if corruption_type = "motivation":
    prompt ← tree.build_prompt("corruptor", edge)
    edge.spurious_motivation ← corruptor_llm.send(prompt)
  elif corruption_type = "spurious_edge":
    prompt ← tree.build_prompt("generateSpuriousEdge", edge)
    {type, expl} ← JSON.parse(corruptor_llm.send(prompt))
    Neo4j: DELETE NONE, CREATE edge.type ← type, motivation ← expl
  elif corruption_type = "polarity_flip":
    prompt ← tree.build_prompt("generateFlippedPolarity", edge)
    {expl} ← JSON.parse(corruptor_llm.send(prompt))
    Neo4j: DELETE old, CREATE edge.type ← flip(type), motivation ← expl
  edge.is_corrupted ← True
save_metadata("corrupted_relationships_{session_id}.json")
return corrupted_count
```

Implementation: `modules.py:1748` (`_corrupt_relationships`), `:1544` (`_corrupt_relationship`), `:1582` (`_convert_none_to_spurious`), `:1645` (`_flip_edge_polarity`)

I.5 Algorithm 4: Corrector

The corrector applies LLM-guided fixes to edges flagged by the judge.

Table I.5: Corrector (`correct_edges_serial`)

Input: session_id, corrector_models[], action_order, rejudge_after
Output: Correction outcomes per edge

```
candidates ← Neo4j.query("WHERE r.verdict IN ['INCORRECT', 'PARTIAL', ...]")
agent ← create_corrector_agent(corrector_models[0], prompt_variant)
available_vars ← get_all_variables(session_id)
outcomes ← []
for each edge ∈ candidates:
  context ← {edge, available_vars, graph_structure}
  for each action ∈ action_order:        // revise_motivation, change_edge_type
    result ← agent.call_tool(action, edge, context)
    if result.success:
      outcomes.append({edge, action, corrected: True}); break
if rejudge_after:
  judge_all_edges_with_citations_serial(approach=rejudge_approach)
return outcomes
```

Implementation: `modules.py:7050` (`correct_edges_serial`)

I.6 Algorithm 5: UQ Metrics Computation

Uncertainty Quantification (UQ) metrics computed from LLM logprobs during generation.

Table I.6: UQ Metrics (`logit_metrics.py`)

Input: logprobs = {tokens[], token_logprobs[], top_logprobs[]}
Output: perplexity, min_prob, max_window_entropy, cosine_sim

// Perplexity (compute_avg_nll + exp)
nll_capped ← [min(-logprob$_i$, 20.0) for logprob$_i$ in token_logprobs]
avg_nll ← mean(nll_capped)
avg_nll_capped ← min(avg_nll, 10.0)
perplexity ← exp(avg_nll_capped)
// Minimum Token Probability (compute_min_prob)
probs ← [exp(logprob$_i$) for logprob$_i$ in token_logprobs]
min_prob ← min(probs)
// Max Window Entropy (compute_max_window_entropy)
entropies ← []
for each top_k_dict ∈ top_logprobs:
  p_vals ← softmax(top_k_dict.values())
  H ← $-\sum_i$ p$_i$ log(p$_i$)
  entropies.append(H)
windowed ← convolve(entropies, window_size=5)
max_window_entropy ← max(windowed)
// Cosine Similarity (get_embedding_local)
e$_{\text{narr}}$ ← SentenceTransformer(motivation, model="all-mpnet-base-v2")
e$_{\text{cit}}$ ← SentenceTransformer(citation_text)
cosine_sim ← max$_c$ $\frac{\mathbf{e}_{\text{narr}} \cdot \mathbf{e}_c}{\|\mathbf{e}_{\text{narr}}\| \|\mathbf{e}_c\|}$
return {perplexity, min_prob, max_window_entropy, cosine_sim}

Implementation: `logit_metrics.py` – `:177` (avg_nll), `:251` (min_prob), `:259` (max_entropy), `:98` (embedding)

I.7 Algorithm 6: Hallucination Classification

Ensemble classifiers trained on UQ metrics to predict hallucinations.

Table I.7: Hallucination Classification (`train_classifiers`)

Input: X = [perplexity, min_prob, max_entropy, cosine_sim], y = hallucination_labels
Output: Trained models, metrics (AUC, Precision, Recall, F1)

```
X_scaled ← StandardScaler.fit_transform(X)
(X_train, X_test, y_train, y_test) ← train_test_split(X_scaled, y, test=0.3)
classifiers ← {LogReg, RandomForest(100), GradBoost(100), MLP(32,16)}
for each (name, clf) ∈ classifiers:
    cv ← StratifiedKFold(n_splits=5)
    cv_auc ← cross_val_score(clf, X_train, y_train, cv, scoring="roc_auc")
    clf.fit(X_train, y_train)
    y_pred_proba ← clf.predict_proba(X_test)[:,1]
    metrics[name].auc ← roc_auc_score(y_test, y_pred_proba)
    metrics[name].{precision, recall, f1} ← precision_recall_fscore(y_test, y_pred)
return classifiers, metrics
```

Implementation: `rq2_ensemble_classifier.py:32` (`train_classifiers`)

I.8 Algorithm 7: Citation Search (Brave)

This procedure retrieves candidate citation URLs for a CLD edge using the Brave Search API, optionally filtered by a trusted-domain whitelist ([103]). It is invoked during citation filling when no usable citations are available from the LLM, and can also be used to obtain candidate sources for citation-based judging.

Trusted Academic Domains (23): Results from the following domains are prioritised ([103]):

- `pubmed.ncbi.nlm.nih.gov`
- `arxiv.org`
- `scholar.google.com`
- `nature.com`
- `science.org`
- `cell.com`
- `nejm.org`
- `bmj.com`
- `plos.org`
- `frontiersin.org`
- `springer.com`
- `wiley.com`
- `sciencedirect.com`
- `tandfonline.com`
- `sage.com`
- `jstor.org`
- `cochranelibrary.com`
- `nih.gov`
- `who.int`
- `cdc.gov`
- `academic.oup.com`
- `cambridge.org`
- `elsevier.com`

Table I.8: Citation Search (`_search_with_brave` + `SearchEngine.search`)

Input: src, rel, tgt, claim_txt, api_key, trusted_domains[] (optional)
Output: citations[] (URLs, up to 5)

// Phase 0: Initialize Brave client (once per run)
1. search_client ← SearchEngine(api_key, trusted_domains)
2. session ← requests.Session() with retries(total=3, backoff=1, status={429,500,502,503,504})

// Phase 1: Build query from edge
3. q ← "src rel tgt claim_txt"
4. **if** words(q) > 50: q ← first_50_words(q) + "..."
5. **if** len(q) > 400: q ← q[0:397] + "..."

// Phase 2: Brave Search API call
6. resp ← GET(`https://api.search.brave.com/res/v1/web/search`,
   headers={Accept: json, X-Subscription-Token: api_key},
   params={q: q, count: 5, search_lang: en, extra_snippets: True})
7. results ← resp.json().web.results

// Phase 3: Domain filter + URL extraction
8. citations ← []
9. **for each** result ∈ results:
10. url ← result.url
11. **if** trusted_domains empty **or** any(domain ∈ url.lower()): citations.append(url)
12. **return** citations

Implementation: `modules.py:2647` (`fill_in_missing_citations`), `:2900` (`_build_brave_search_client`), `:2967` (`_search_with_brave`), `search_engine_copy.py:47` (`SearchEngine.search`), `citation_whitelist_copy.py:250` (`CitationWhitelist.get_`

I.9 Algorithm 8: Citation Fetcher

The citation fetcher retrieves scientific literature content from URLs using a two-tier architecture: Jina AI Reader as primary with ContentScraper fallback for PDF extraction.

Table I.9: Citation Fetcher (`fetch_citations_with_jina`)

Input: citations[] (URLs), timeout, max_retries, api_key (optional)
Output: Dict[URL → text_content]

// Phase 1: Jina AI Reader (Primary)
max_workers ← 5 **if** api_key **else** 2 *// Rate limit tiers*
citation_text_map ← {}
failed_urls ← []
parallel for each url ∈ citations (ThreadPoolExecutor):
  jina_url ← "https://r.jina.ai/" + url
  delay ← 0.3s **if** api_key **else** 1.0s *// Respect rate limits*
  sleep(delay)
  headers ← {"Authorization": "Bearer " + api_key} **if** api_key
  for attempt ∈ [0..max_retries]:
    response ← GET(jina_url, headers, timeout)
    if response.ok ∧ len(content) > 200 ∧ "[ERROR]" ∉ content[:500]:
      citation_text_map[url] ← content; **break**
    else: sleep(2^{attempt}) *// Exponential backoff*
  if url ∉ citation_text_map: failed_urls.append(url)

// Phase 2: ContentScraper Fallback (PDF-capable)
if len(failed_urls) > 0:
  scraper ← ContentScraper(timeout=15, whitelist=True) *// PyMuPDF for PDFs*
  results ← scraper.process_citations(failed_urls, fetch_content=True)
  for each (url, content) ∈ results:
    if len(content) > 200 ∧ "[ERROR]" ∉ content[:100]:
      citation_text_map[url] ← content
return citation_text_map

*** Thesis configuration note:** In the experiments reported in this thesis, citation fetching was run in **ContentScraper-only** mode (Jina AI Reader disabled).

Implementation: `modules.py:265` (`fetch_citations_with_jina`), `:302` (`fetch_single_citation`), `:378-410` (ContentScraper fallback)

I.10 Deep Research Multi-Agent Components

Table I.10 details the complete multi-agent pipeline components used in the Deep Research system.

Table I.10: Deep Research Multi-Agent Pipeline

Component	Model	Input	Output	Purpose
Orchestrator				
`DeepResearchManager`	Python	Causal claim	FinalVerdict	Coordinate agent execution: iteration loop (max 3), early stopping on direct evidence, time limit (2 min), parallelization via `asyncio.gather()`
LLM Agents				
`hypothesis_evaluator`	GPT-5	Causal claim	SearchPlan	Generate 3–4 targeted search queries, keywords, and potential confounders
`search_agent`	GPT-5-mini	Search query	List[SearchResult]	Execute iterative web searches (4–6 calls per query thread, max 5)
`evidence_scorer`	GPT-5-mini	Evidence content	EvidenceScore	Score relevance (1–10) and methodological quality (1–10)
`article_selector`	GPT-5	Snippets + scores	ArticleSelection	Select 2–3 articles most likely to contain causal evidence for full-text fetch
`judgement_agent`	GPT-5	Evidence + scores	EvidenceJudgement	Judge whether accumulated evidence supports the claim
`causal_evidence_detector`	GPT-5	All search results	CausalEvidenceEvaluation	Detect *direct* causal evidence (RCTs, experiments, meta-analyses only)
`verdict_agent`	GPT-5	All data	FinalVerdict	Synthesize final verdict with evidence summary and limitations

I.11 Algorithm 9: Deep Research Multi-Agent Pipeline

The Deep Research system validates causal claims through iterative literature search using a 7-agent pipeline with parallel execution.

Table I.11: Deep Research Pipeline (`pydantic_ai_deepresearch_orchestration.py`)

Input: causal_claim = "{Source} causes {Target}. {Motivation}"
Output: FinalVerdict (verdict, confidence, direct_causal_found, evidence_urls)

```
// Phase 1: Search Plan Generation (hypothesis_evaluator, GPT-5)
 1.  search_plan ← hypothesis_evaluator.run(causal_claim)
 2.  queries[] ← [motivation] + search_plan.queries                    // 3–4 queries
 3.  keywords ← search_plan.keywords
 4.  confounders ← search_plan.potential_confounders
// Phase 2: Parallel Search Execution (search_agent, GPT-5-mini)
 5.  all_results ← []
 6.  parallel for each query ∈ queries (asyncio.gather):
 7.    history ← fresh_conversation()                  // Prevent token accumulation
 8.    search_count ← 0
 9.    while search_count < MAX_SEARCHES (5):
10.      results ← search_agent.run(query, tool=web_search)
11.      all_results.extend(results)
12.      search_count += 1
// Phase 3: Parallel Evidence Scoring (evidence_scorer, GPT-5-mini)
13.  parallel for each result ∈ all_results:
14.    score ← evidence_scorer.run(result.content, causal_claim)
15.    result.relevance ← score.relevance                              // 1–10
16.    result.quality ← score.quality          // 1–10: RCT > cohort > case-control
// Phase 4: Smart Article Selection (article_selector, GPT-5)
17.  ranked ← sort(all_results, key=relevance × quality)
18.  selected ← article_selector.run(ranked[:10])              // Select 2–3 articles
19.  for each article ∈ selected:
20.    article.full_text ← fetch_full_text(article.url, max=50K chars)
// Phase 5: Direct Causal Evidence Detection (causal_evidence_detector, GPT-5)
21.  causal_eval ← causal_evidence_detector.run(all_results)
22.  direct_found ← causal_eval.direct_evidence_found // RCT/experiment/meta-analysis only
23.  causal_confidence ← causal_eval.confidence
24.  cited_passage ← causal_eval.cited_passage                      // Verbatim quote
25.  study_type ← causal_eval.study_type
// Phase 6: Judgement with Early Stopping
26.  high_quality ← {r ∈ all_results : r.relevance ≥ 8 ∧ r.quality ≥ 7}
27.  if direct_found ∧ |high_quality| ≥ 3:
28.    early_stop                                  // Strong evidence found
29.  judgement ← judgement_agent.run(all_results, scores)
30.  if judgement.verdict = "supported" ∧ judgement.confidence ≥ 8:
31.    early_stop                                  // High confidence support
// Phase 7: Final Verdict Synthesis (verdict_agent, GPT-5)
32.  verdict ← verdict_agent.run(causal_claim, all_results, scores, judgement, causal_eval)
33.  return FinalVerdict{
34.    verdict: {supported, partially_supported, unsupported, inconclusive},
35.    confidence: 1–10,
36.    direct_causal_evidence_found: direct_found,
37.    strongest_evidence: (url, cited_passage, study_type),
38.    evidence_urls: [all supporting URLs]
39.  }
```

Key Implementation Details:

- **Search API:** Brave Search with 23 trusted academic domains (PubMed, arXiv, Nature, etc.; [103])
- **Query enhancement:** All queries appended with "scientific study research evidence"
- **Retry strategy:** 3 retries with exponential backoff for transient failures

- **Timeouts:** 10s connect, 40s read; 900s (15 min) total time limit per claim
- **Full-text fetching:** BeautifulSoup HTML parsing, max 50K characters

Implementation: `pydantic_ai_deepresearch_orchestration.py` – agents defined at lines 166–248, search execution at :953, evidence scoring at :1006, early stopping at :784–795

I.12 Algorithm 10: Node Concept Similarity (Similarity-Weighted Node F1)

This metric quantifies conceptual overlap between a generated CLD's node set and a validation CLD's node set. It is implemented as (i) a node mapping phase (LLM- or cosine-driven, depending on mode) and (ii) a similarity-weighted score that assigns **1.0** to nodes matched in Stage 1 and assigns cosine similarity scores to a **greedy bipartite matching** over remaining unmatched nodes.

Table I.12: Node Concept Similarity (`compare_session_graph_to_validation`, similarity-weighted node metrics)

Input: session_nodes S, validation_nodes V, node_match_mode, cosine_percentile p (default 85), embedding_model (default `text-embeddin`
Output: binary node mapping + binary node F1, cosine-only node F1, similarity-weighted (hybrid) node F1

// Step 0: Build node texts for embeddings
for each node $n \in (S \cup V)$:
  text(n) ← join({name(n), description/definition(n) if present, units(n) if present}, " | ")

// Step 1: Compute cosine similarities
$E_S \leftarrow \text{embed}(\text{text}(S))$; $E_V \leftarrow \text{embed}(\text{text}(V))$
$C \leftarrow \text{cosine}(E_S, E_V)$ *// via normalized dot product*
$\tau \leftarrow \text{percentile}(C.\text{flatten}(), p)$
candidates $\leftarrow \{(s, v, C[s, v]) : C[s, v] \geq \tau\}$, sorted desc.

// Step 2: Stage 1 mapping (mode-dependent)
if node_match_mode = `cosine`:
  node_mapping ← greedy_max_weight_matching(candidates) *// 1:1 matching*
elif node_match_mode ∈ {`hybrid`, `llm`}:
  to_check ← (all pairs $S \times V$ if `llm` else candidates)
  positives ← $\{(s, v, C[s, v]) : \texttt{LLM\_equivalent}(s, v) = \texttt{True}$ for $(s, v) \in$ `to_check`$\}$
  node_mapping ← greedy_max_weight_matching(positives)
elif node_match_mode = `llm_batch`:
  // Single LLM call proposes best match (or NO_MATCH) with confidence
  for each $s \in S$: (v_llm, conf) ← LLM_best_match(s, V)
    if v_llm ≠ NO_MATCH and conf ≥ 0.7: node_mapping[s] ← v_llm
    else: $v^* \leftarrow \arg\max_{v \in V} C[s, v]$; **if** $C[s, v^*] \geq 0.7$ then node_mapping[s] ← v^*

// Step 3: Binary node metrics (mapping cardinality)
$P_{\text{bin}} \leftarrow |\text{node\_mapping}|/|S|$; $R_{\text{bin}} \leftarrow |\text{node\_mapping}|/|V|$
$F1_{\text{bin}} \leftarrow 2PR/(P + R)$

// Step 4: Cosine-only node metrics (max similarity per node)
$P_{\cos} \leftarrow \text{mean}_{s \in S}\left[\max_{v \in V} C[s, v]\right]$
$R_{\cos} \leftarrow \text{mean}_{v \in V}\left[\max_{s \in S} C[s, v]\right]$
$F1_{\cos} \leftarrow 2PR/(P + R)$

// Step 5: Similarity-weighted (hybrid) node metrics via two-stage bipartite matching
matched_S ← keys(node_mapping); matched_V ← values(node_mapping)
$S_u \leftarrow S \setminus$ matched_S; $V_u \leftarrow V \setminus$ matched_V
cosine_mapping ← `greedy_max_weight_matching`$(\{(s, v, C[s, v]) : s \in S_u, v \in V_u\})$

$$\text{score\_gen}(s) \leftarrow \begin{cases} 1.0 & s \in \text{matched\_S} \\ C[s, \text{cosine\_mapping}(s)] & s \in \text{cosine\_mapping} \\ 0.0 & \text{otherwise} \end{cases}$$

$$\text{score\_val}(v) \leftarrow \begin{cases} 1.0 & v \in \text{matched\_V} \\ C[\text{cosine\_rev}(v), v] & v \in \text{range(cosine\_mapping)} \\ 0.0 & \text{otherwise} \end{cases}$$

$P_{\text{hyb}} \leftarrow \text{mean}_{s \in S}[\text{score\_gen}(s)]$; $R_{\text{hyb}} \leftarrow \text{mean}_{v \in V}[\text{score\_val}(v)]$
$F1_{\text{hyb}} \leftarrow 2PR/(P + R)$

Adjusted (seed-excluding) variant: If the target variable was *seeded* from the validation CLD (e.g., the inferred target), the implementation additionally reports "adjusted" node metrics by excluding that variable from both S and V prior to metric computation.

Implementation: `modules.py:8242` (`compare_session_graph_to_validation`), `:8098` (`_compute_node_cosine_canc` `:8151` (`_maximum_weight_bipartite_matching`)

Appendix J

ROC Curves

This section presents ROC curves for all RQ1a experiments. **Important methodological note:** The aggregate scores used for classification are discrete, taking only three values (0, 0.5, and 1.0). Consequently, only three meaningful operating points exist on each ROC curve, and the apparent smoothness results from linear interpolation between these points. The fixed classification threshold of 0.5 is used throughout the main analysis, making these curves primarily illustrative of the AUC metric rather than a continuous threshold trade-off.

J.1 Corruption Detection Correctness ROC Curves

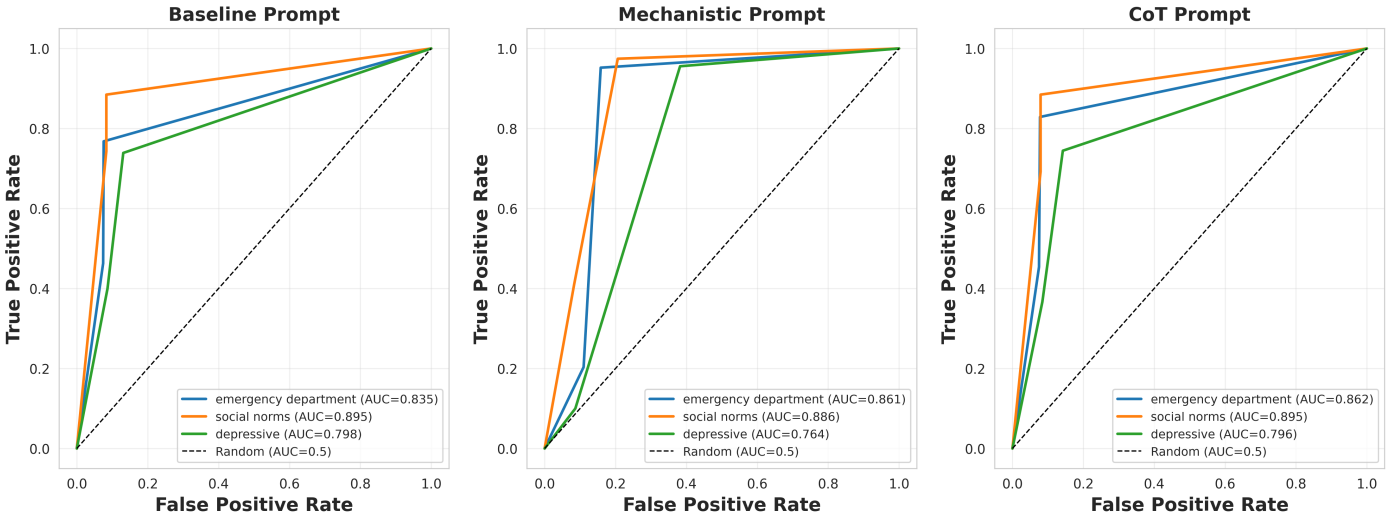


Figure J.1: ROC curves for correctness-based corruption detection by prompt. Each subplot shows performance for one prompt variant across all CLDs. **Note:** Scores are discrete (0, 0.5, 1), so curves represent interpolation between only 3 operating points.

J.2 Corruption Detection Citation ROC Curves

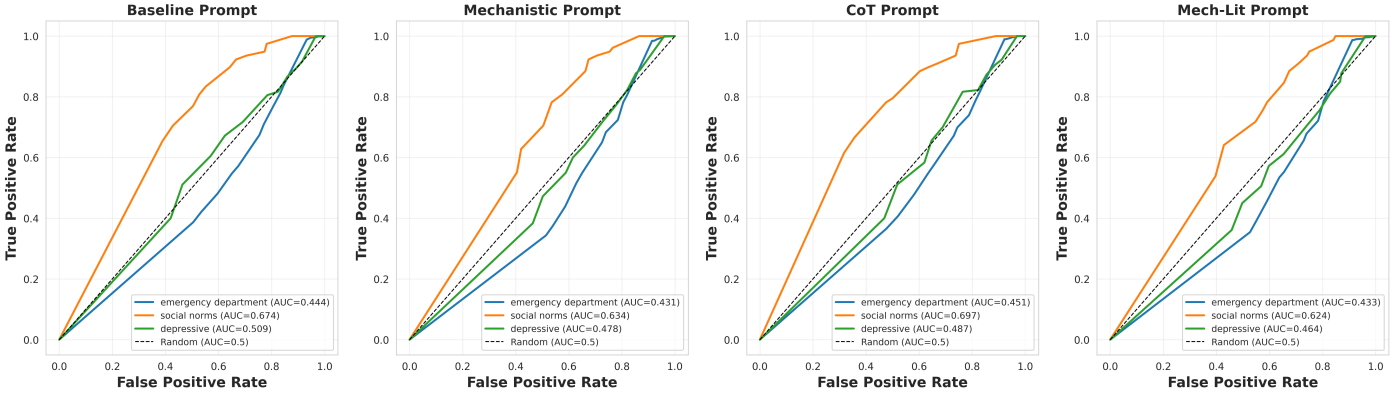


Figure J.2: ROC curves for citation-based corruption detection by prompt. Each subplot shows performance for one of four prompt variants (Baseline, Mechanistic, CoT, Mechanistic-Lit) across all three CLDs. **Note:** Scores are discrete (0, 0.5, 1), so curves represent interpolation between only 3 operating points.

J.3 Ground Truth Correctness ROC Curves

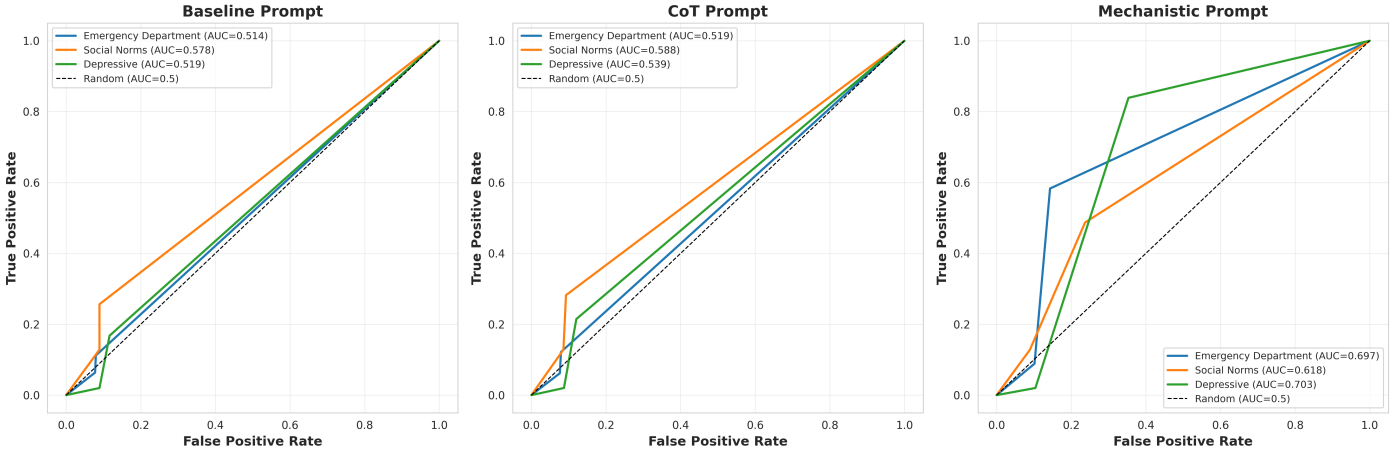


Figure J.3: ROC curves for ground truth correctness validation. Shows discrimination ability per prompt variant across CLDs, with AUC values for each CLD.

J.4 Ground Truth Citation ROC Curves

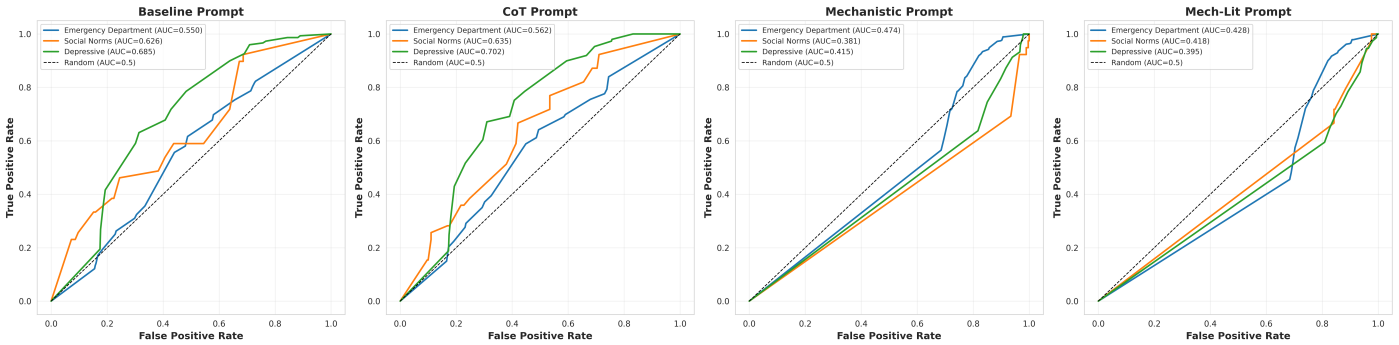


Figure J.4: ROC curves for ground truth citation validation. Shows discrimination ability per prompt variant across CLDs, with AUC values for each CLD.

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